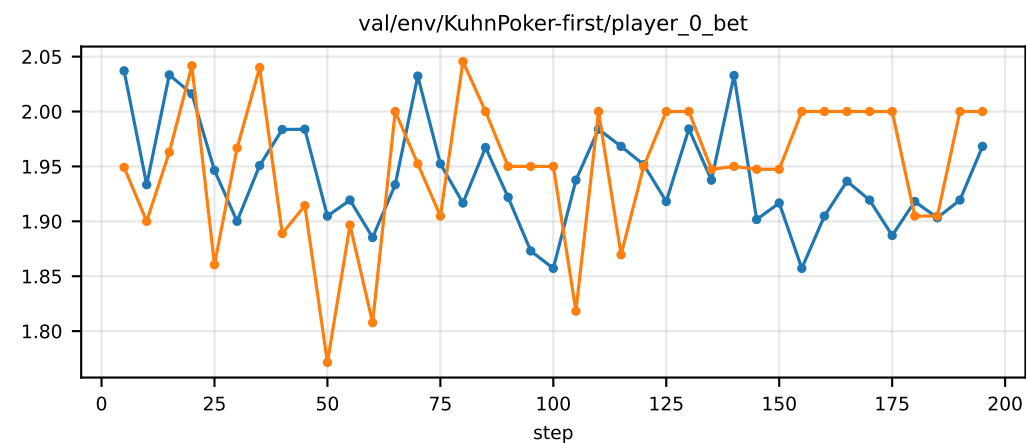
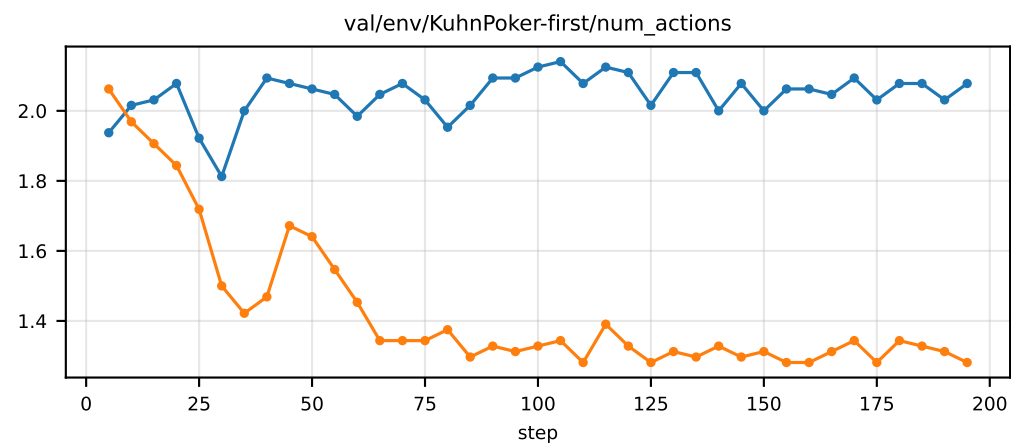
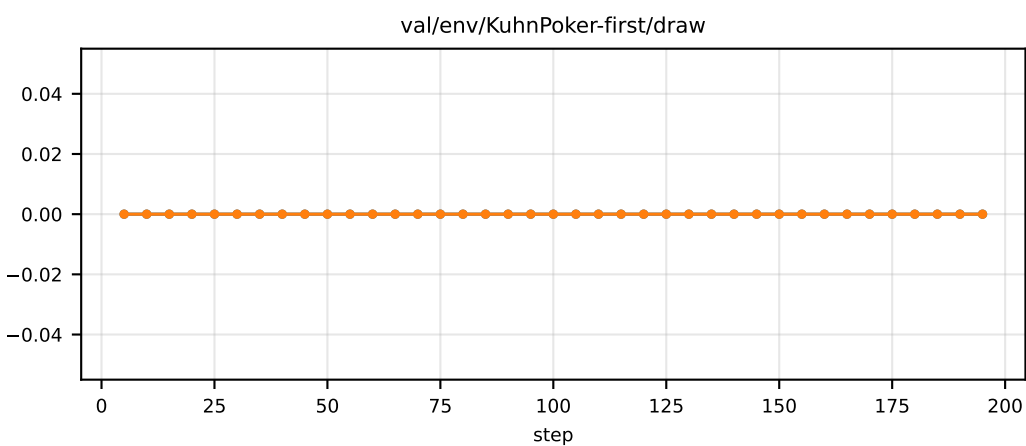
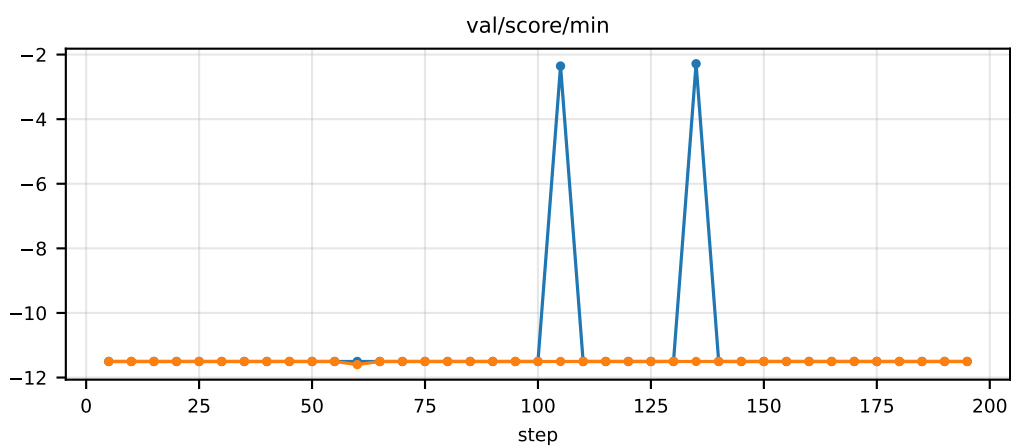
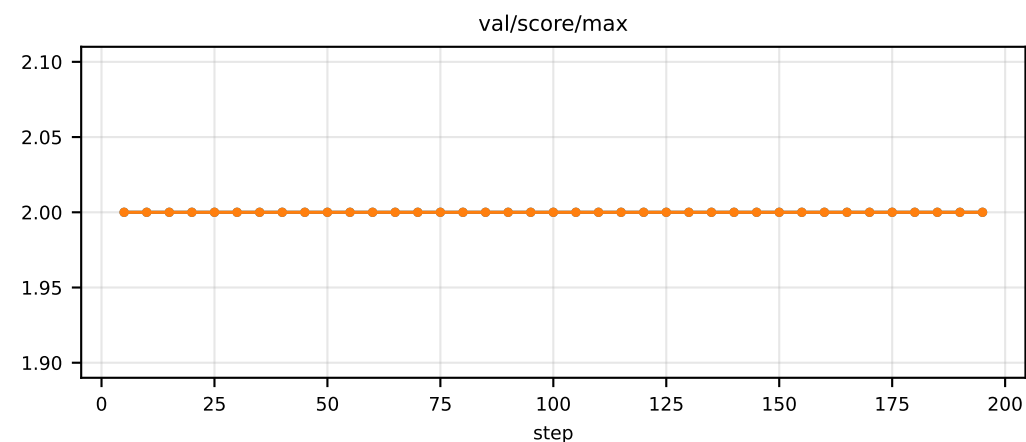
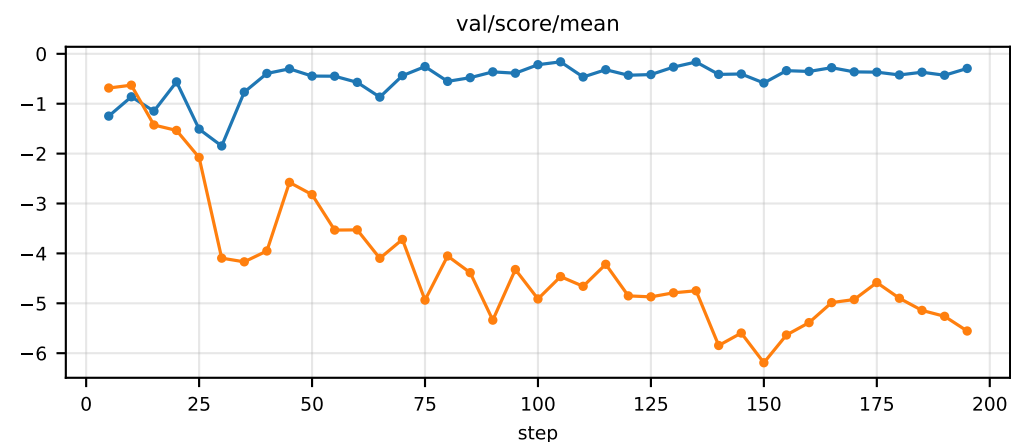


TensorBoard report: /scratch/workspace/eunbiyoon_umass_edu-paper/megatron/runs
generated 2026-10-06 23:51

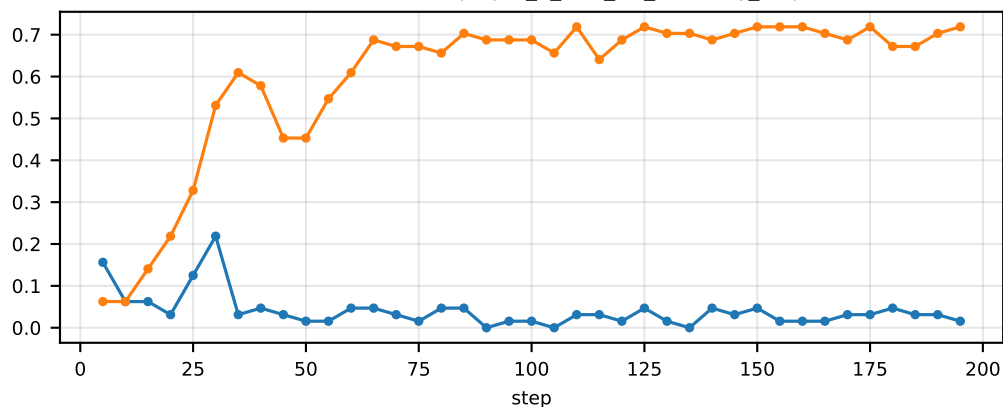
run	path (under runs/)	game	last step	last log	status
k1b (kuhn)	kuhn_poker_selfplay/20261001-152723	KuhnPoker	199	2026-10-05 18:2	finished
k2b (kuhn quickcheck)	kuhn_poker_selfplay_quickcheck/20261002-0913	KuhnPoker	199	2026-10-03 00:3	finished
m1b (ttt, lr1e-6, no min_lr)	tictactoe_selfplay/20260924-001348	TicTacToe	199	2026-09-29 09:3	finished
opt1 (ttt, lr5e-6)	tictactoe_selfplay_opt1/20261004-004234	TicTacToe	111	2026-10-06 19:4	stopped/running at 111
opt2 (ttt, lr1e-6, min_lr1e-7)	tictactoe_selfplay_opt2/20261006-004511	TicTacToe	34	2026-10-06 23:1	stopped/running at 34
m2b (ttt quickcheck)	tictactoe_selfplay_quickcheck/20260924-001348	TicTacToe	199	2026-09-24 15:5	finished

[KuhnPoker] Validation (vs built-in opponents) (1/8)

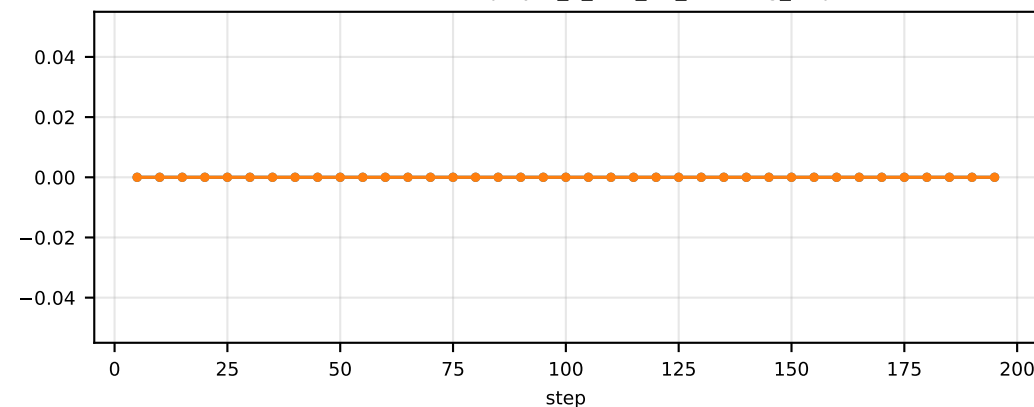


[KuhnPoker] Validation (vs built-in opponents) (2/8)

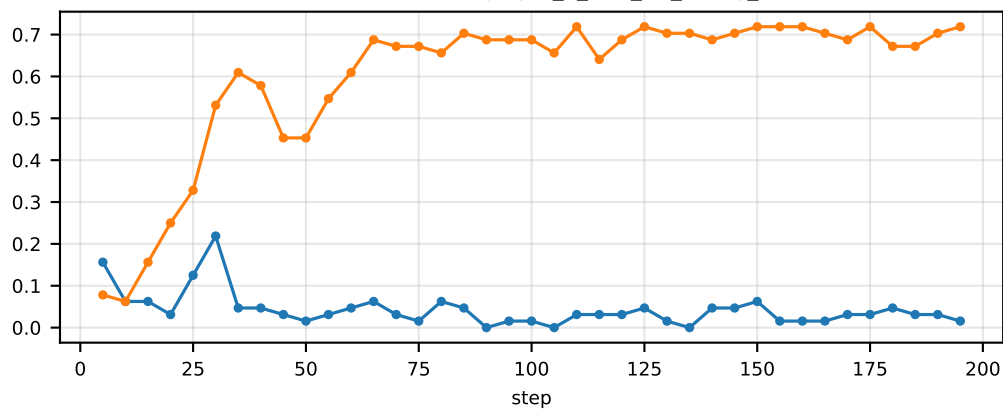
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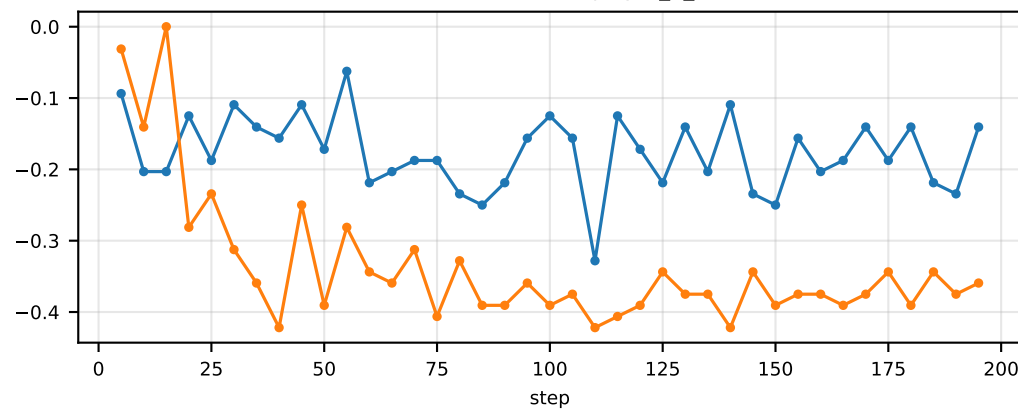
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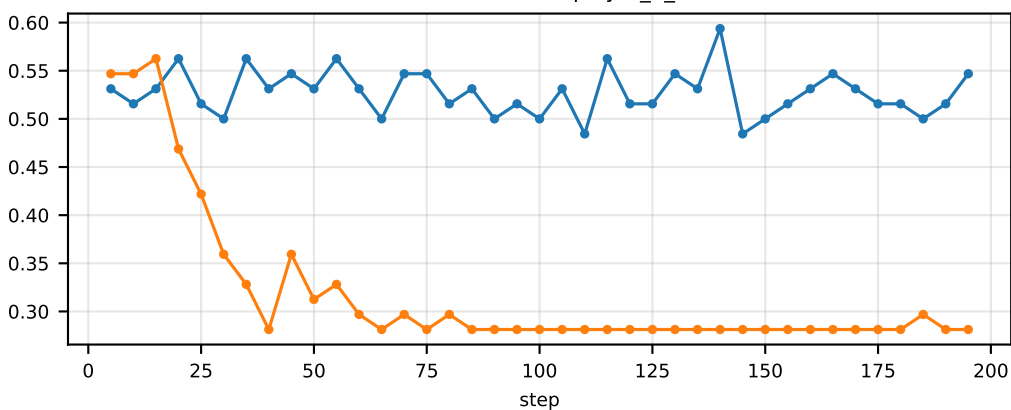
val/env/KuhnPoker-first/player_0_lose_for_wrong_format



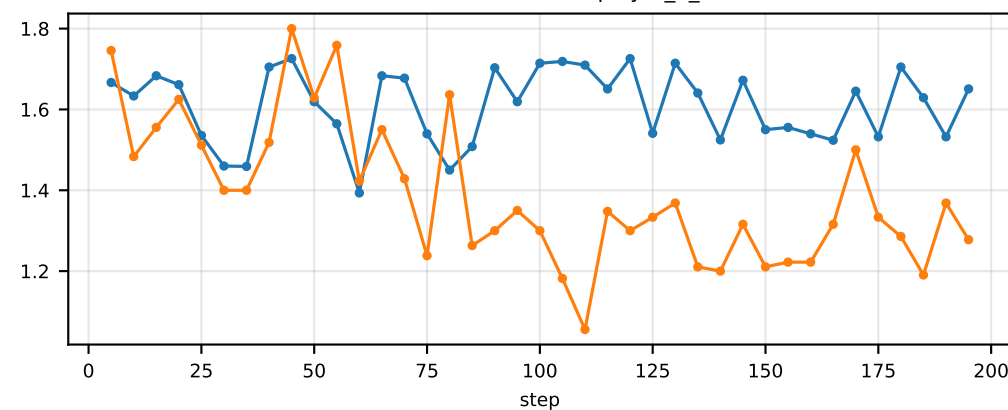
val/env/KuhnPoker-first/player_0_return



val/env/KuhnPoker-first/player_0_success

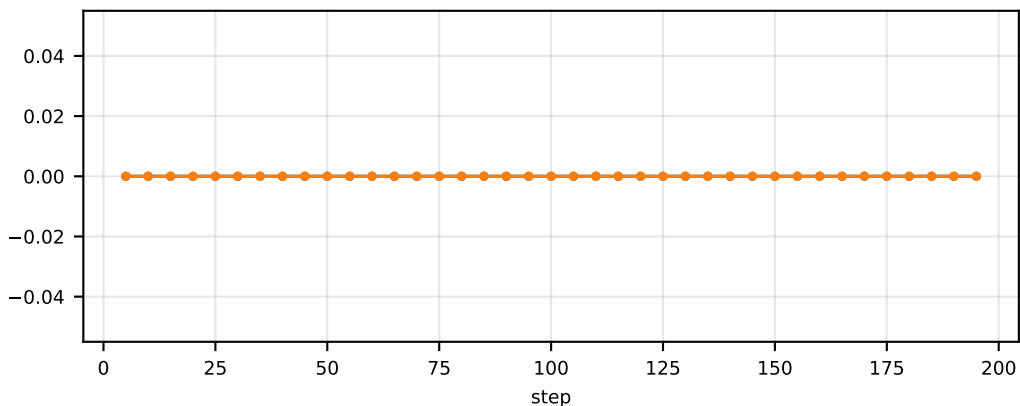


val/env/KuhnPoker-first/player_1_bet

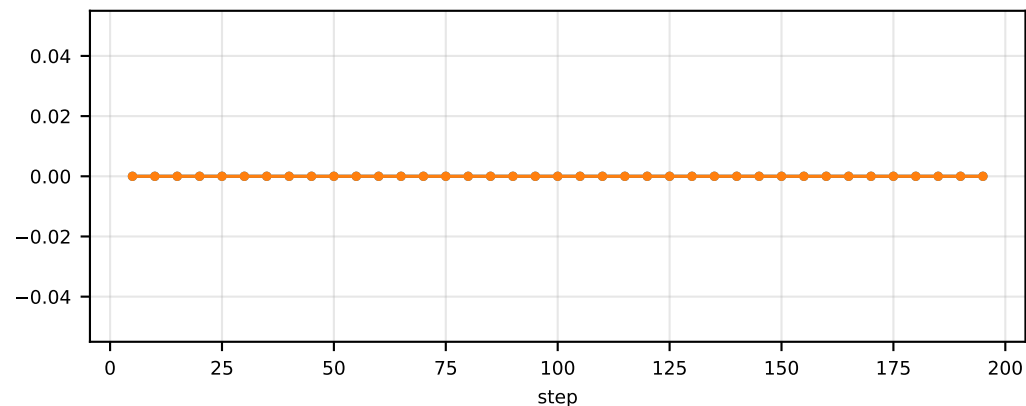


[KuhnPoker] Validation (vs built-in opponents) (3/8)

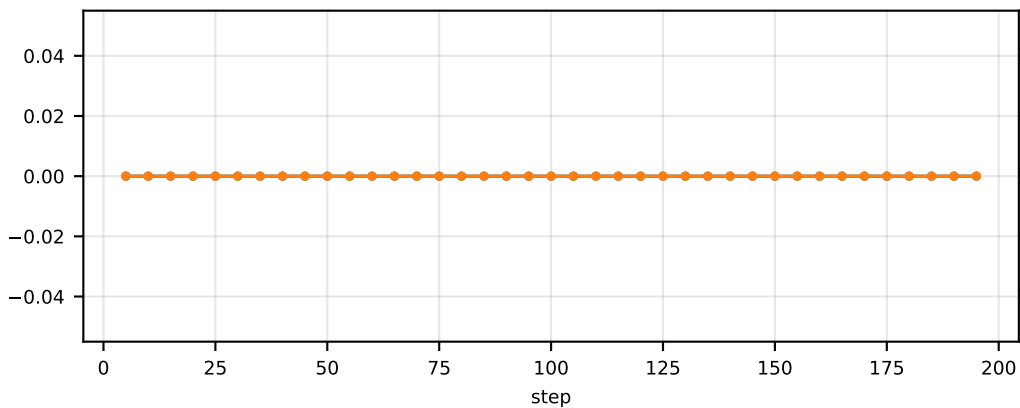
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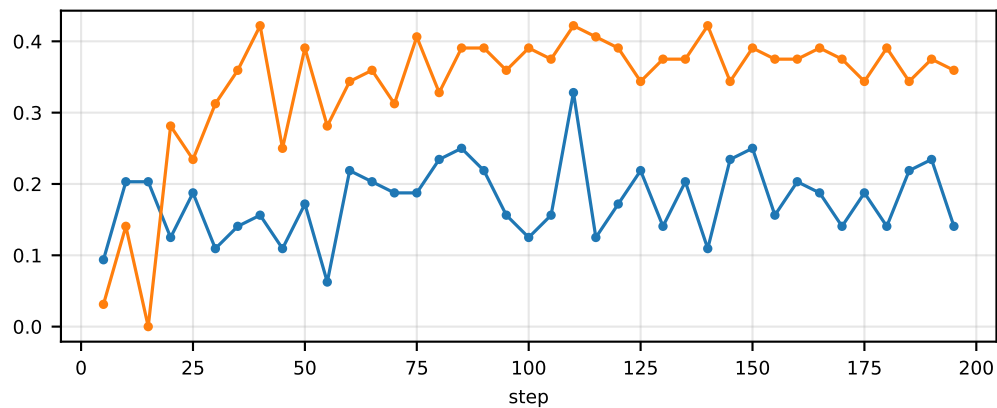
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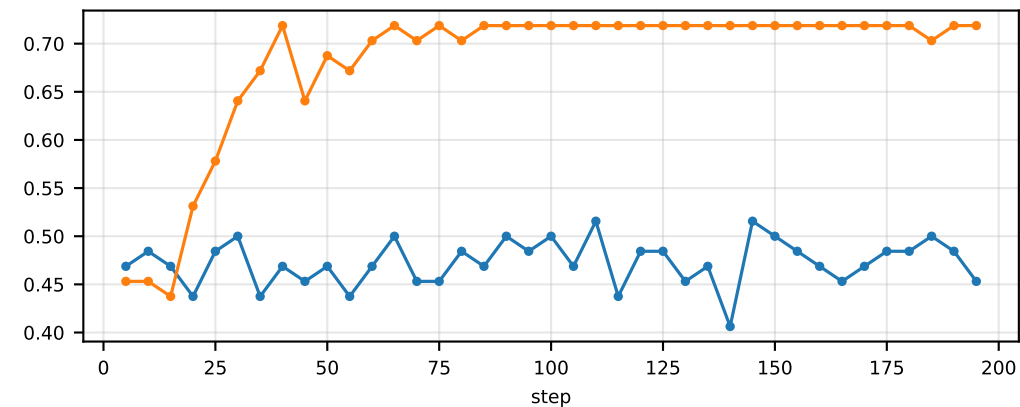
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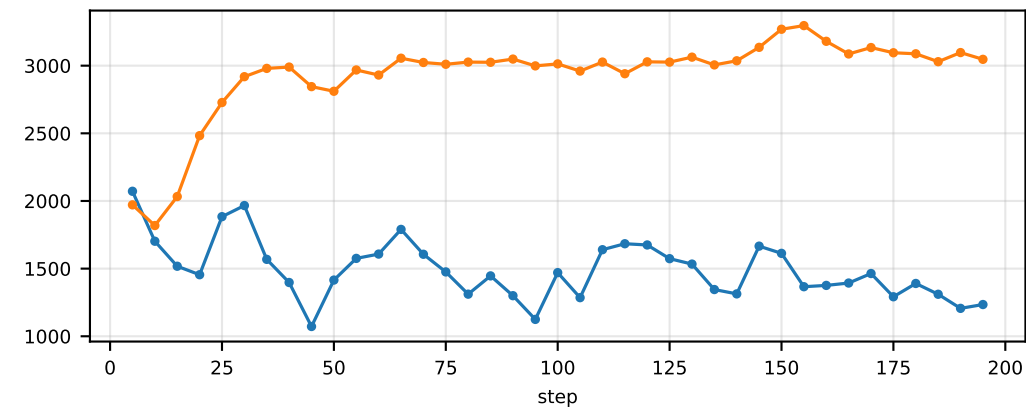
val/env/KuhnPoker-first/player_1_return



val/env/KuhnPoker-first/player_1_success

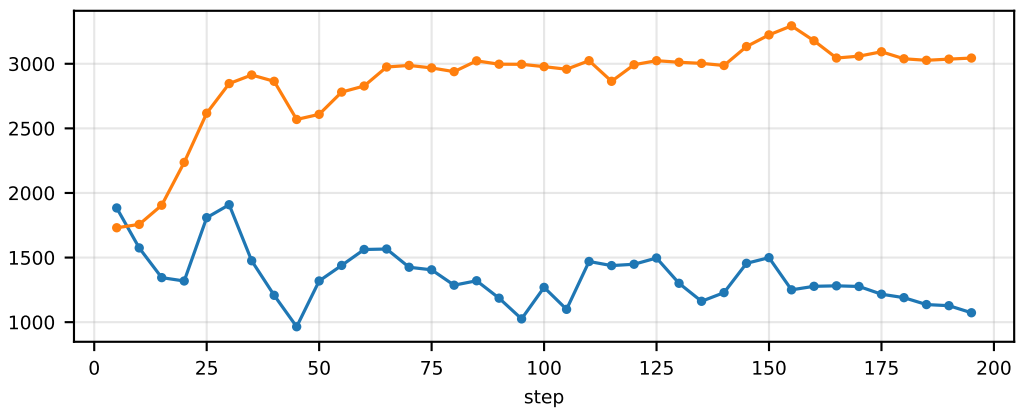


val/env/KuhnPoker-first/response_length

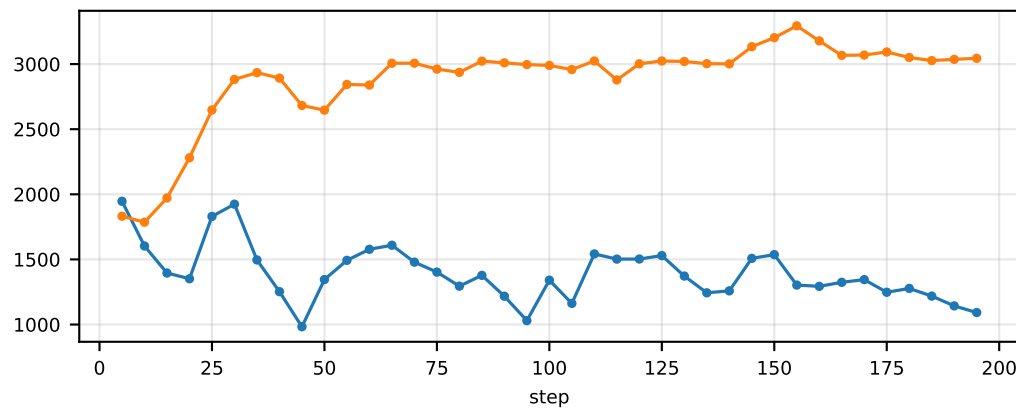


[KuhnPoker] Validation (vs built-in opponents) (4/8)

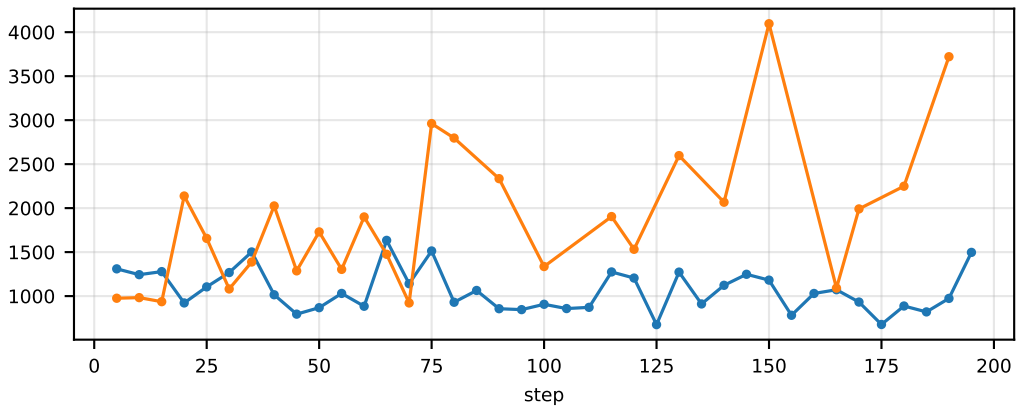
val/env/KuhnPoker-first/response_length_per_turn



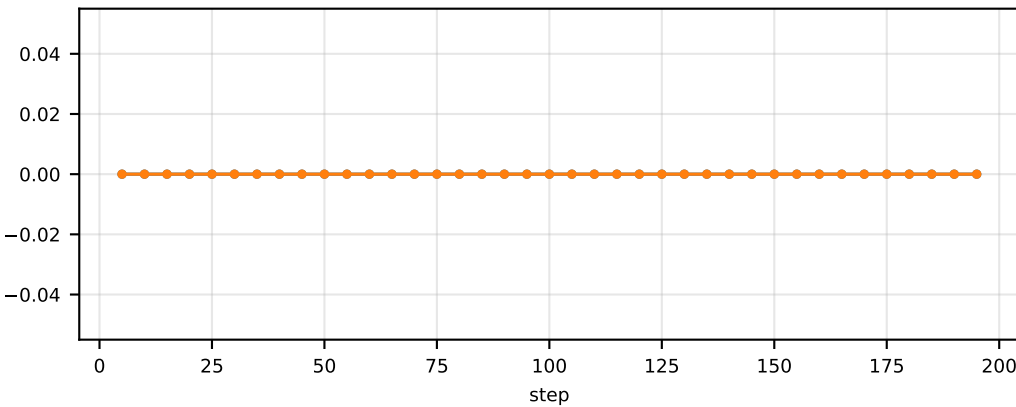
val/env/KuhnPoker-first/response_length_turn_0



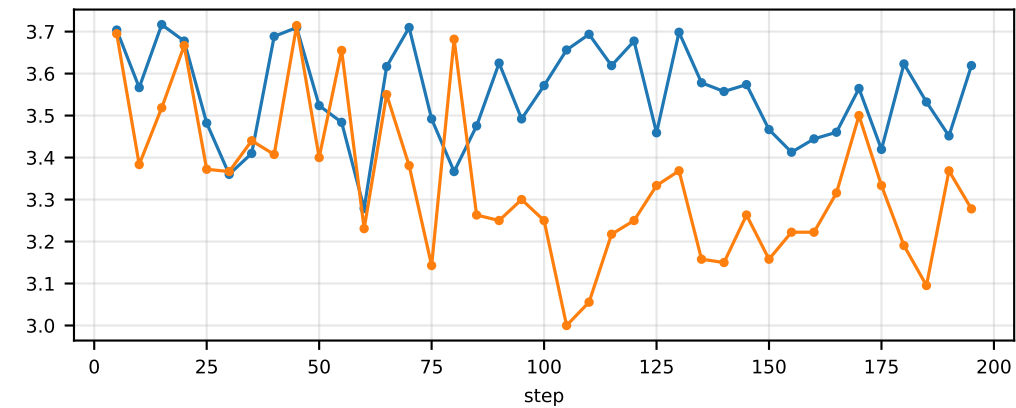
val/env/KuhnPoker-first/response_length_turn_1



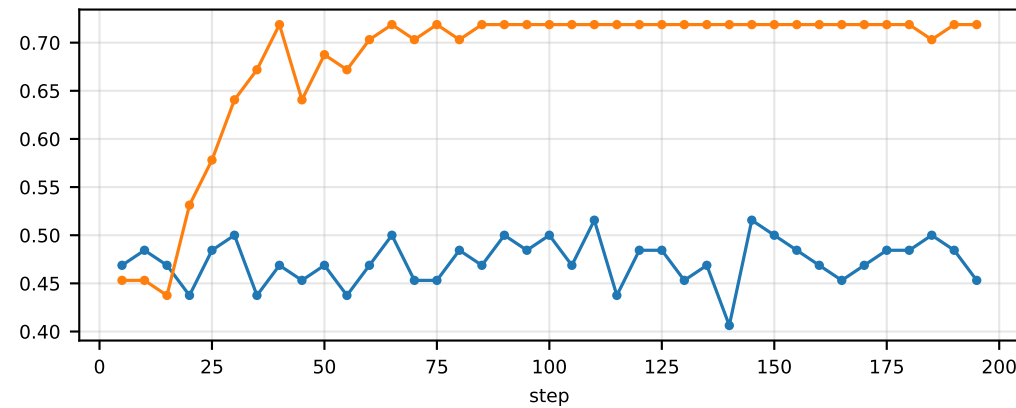
val/env/KuhnPoker-first/success



val/env/KuhnPoker-first/total_pot

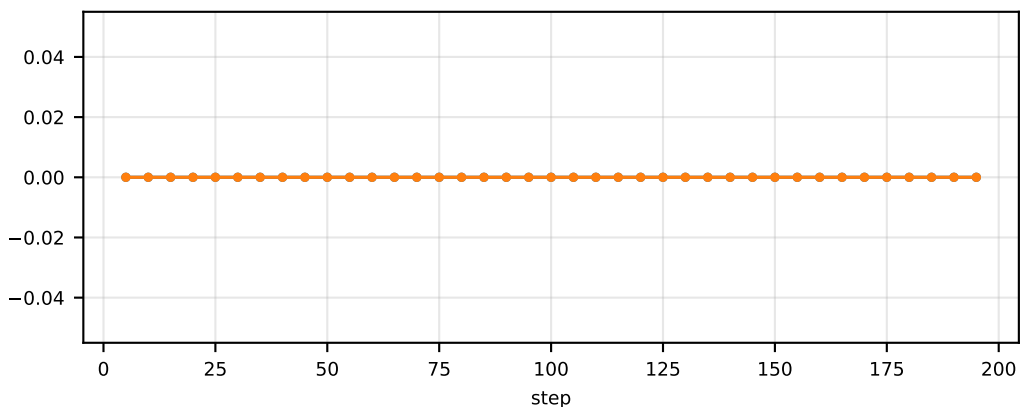


val/env/KuhnPoker-first/winner

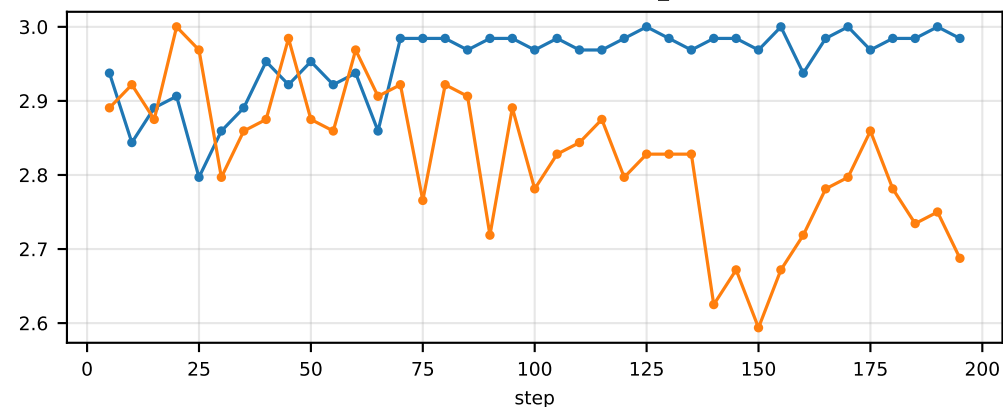


[KuhnPoker] Validation (vs built-in opponents) (5/8)

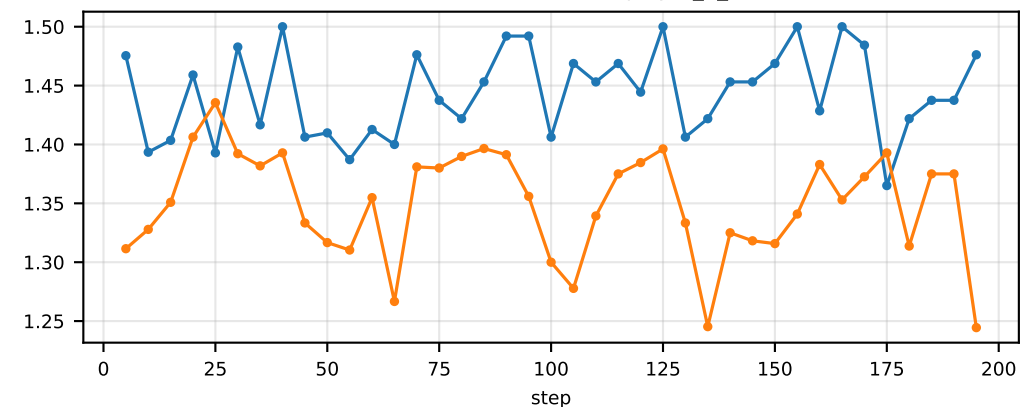
val/env/KuhnPoker-second/draw



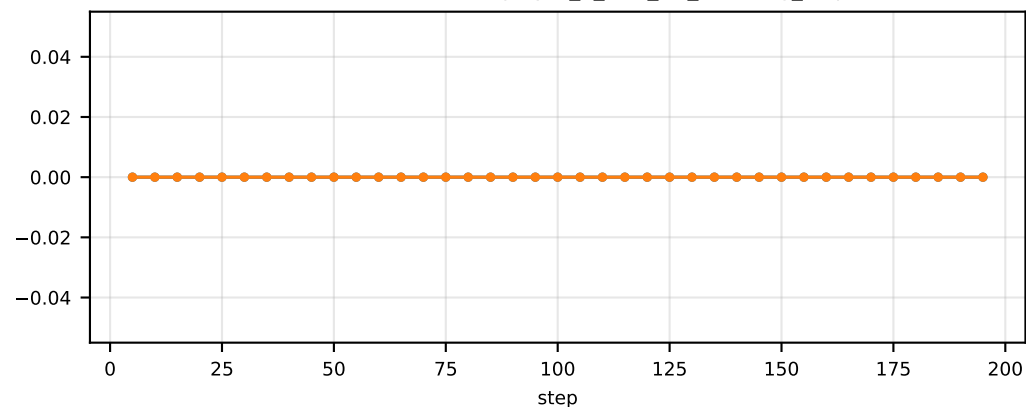
val/env/KuhnPoker-second/num_actions



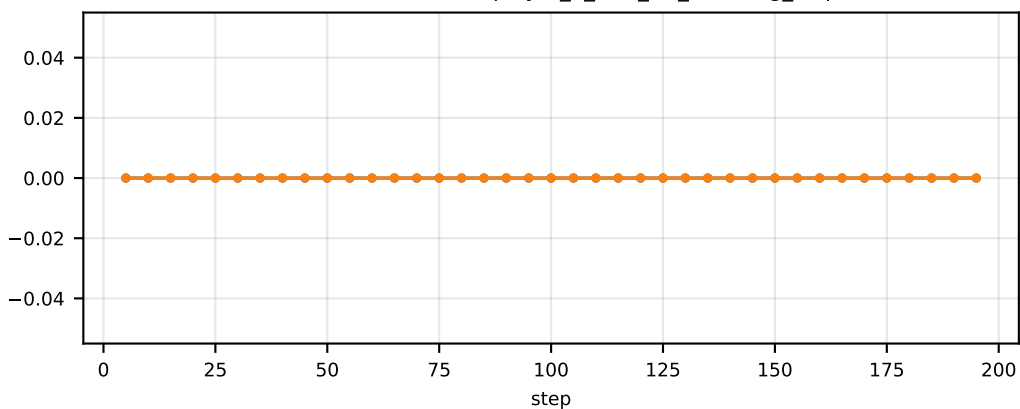
val/env/KuhnPoker-second/player_0_bet



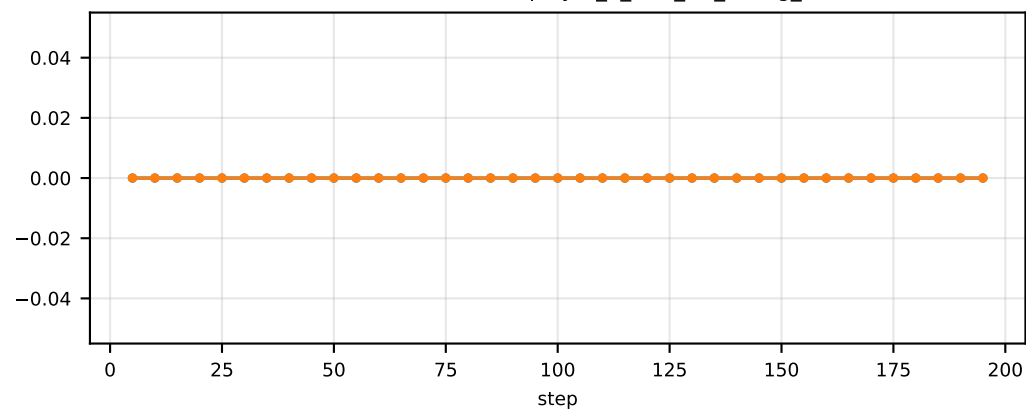
val/env/KuhnPoker-second/player_0_lose_for_overlong_response



val/env/KuhnPoker-second/player_0_lose_for_overlong_sequence

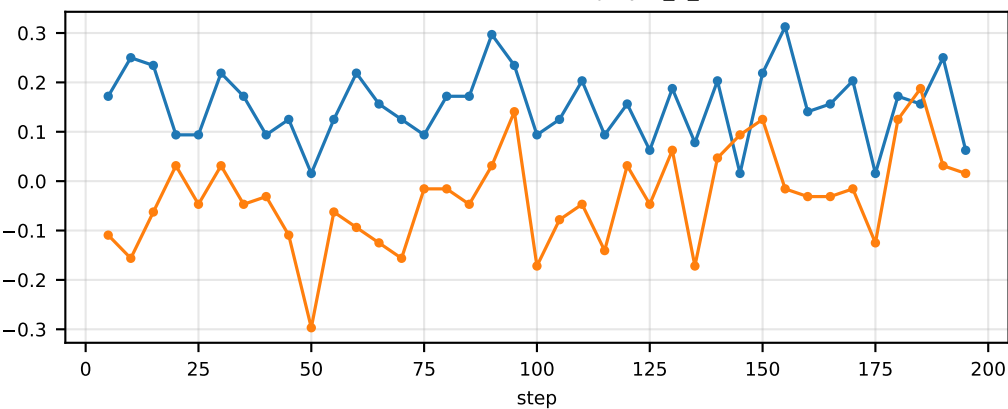


val/env/KuhnPoker-second/player_0_lose_for_wrong_format

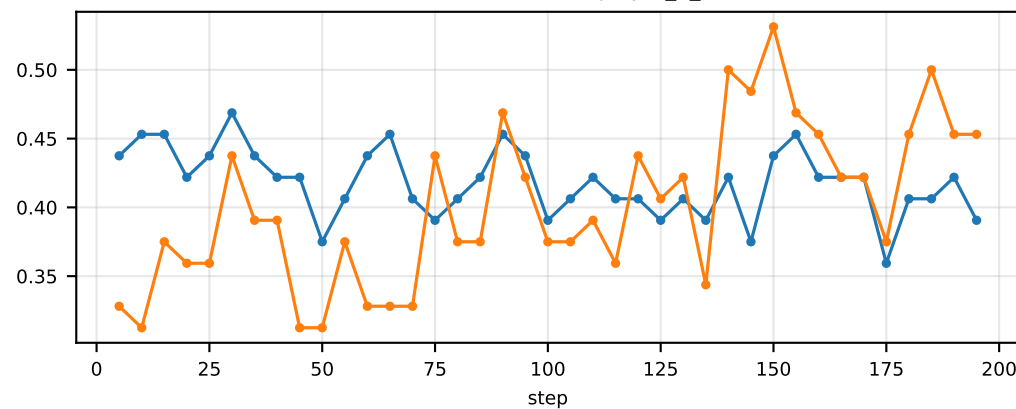


[KuhnPoker] Validation (vs built-in opponents) (6/8)

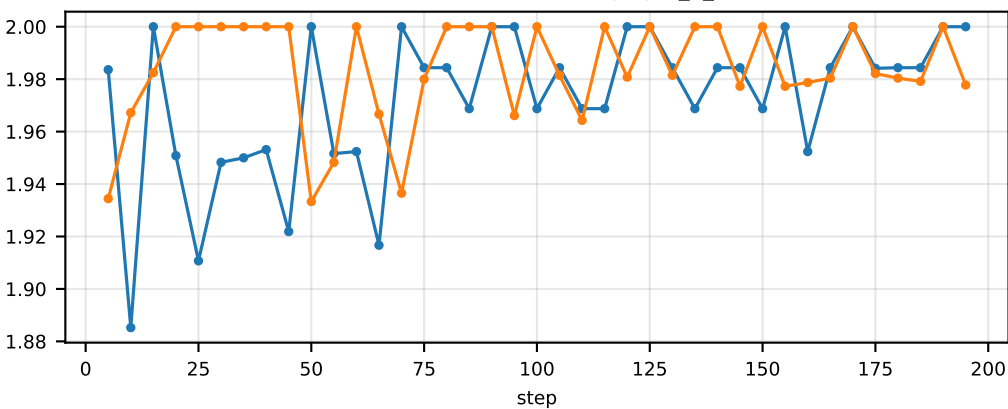
val/env/KuhnPoker-second/player_0_return



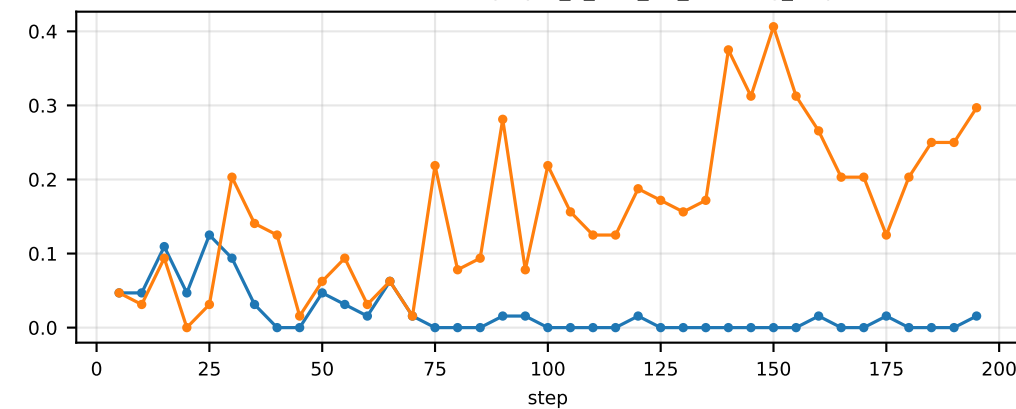
val/env/KuhnPoker-second/player_0_success



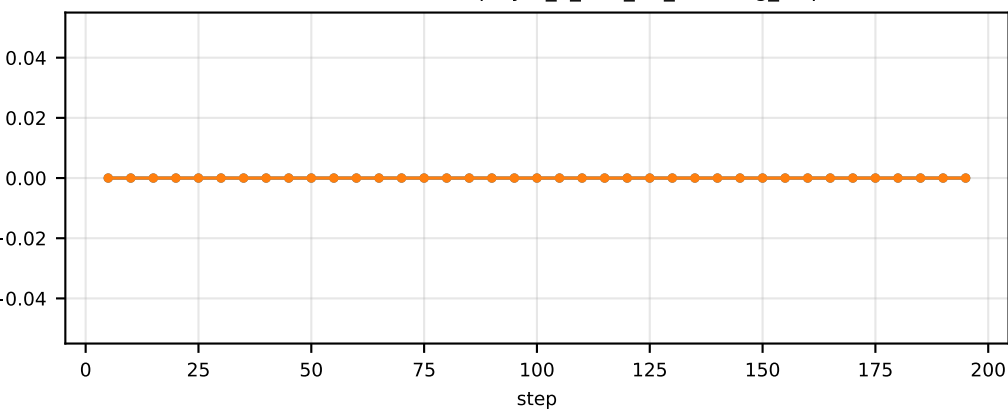
val/env/KuhnPoker-second/player_1_bet



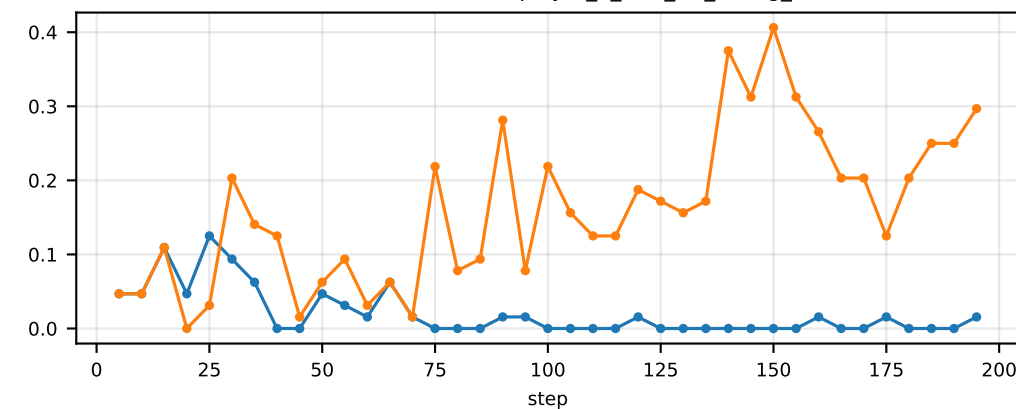
val/env/KuhnPoker-second/player_1_lose_for_overlong_response



val/env/KuhnPoker-second/player_1_lose_for_overlong_sequence

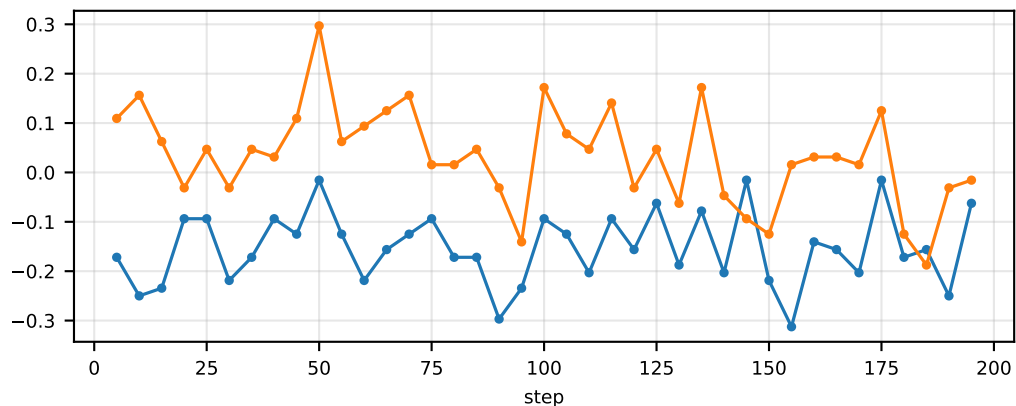


val/env/KuhnPoker-second/player_1_lose_for_wrong_format

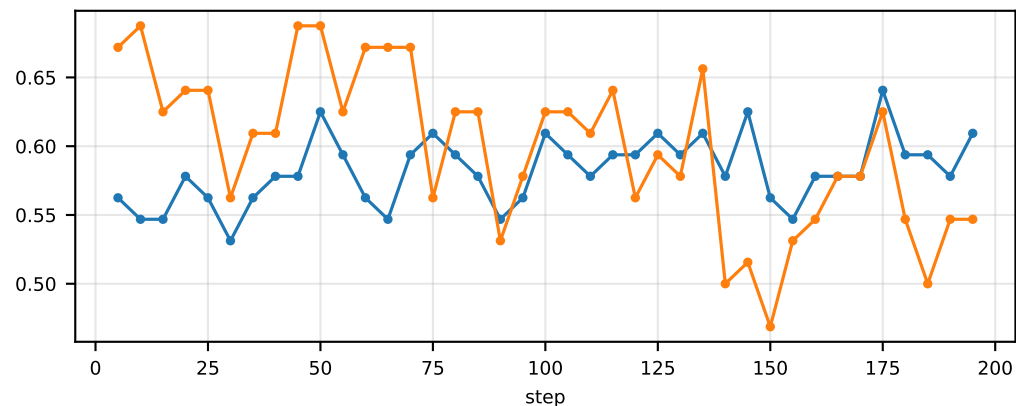


[KuhnPoker] Validation (vs built-in opponents) (7/8)

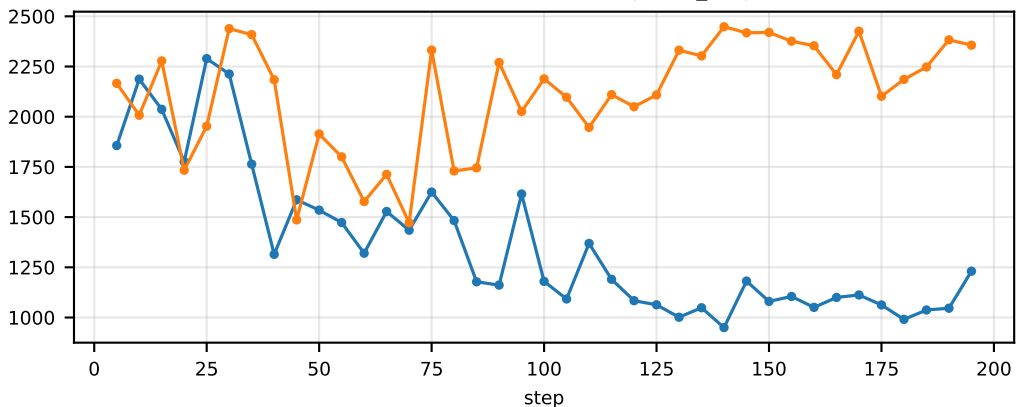
val/env/KuhnPoker-second/player_1_return



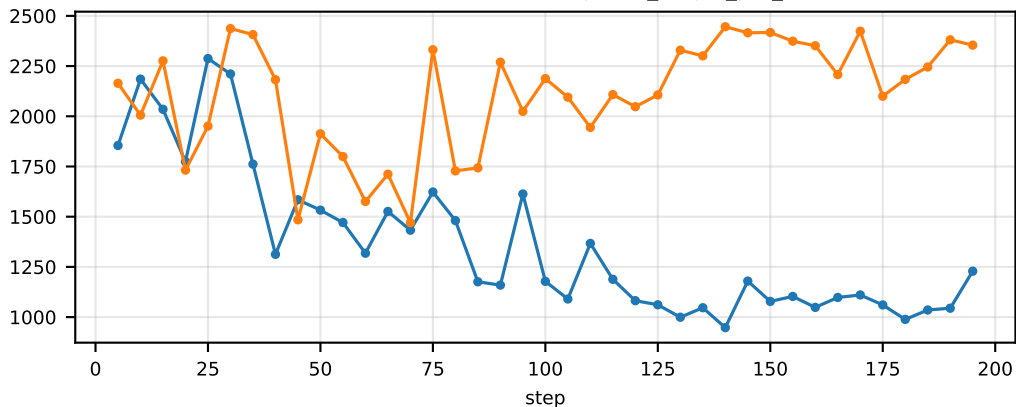
val/env/KuhnPoker-second/player_1_success



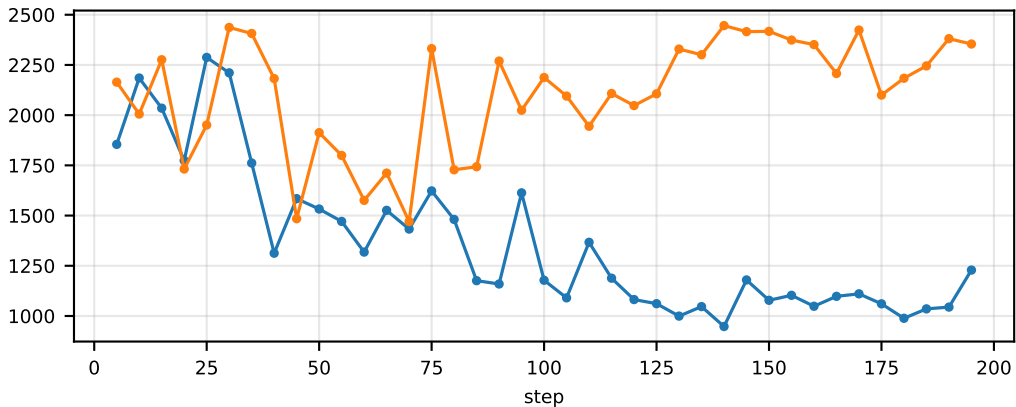
val/env/KuhnPoker-second/response_length



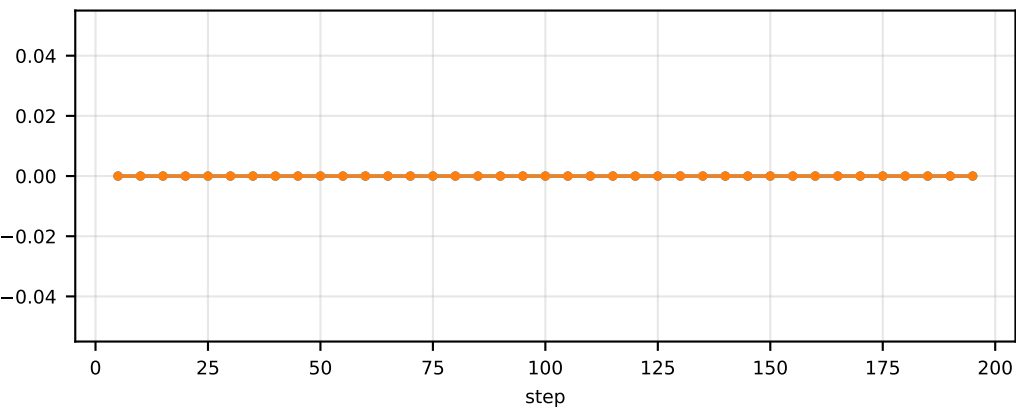
val/env/KuhnPoker-second/response_length_per_turn



val/env/KuhnPoker-second/response_length_turn_0

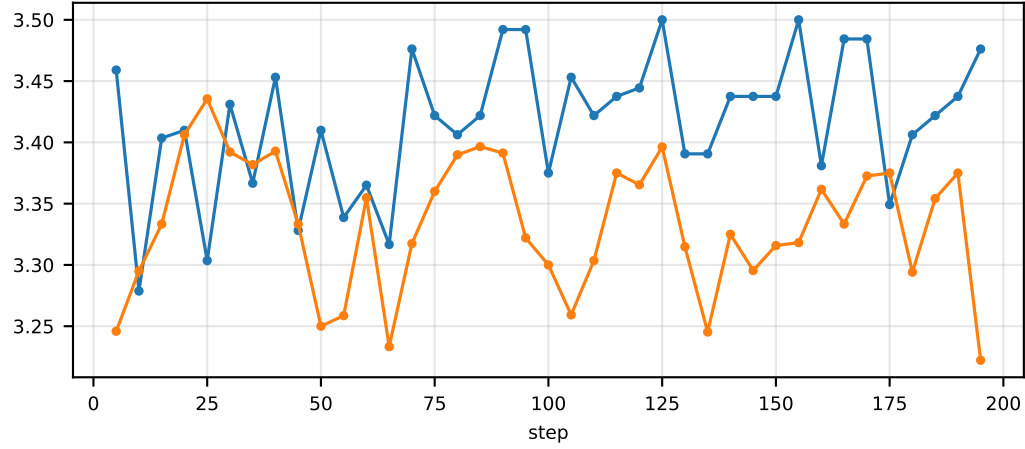


val/env/KuhnPoker-second/success

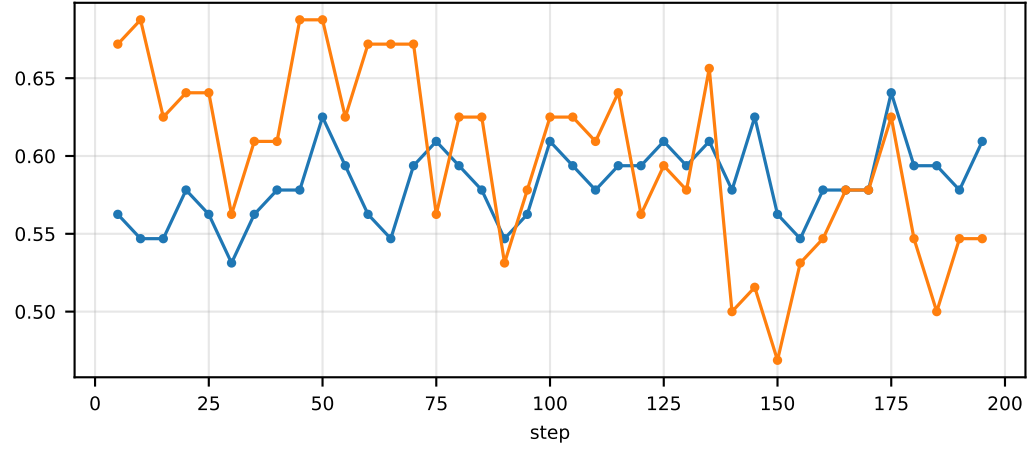


[KuhnPoker] Validation (vs built-in opponents) (8/8)

val/env/KuhnPoker-second/total_pot

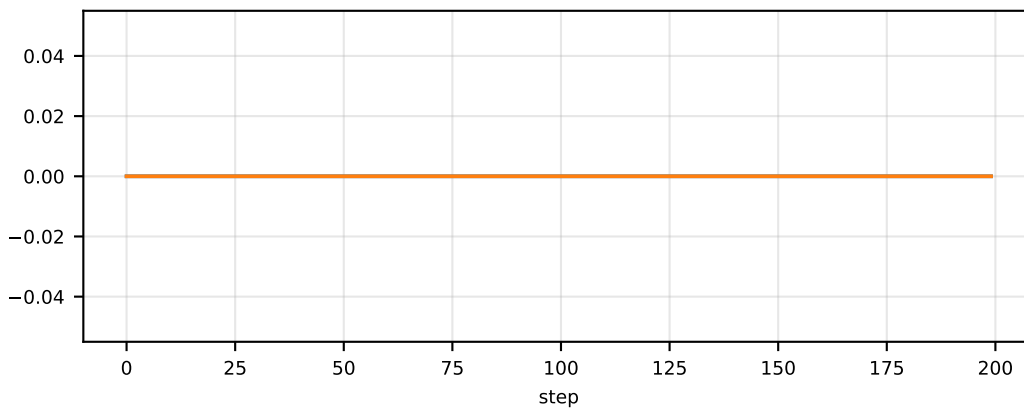


val/env/KuhnPoker-second/winner

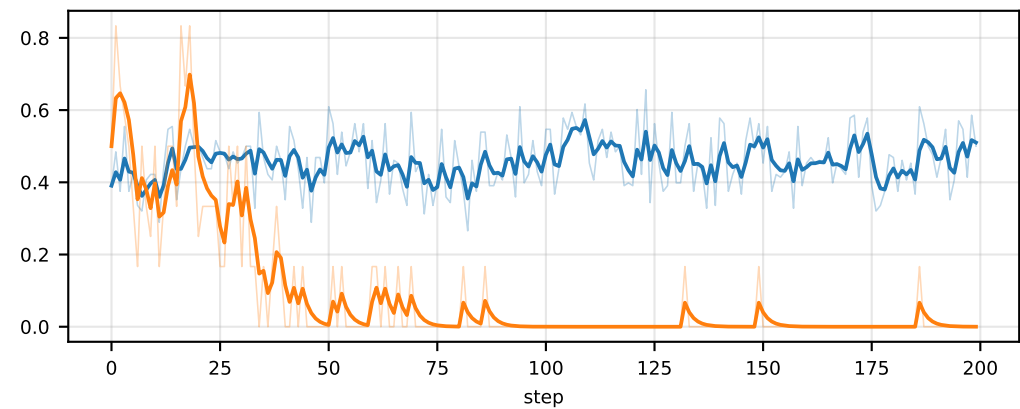


[KuhnPoker] Self-play outcome (train) (1/1)

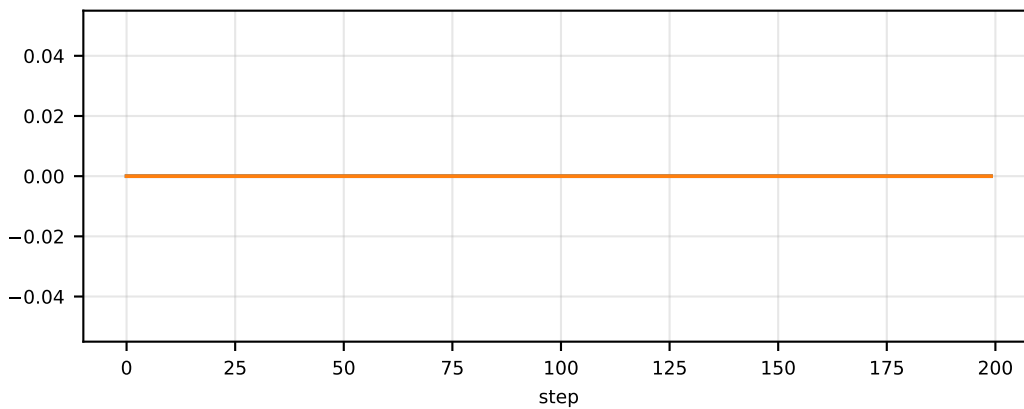
env/KuhnPoker/success



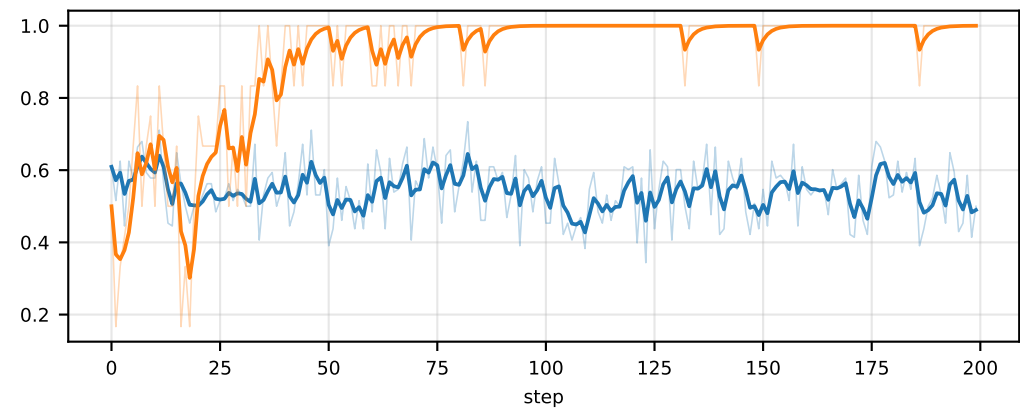
env/KuhnPoker/winner



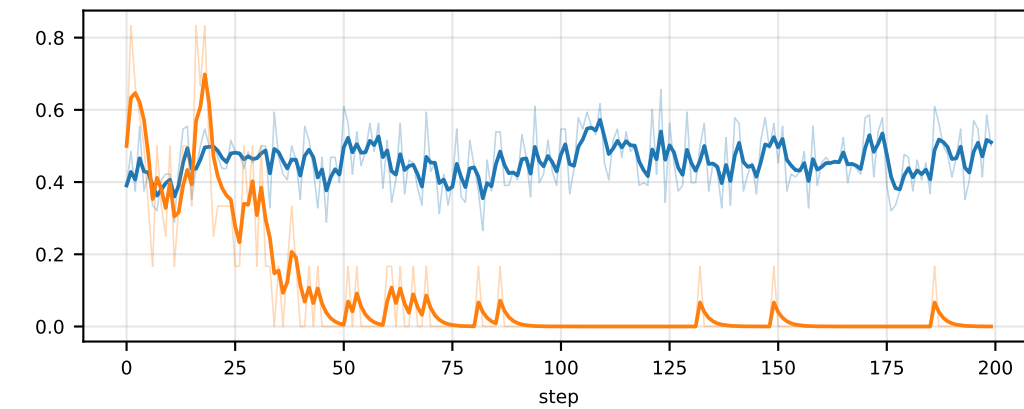
env/KuhnPoker/draw



env/KuhnPoker/player_0_success

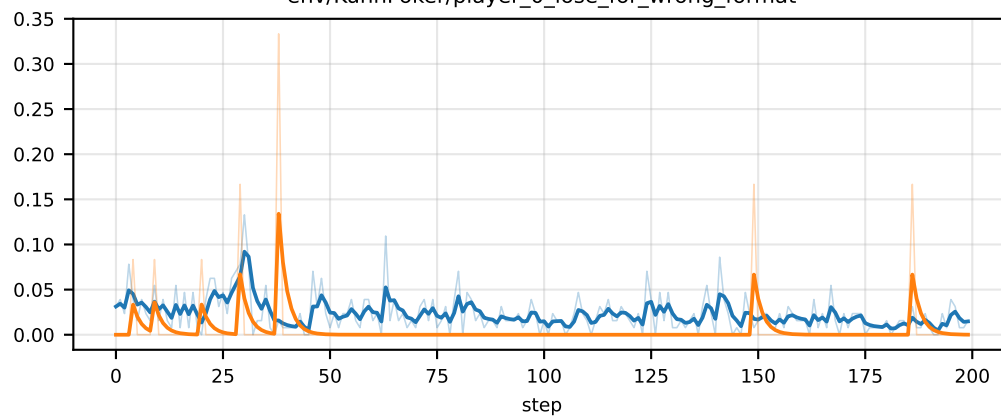


env/KuhnPoker/player_1_success

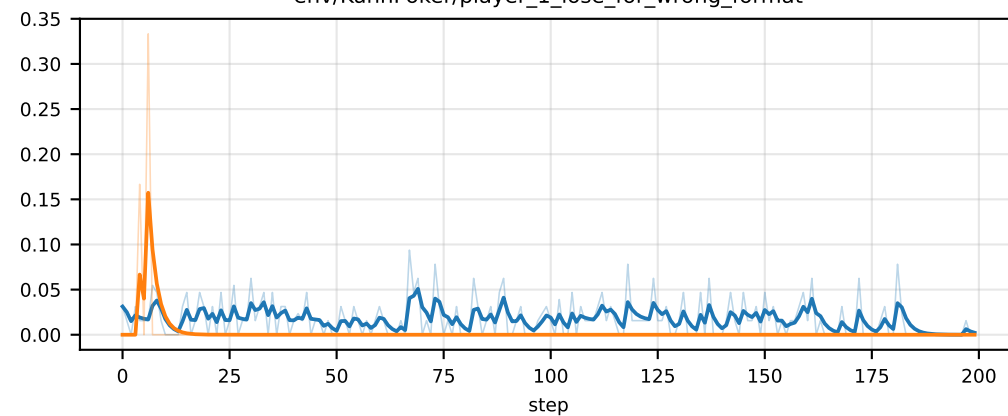


[KuhnPoker] Losses from format / length (1/1)

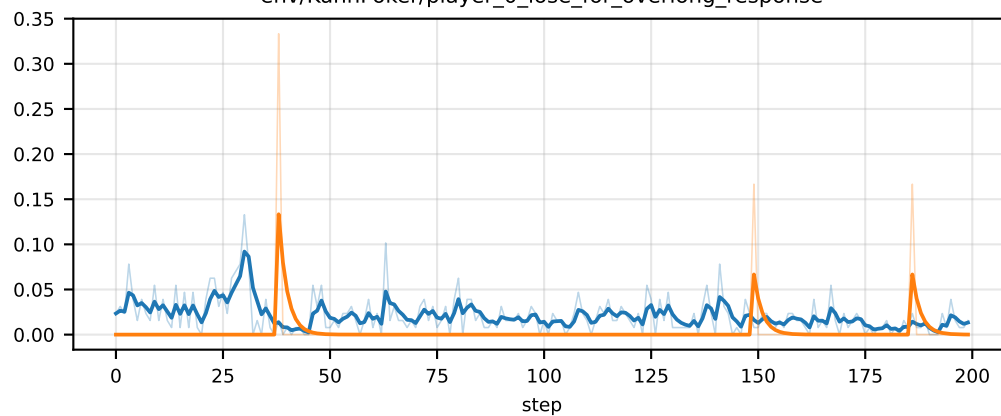
env/KuhnPoker/player_0_lose_for_wrong_format



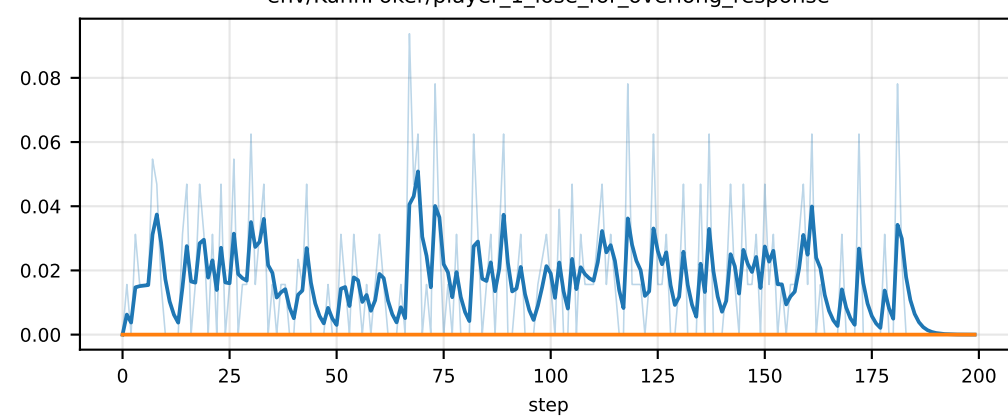
env/KuhnPoker/player_1_lose_for_wrong_format



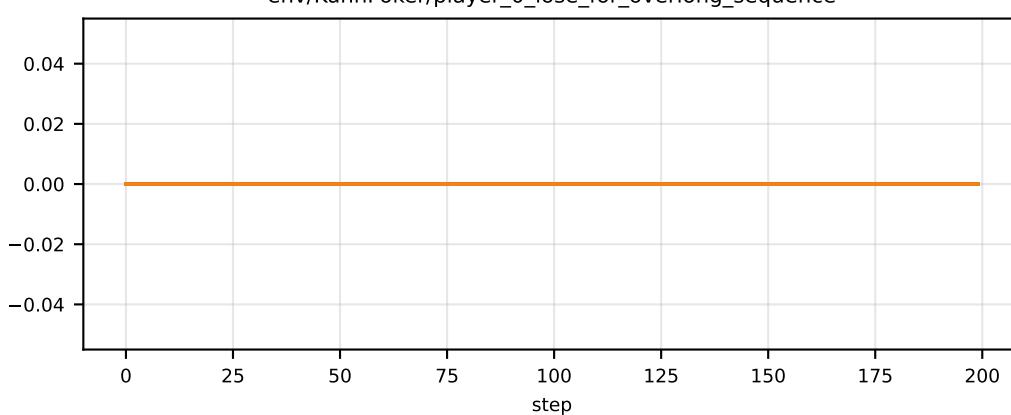
env/KuhnPoker/player_0_lose_for_overlong_response



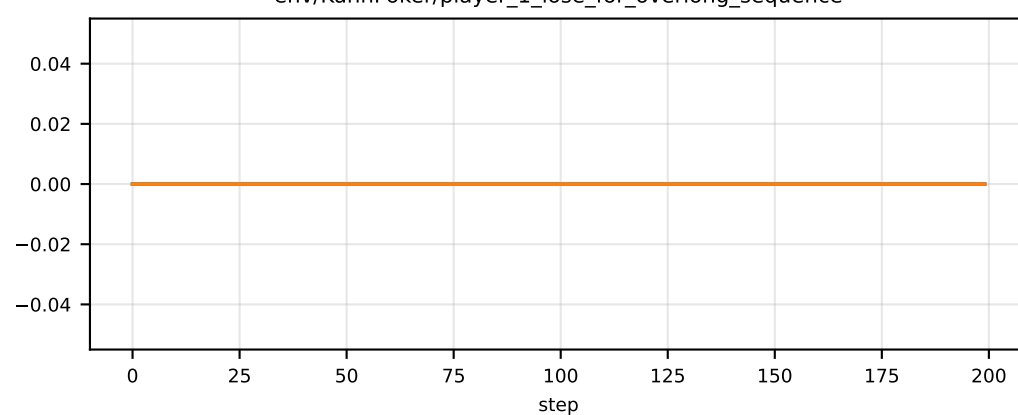
env/KuhnPoker/player_1_lose_for_overlong_response



env/KuhnPoker/player_0_lose_for_overlong_sequence

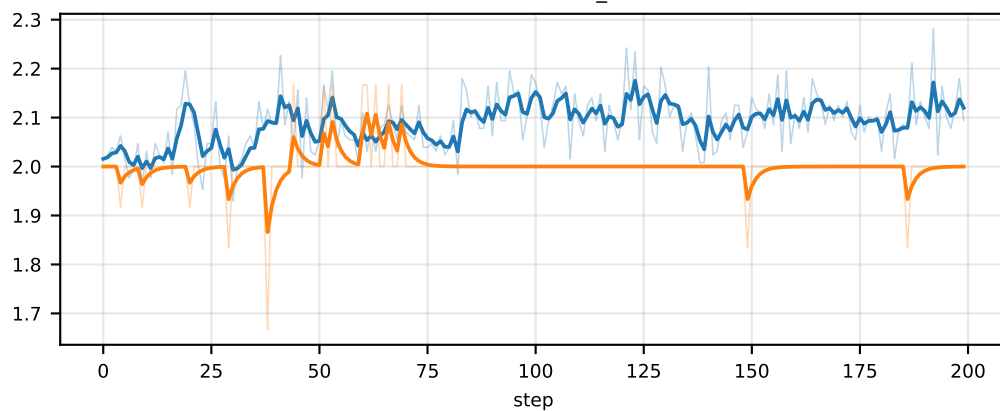


env/KuhnPoker/player_1_lose_for_overlong_sequence

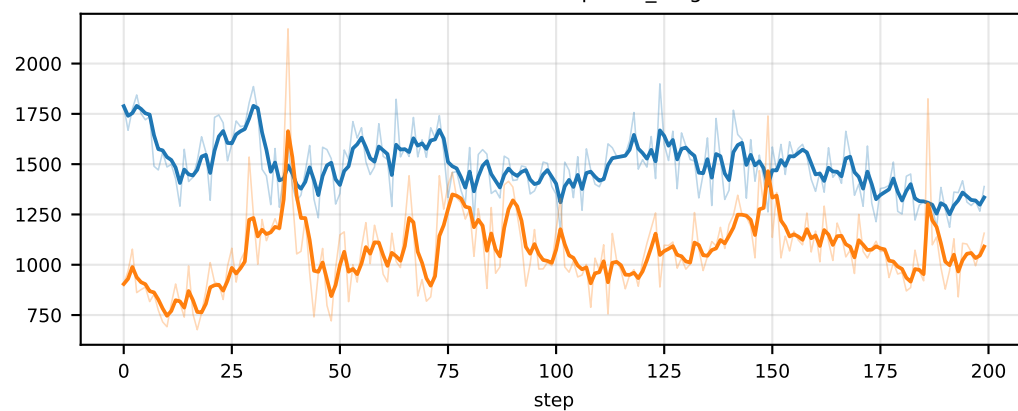


[KuhnPoker] Game length & response length (1/3)

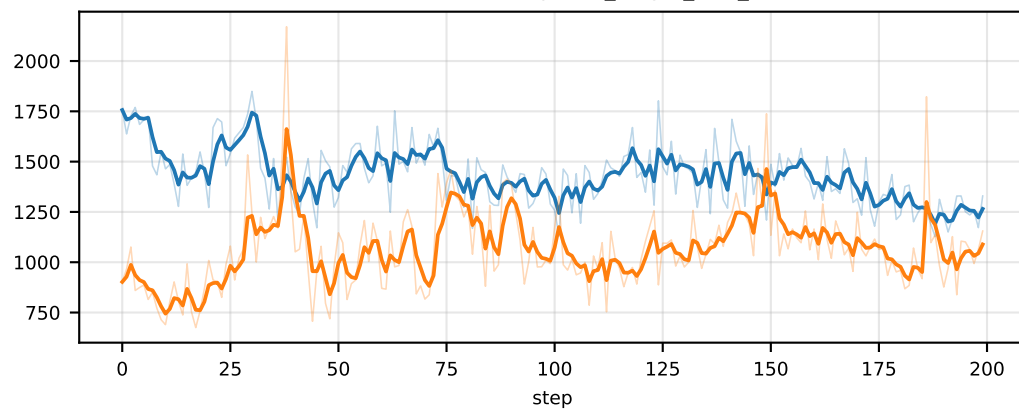
env/KuhnPoker/num_actions



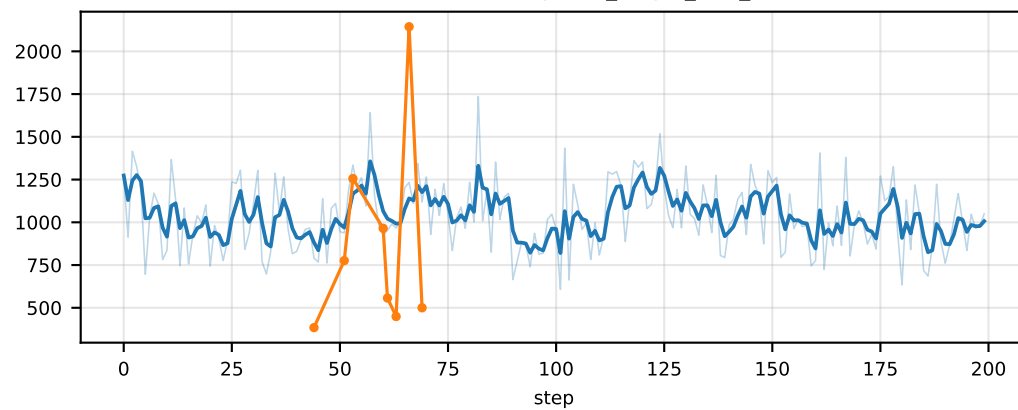
env/KuhnPoker/response_length



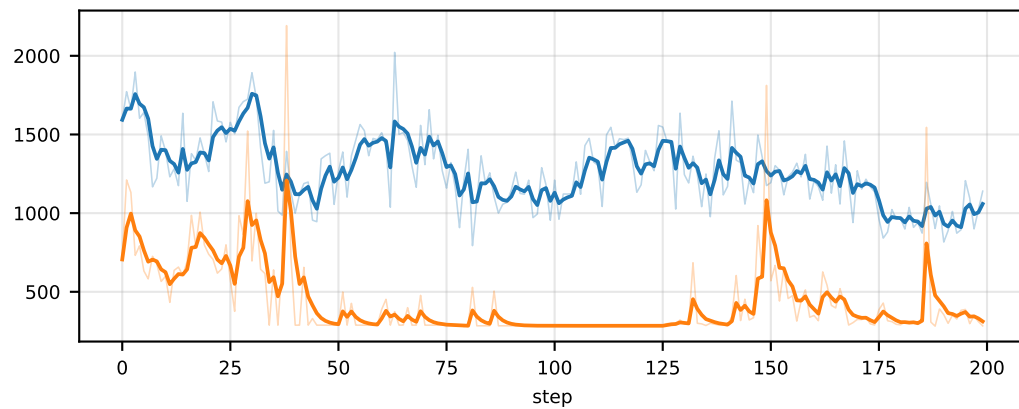
env/KuhnPoker/response_length_turn_0



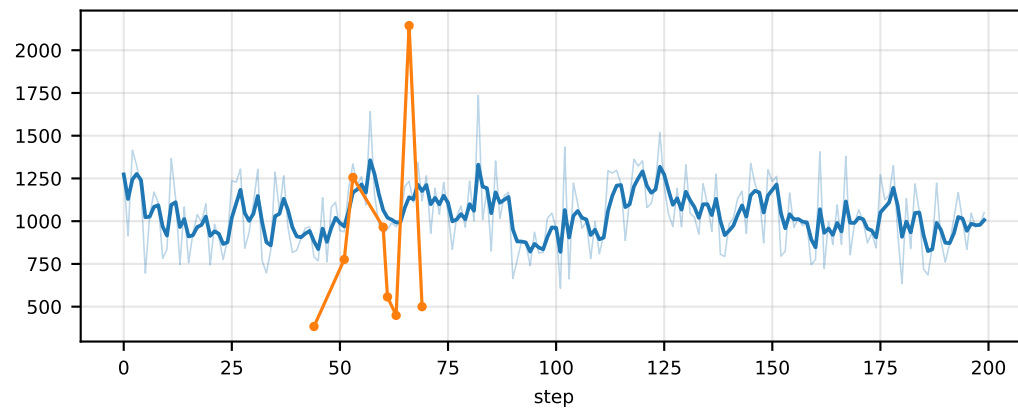
env/KuhnPoker/response_length_turn_1



env/KuhnPoker/response_length_player_0_turn_0

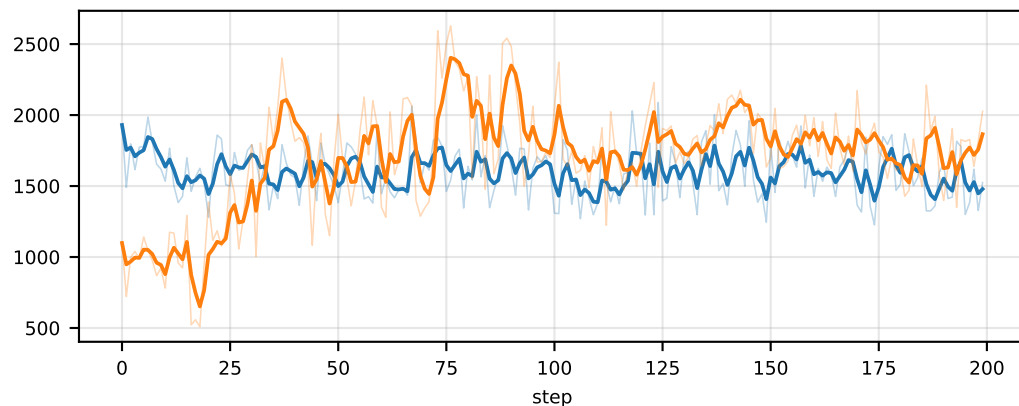


env/KuhnPoker/response_length_player_0_turn_1

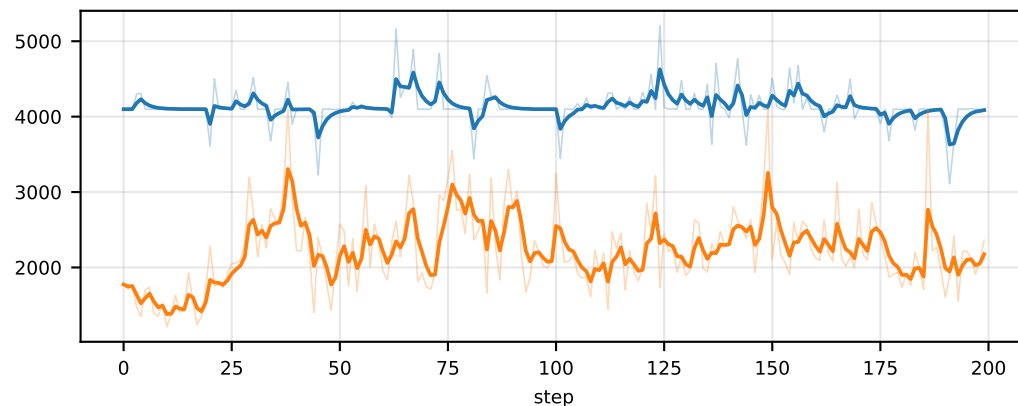


[KuhnPoker] Game length & response length (2/3)

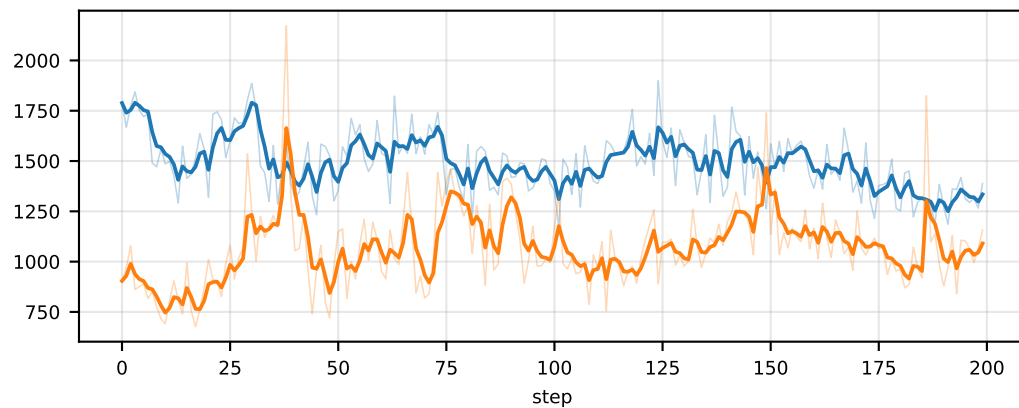
env/KuhnPoker/response_length_player_1_turn_0



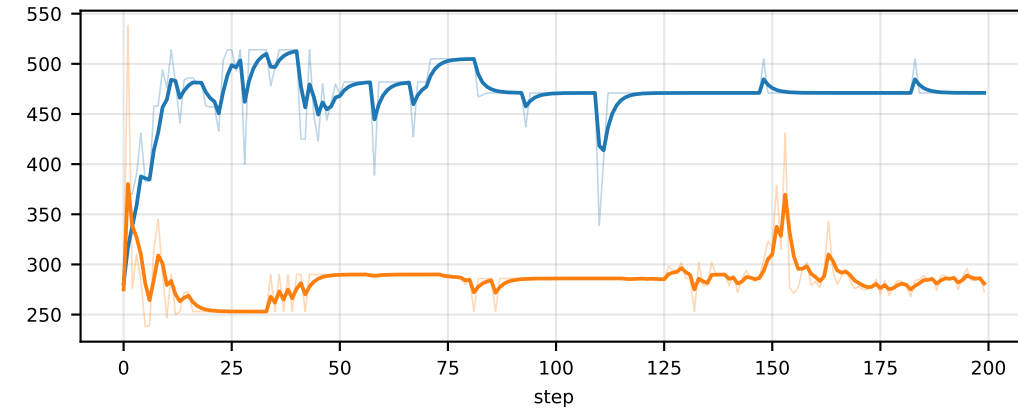
tokens/response_length/max



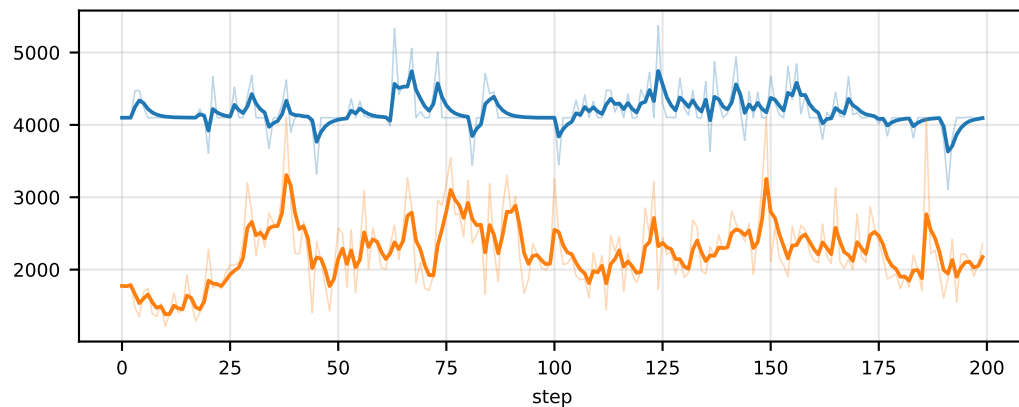
tokens/response_length/mean



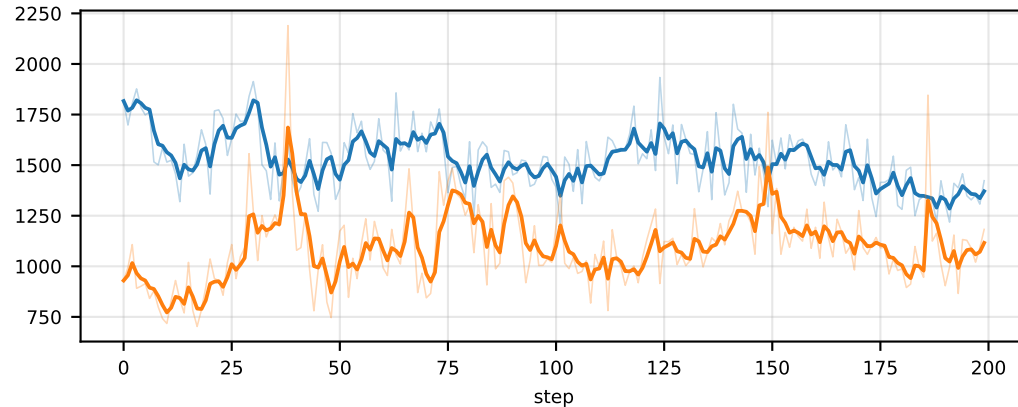
tokens/response_length/min



tokens/non_prompt_length/max



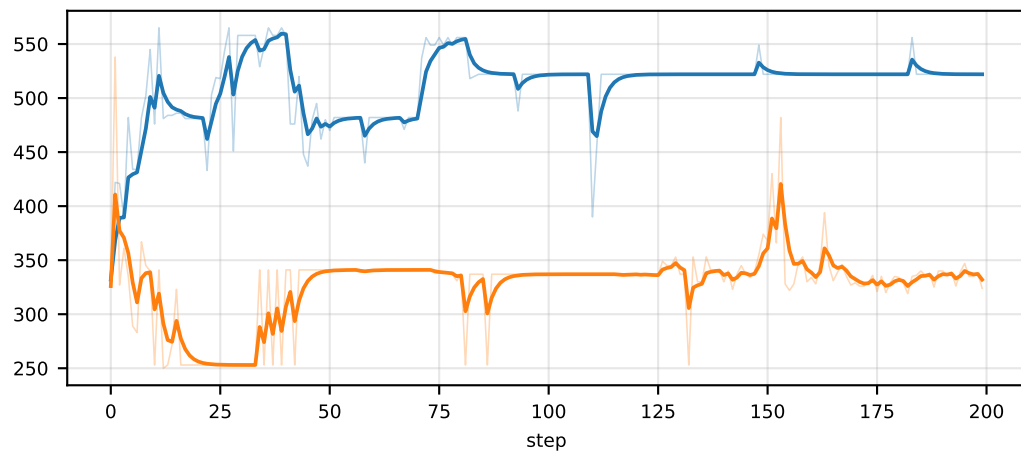
tokens/non_prompt_length/mean



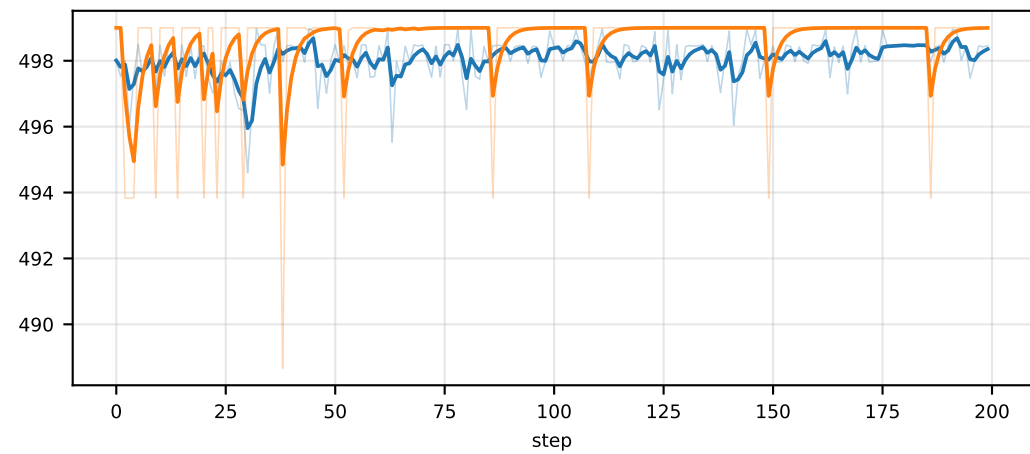
— k1b (kuhn) — k2b (kuhn quickcheck)

[KuhnPoker] Game length & response length (3/3)

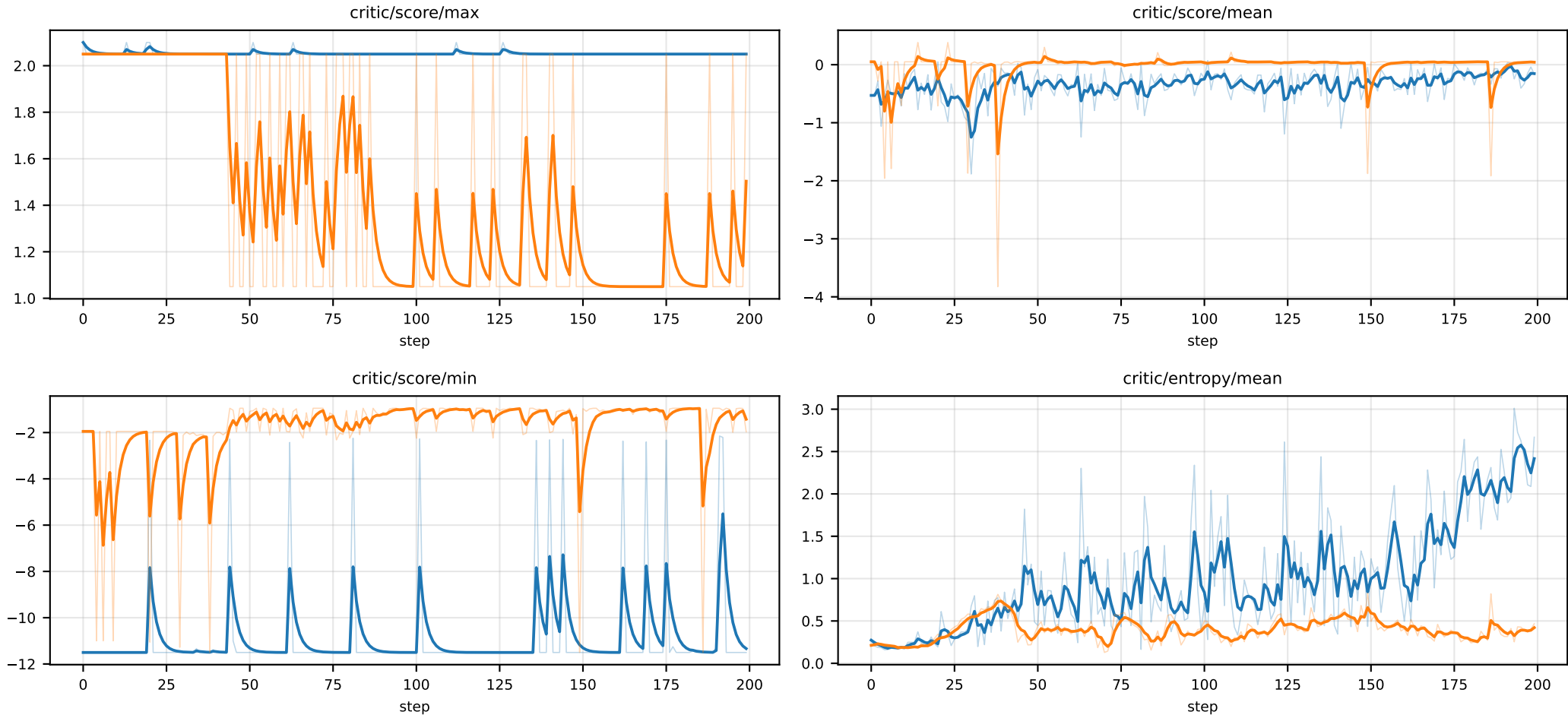
tokens/non_prompt_length/min



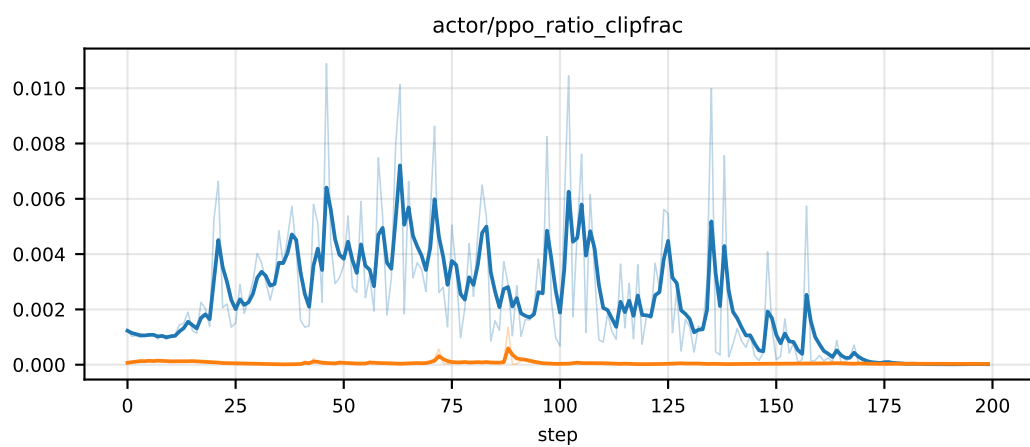
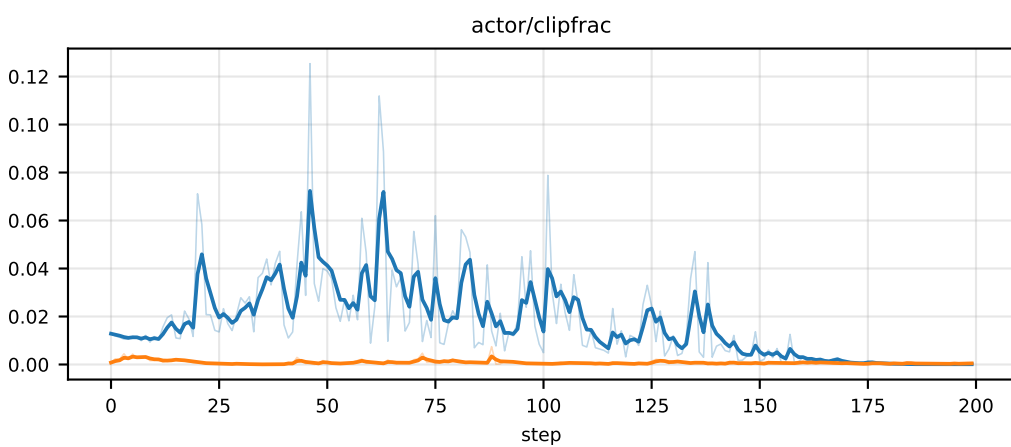
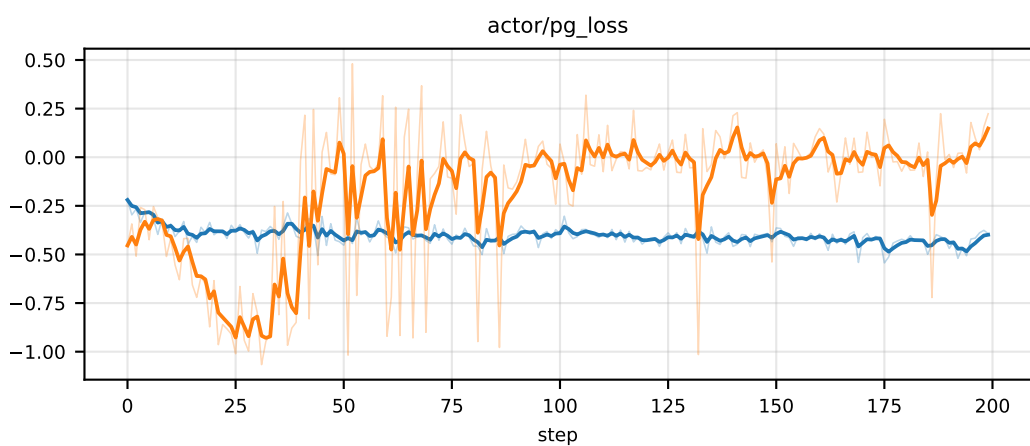
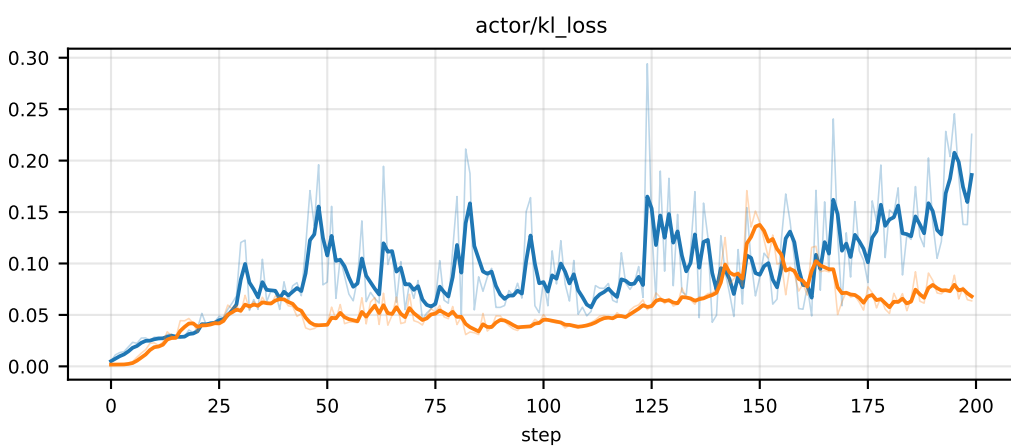
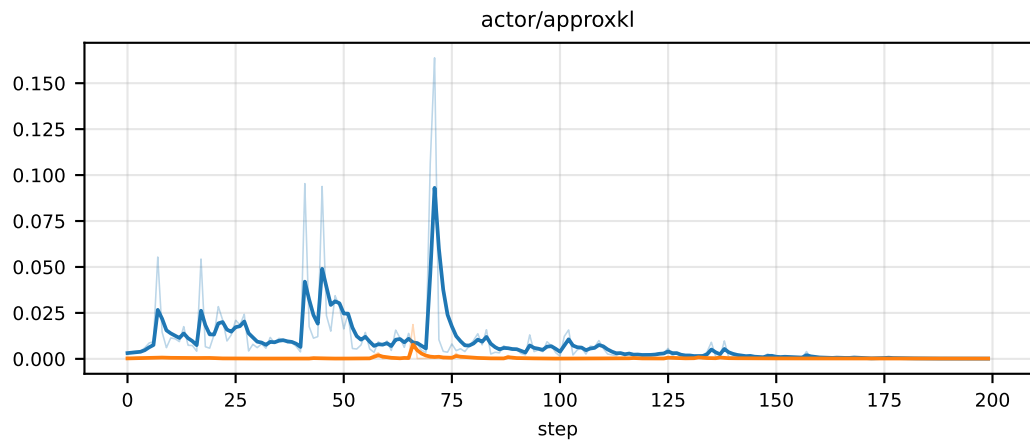
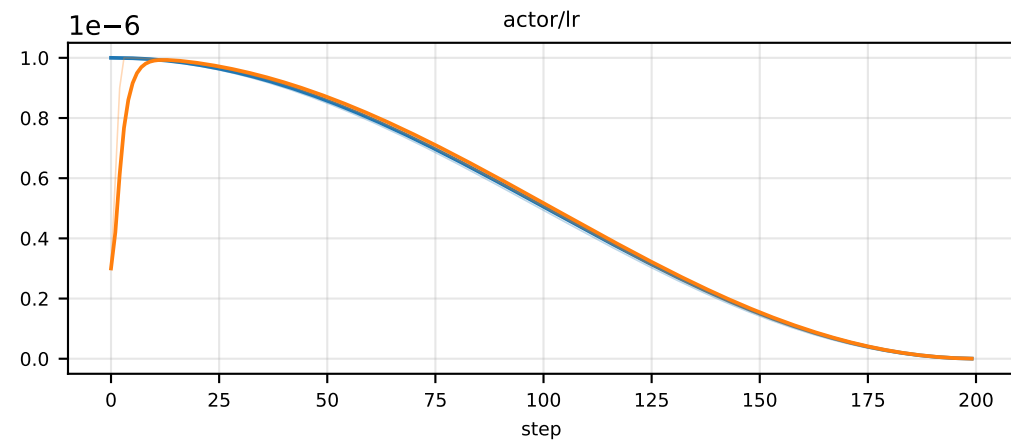
tokens/prompt_length/mean



[KuhnPoker] Reward & entropy (1/1)



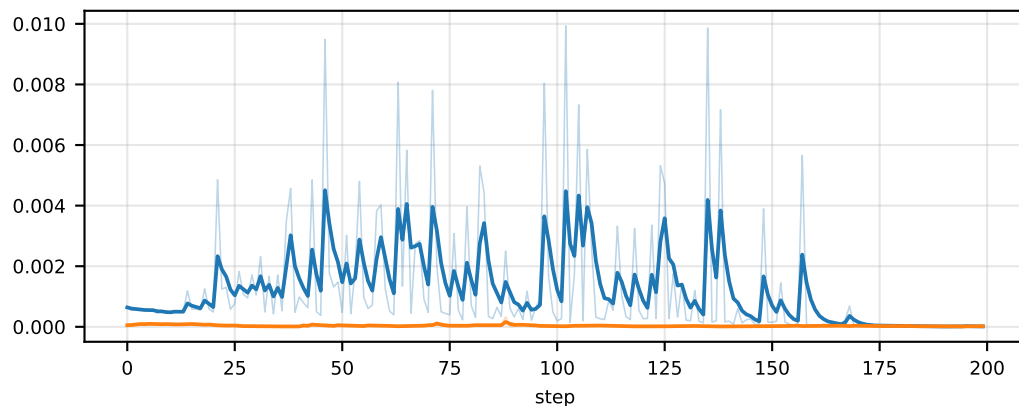
[KuhnPoker] Policy update (PPO / KL) (1/2)



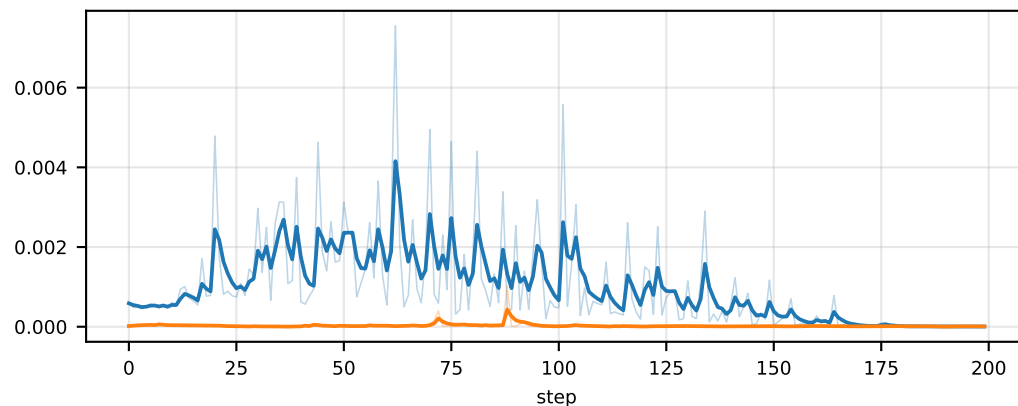
— k1b (kuhn) — k2b (kuhn quickcheck)

[KuhnPoker] Policy update (PPO / KL) (2/2)

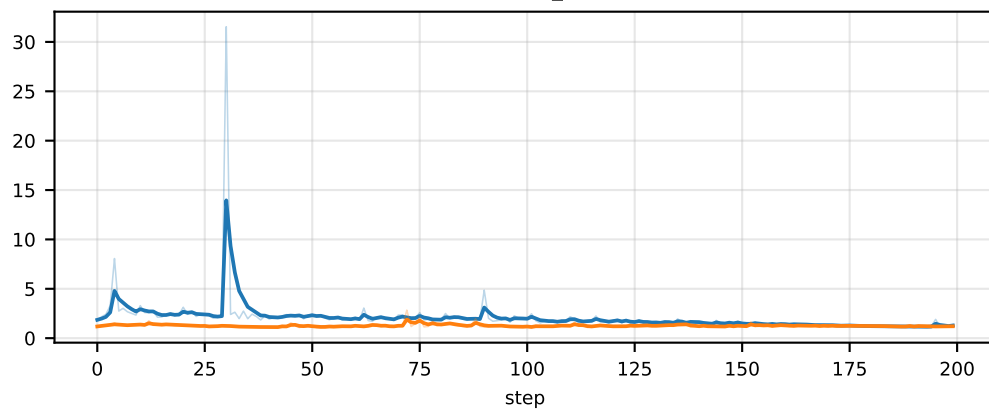
actor/ppo_ratio_high_clipfrac



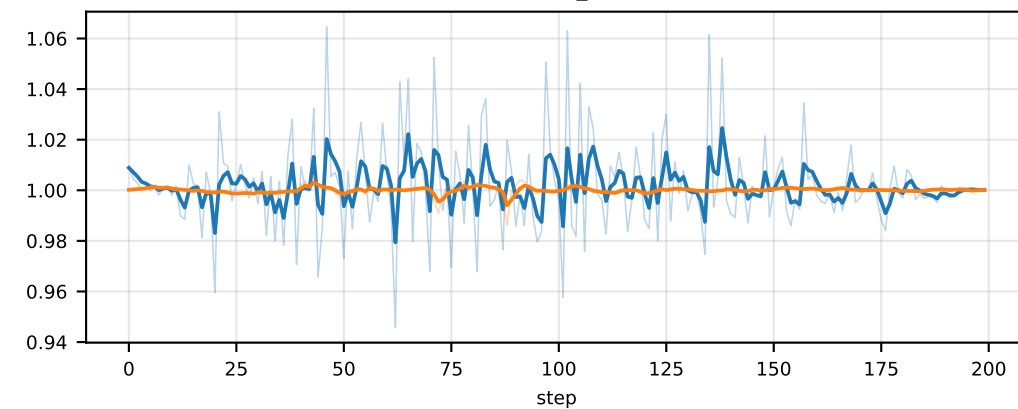
actor/ppo_ratio_low_clipfrac



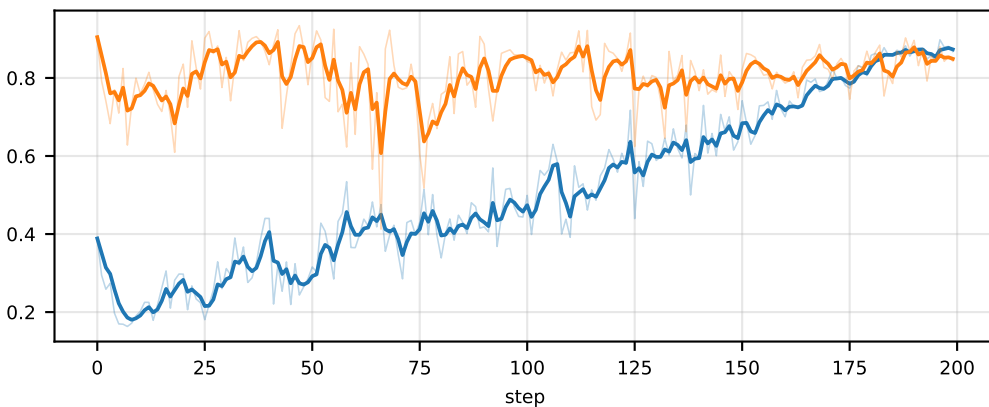
actor/ratio_max



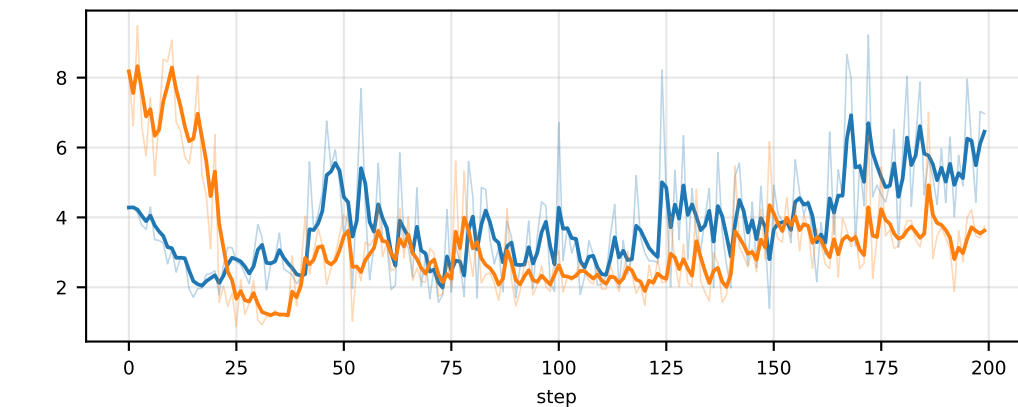
actor/ratio_mean



actor/ratio_min

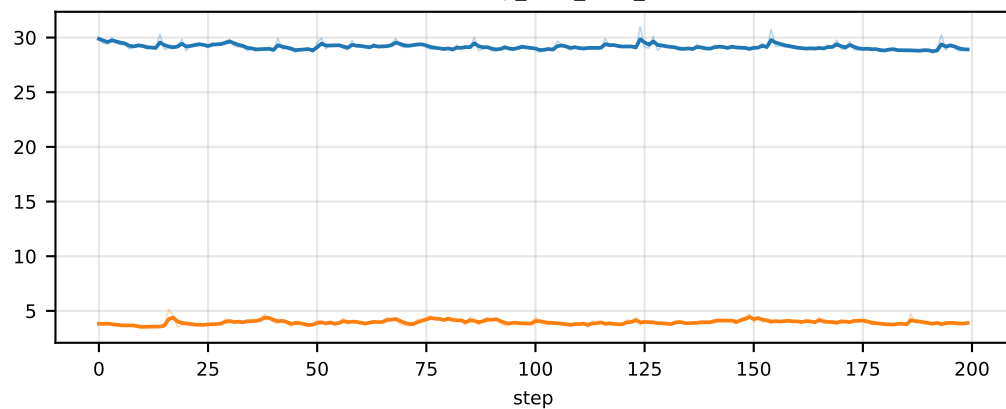


actor_train/grad_norm

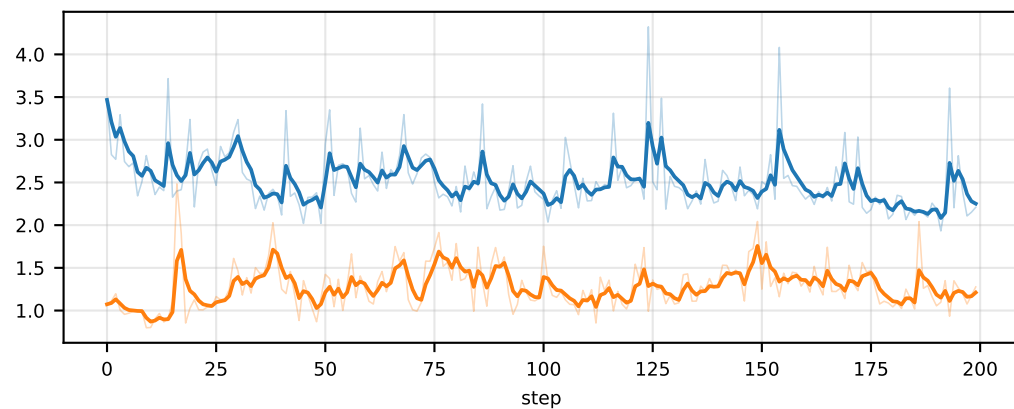


[KuhnPoker] Step time breakdown (minutes) (1/1)

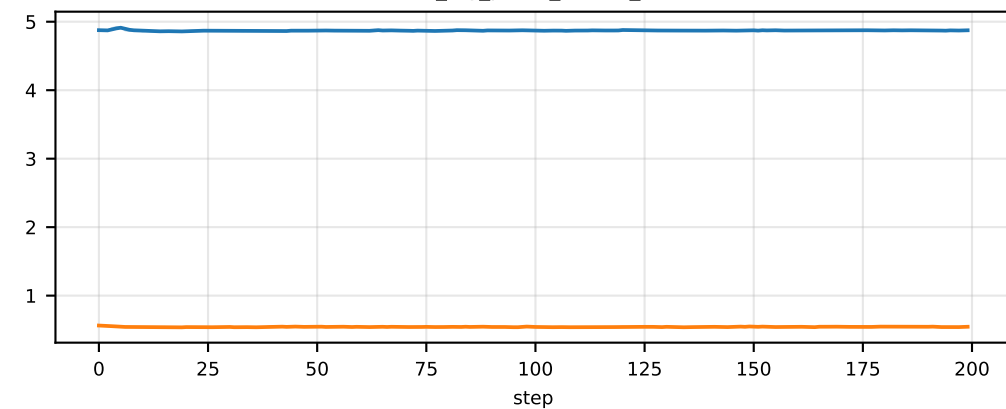
derived/step_time_total_min



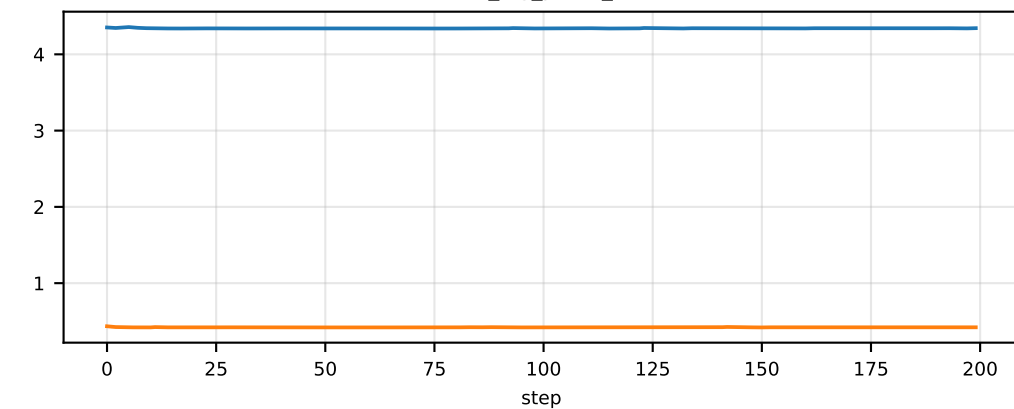
time/rollout



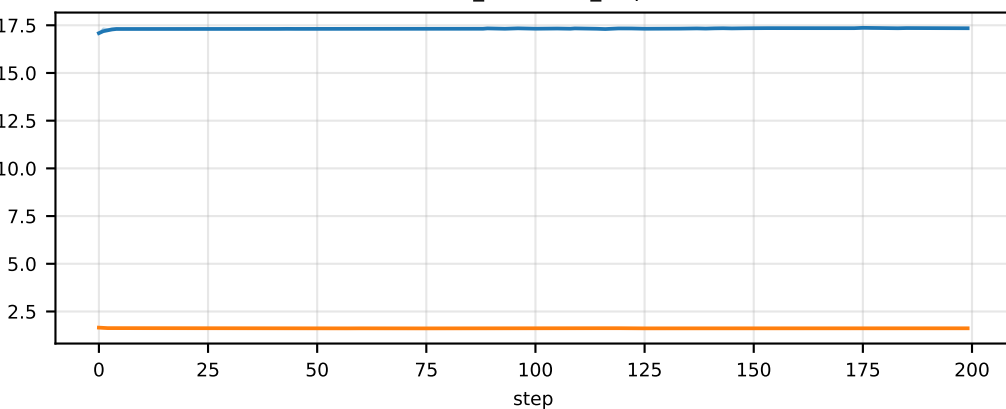
time/ref_log_probs_values_reward



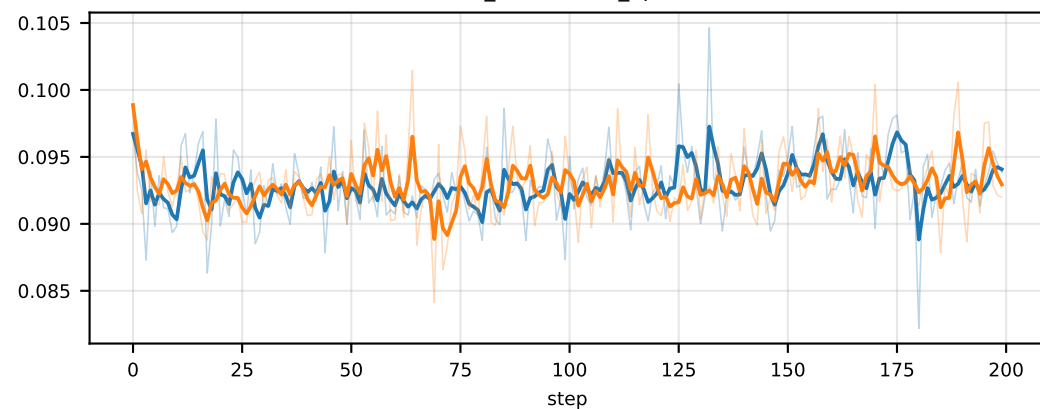
time/old_log_probs_values



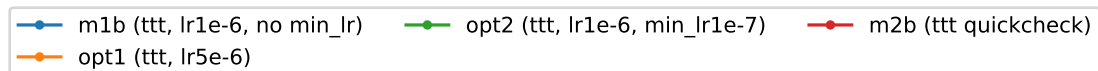
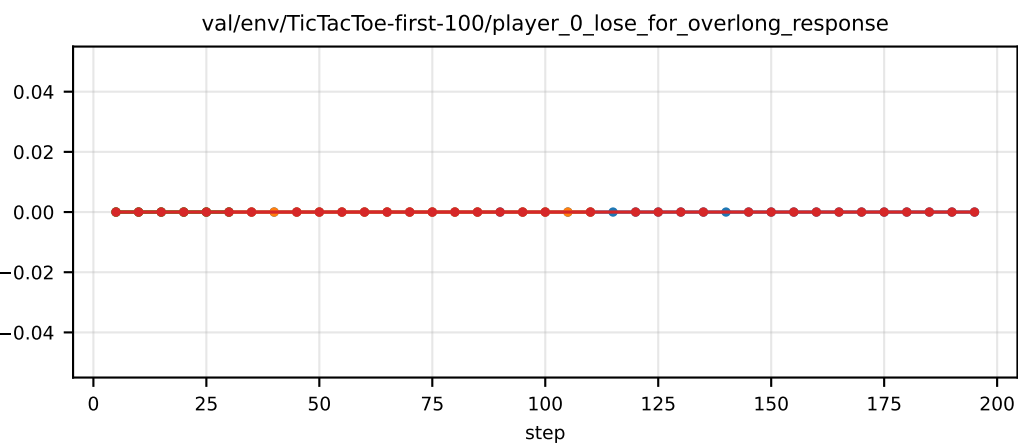
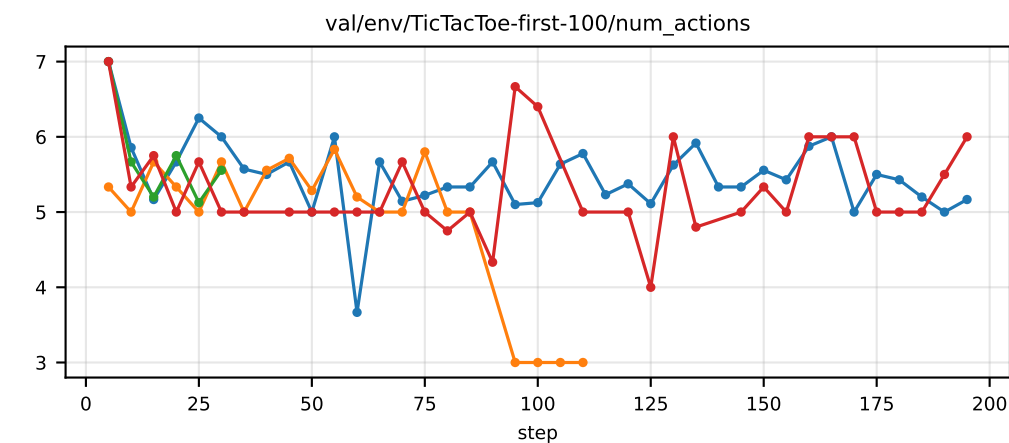
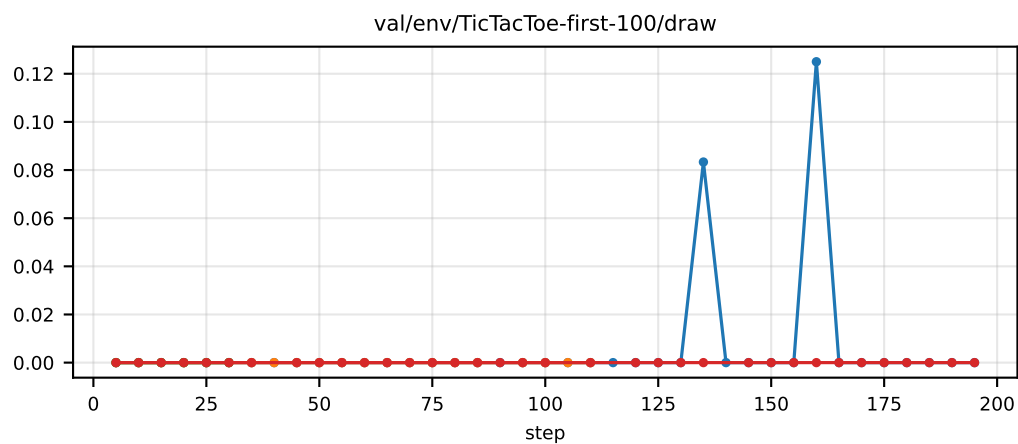
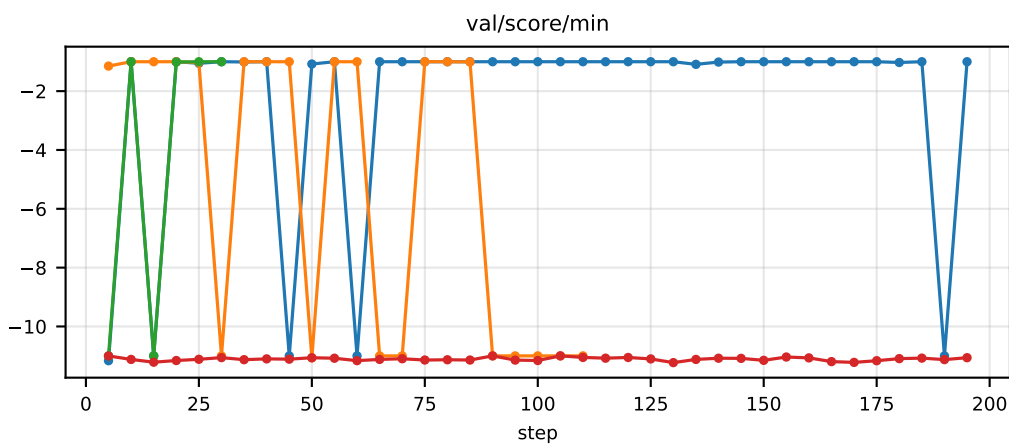
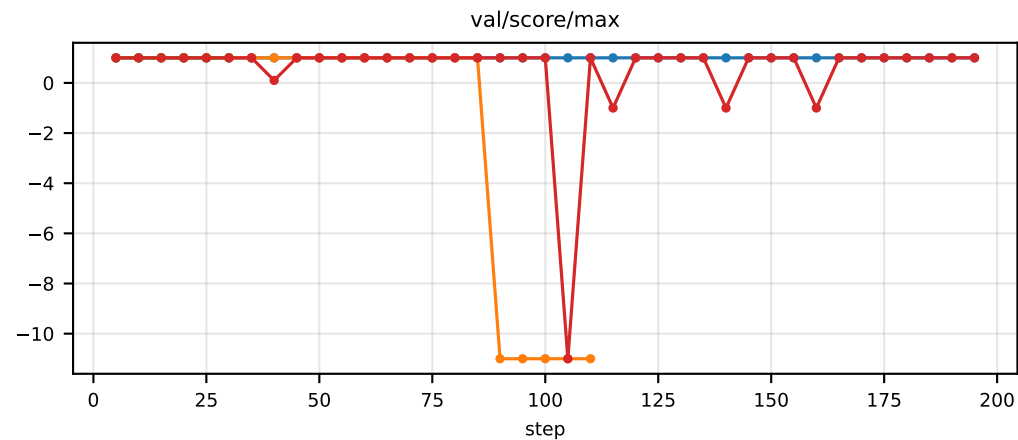
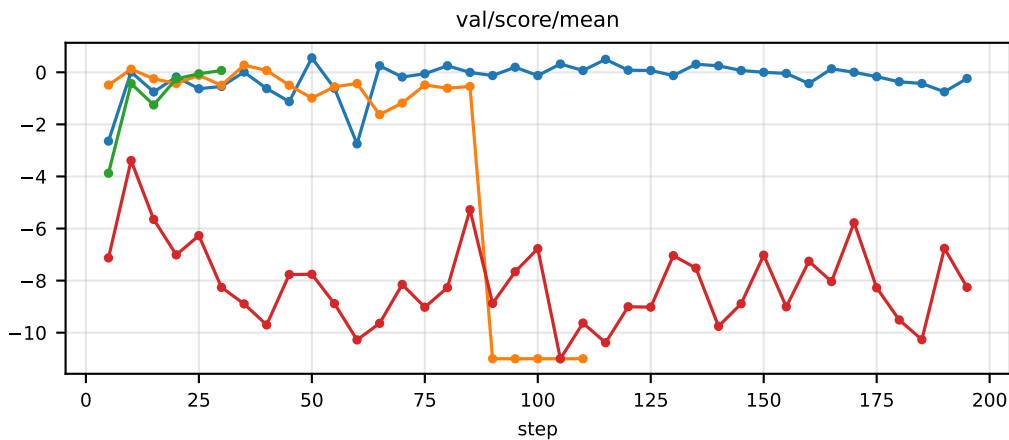
time/actor_train/train_step/total



time/actor_train/model_update/total

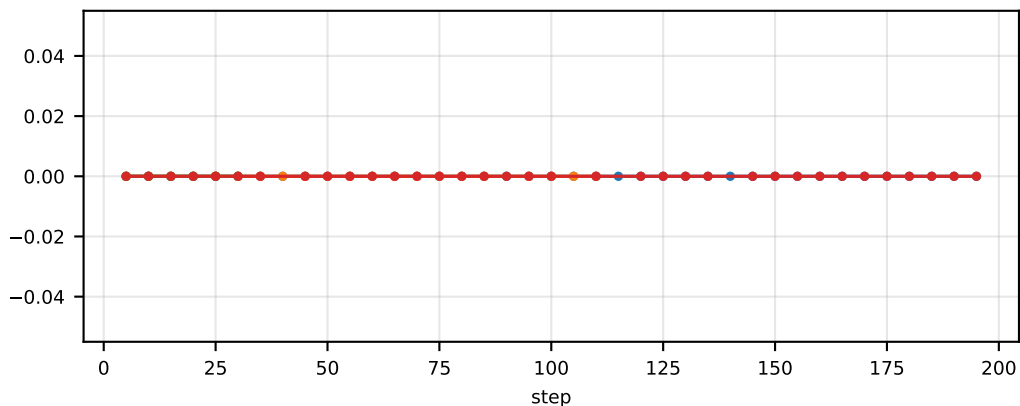


[TicTacToe] Validation (vs built-in opponents) (1/14)

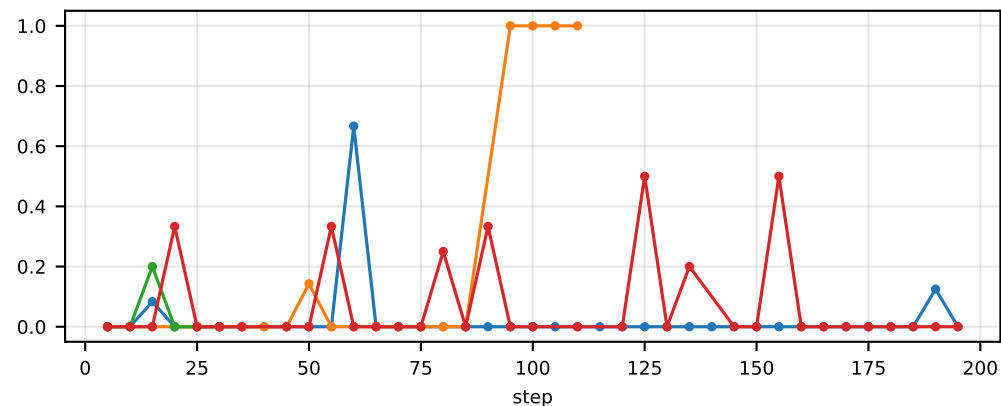


[TicTacToe] Validation (vs built-in opponents) (2/14)

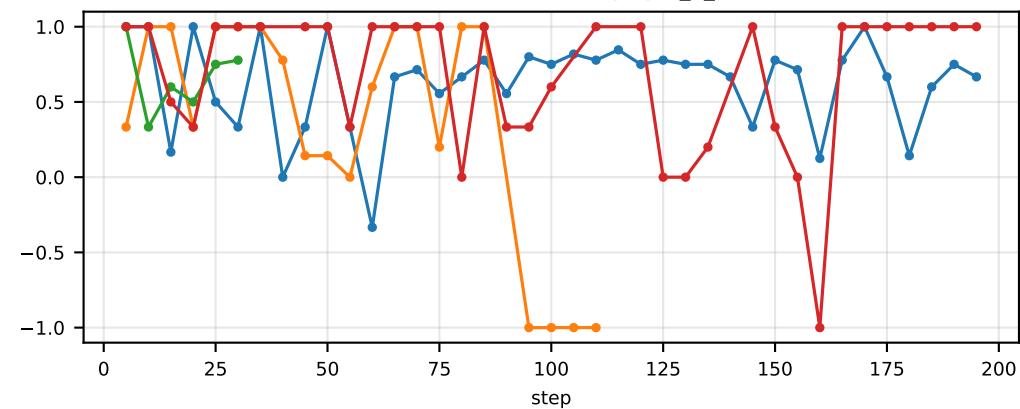
val/env/TicTacToe-first-100/player_0_lose_for_overlong_sequence



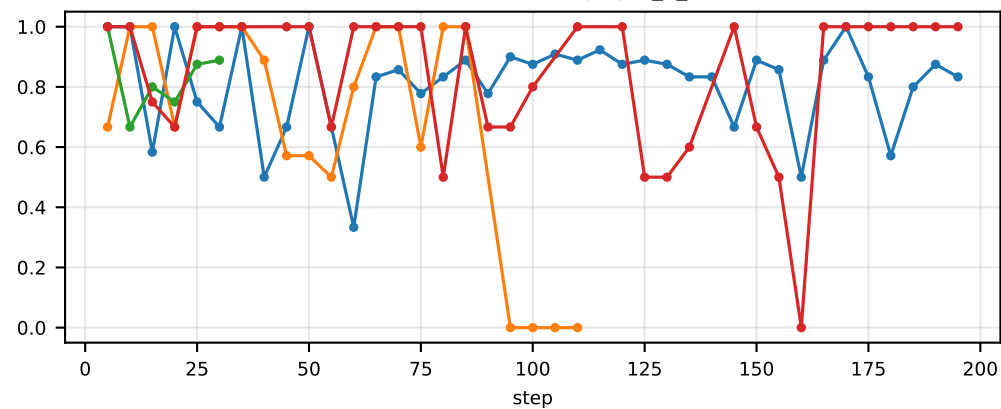
val/env/TicTacToe-first-100/player_0_lose_for_wrong_format



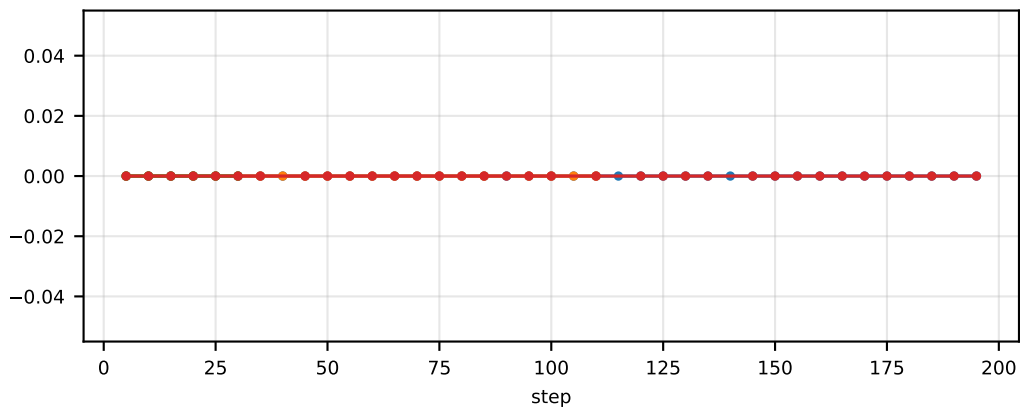
val/env/TicTacToe-first-100/player_0_return



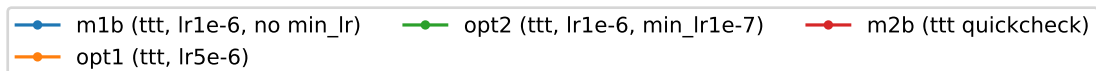
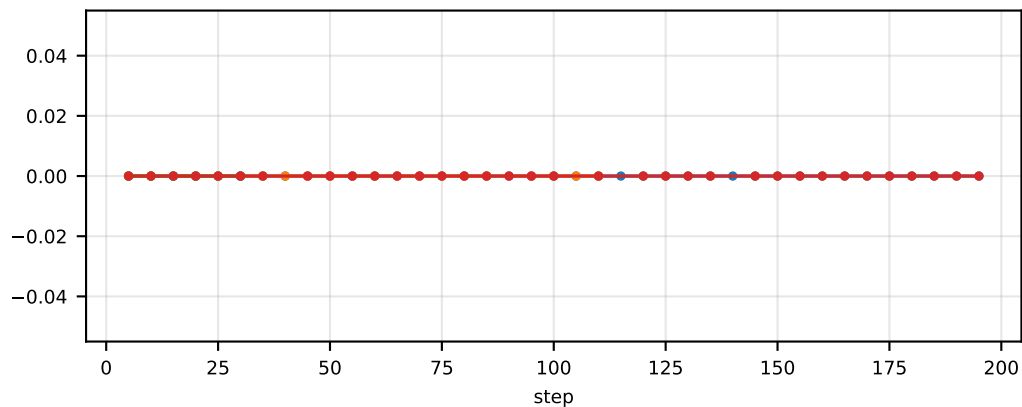
val/env/TicTacToe-first-100/player_0_success



val/env/TicTacToe-first-100/player_1_lose_for_overlong_response

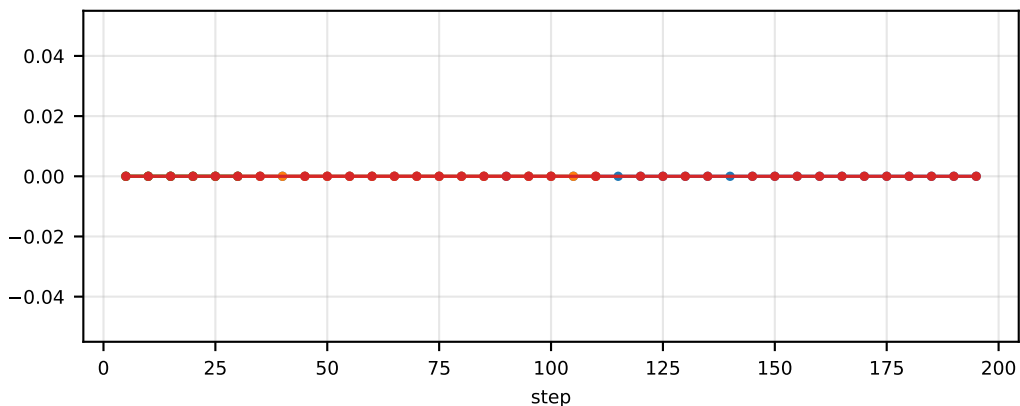


val/env/TicTacToe-first-100/player_1_lose_for_overlong_sequence

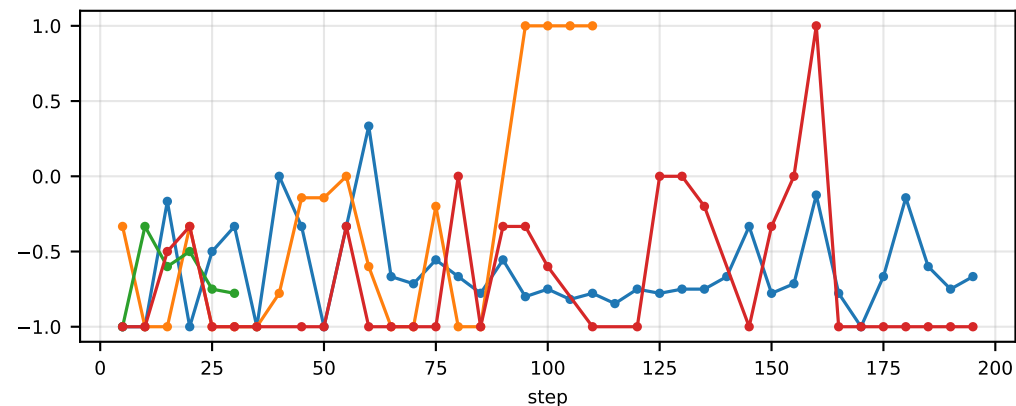


[TicTacToe] Validation (vs built-in opponents) (3/14)

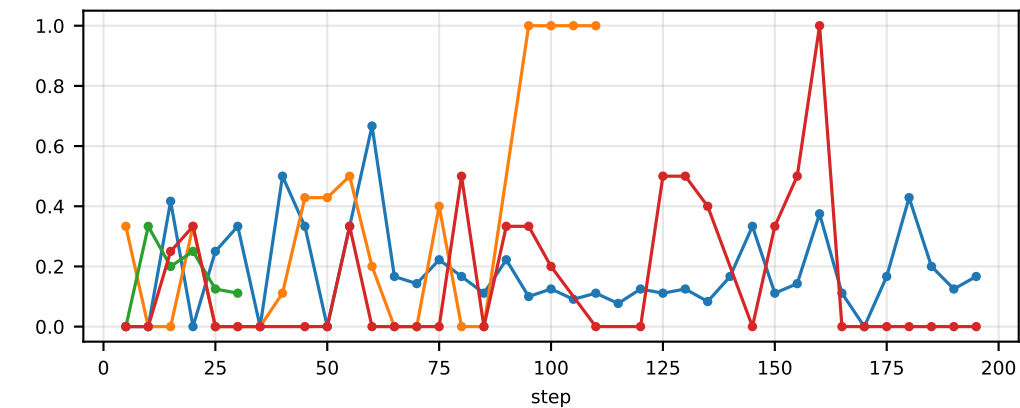
val/env/TicTacToe-first-100/player_1_lose_for_wrong_format



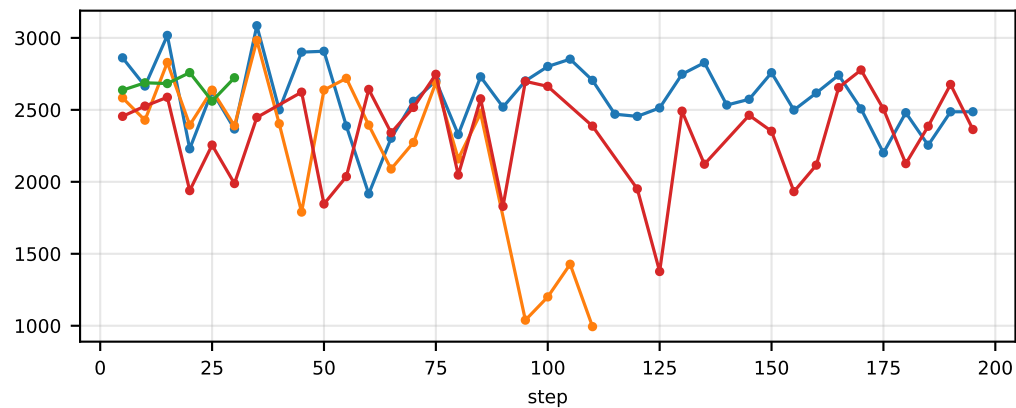
val/env/TicTacToe-first-100/player_1_return



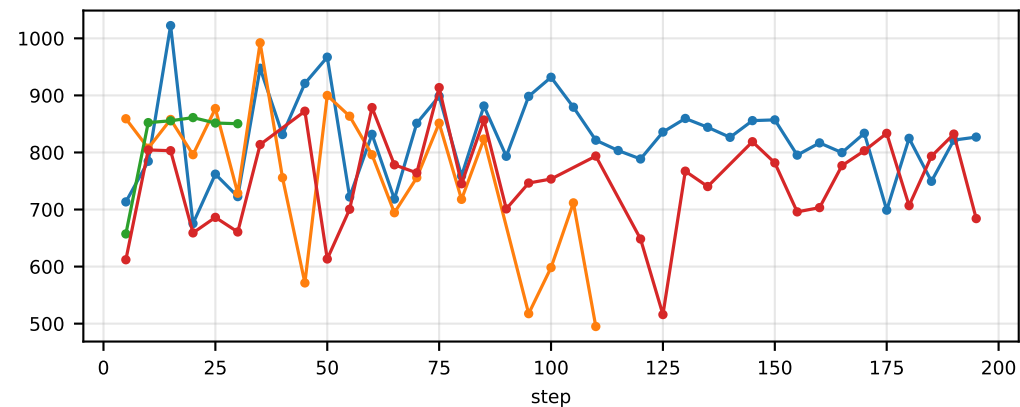
val/env/TicTacToe-first-100/player_1_success



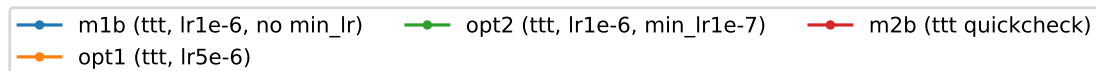
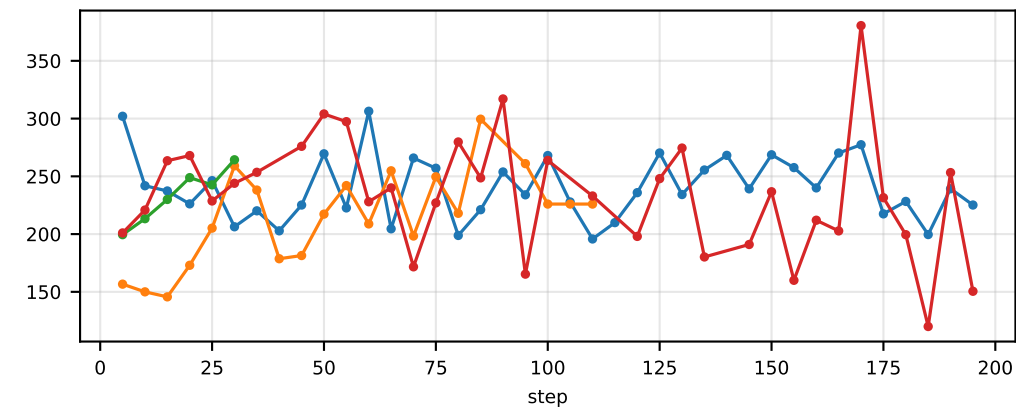
val/env/TicTacToe-first-100/response_length



val/env/TicTacToe-first-100/response_length_per_turn

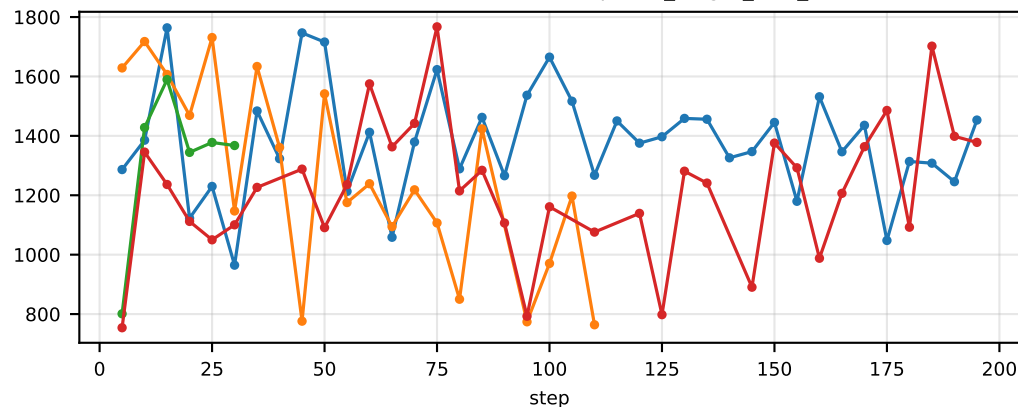


val/env/TicTacToe-first-100/response_length_turn_0

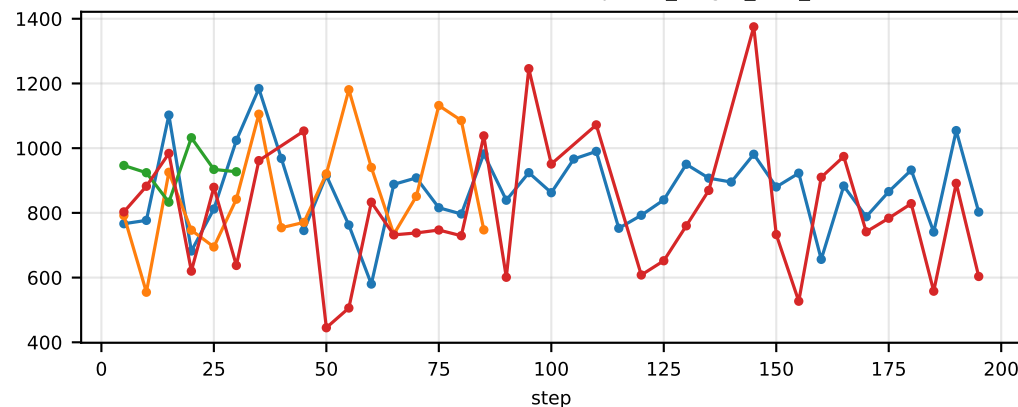


[TicTacToe] Validation (vs built-in opponents) (4/14)

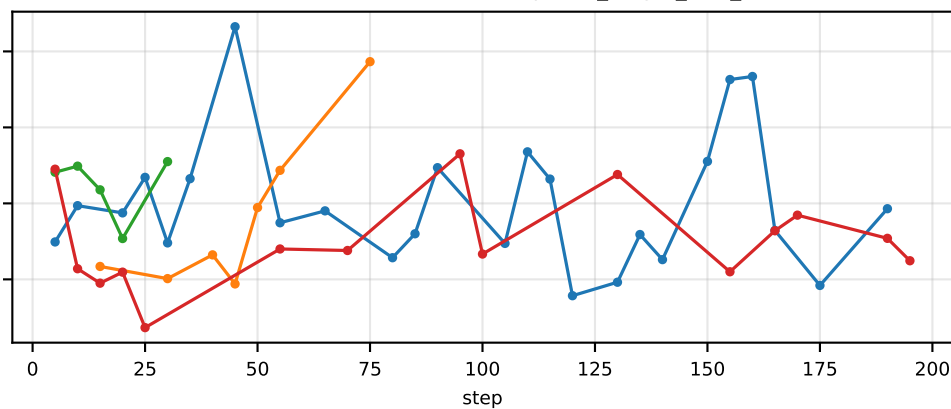
val/env/TicTacToe-first-100/response_length_turn_1



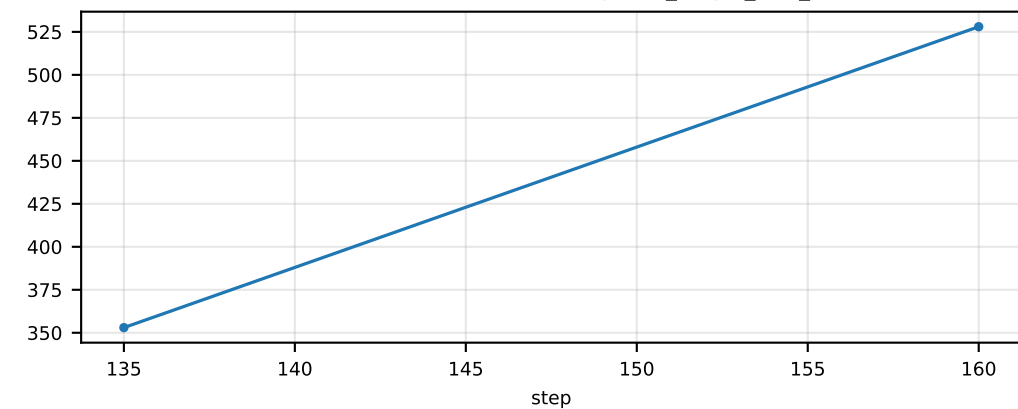
val/env/TicTacToe-first-100/response_length_turn_2



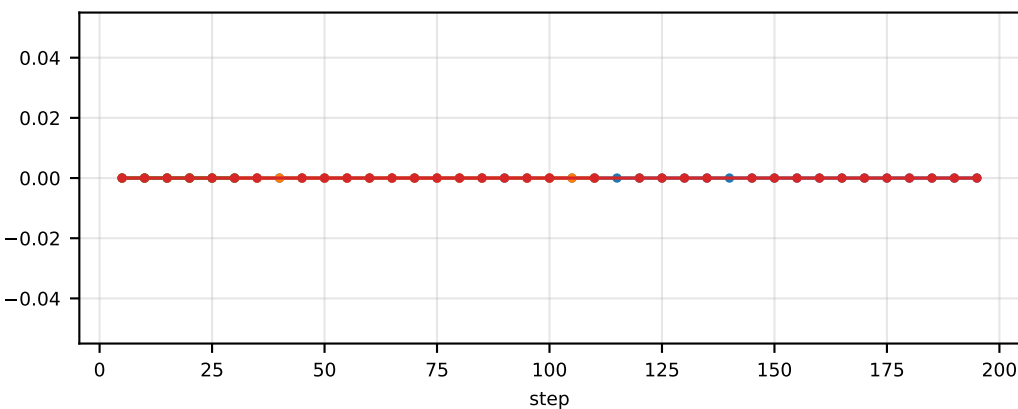
val/env/TicTacToe-first-100/response_length_turn_3



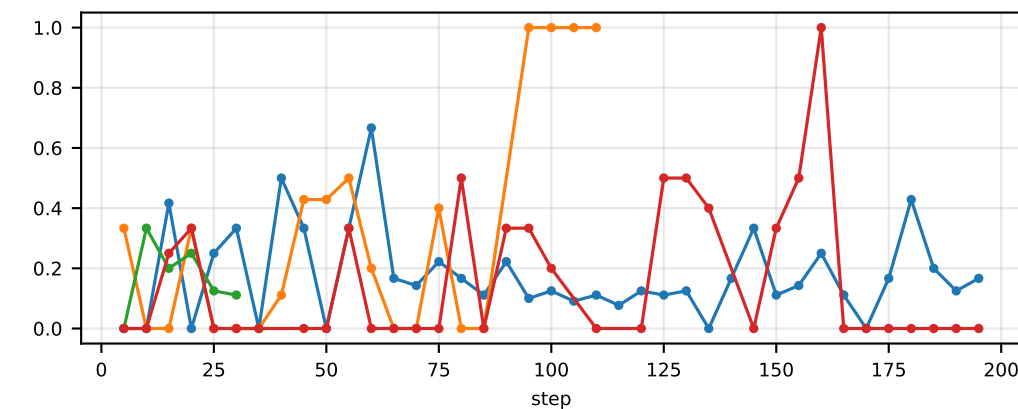
val/env/TicTacToe-first-100/response_length_turn_4



val/env/TicTacToe-first-100/success



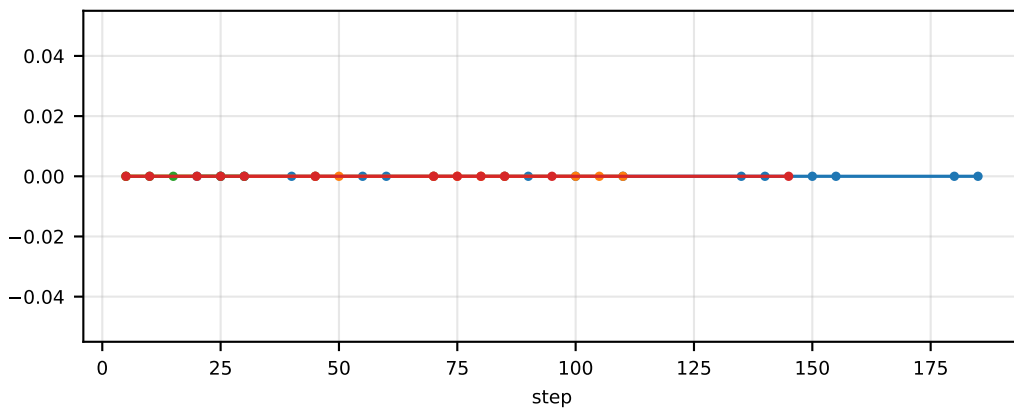
val/env/TicTacToe-first-100/winner



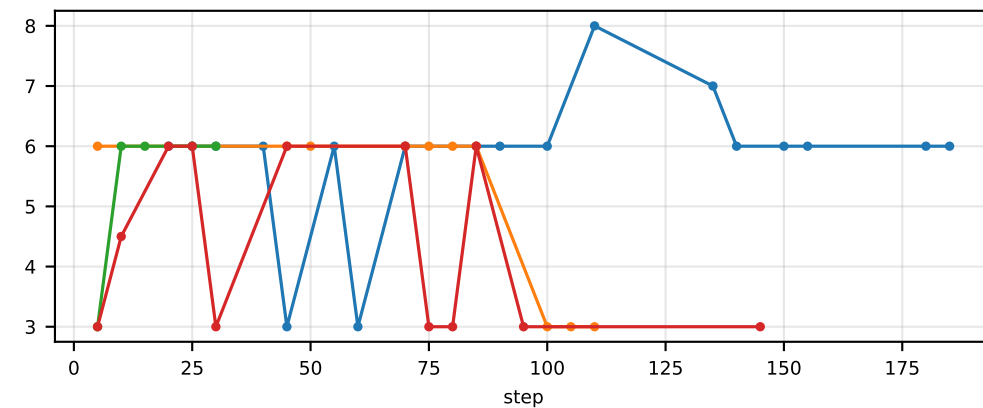
—●— m1b (ttt, lr1e-6, no min_lr)
 —●— opt2 (ttt, lr1e-6, min_lr1e-7)
 —●— m2b (ttt quickcheck)
 —●— opt1 (ttt, lr5e-6)

[TicTacToe] Validation (vs built-in opponents) (5/14)

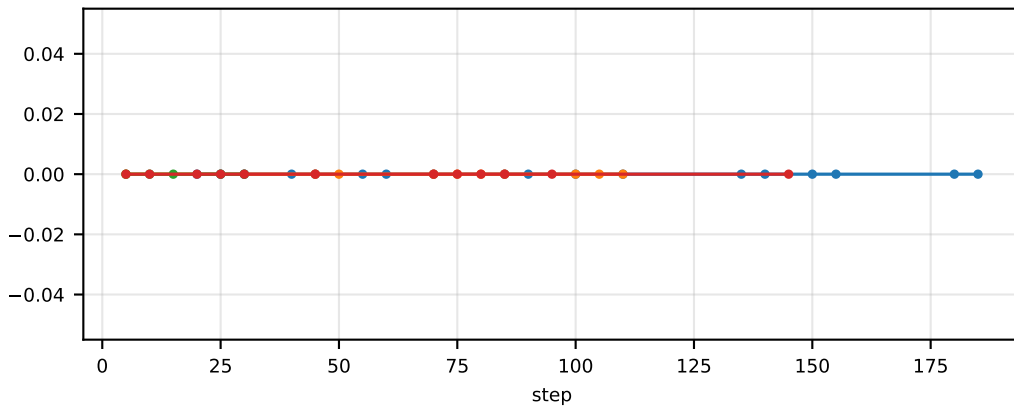
val/env/TicTacToe-first-1000/draw



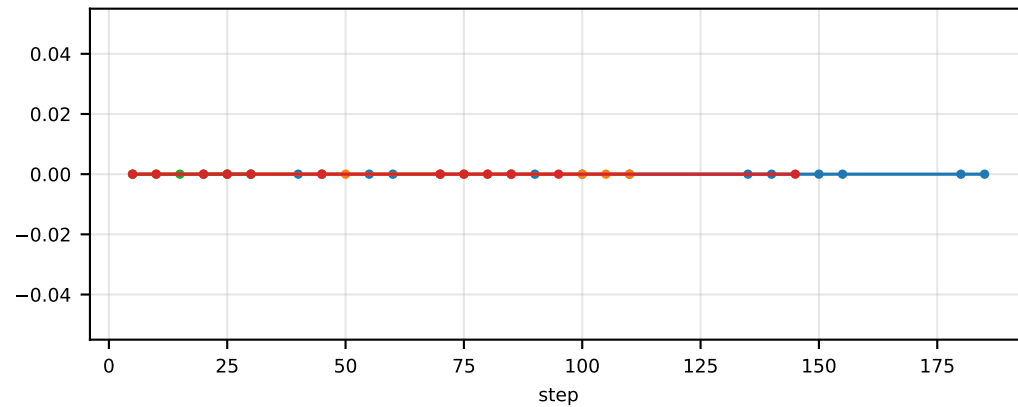
val/env/TicTacToe-first-1000/num_actions



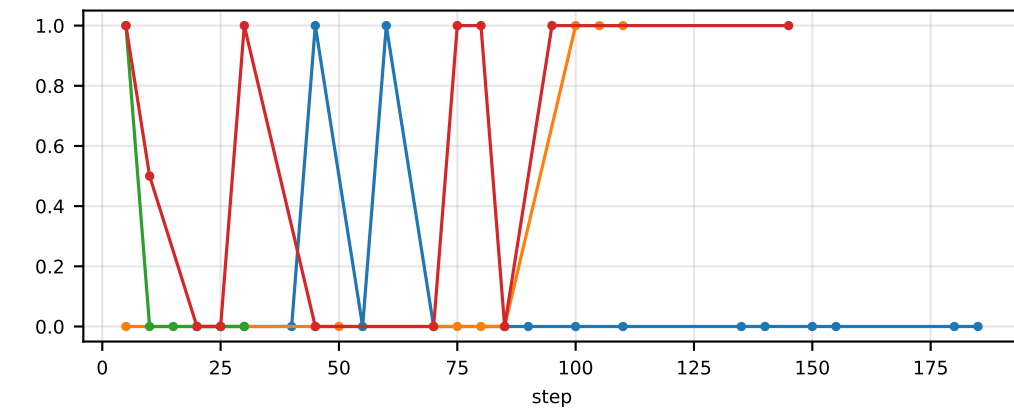
val/env/TicTacToe-first-1000/player_0_lose_for_overlong_response



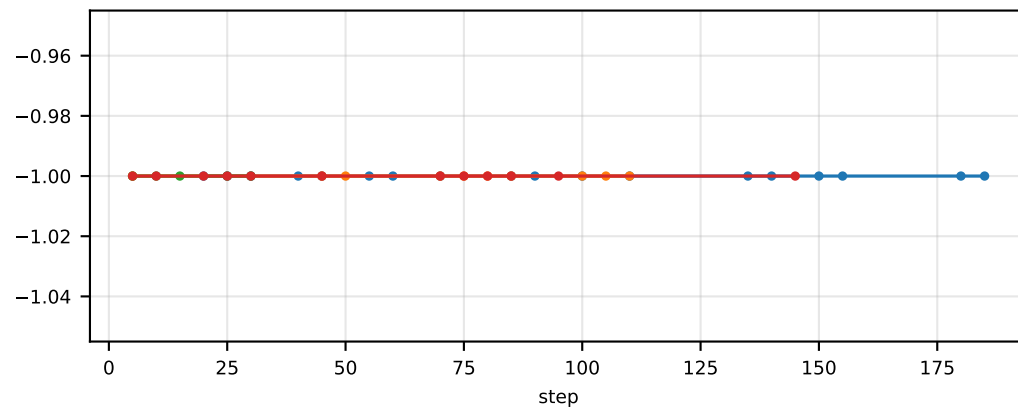
val/env/TicTacToe-first-1000/player_0_lose_for_overlong_sequence



val/env/TicTacToe-first-1000/player_0_lose_for_wrong_format



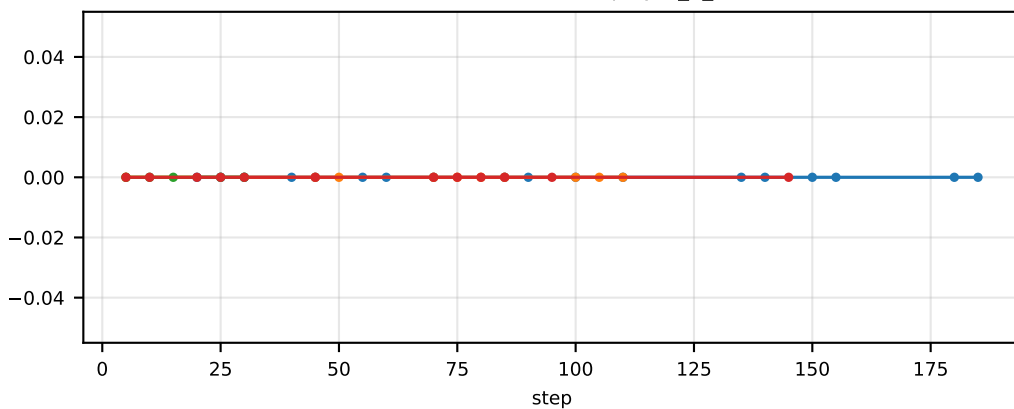
val/env/TicTacToe-first-1000/player_0_return



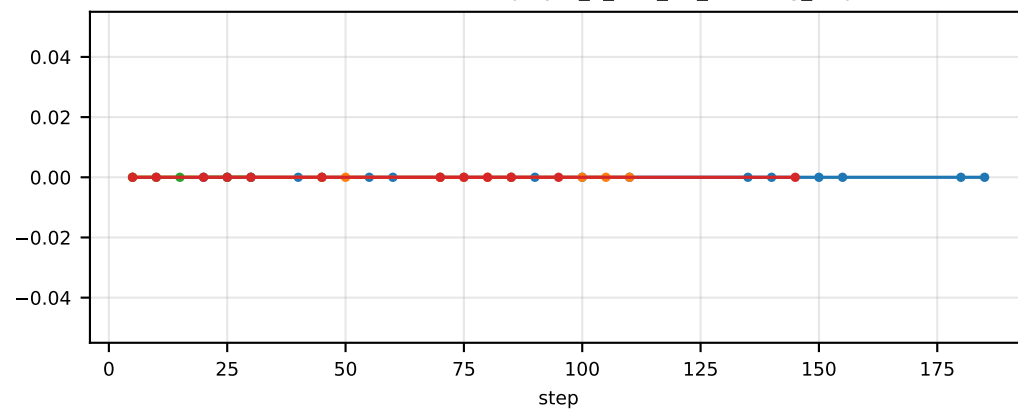
—●— m1b (ttd, lr1e-6, no min_lr)
 —●— opt2 (ttd, lr1e-6, min_lr1e-7)
 —●— m2b (ttd quickcheck)
 —●— opt1 (ttd, lr5e-6)

[TicTacToe] Validation (vs built-in opponents) (6/14)

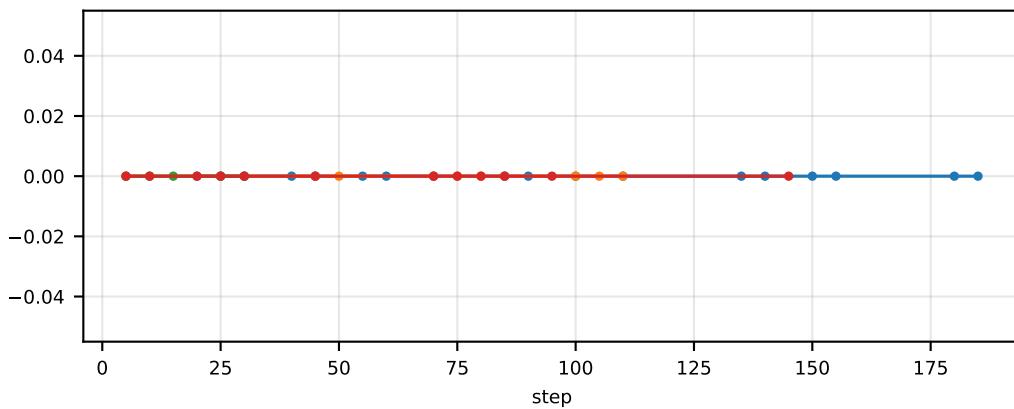
val/env/TicTacToe-first-1000/player_0_success



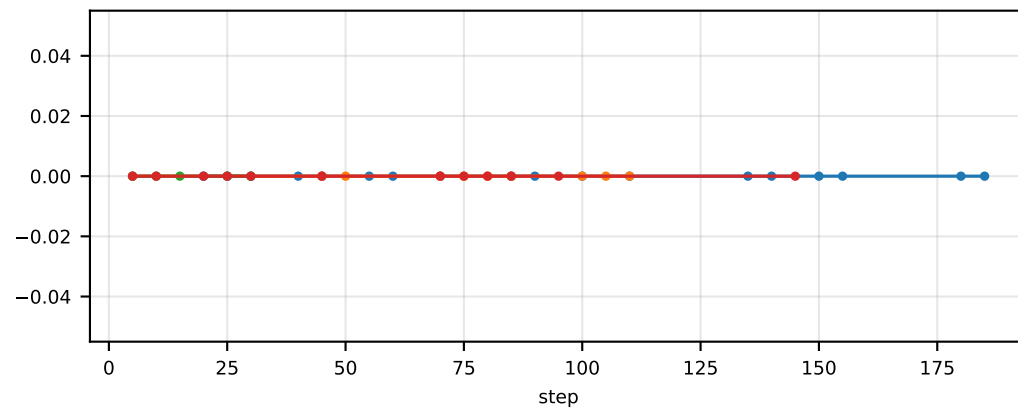
val/env/TicTacToe-first-1000/player_1_lose_for_overlong_response



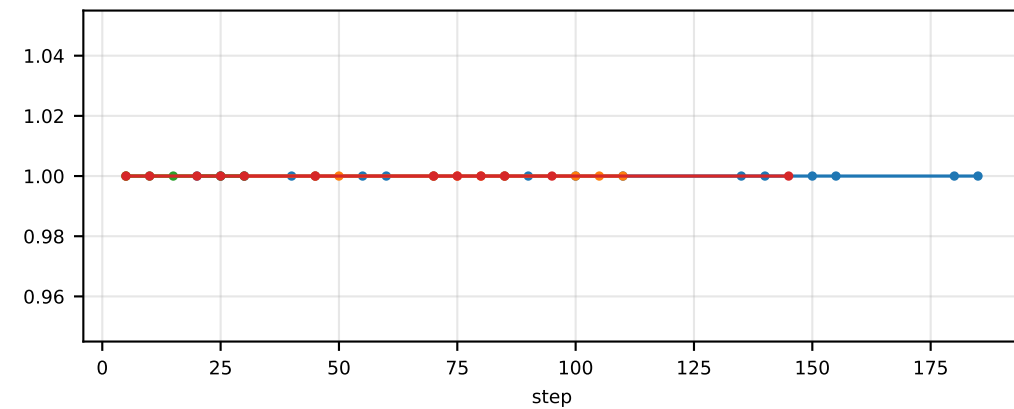
val/env/TicTacToe-first-1000/player_1_lose_for_overlong_sequence



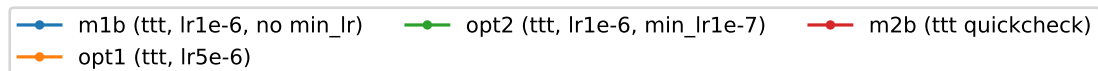
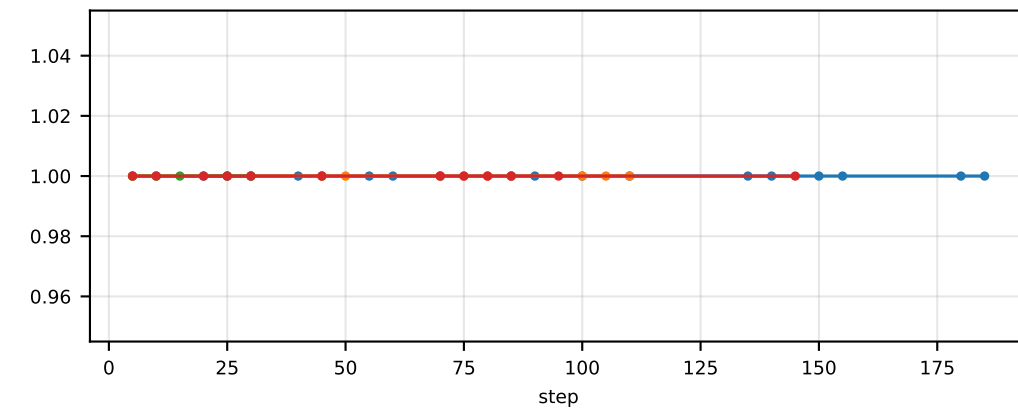
val/env/TicTacToe-first-1000/player_1_lose_for_wrong_format



val/env/TicTacToe-first-1000/player_1_return

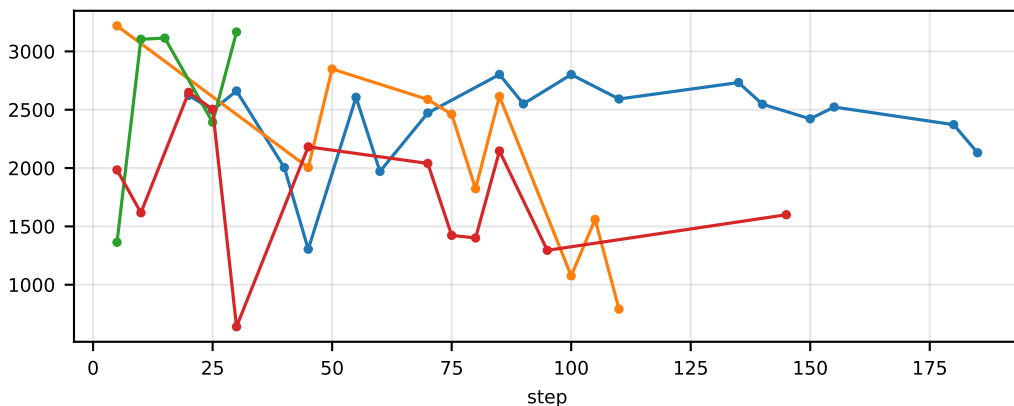


val/env/TicTacToe-first-1000/player_1_success

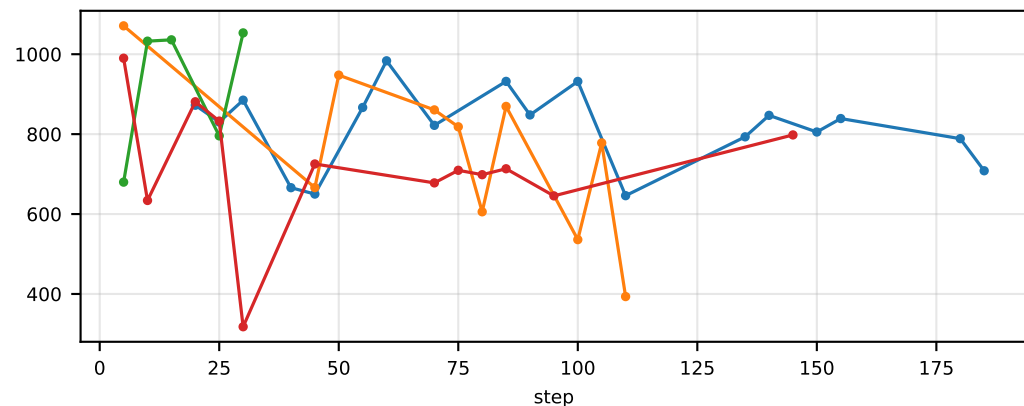


[TicTacToe] Validation (vs built-in opponents) (7/14)

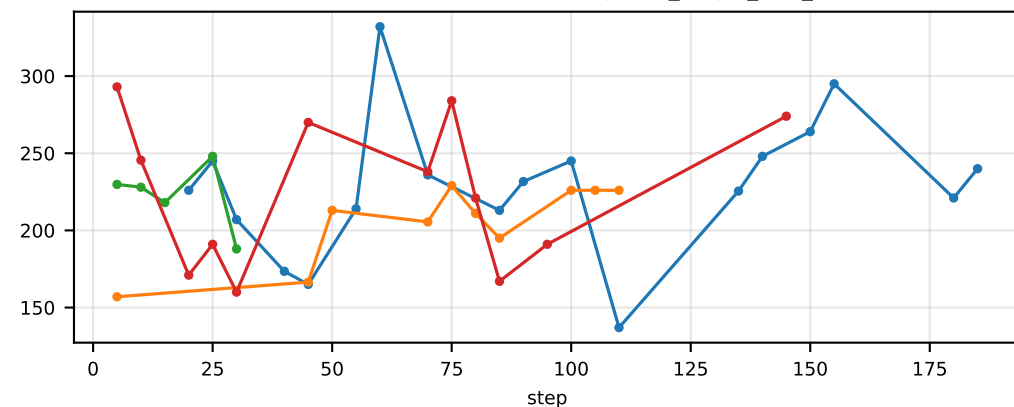
val/env/TicTacToe-first-1000/response_length



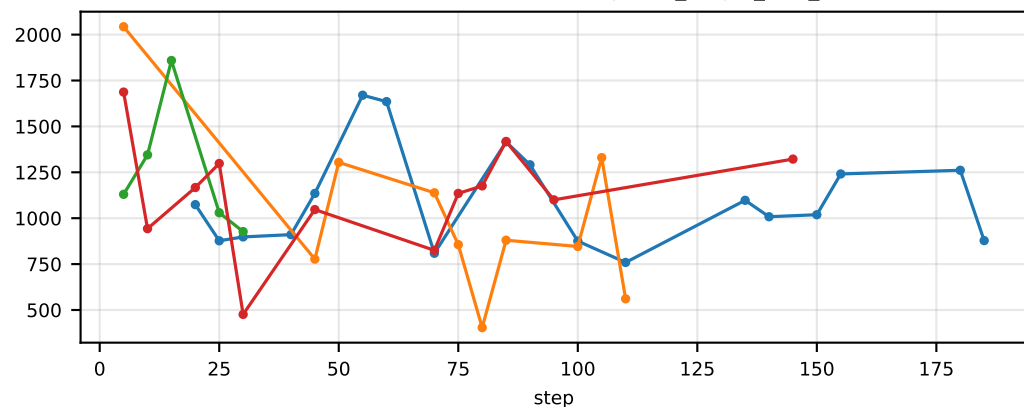
val/env/TicTacToe-first-1000/response_length_per_turn



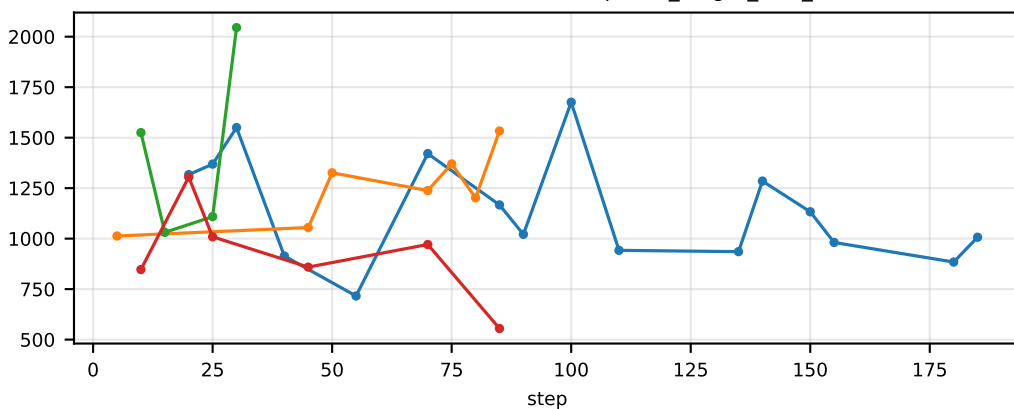
val/env/TicTacToe-first-1000/response_length_turn_0



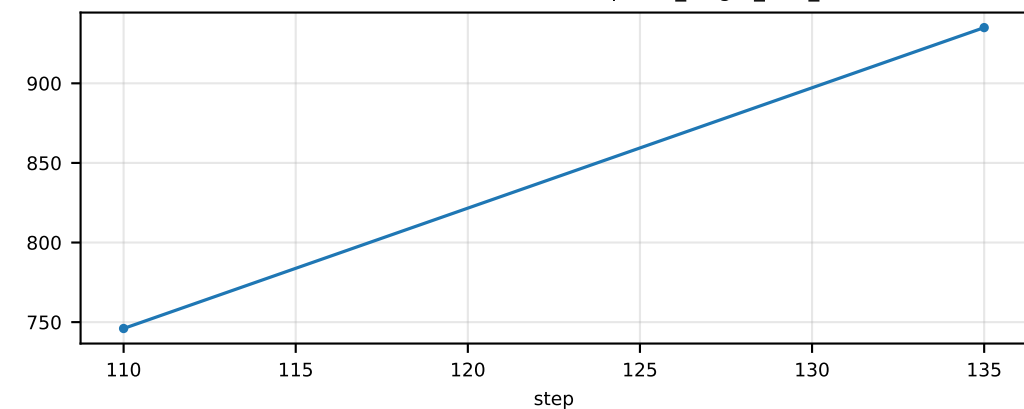
val/env/TicTacToe-first-1000/response_length_turn_1



val/env/TicTacToe-first-1000/response_length_turn_2



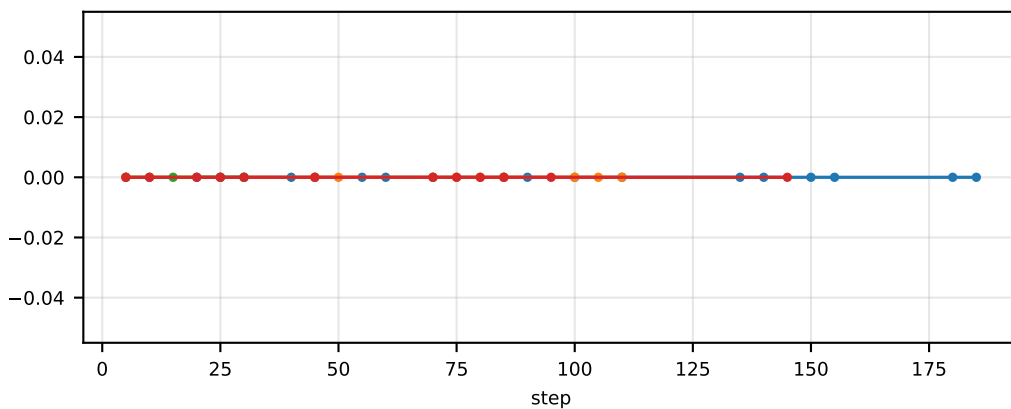
val/env/TicTacToe-first-1000/response_length_turn_3



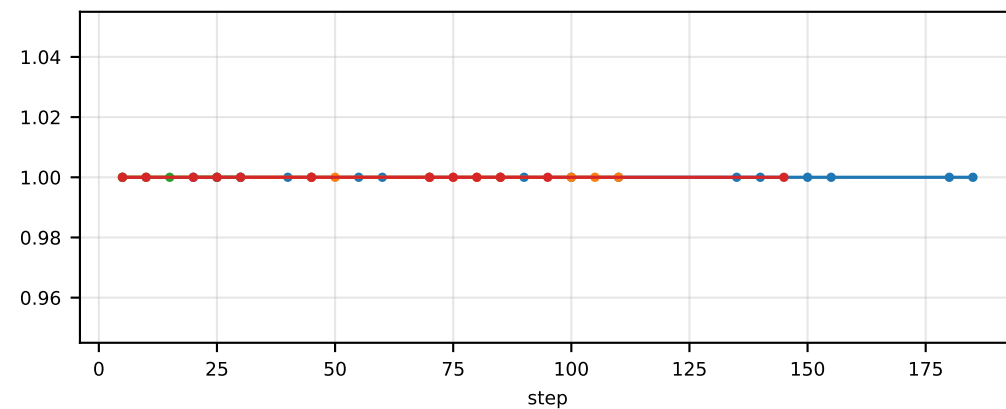
—●— m1b (ttt, lr1e-6, no min_lr)
 —●— opt2 (ttt, lr1e-6, min_lr1e-7)
 —●— m2b (ttt quickcheck)
 —●— opt1 (ttt, lr5e-6)

[TicTacToe] Validation (vs built-in opponents) (8/14)

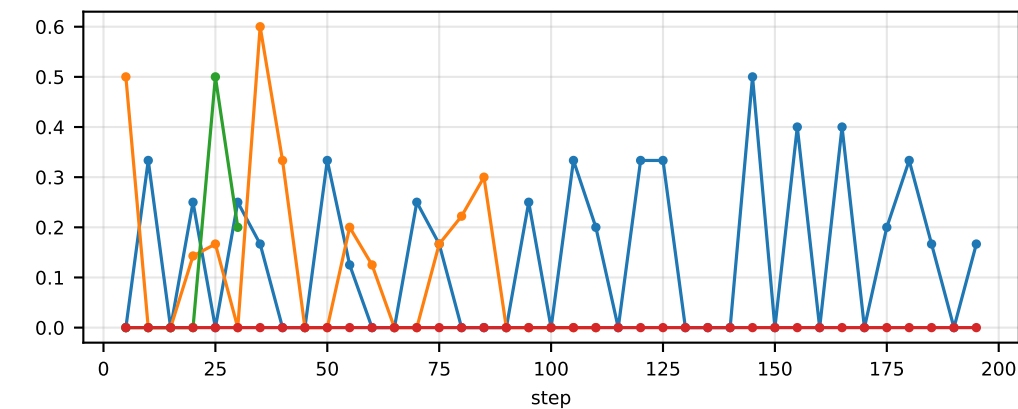
val/env/TicTacToe-first-1000/success



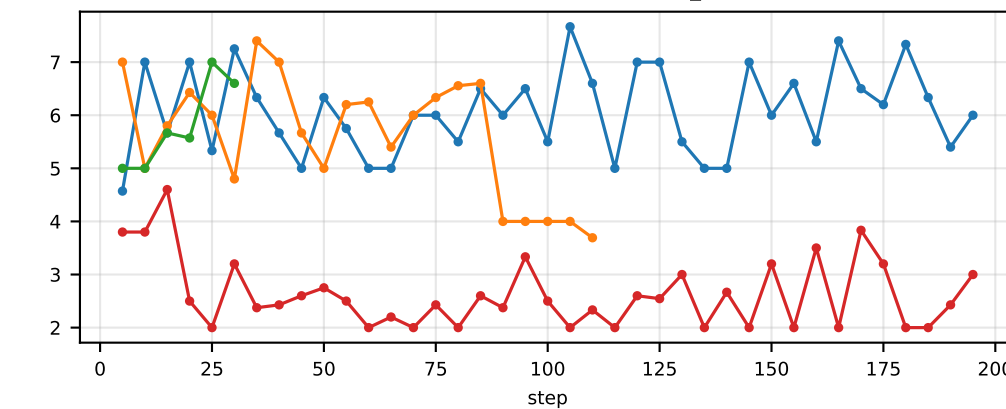
val/env/TicTacToe-first-1000/winner



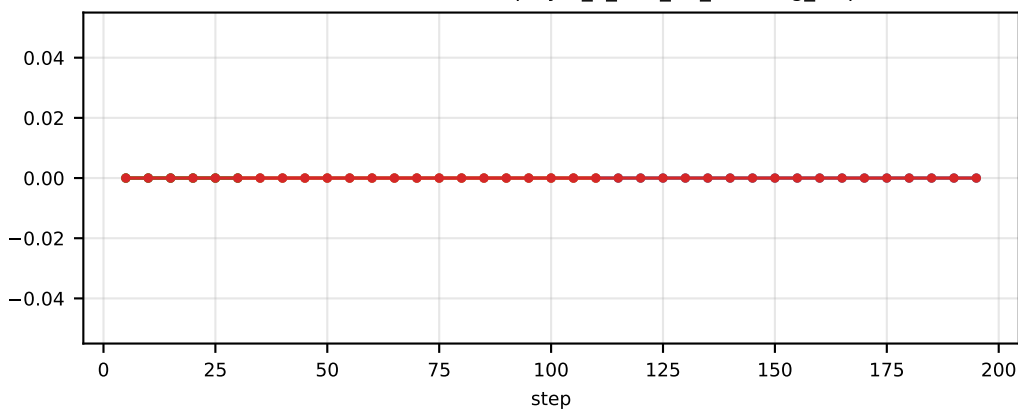
val/env/TicTacToe-second-100/draw



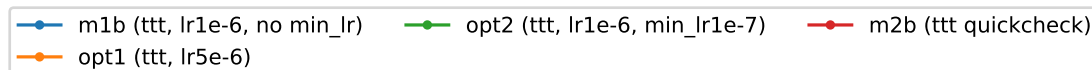
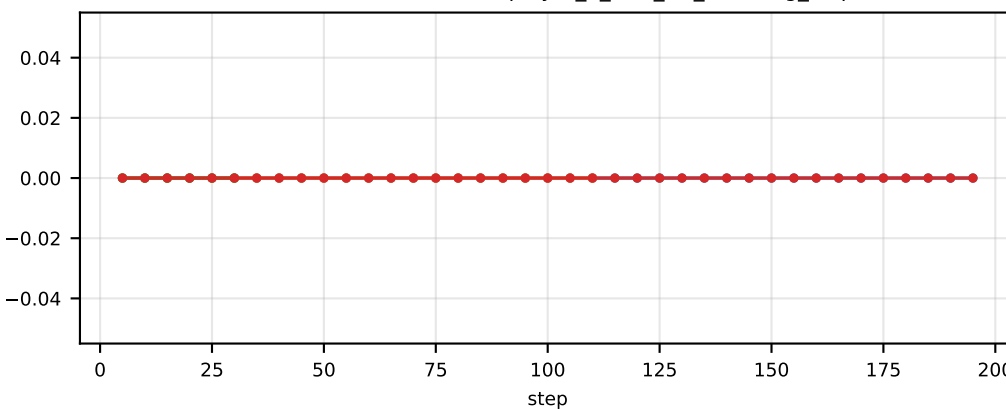
val/env/TicTacToe-second-100/num_actions



val/env/TicTacToe-second-100/player_0_lose_for_overlong_response

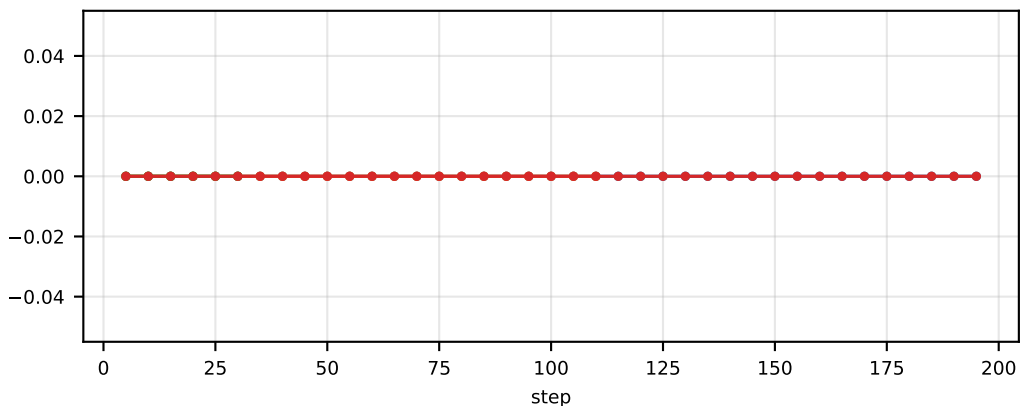


val/env/TicTacToe-second-100/player_0_lose_for_overlong_sequence

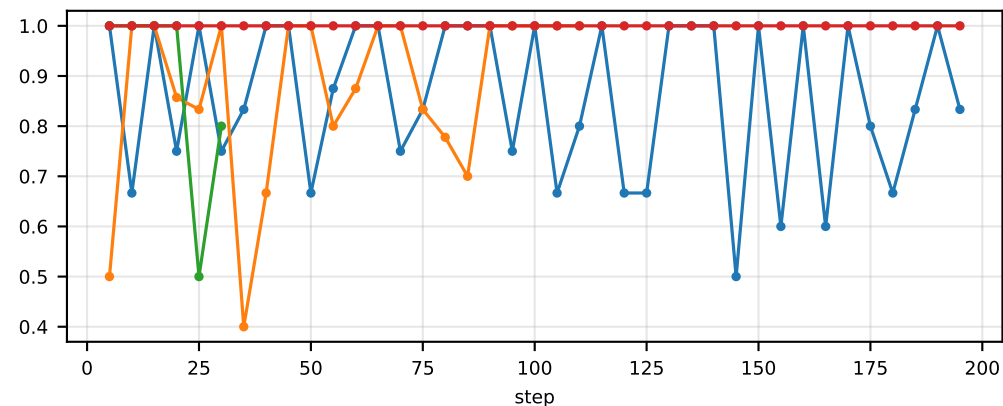


[TicTacToe] Validation (vs built-in opponents) (9/14)

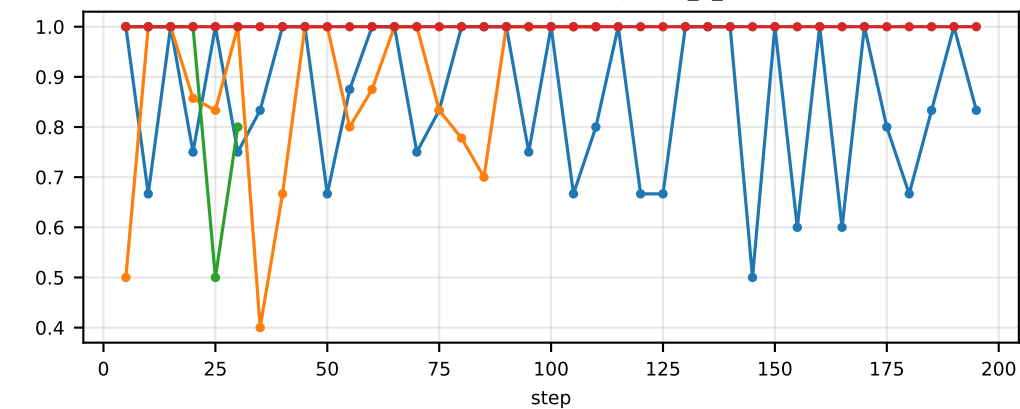
val/env/TicTacToe-second-100/player_0_lose_for_wrong_format



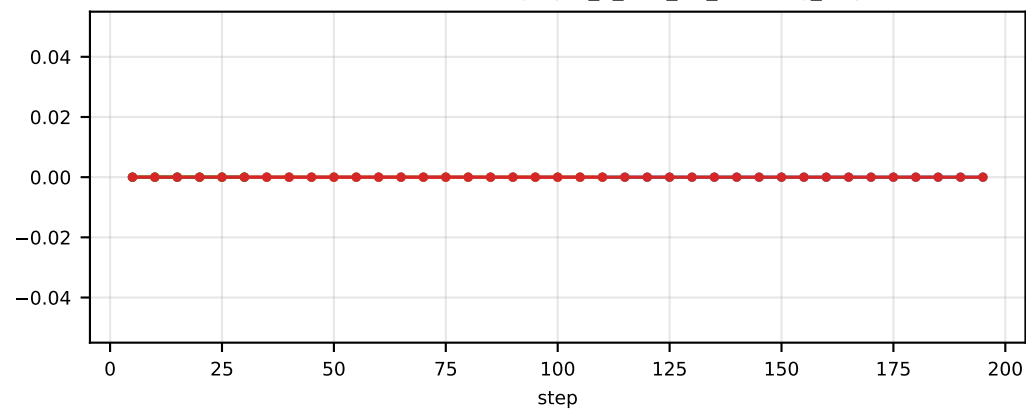
val/env/TicTacToe-second-100/player_0_return



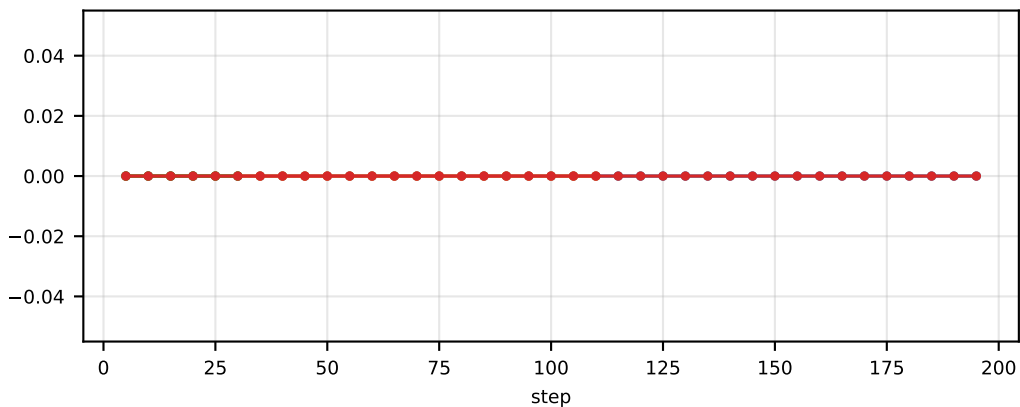
val/env/TicTacToe-second-100/player_0_success



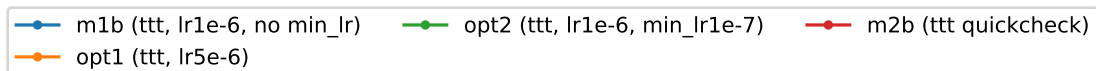
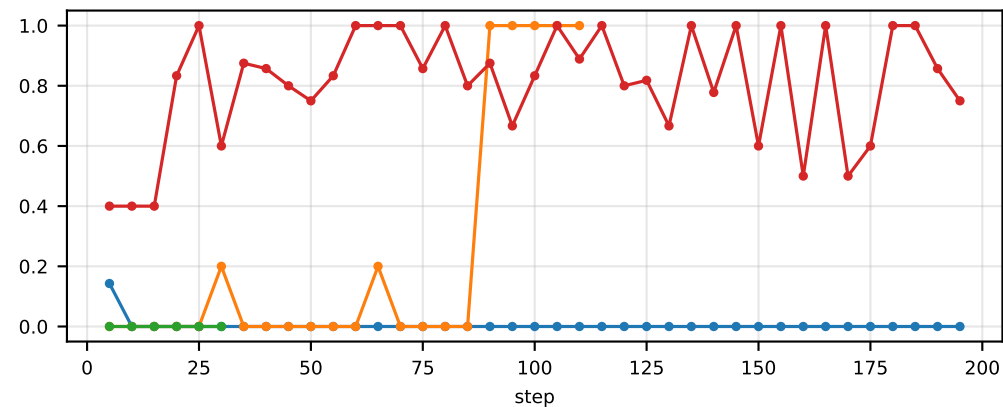
val/env/TicTacToe-second-100/player_1_lose_for_overlong_response



val/env/TicTacToe-second-100/player_1_lose_for_overlong_sequence

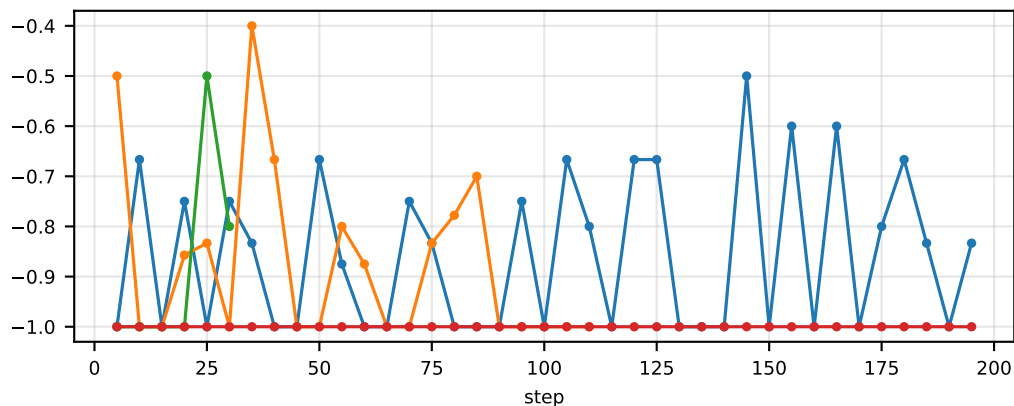


val/env/TicTacToe-second-100/player_1_lose_for_wrong_format

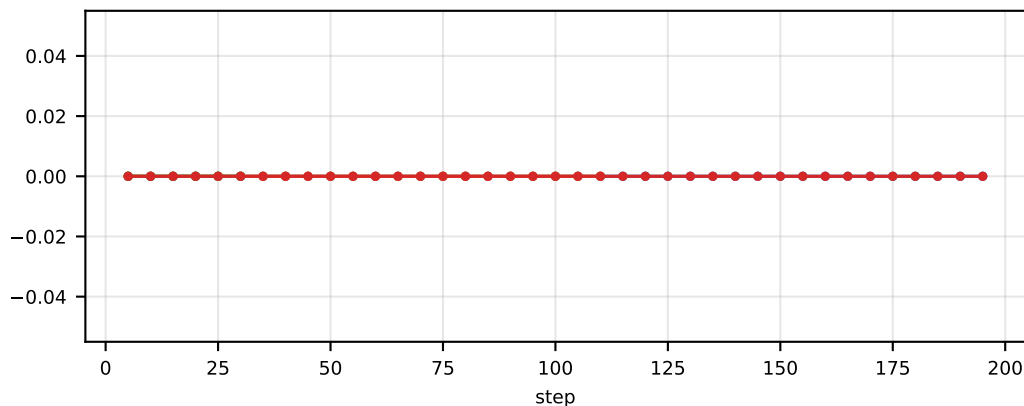


[TicTacToe] Validation (vs built-in opponents) (10/14)

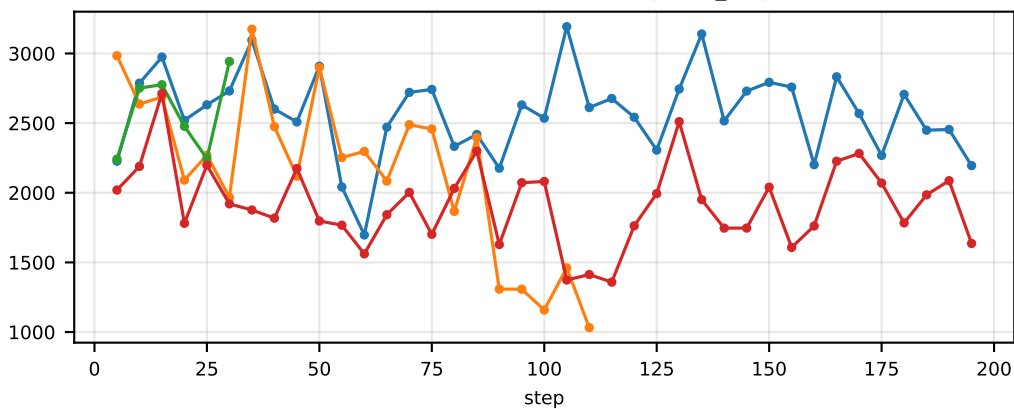
val/env/TicTacToe-second-100/player_1_return



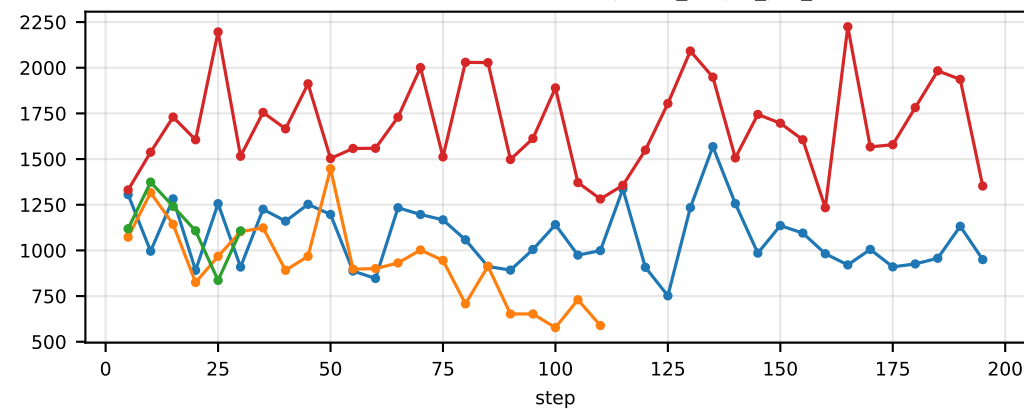
val/env/TicTacToe-second-100/player_1_success



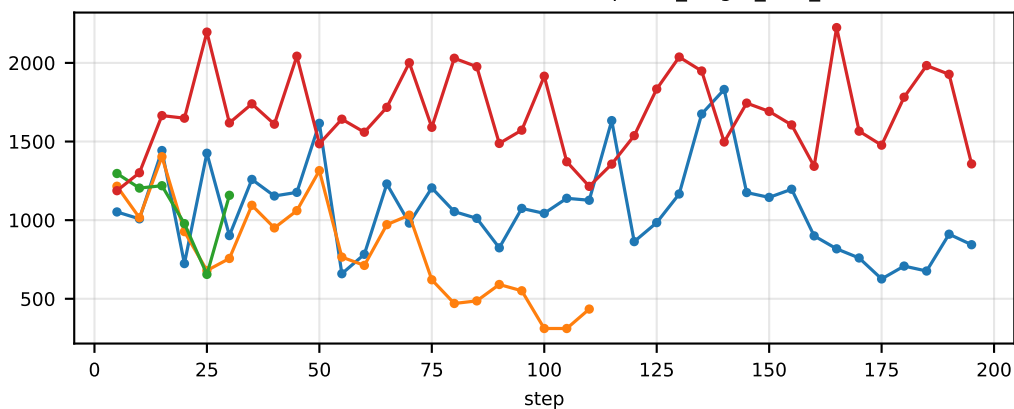
val/env/TicTacToe-second-100/response_length



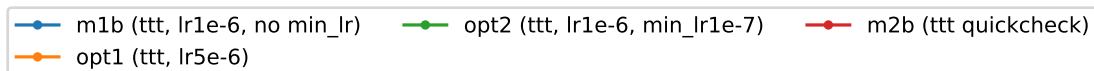
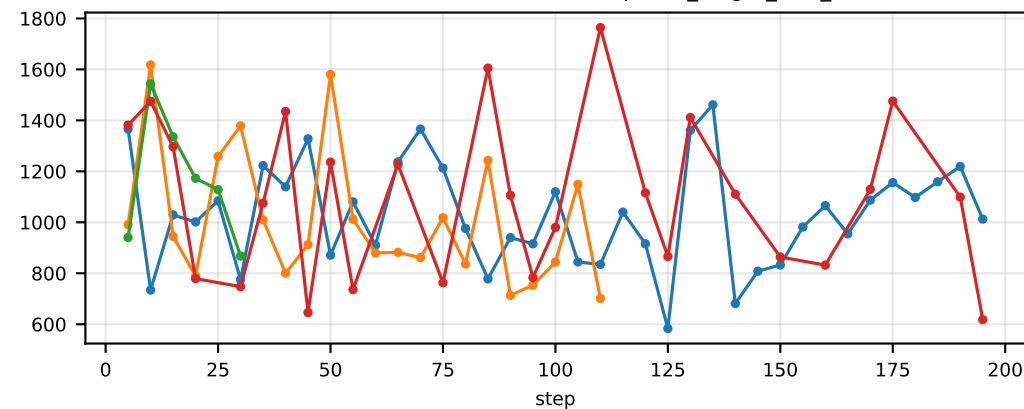
val/env/TicTacToe-second-100/response_length_per_turn



val/env/TicTacToe-second-100/response_length_turn_0

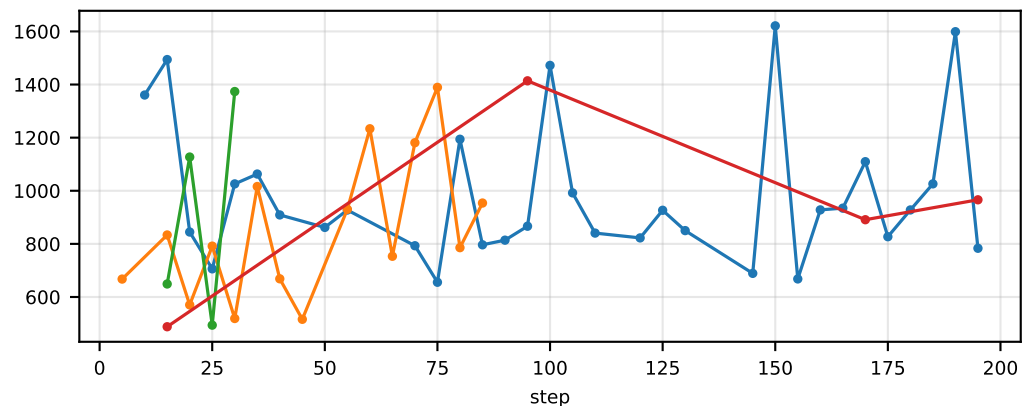


val/env/TicTacToe-second-100/response_length_turn_1

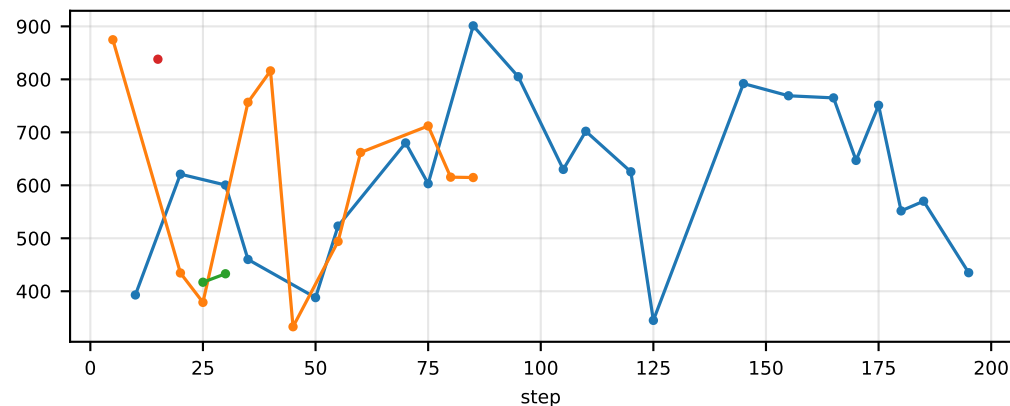


[TicTacToe] Validation (vs built-in opponents) (11/14)

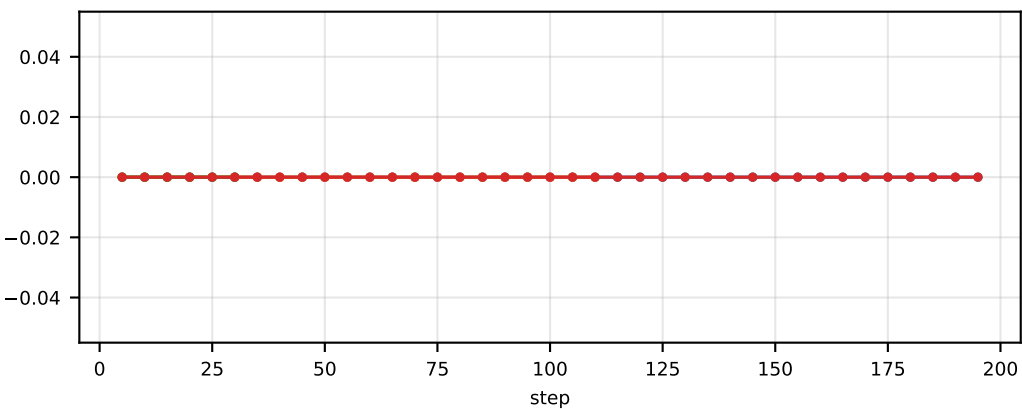
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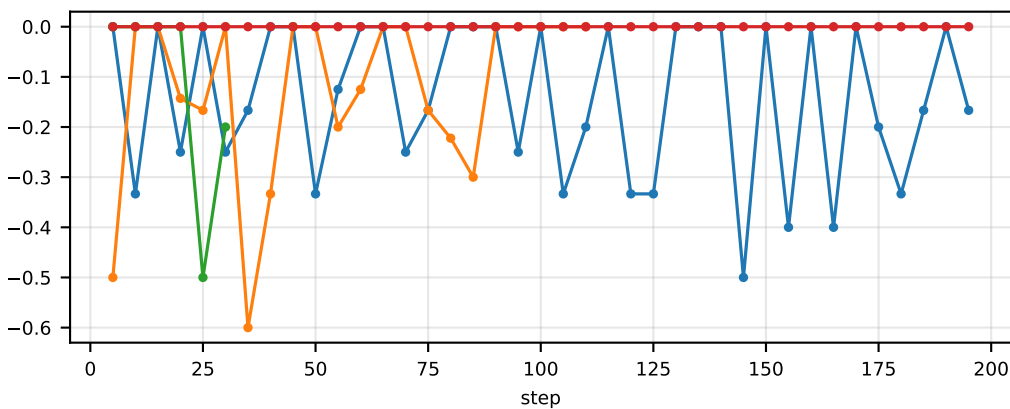
val/env/TicTacToe-second-100/response_length_turn_3



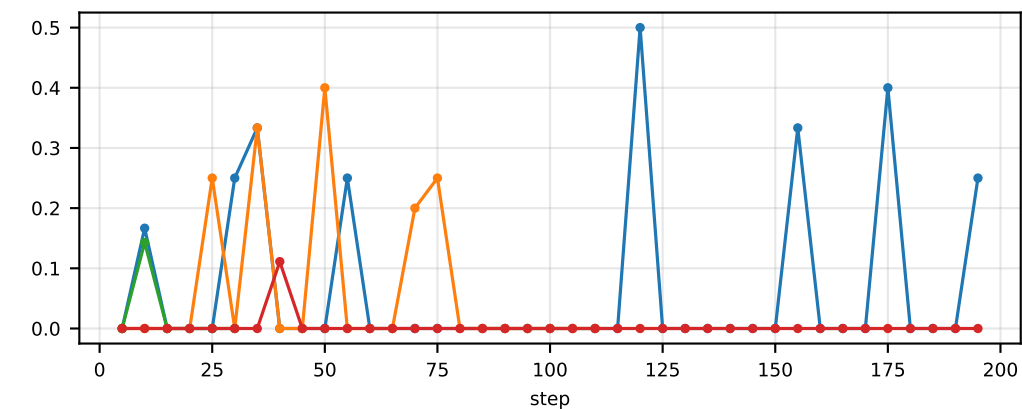
val/env/TicTacToe-second-100/success



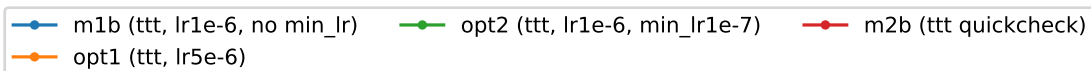
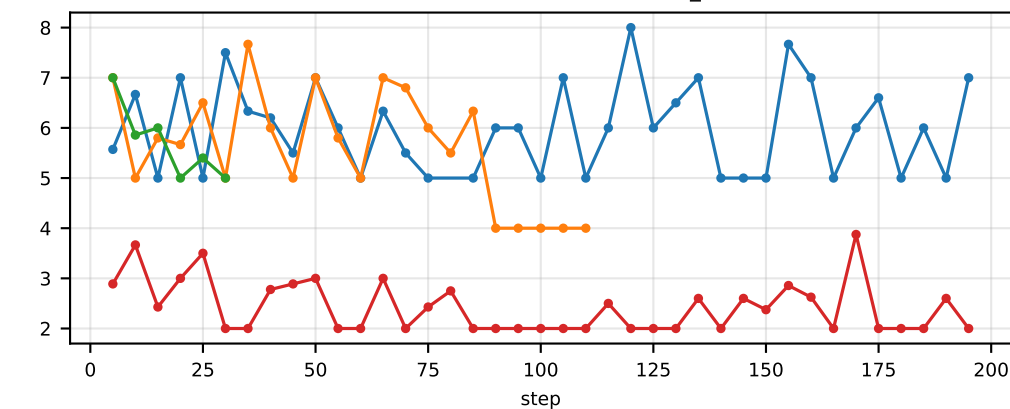
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val/env/TicTacToe-second-1000/draw

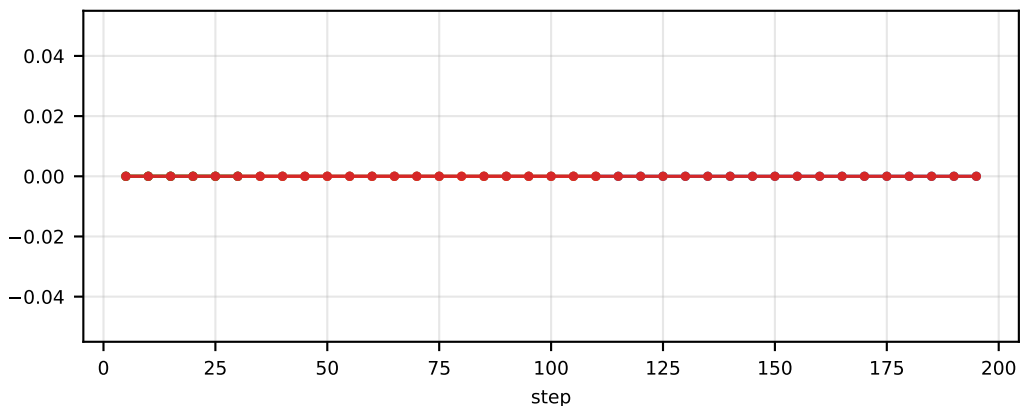


val/env/TicTacToe-second-1000/num_actions

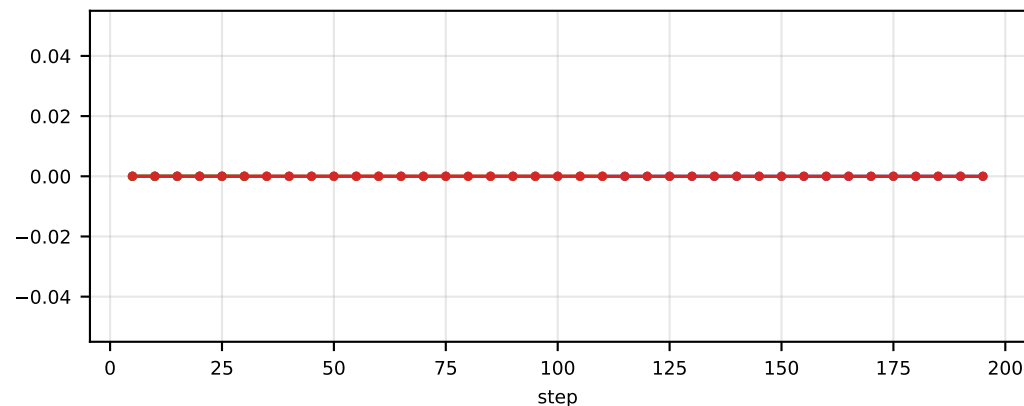


[TicTacToe] Validation (vs built-in opponents) (12/14)

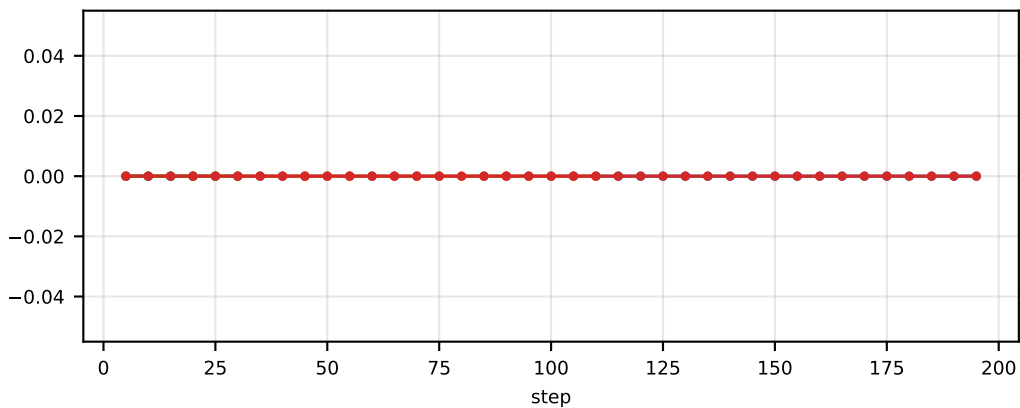
val/env/TicTacToe-second-1000/player_0_lose_for_overlong_response



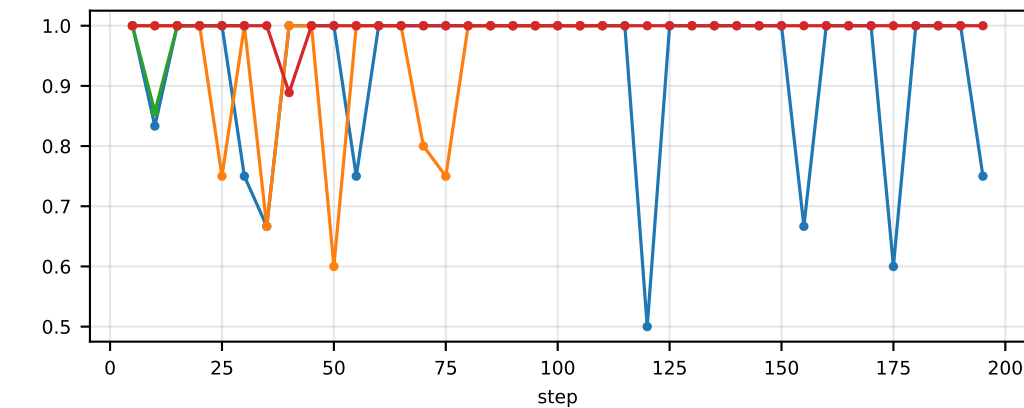
val/env/TicTacToe-second-1000/player_0_lose_for_overlong_sequence



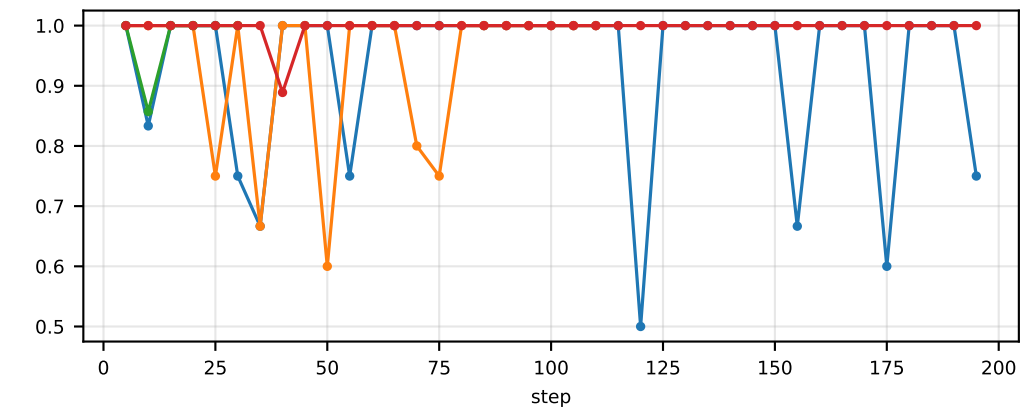
val/env/TicTacToe-second-1000/player_0_lose_for_wrong_format



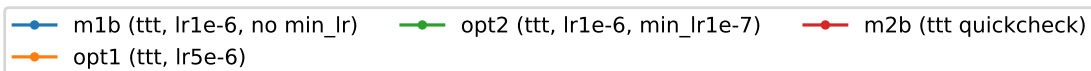
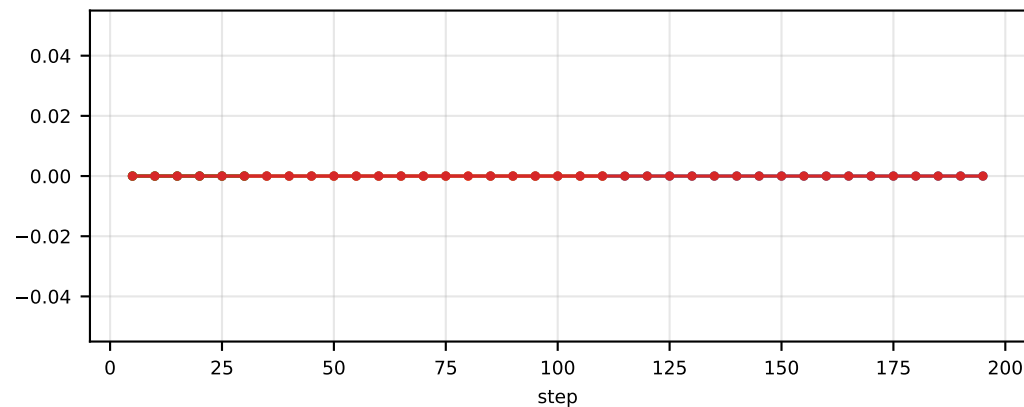
val/env/TicTacToe-second-1000/player_0_return



val/env/TicTacToe-second-1000/player_0_success

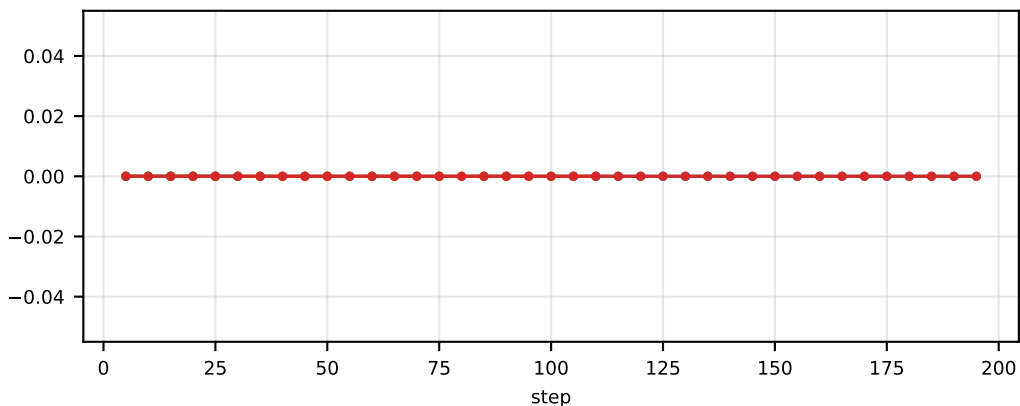


val/env/TicTacToe-second-1000/player_1_lose_for_overlong_response

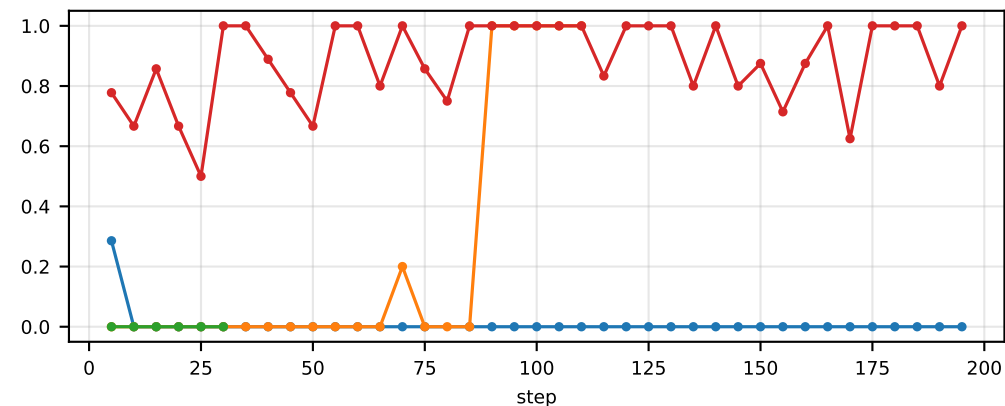


[TicTacToe] Validation (vs built-in opponents) (13/14)

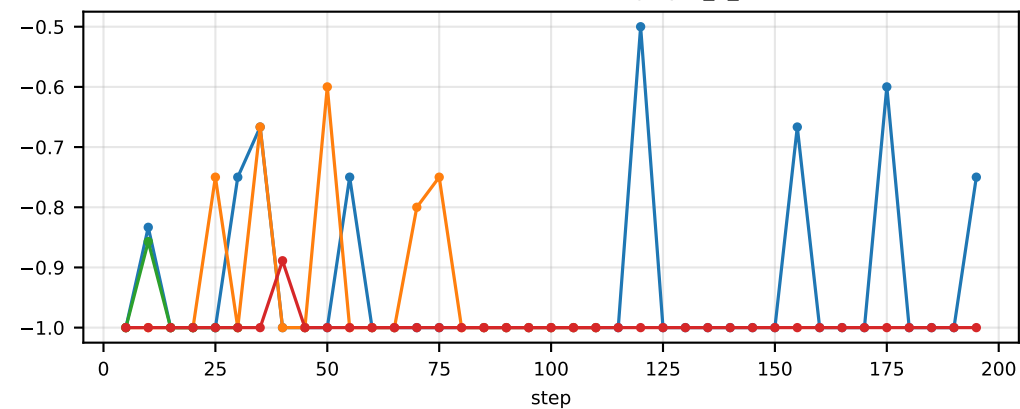
val/env/TicTacToe-second-1000/player_1_lose_for_overlong_sequence



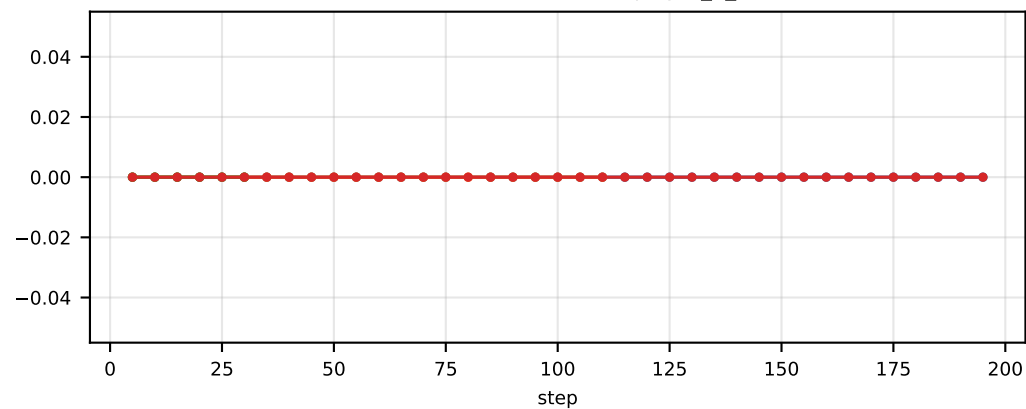
val/env/TicTacToe-second-1000/player_1_lose_for_wrong_format



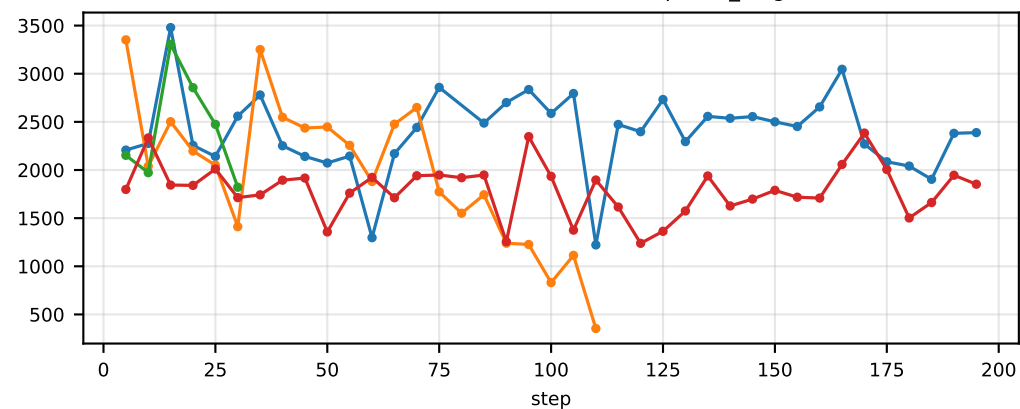
val/env/TicTacToe-second-1000/player_1_return



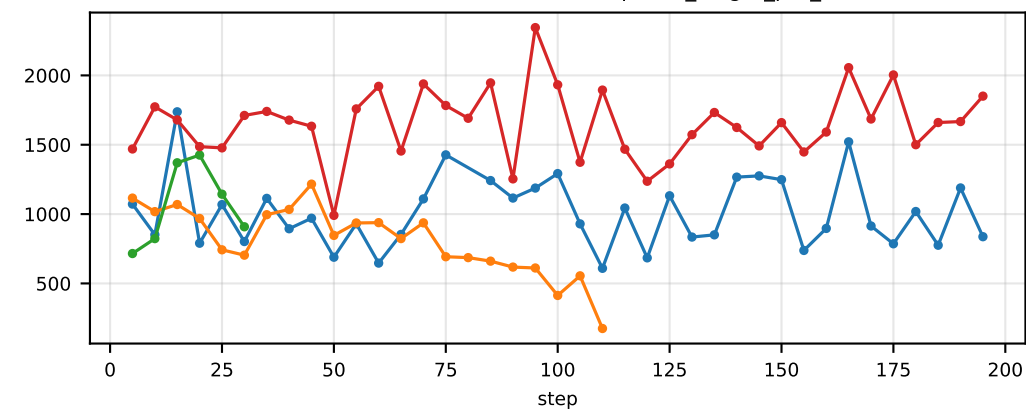
val/env/TicTacToe-second-1000/player_1_success



val/env/TicTacToe-second-1000/response_length



val/env/TicTacToe-second-1000/response_length_per_turn

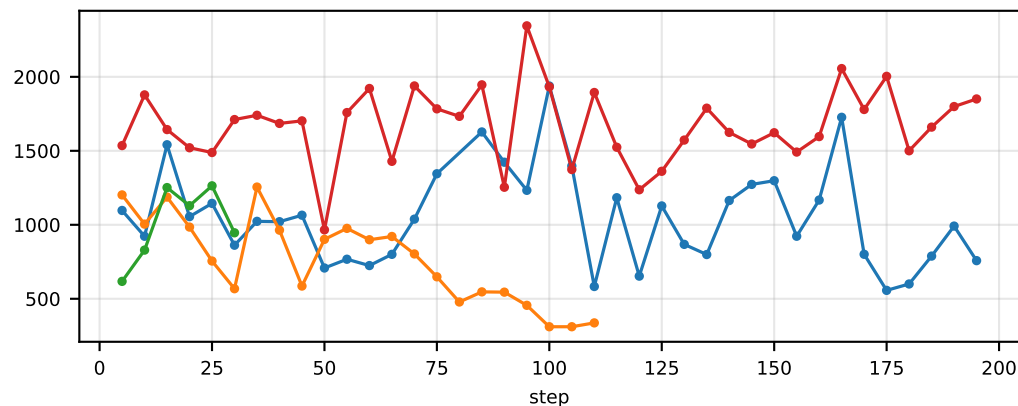


—●— m1b (ttt, lr1e-6, no min_lr)
 —●— opt2 (ttt, lr1e-6, min_lr1e-7)
 —●— m2b (ttt quickcheck)

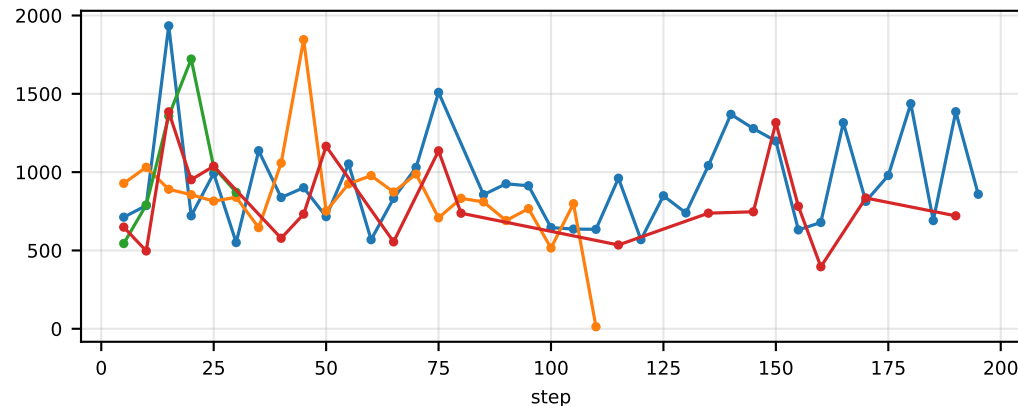
—●— opt1 (ttt, lr5e-6)

[TicTacToe] Validation (vs built-in opponents) (14/14)

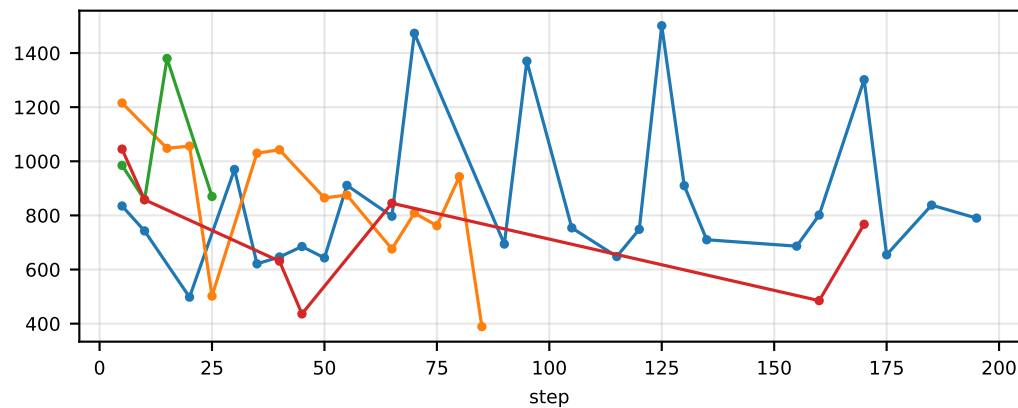
val/env/TicTacToe-second-1000/response_length_turn_0



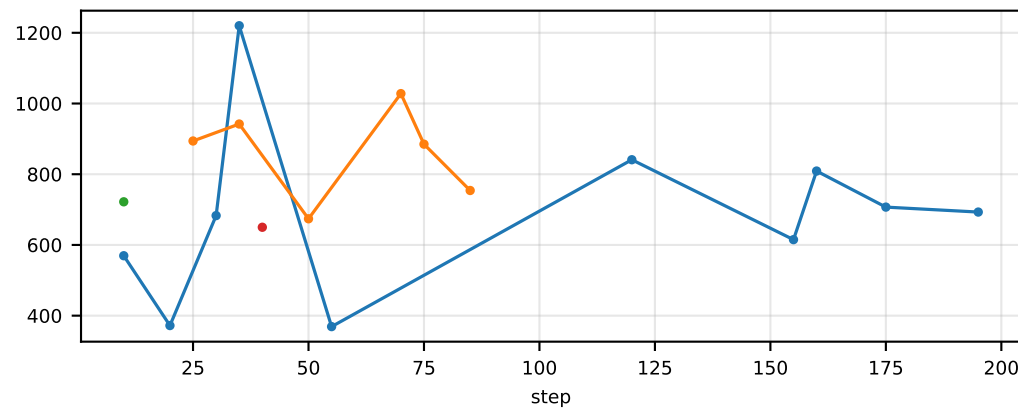
val/env/TicTacToe-second-1000/response_length_turn_1



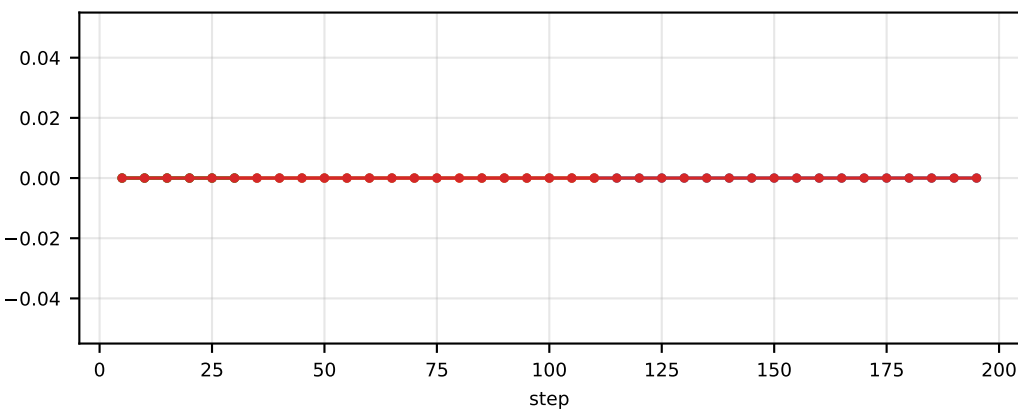
val/env/TicTacToe-second-1000/response_length_turn_2



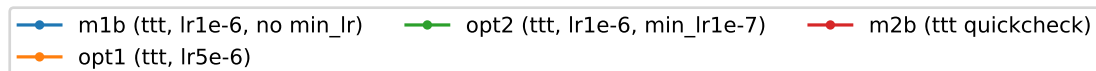
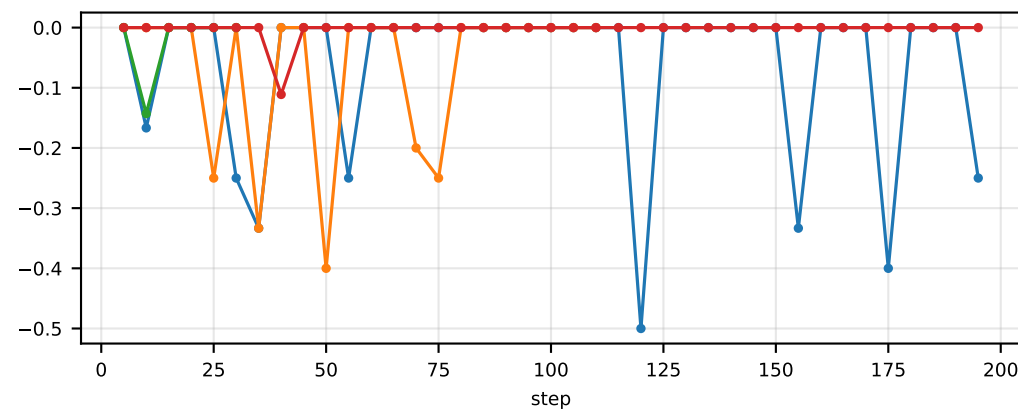
val/env/TicTacToe-second-1000/response_length_turn_3



val/env/TicTacToe-second-1000/success



val/env/TicTacToe-second-1000/winner

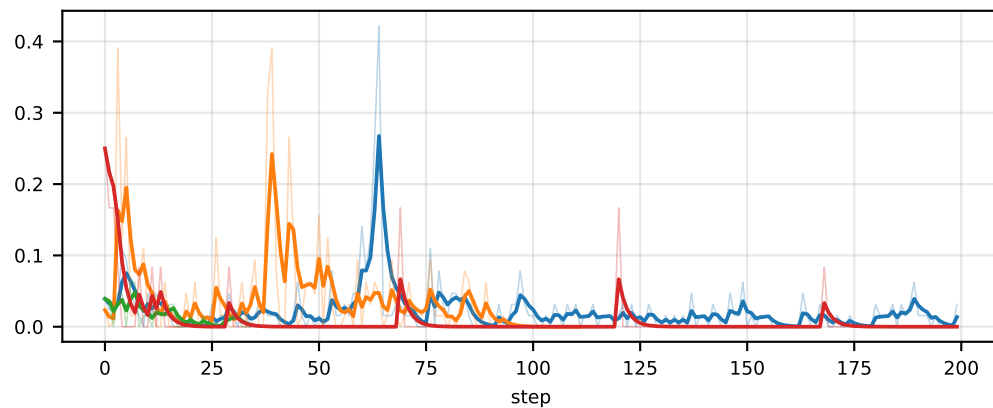


[TicTacToe] Self-play outcome (train) (1/1)

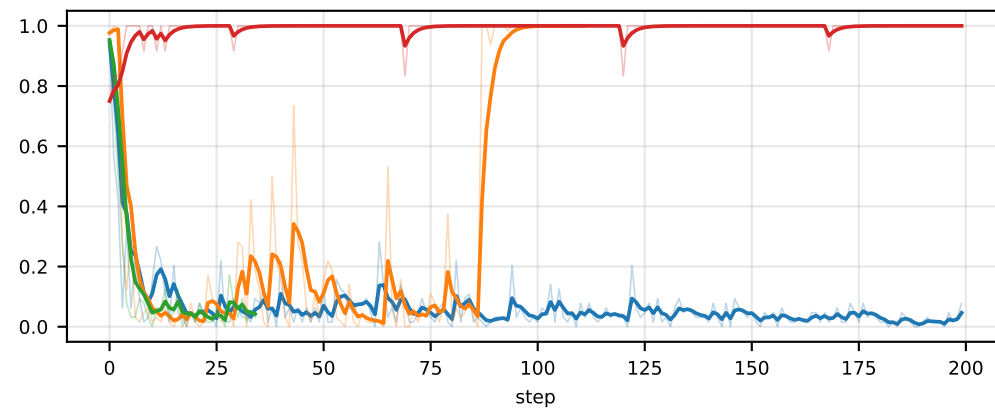


[TicTacToe] Losses from format / length (1/1)

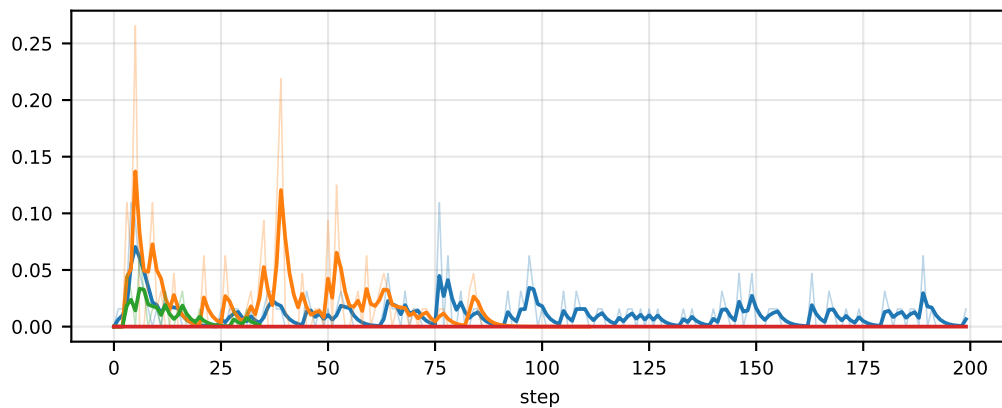
env/TicTacToe/player_0_lose_for_wrong_format



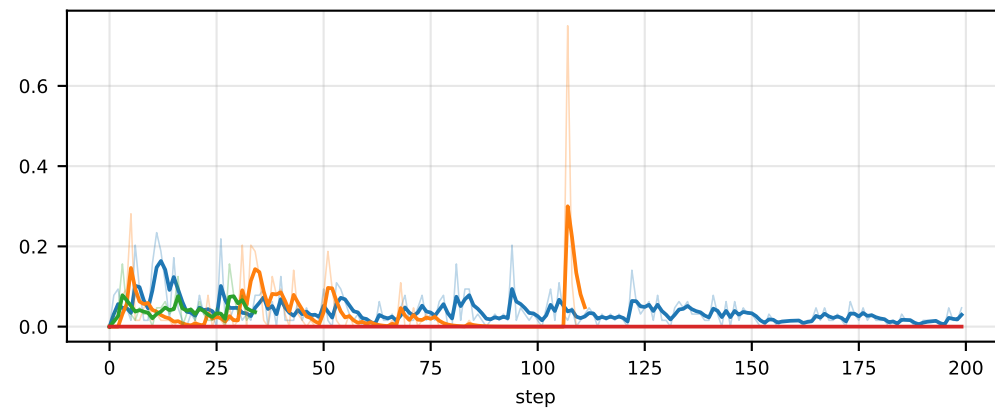
env/TicTacToe/player_1_lose_for_wrong_format



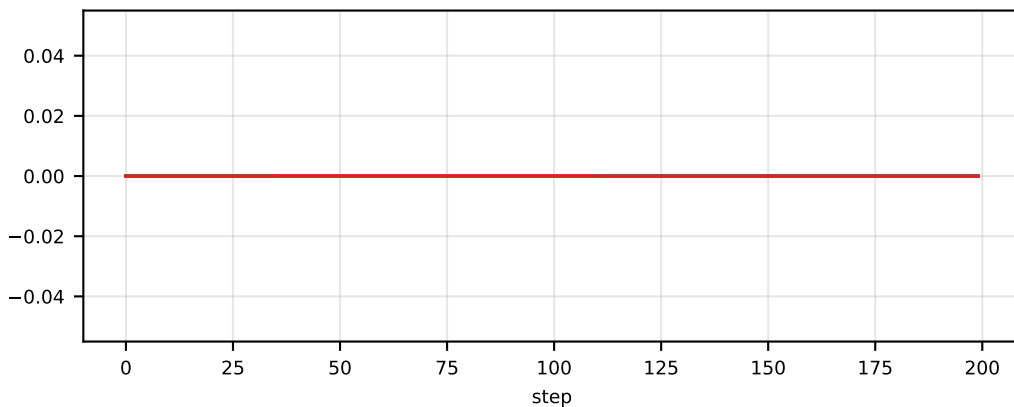
env/TicTacToe/player_0_lose_for_overlong_response



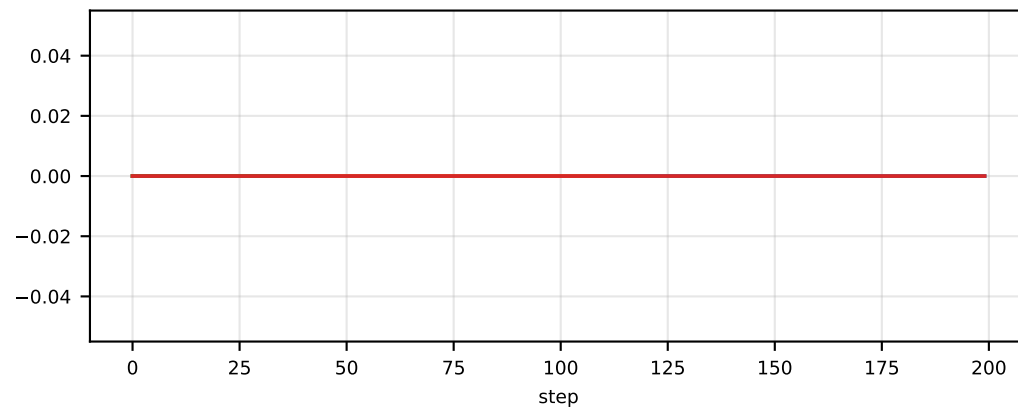
env/TicTacToe/player_1_lose_for_overlong_response



env/TicTacToe/player_0_lose_for_overlong_sequence



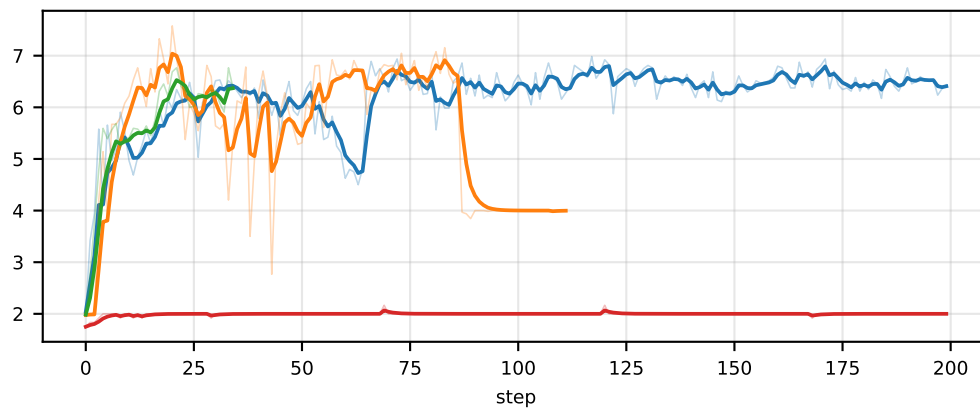
env/TicTacToe/player_1_lose_for_overlong_sequence



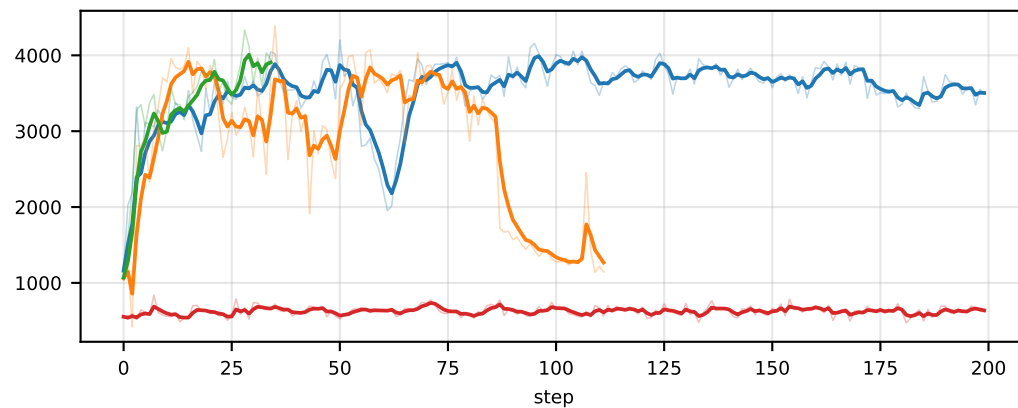
— m1b (ttt, lr1e-6, no min_lr) — opt2 (ttt, lr1e-6, min_lr1e-7) — m2b (ttt quickcheck)
— opt1 (ttt, lr5e-6)

[TicTacToe] Game length & response length (1/4)

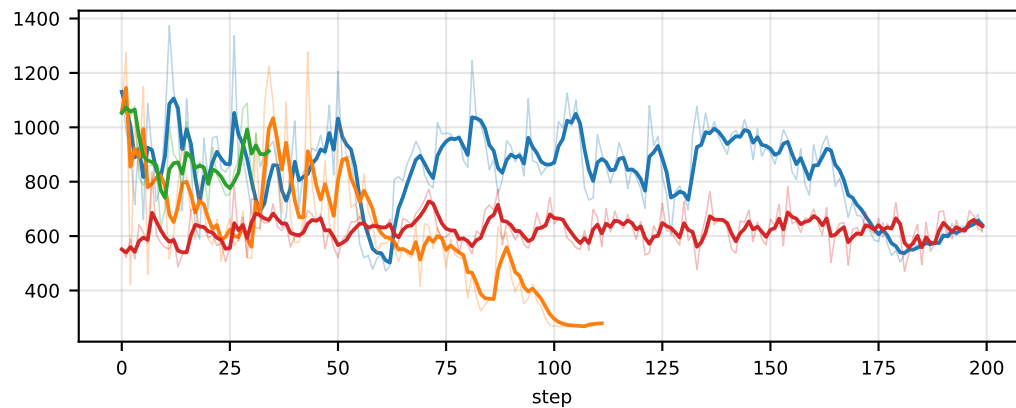
env/TicTacToe/num_actions



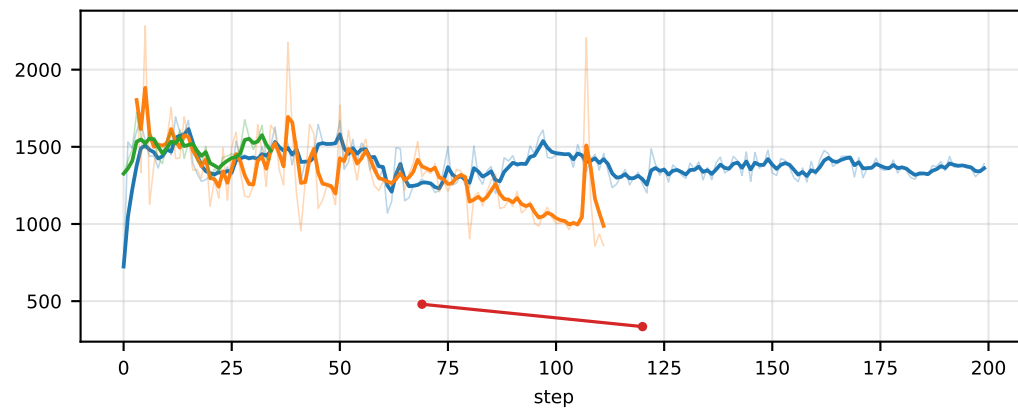
env/TicTacToe/response_length



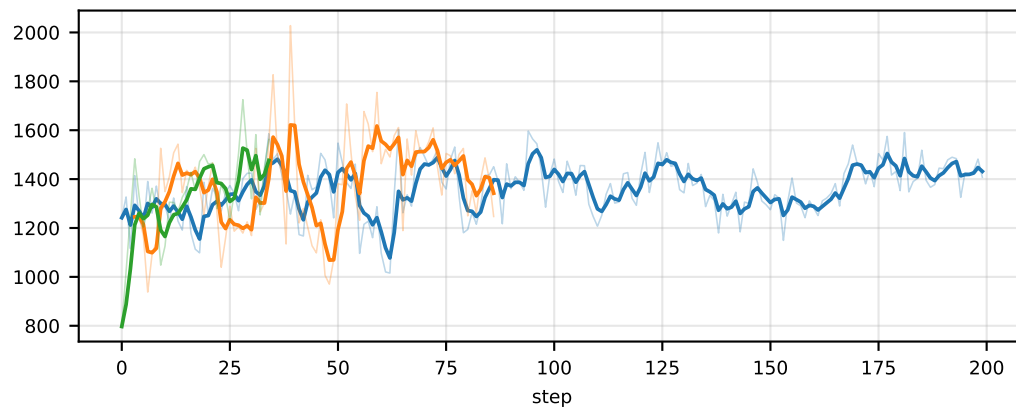
env/TicTacToe/response_length_turn_0



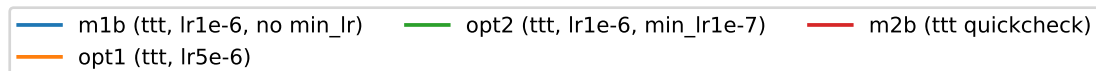
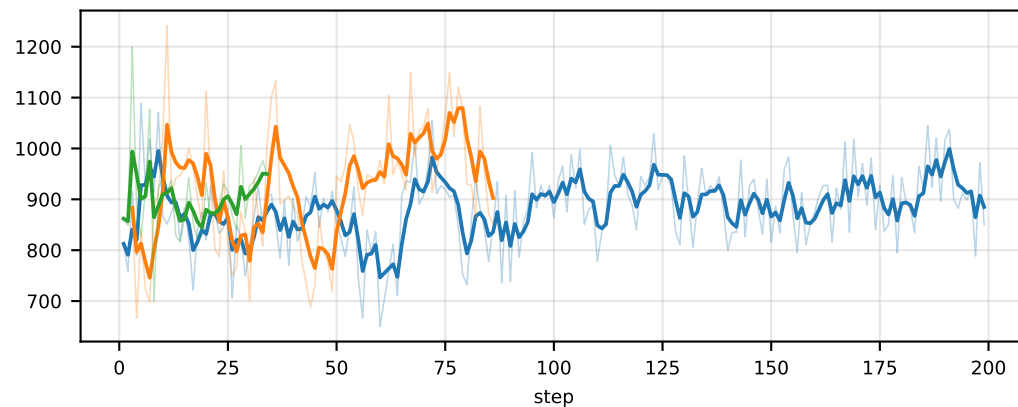
env/TicTacToe/response_length_turn_1



env/TicTacToe/response_length_turn_2

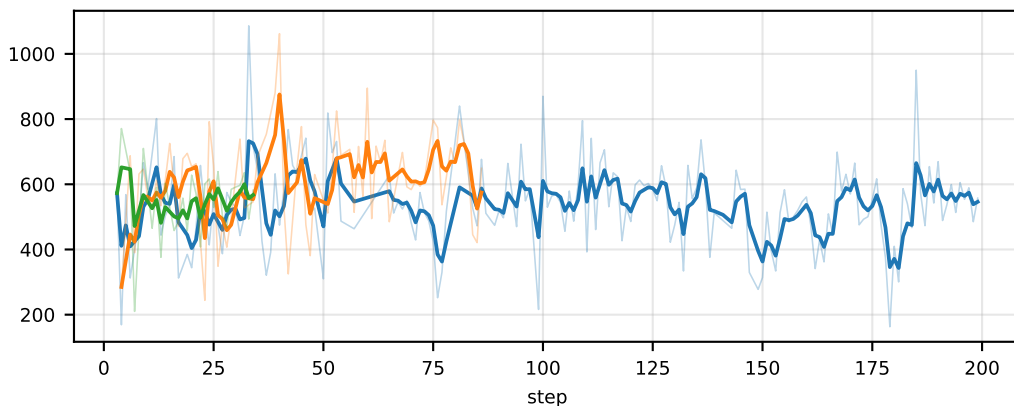


env/TicTacToe/response_length_turn_3

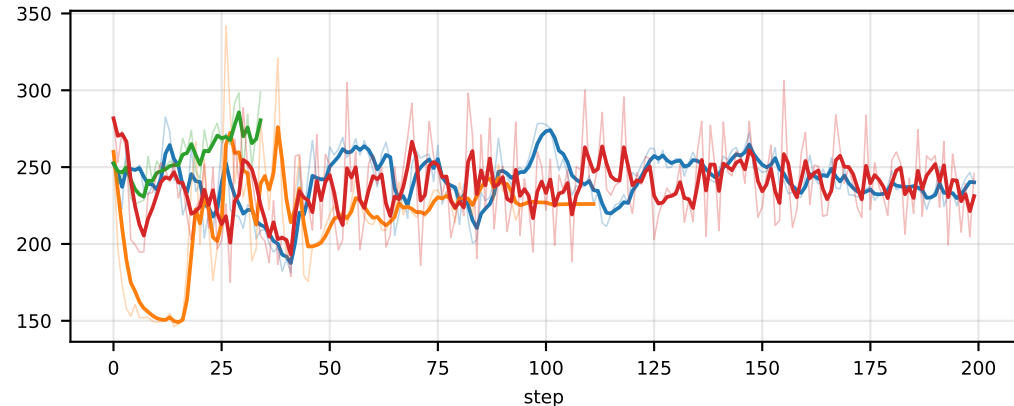


[TicTacToe] Game length & response length (2/4)

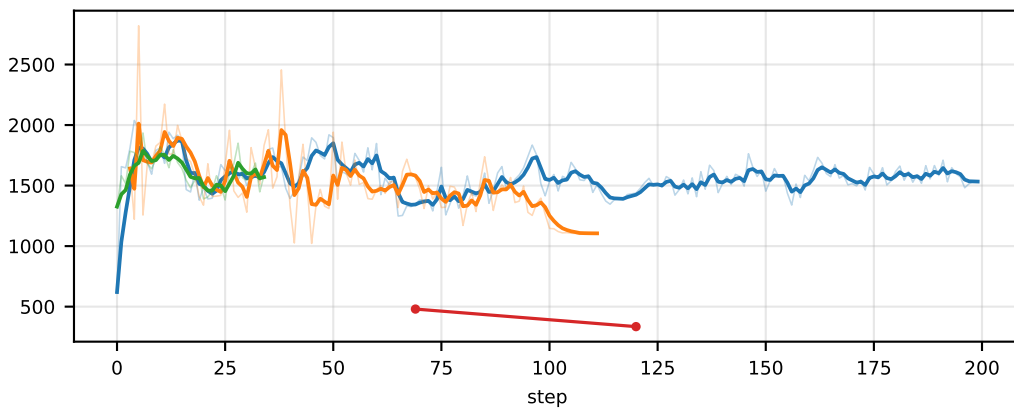
env/TicTacToe/response_length_turn_4



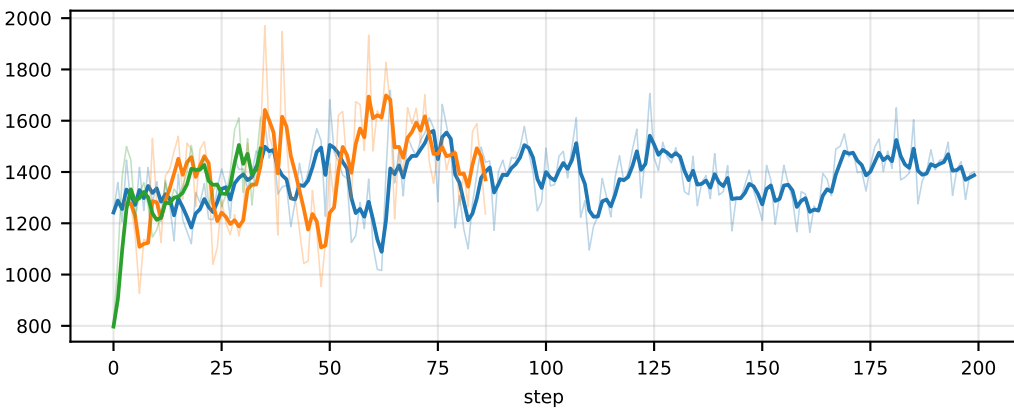
env/TicTacToe/response_length_player_0_turn_0



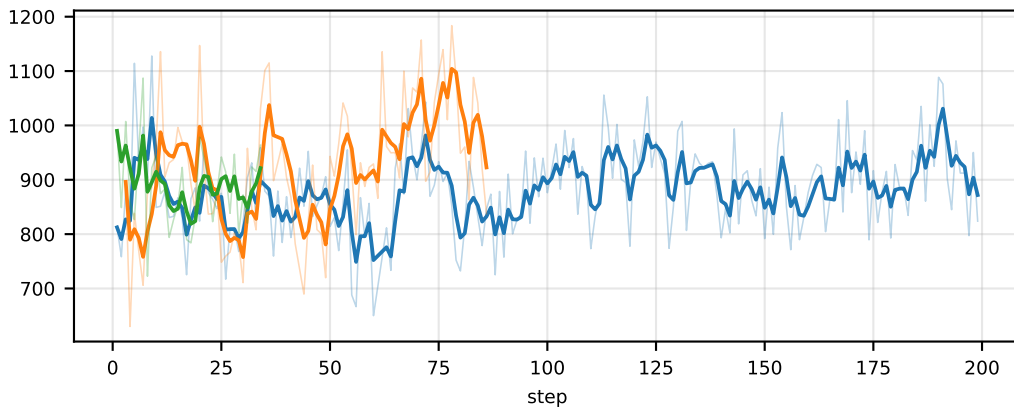
env/TicTacToe/response_length_player_0_turn_1



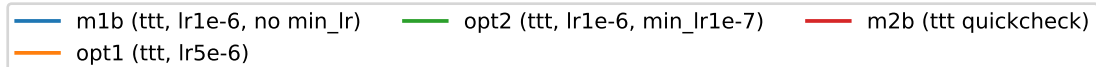
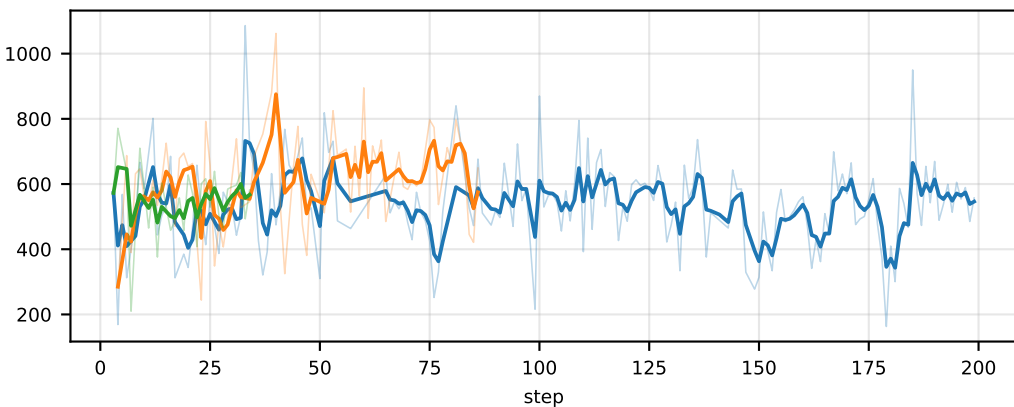
env/TicTacToe/response_length_player_0_turn_2



env/TicTacToe/response_length_player_0_turn_3

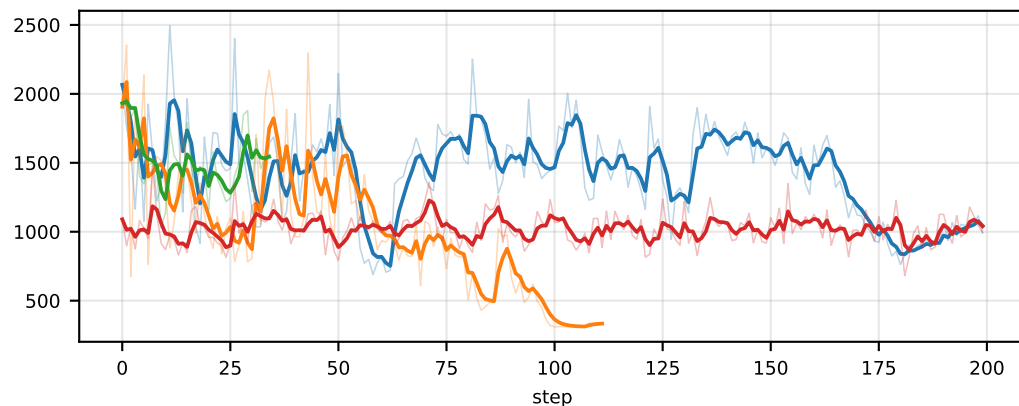


env/TicTacToe/response_length_player_0_turn_4

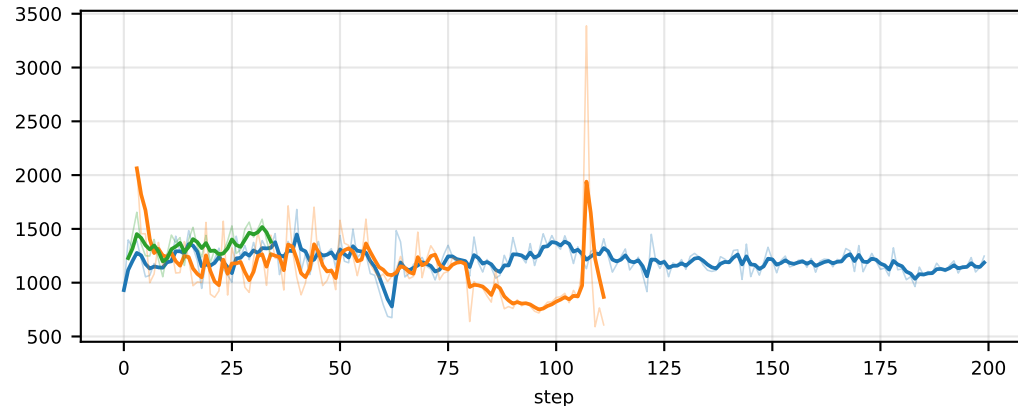


[TicTacToe] Game length & response length (3/4)

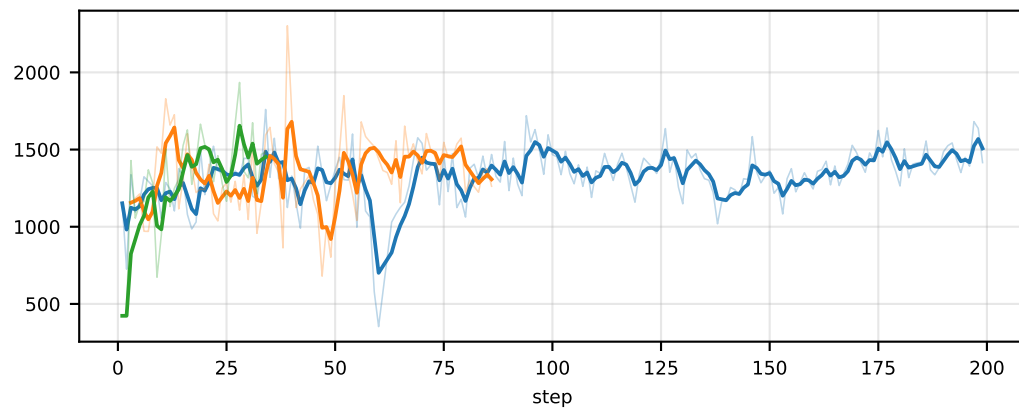
env/TicTacToe/response_length_player_1_turn_0



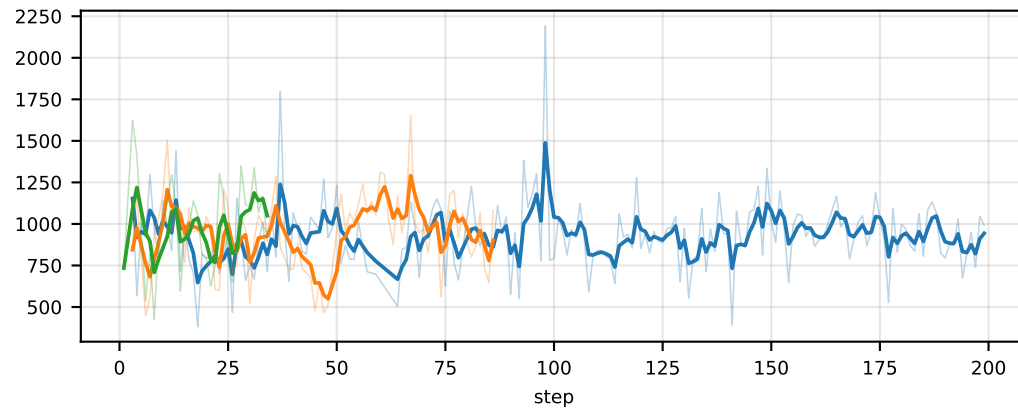
env/TicTacToe/response_length_player_1_turn_1



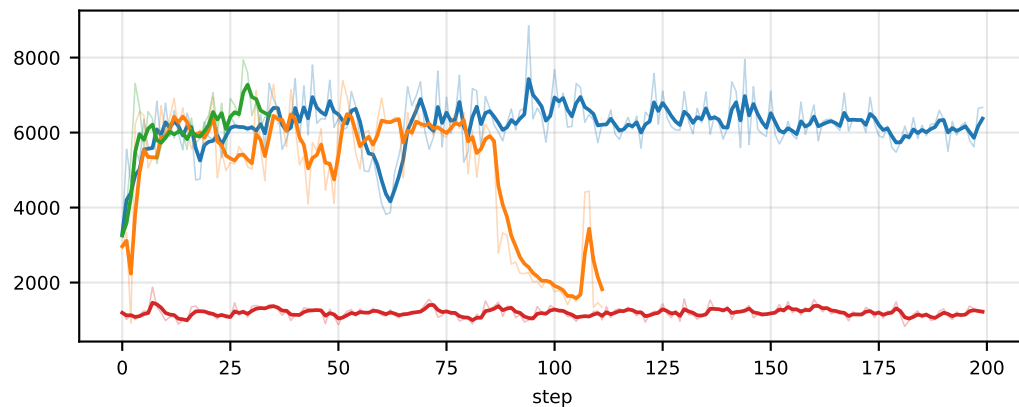
env/TicTacToe/response_length_player_1_turn_2



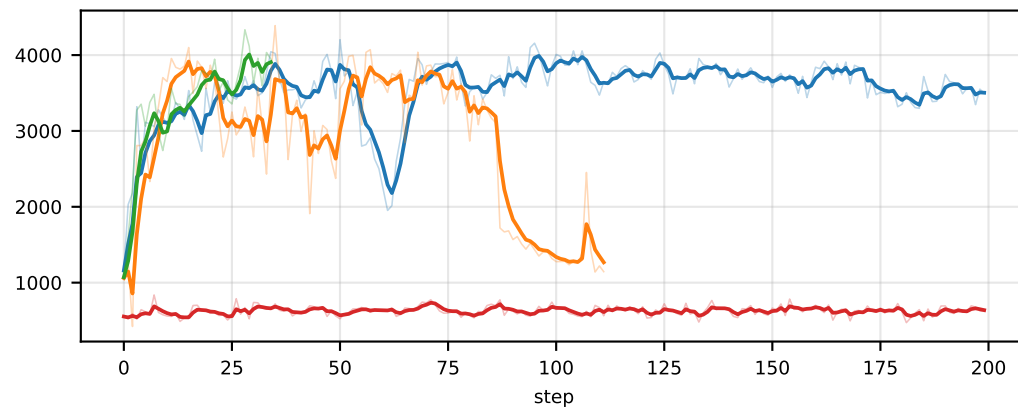
env/TicTacToe/response_length_player_1_turn_3



tokens/response_length/max



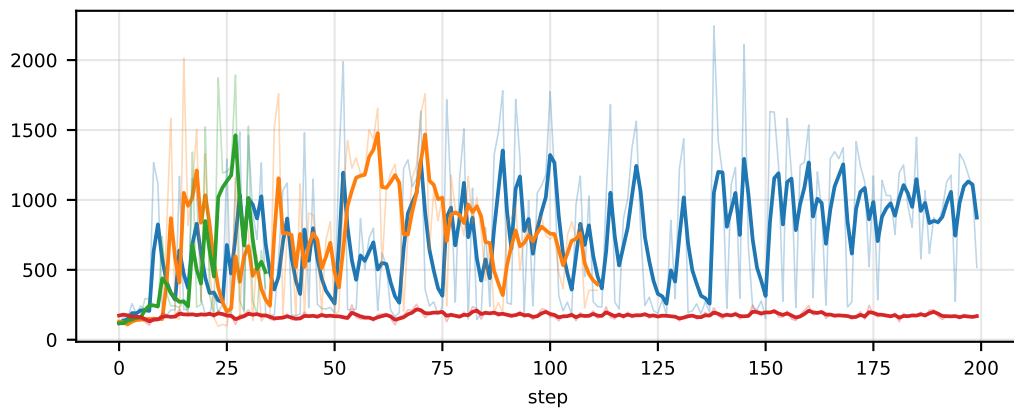
tokens/response_length/mean



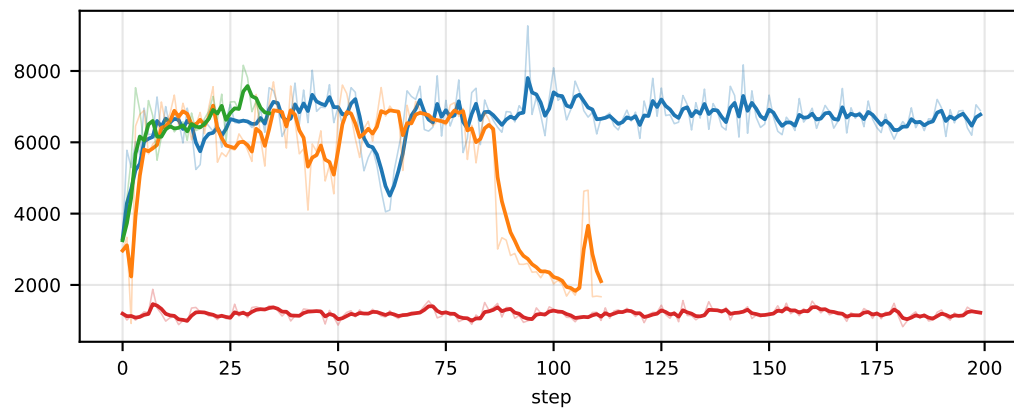
— m1b (ttr, lr1e-6, no min_lr)
 — opt2 (ttr, lr1e-6, min_lr1e-7)
 — m2b (ttr quickcheck)

[TicTacToe] Game length & response length (4/4)

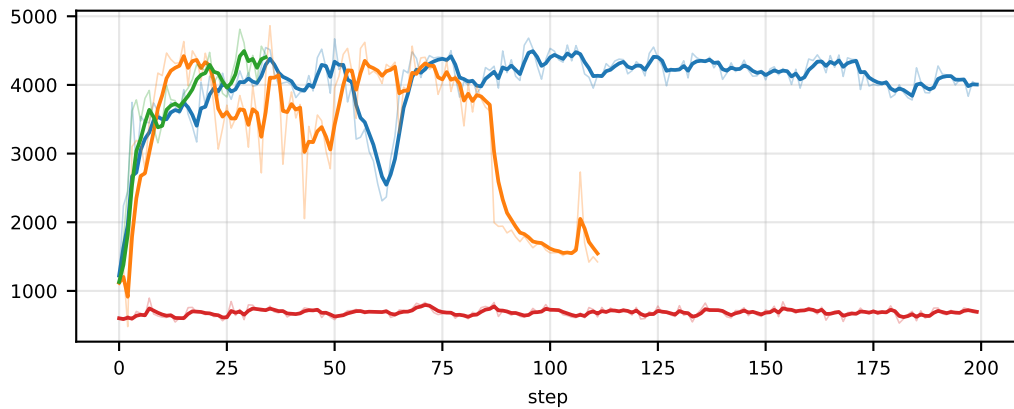
tokens/response_length/min



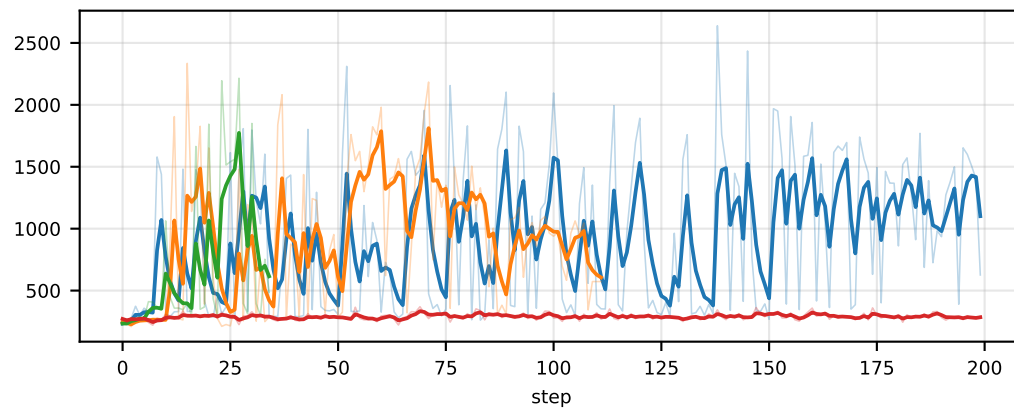
tokens/non_prompt_length/max



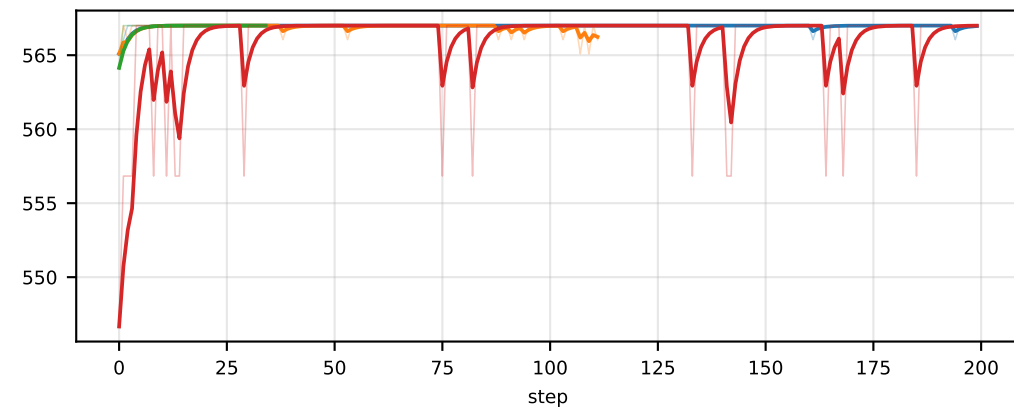
tokens/non_prompt_length/mean



tokens/non_prompt_length/min

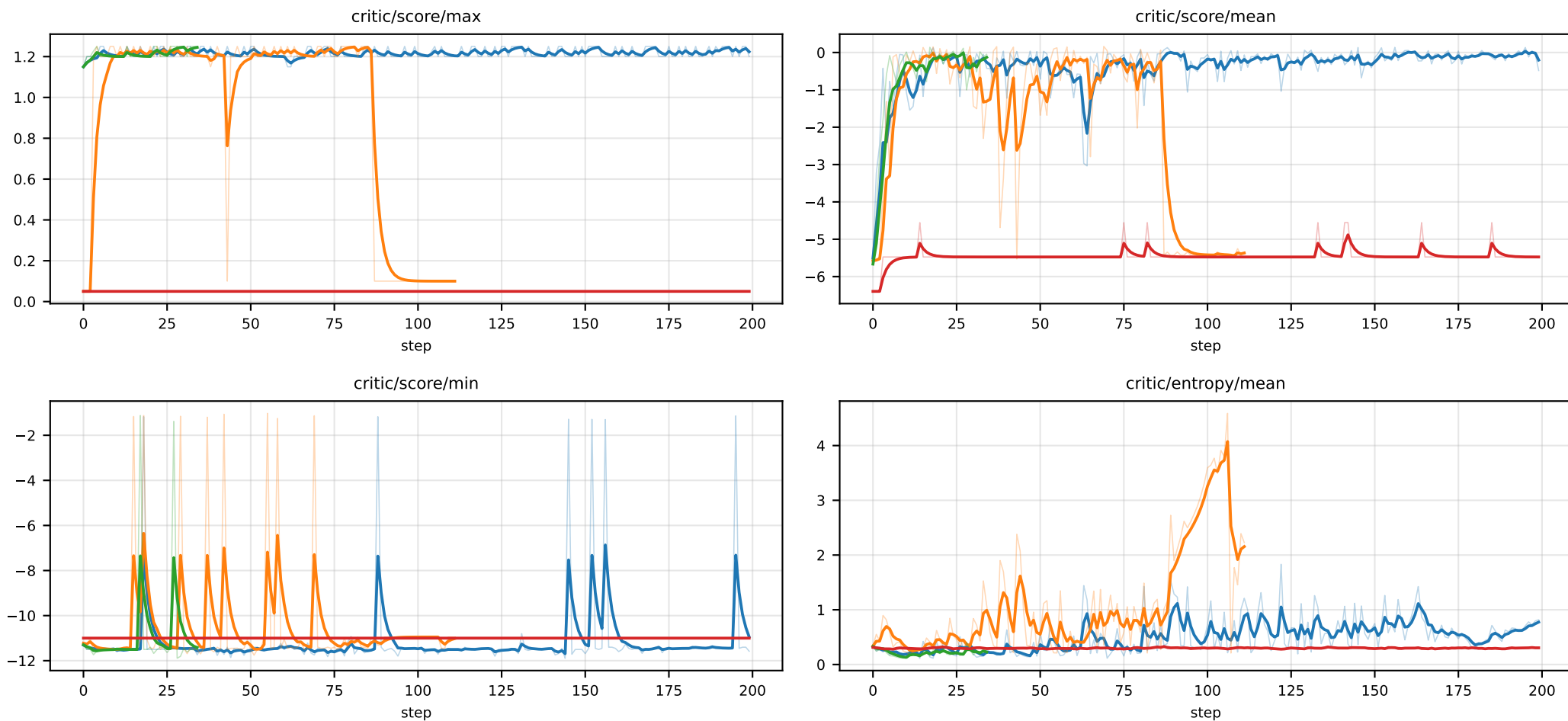


tokens/prompt_length/mean



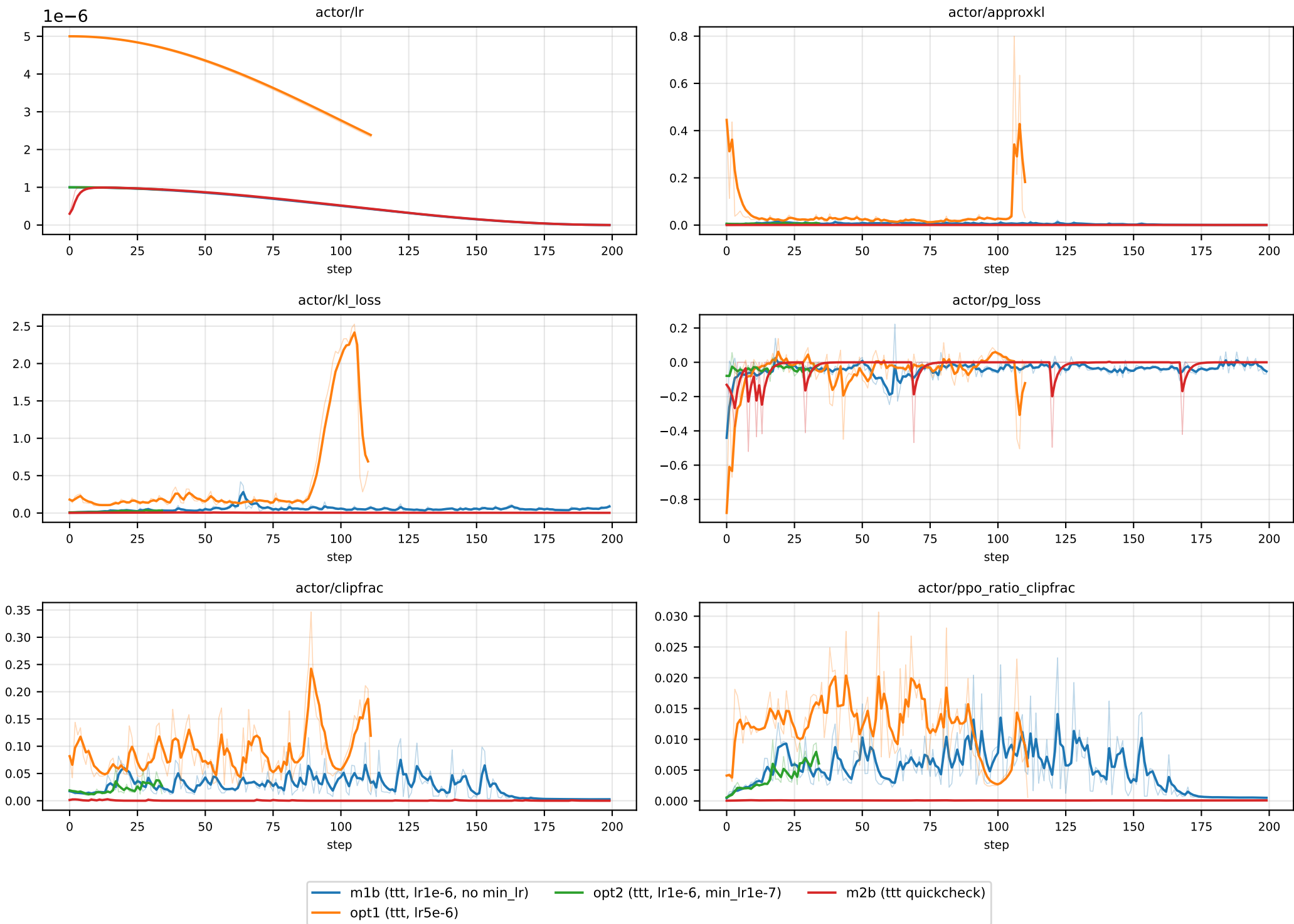
— m1b (ttr, lr1e-6, no min_lr) — opt2 (ttr, lr1e-6, min_lr1e-7) — m2b (ttr quickcheck)
— opt1 (ttr, lr5e-6)

[TicTacToe] Reward & entropy (1/1)



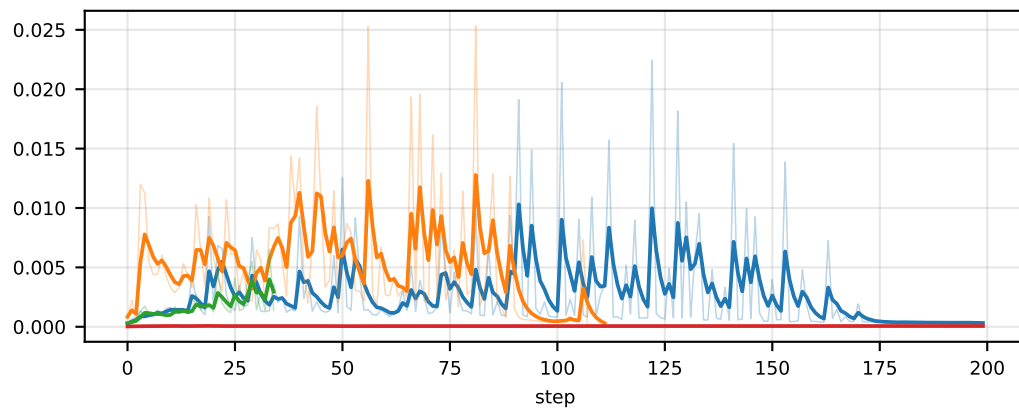
— m1b (ttt, lr1e-6, no min_lr) — opt2 (ttt, lr1e-6, min_lr1e-7) — m2b (ttt quickcheck)
— opt1 (ttt, lr5e-6)

[TicTacToe] Policy update (PPO / KL) (1/2)

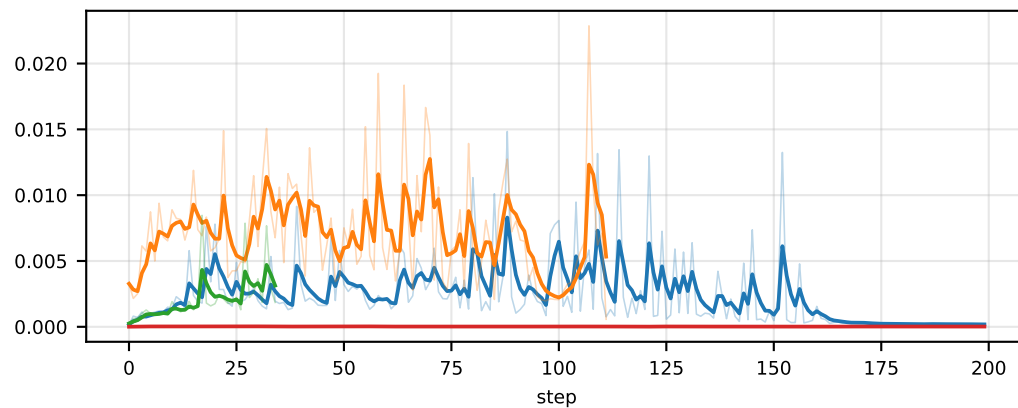


[TicTacToe] Policy update (PPO / KL) (2/2)

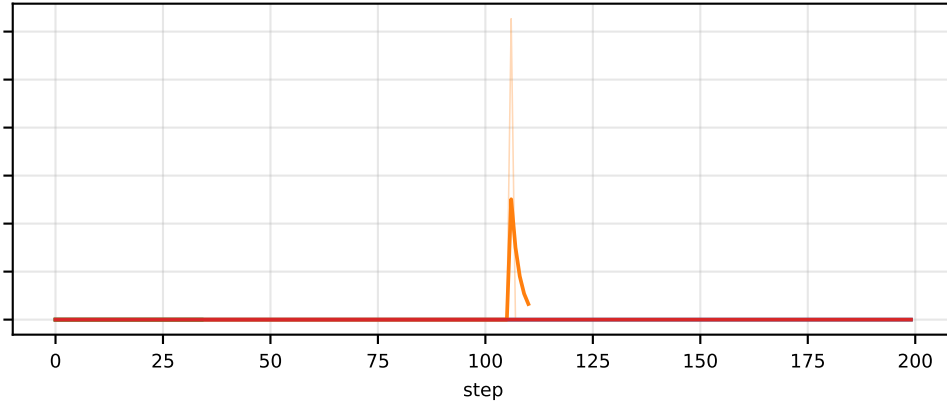
actor/ppo_ratio_high_clipfrac



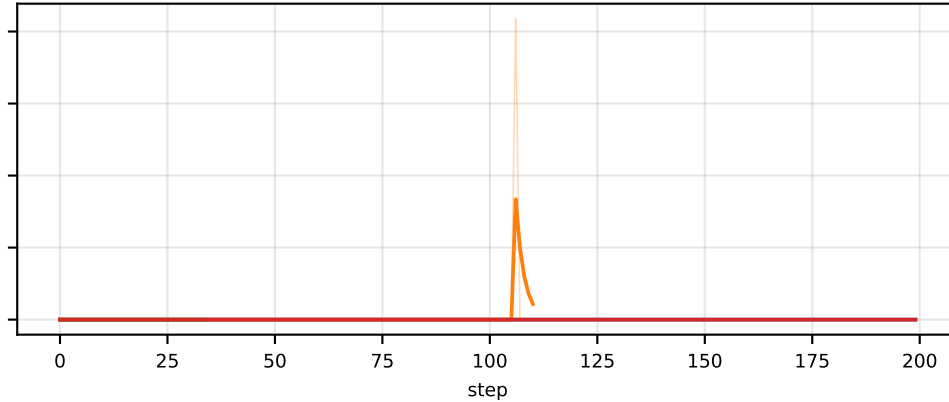
actor/ppo_ratio_low_clipfrac



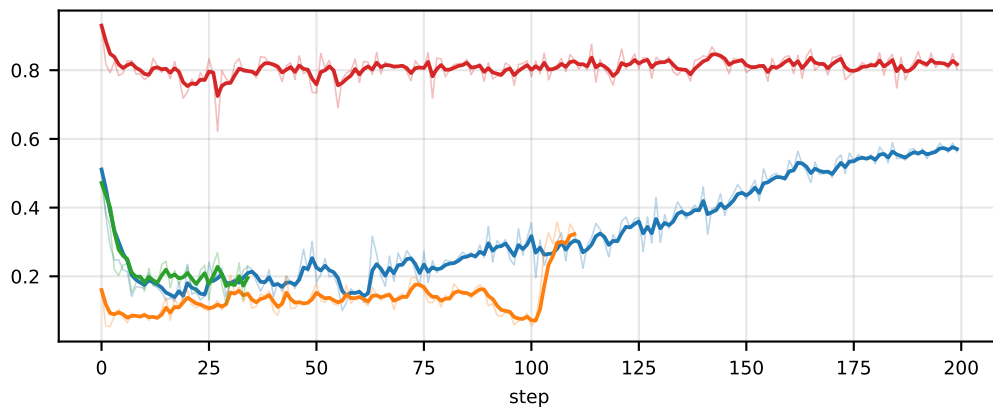
1e12 actor/ratio_max



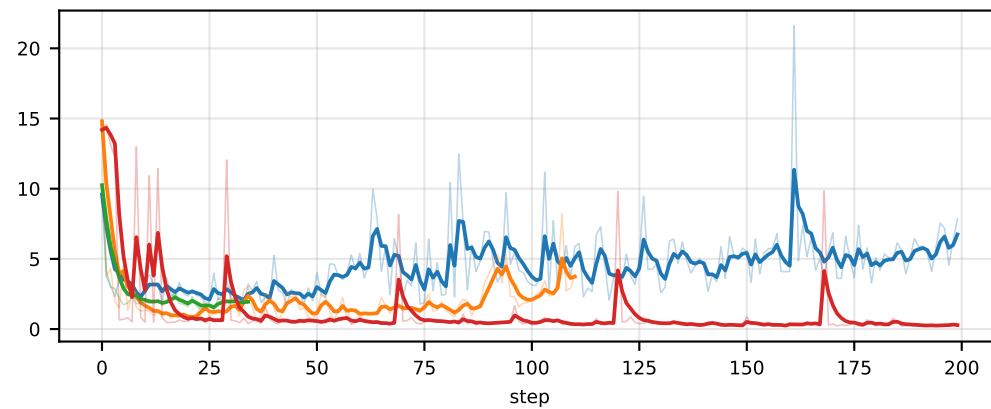
1e8 actor/ratio_mean



actor/ratio_min



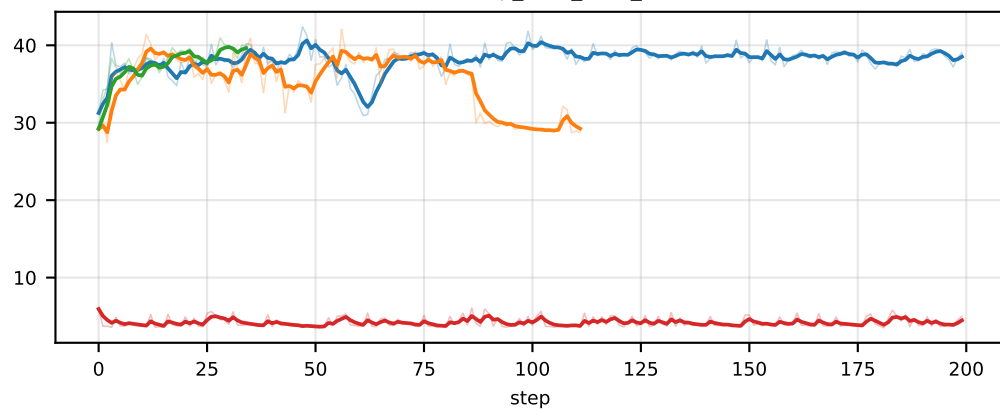
actor_train/grad_norm



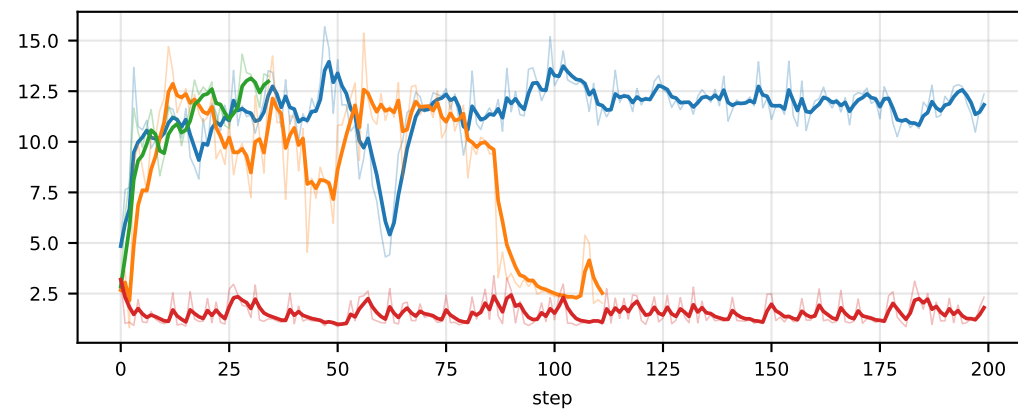
— m1b (ttd, lr1e-6, no min_lr) — opt2 (ttd, lr1e-6, min_lr1e-7) — m2b (ttd quickcheck)
— opt1 (ttd, lr5e-6)

[TicTacToe] Step time breakdown (minutes) (1/1)

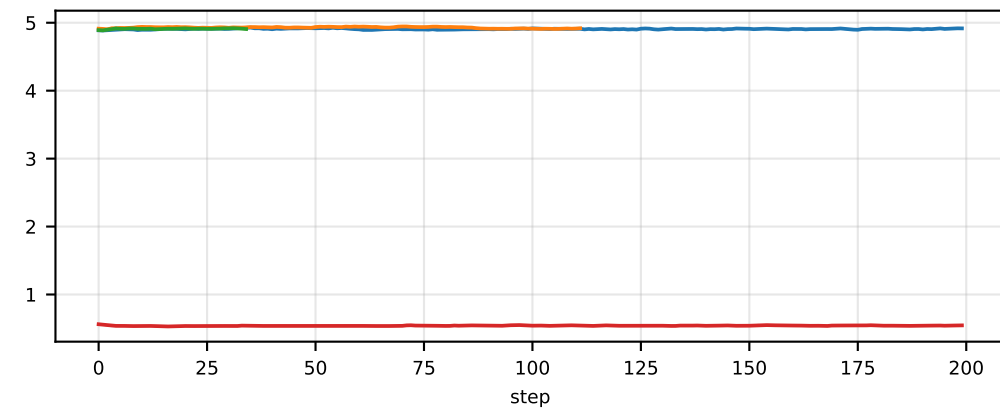
derived/step_time_total_min



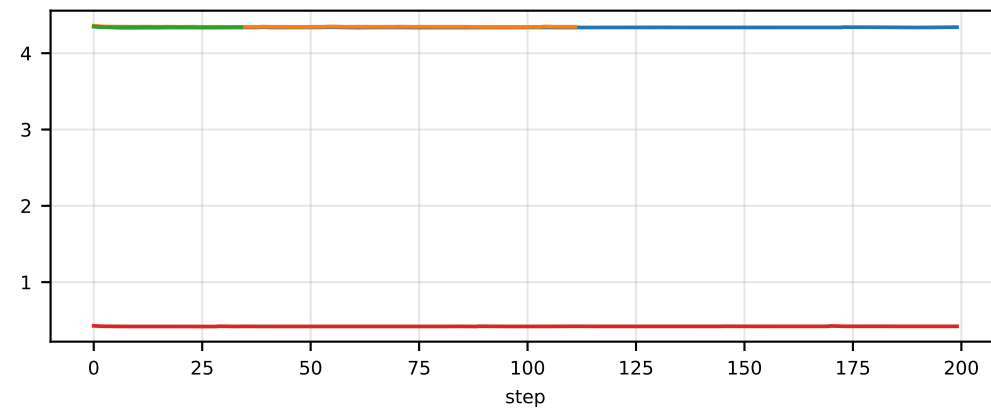
time/rollout



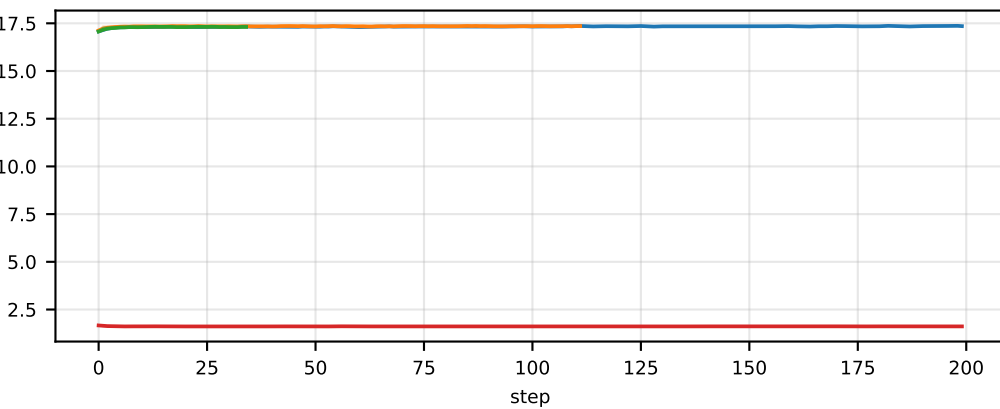
time/ref_log_probs_values_reward



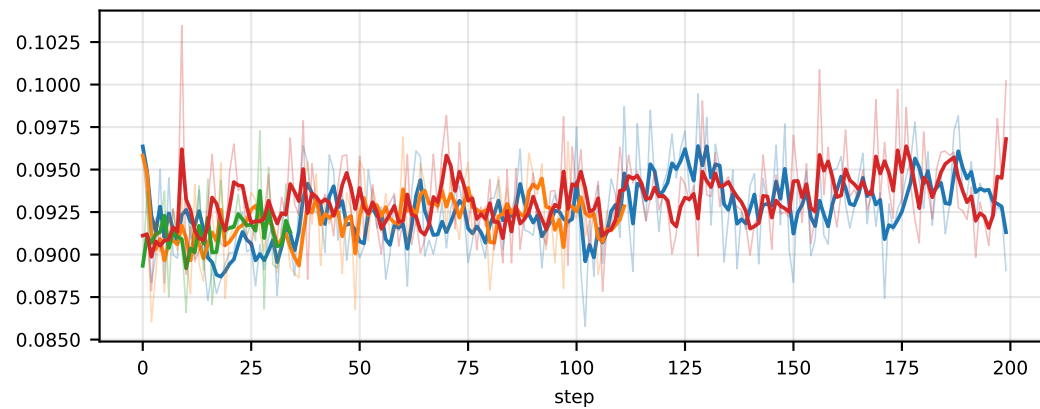
time/old_log_probs_values



time/actor_train/train_step/total



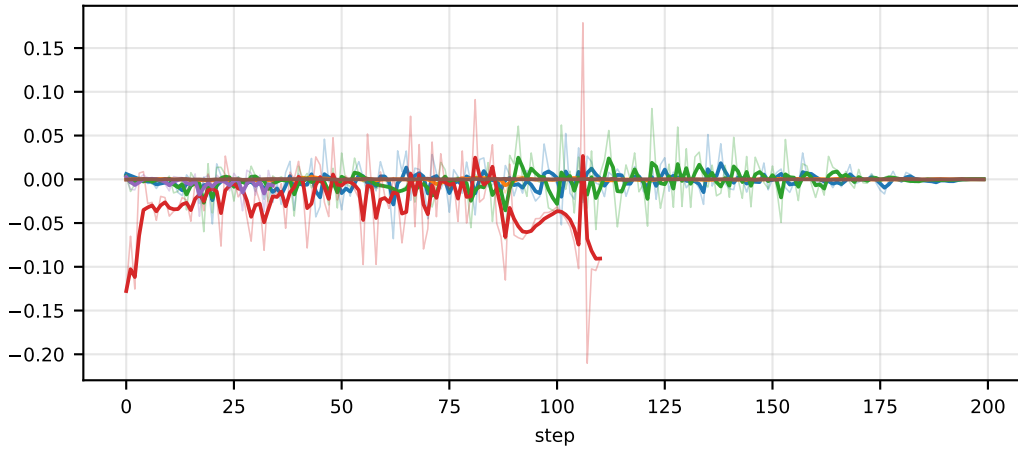
time/actor_train/model_update/total



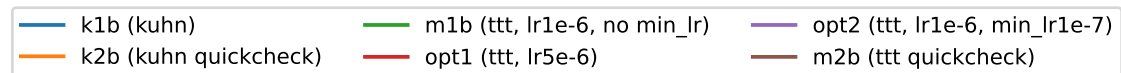
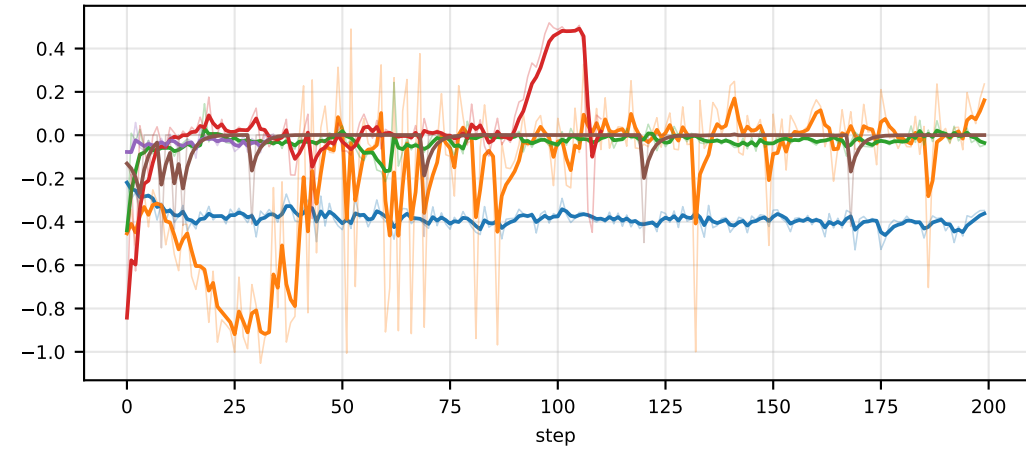
— m1b (ttd, lr1e-6, no min_lr) — opt2 (ttd, lr1e-6, min_lr1e-7) — m2b (ttd quickcheck)
— opt1 (ttd, lr5e-6)

[Appendix] actor/* (1/1)

actor/policykl

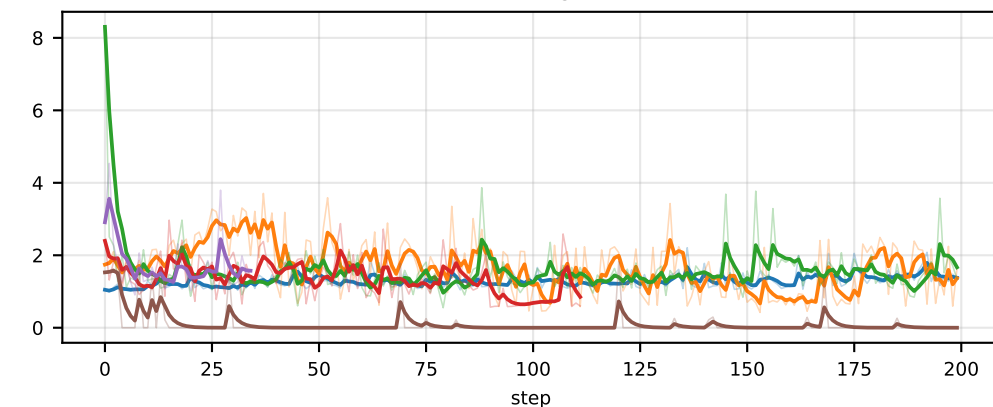


actor/total_loss



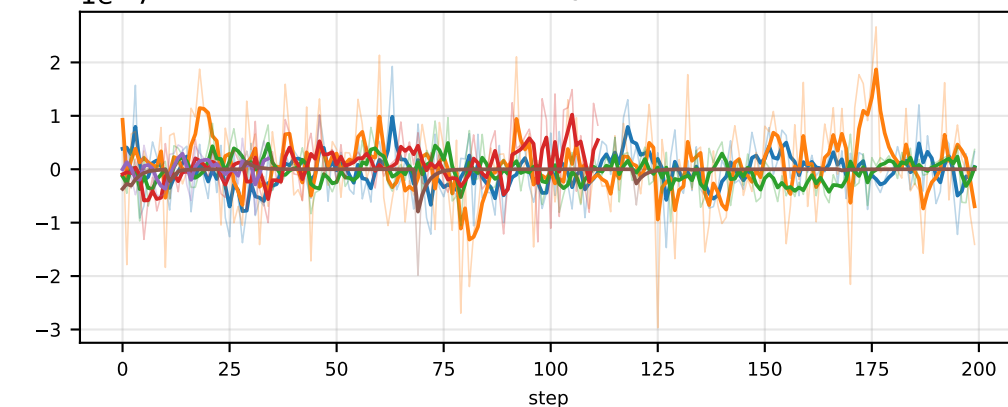
[Appendix] critic/* (1/3)

critic/advantages/max

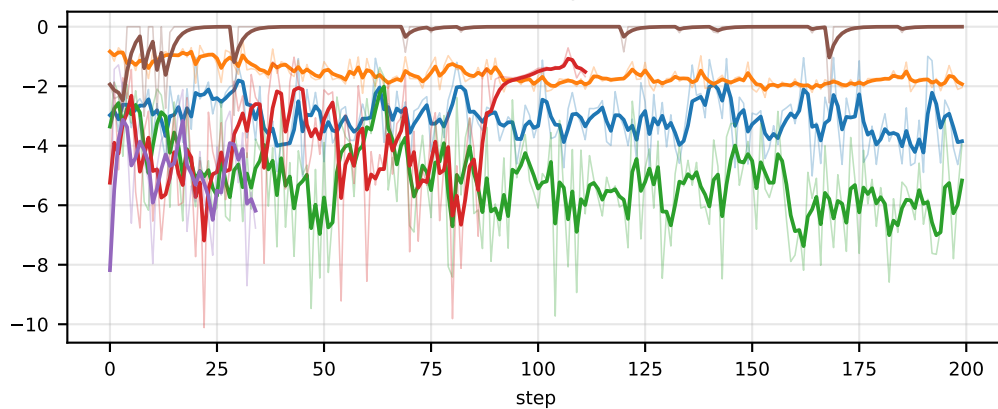


1e-7

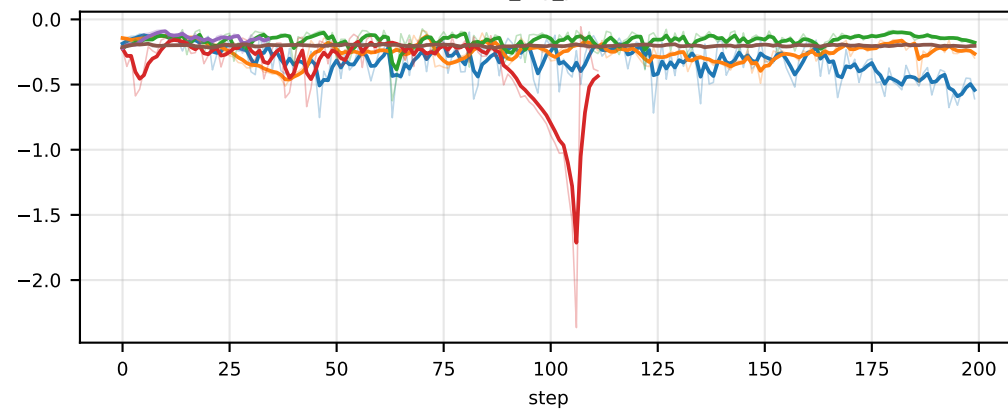
critic/advantages/mean



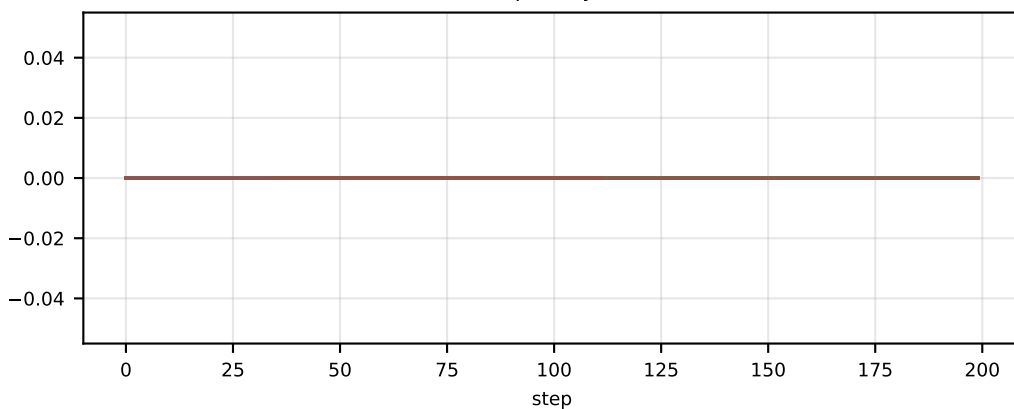
critic/advantages/min



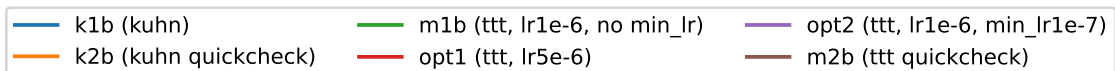
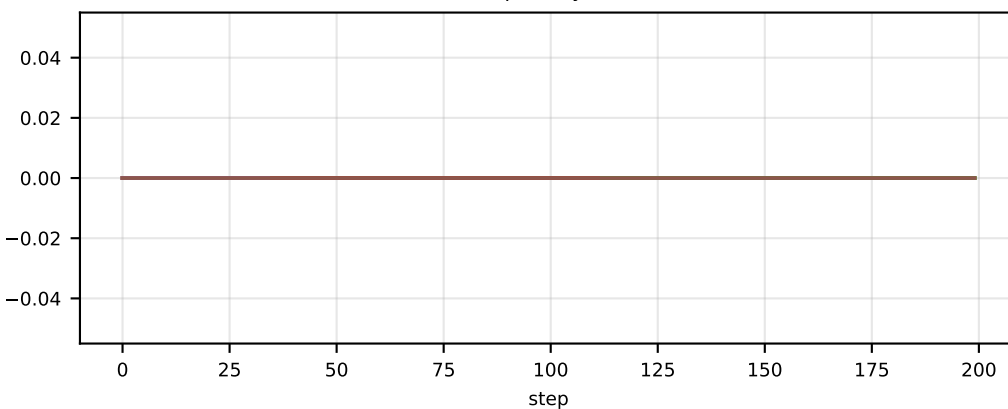
critic/old_log_prob/mean



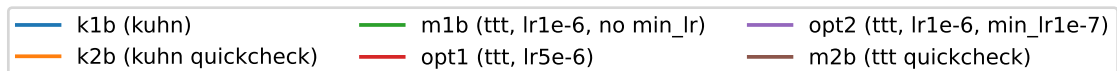
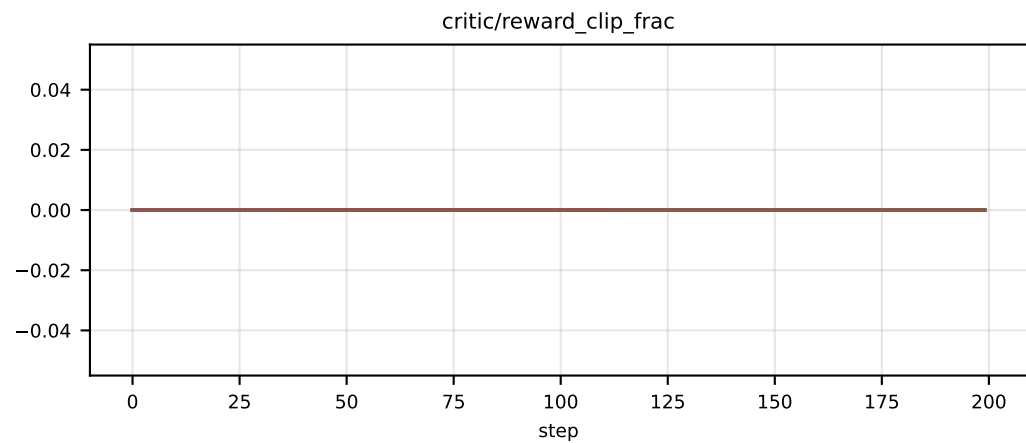
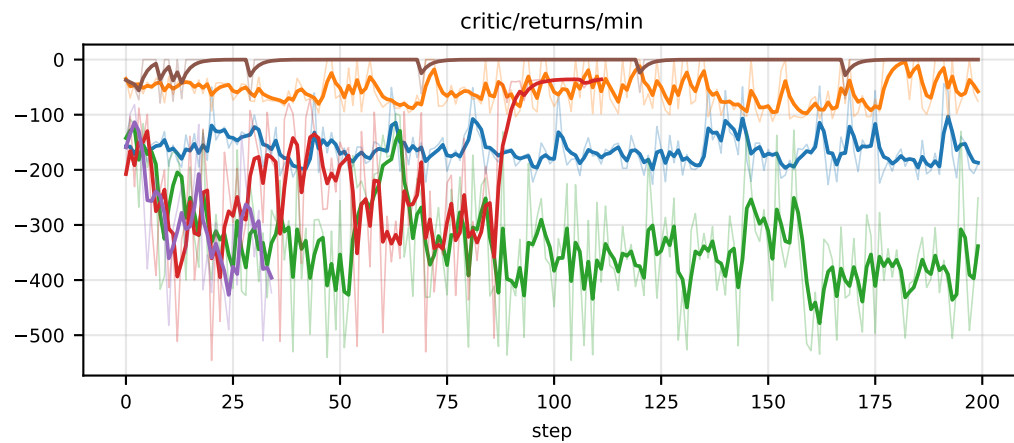
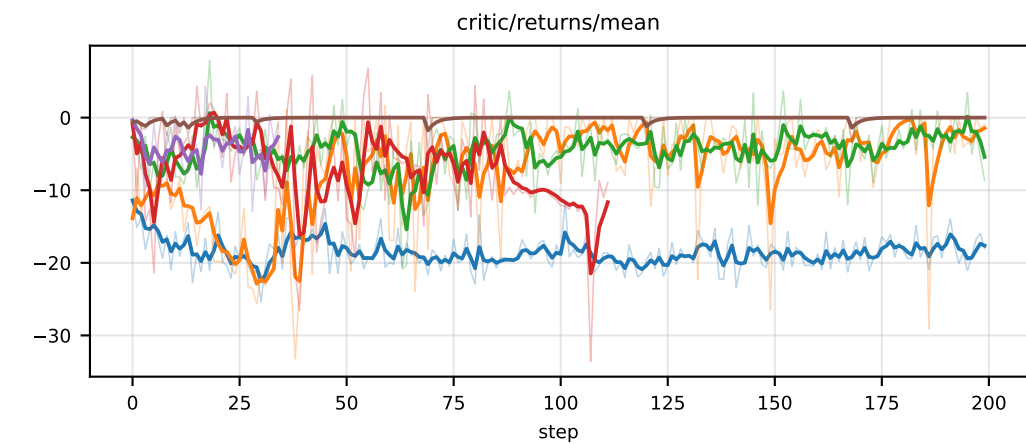
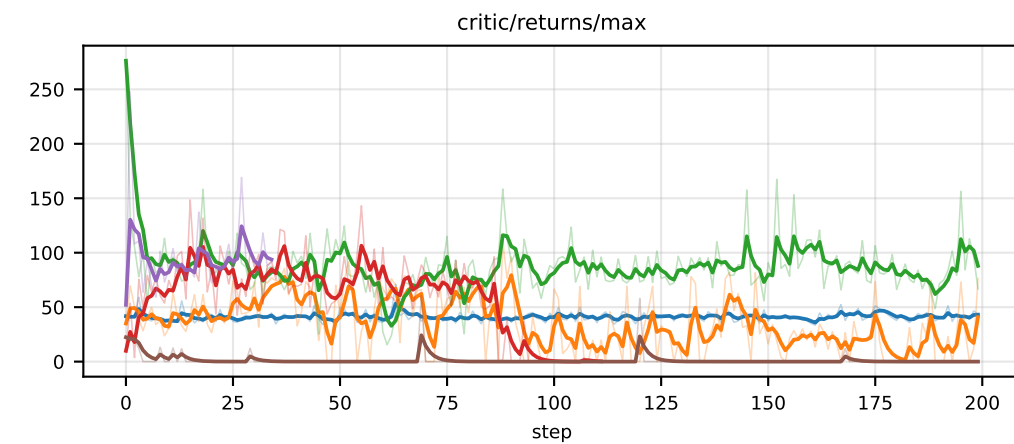
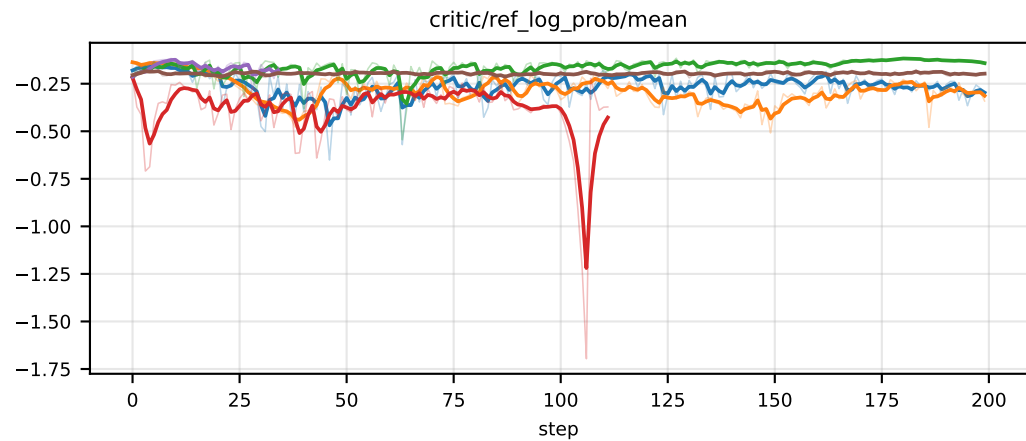
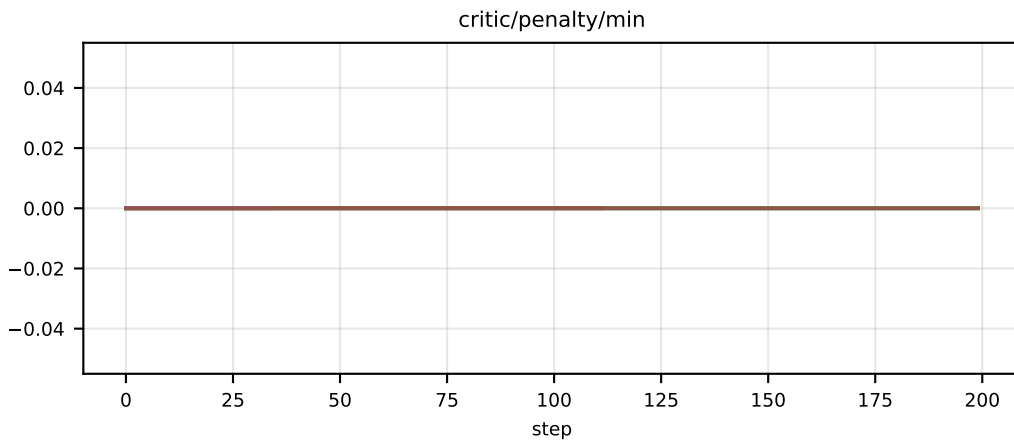
critic/penalty/max



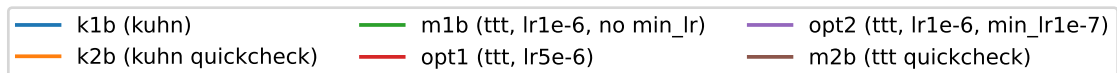
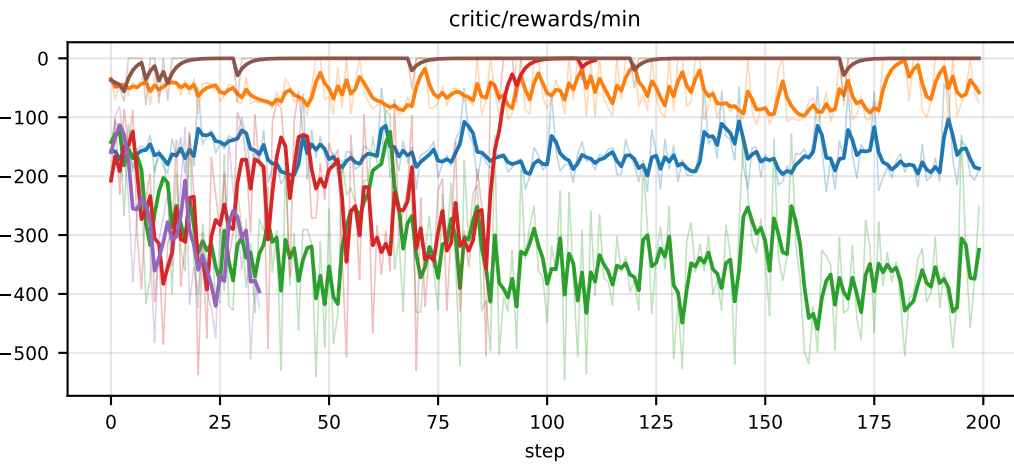
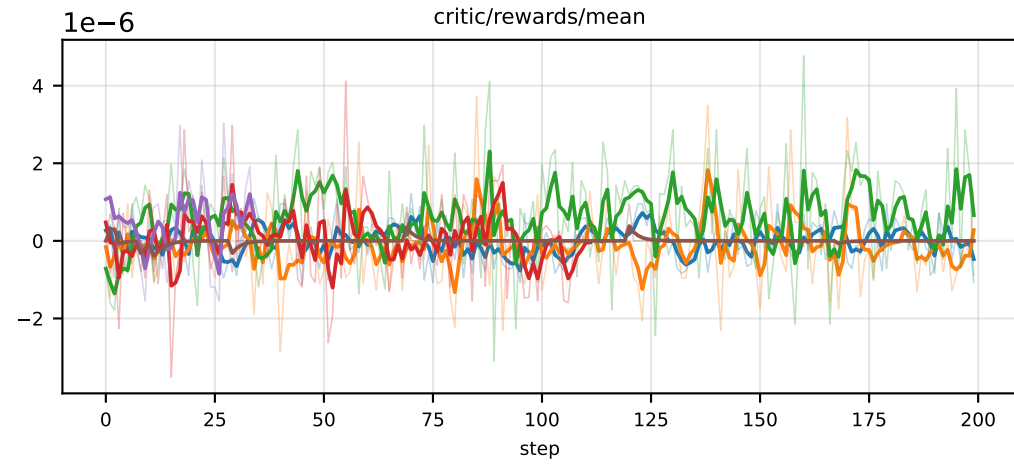
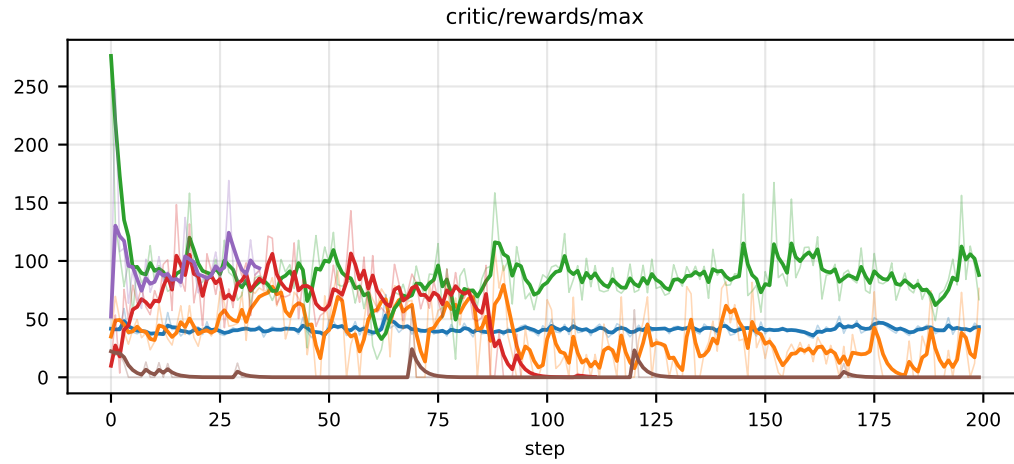
critic/penalty/mean



[Appendix] critic/* (2/3)

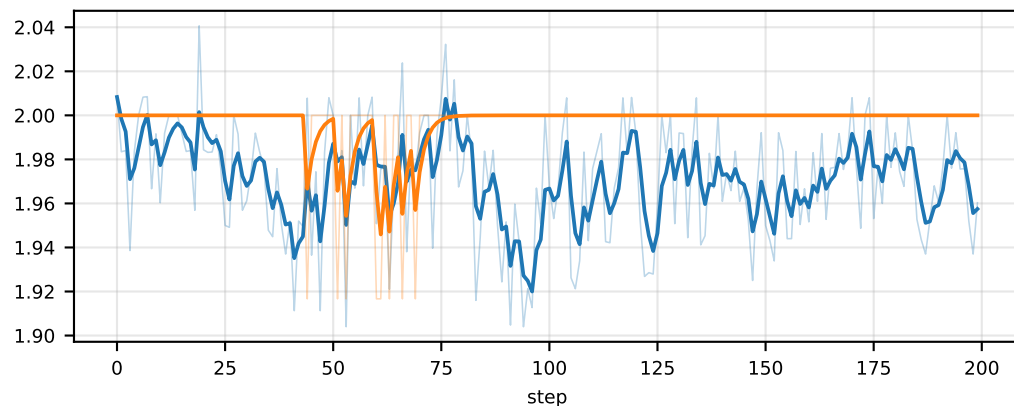


[Appendix] critic/* (3/3)

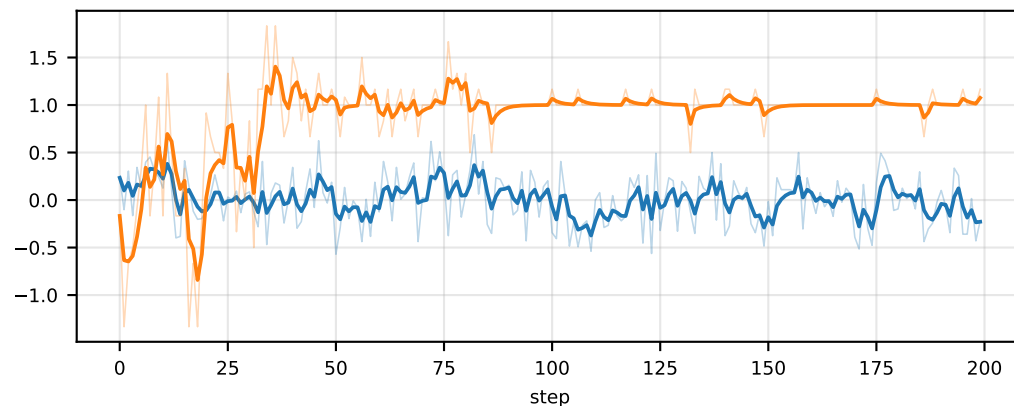


[Appendix] env/* (1/3)

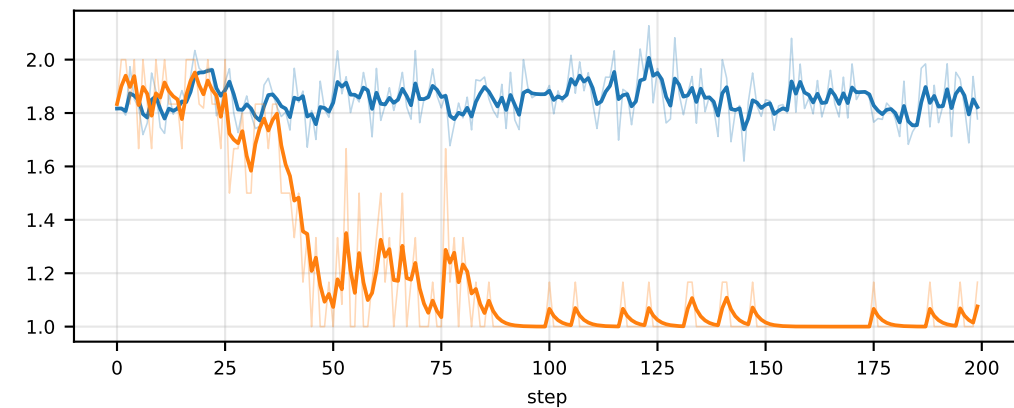
env/KuhnPoker/player_0_bet



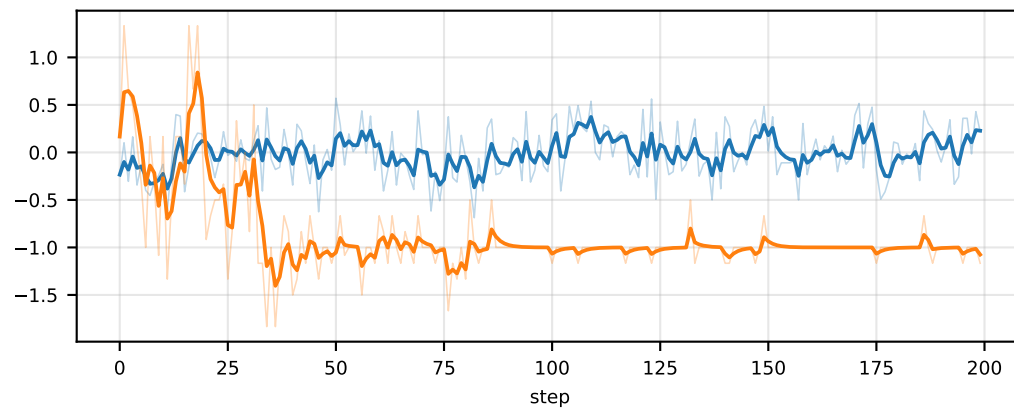
env/KuhnPoker/player_0_return



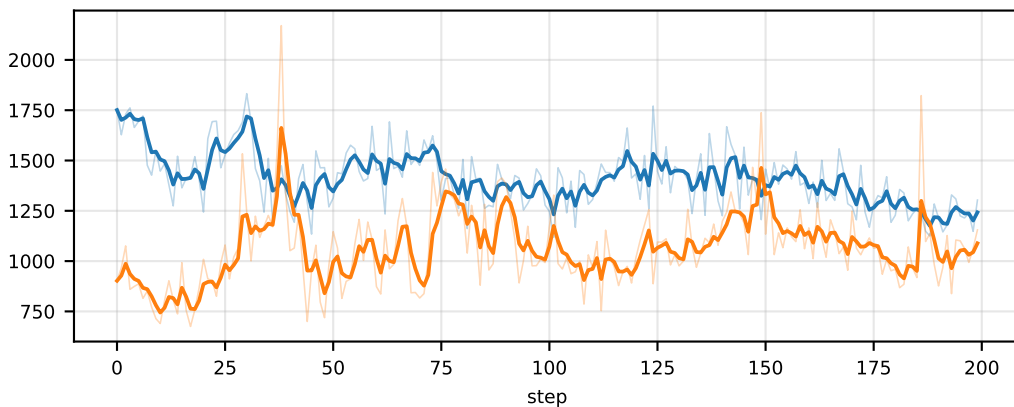
env/KuhnPoker/player_1_bet



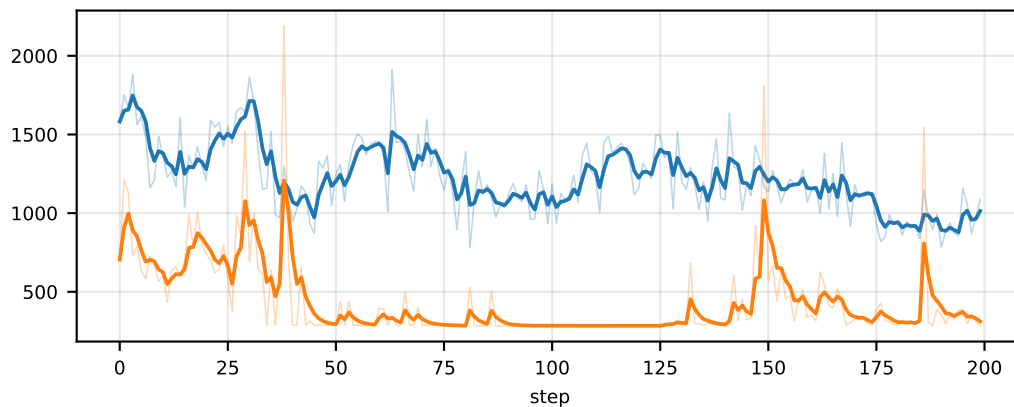
env/KuhnPoker/player_1_return



env/KuhnPoker/response_length_per_turn

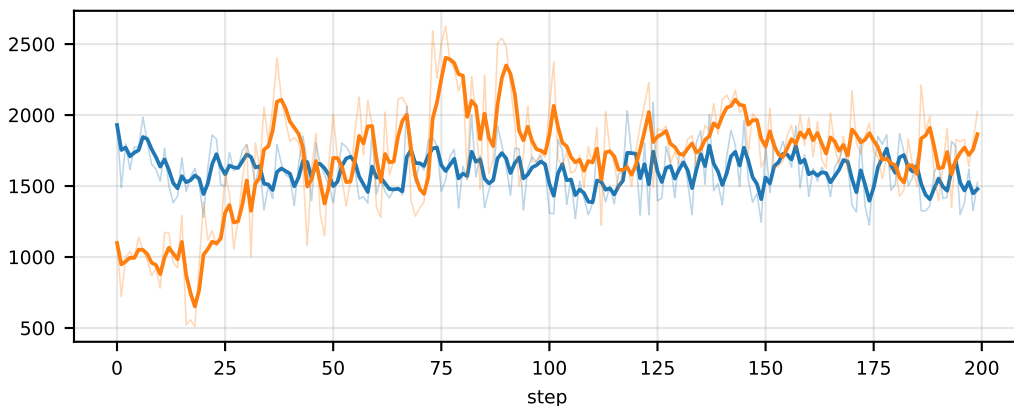


env/KuhnPoker/response_length_per_turn_player_0

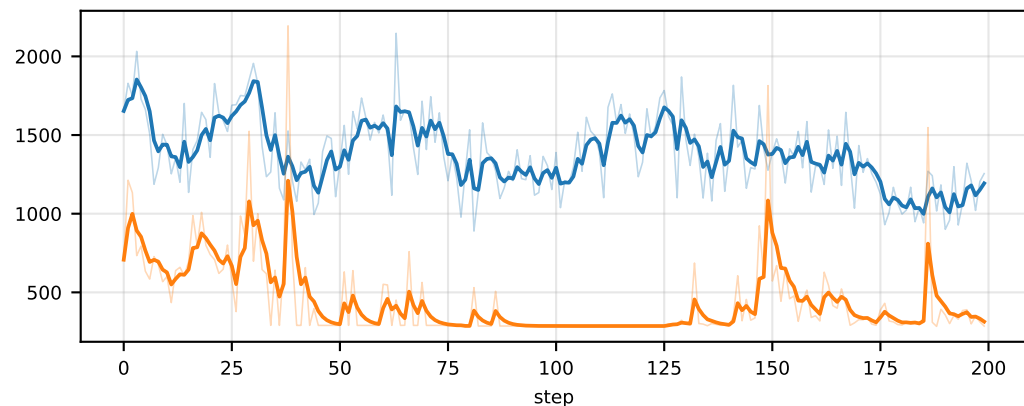


[Appendix] env/* (2/3)

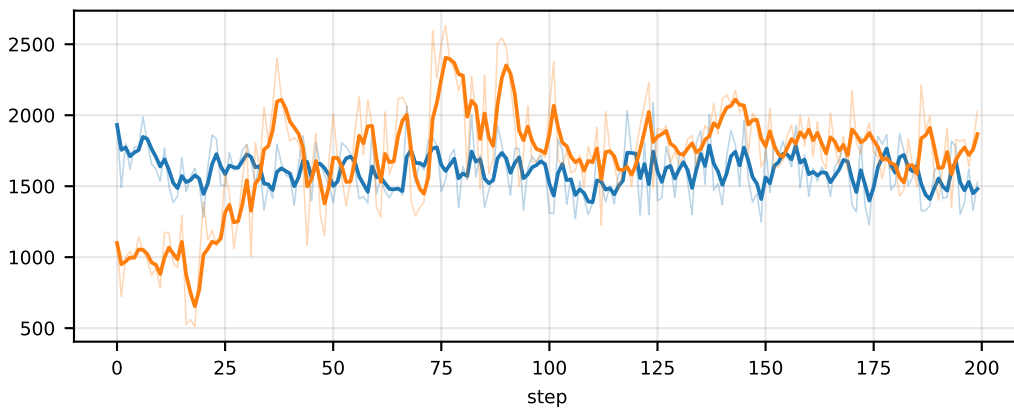
env/KuhnPoker/response_length_per_turn_player_1



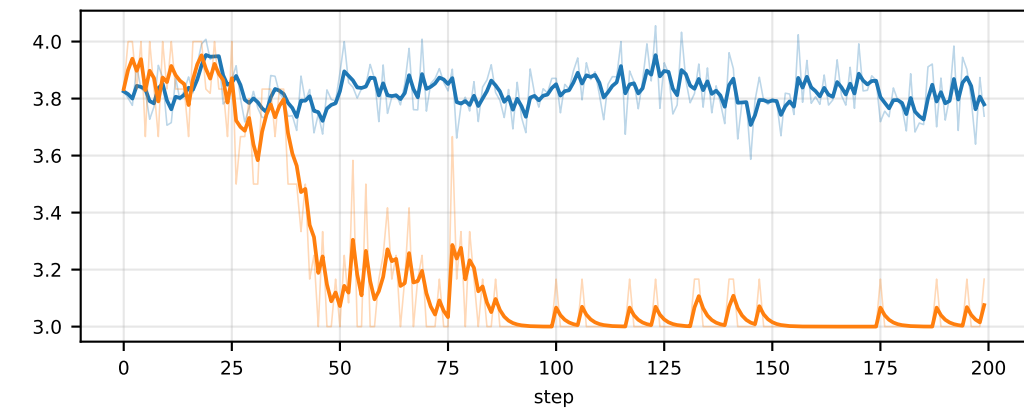
env/KuhnPoker/response_length_player_0



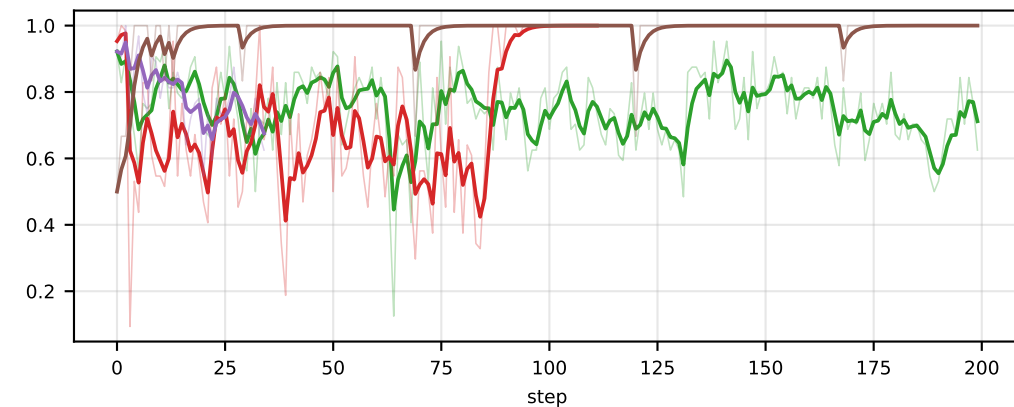
env/KuhnPoker/response_length_player_1



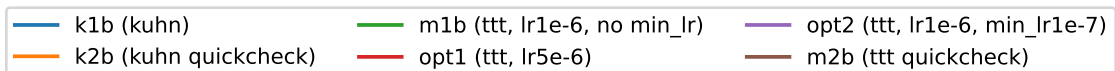
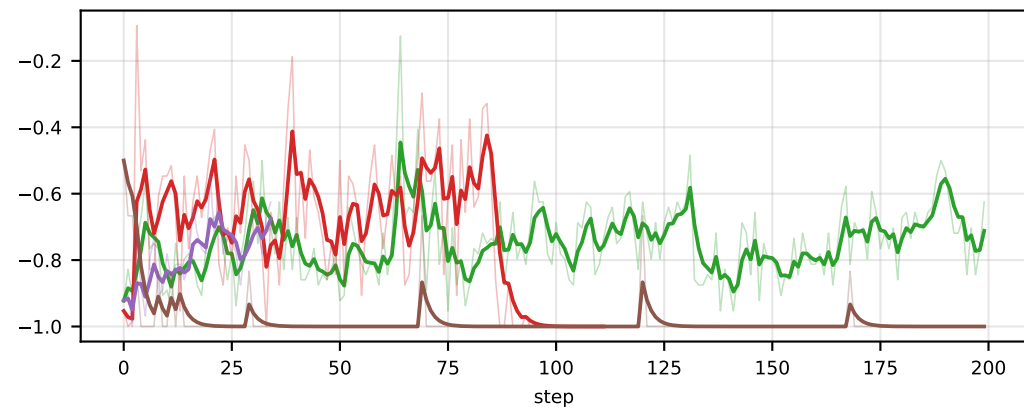
env/KuhnPoker/total_pot



env/TicTacToe/player_0_return

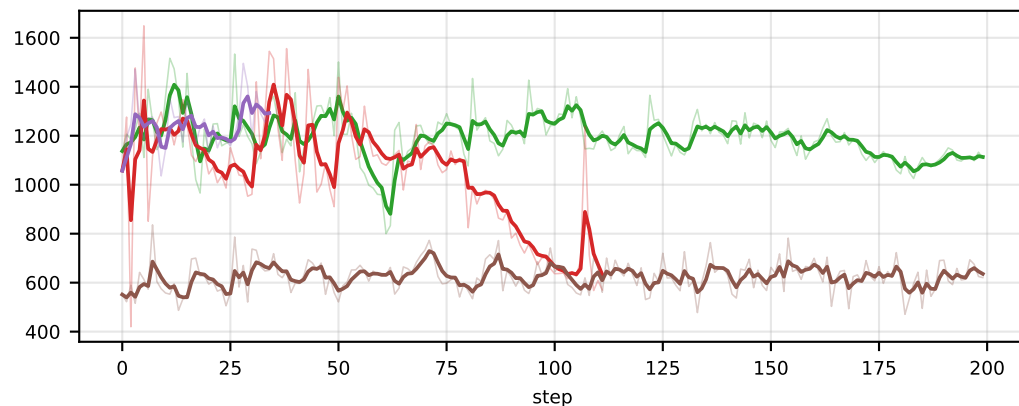


env/TicTacToe/player_1_return

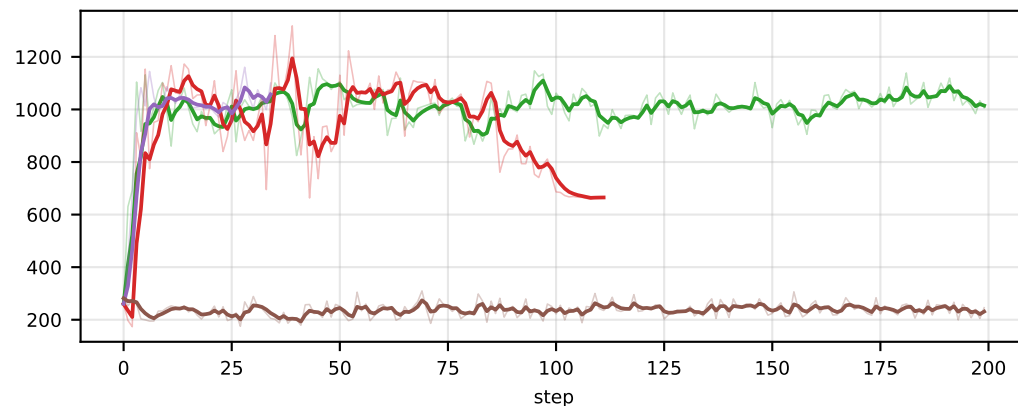


[Appendix] env/* (3/3)

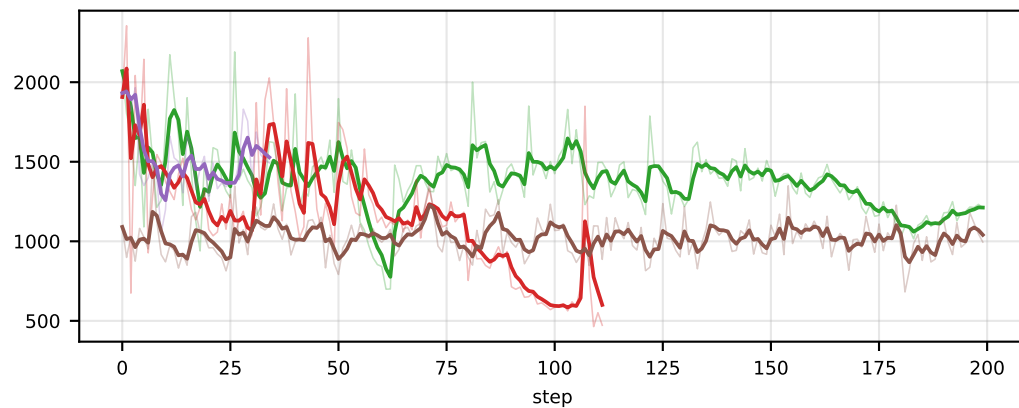
env/TicTacToe/response_length_per_turn



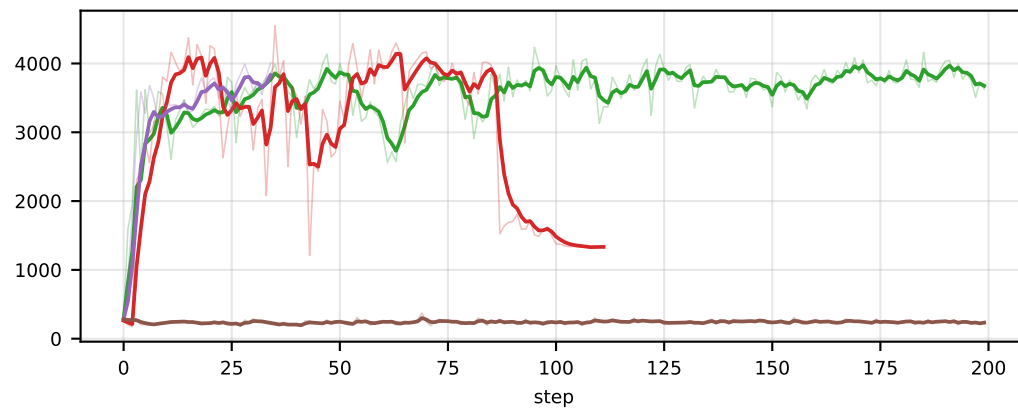
env/TicTacToe/response_length_per_turn_player_0



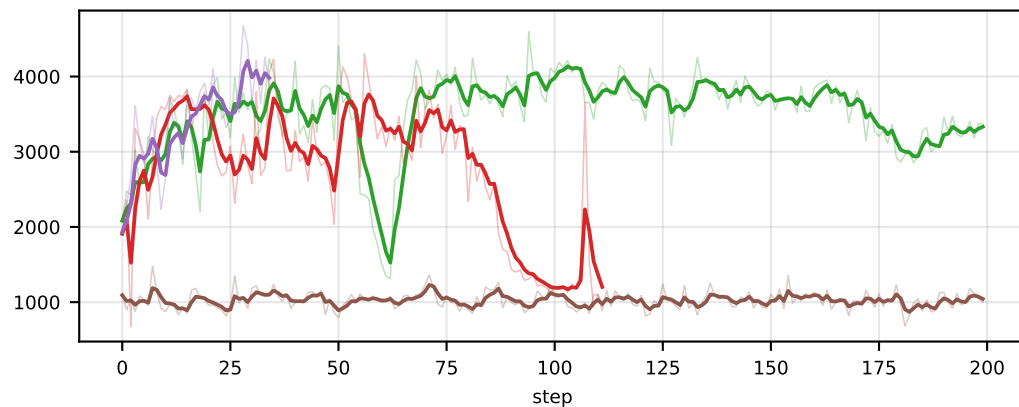
env/TicTacToe/response_length_per_turn_player_1



env/TicTacToe/response_length_player_0



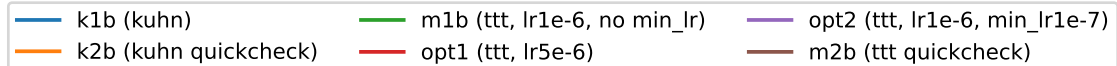
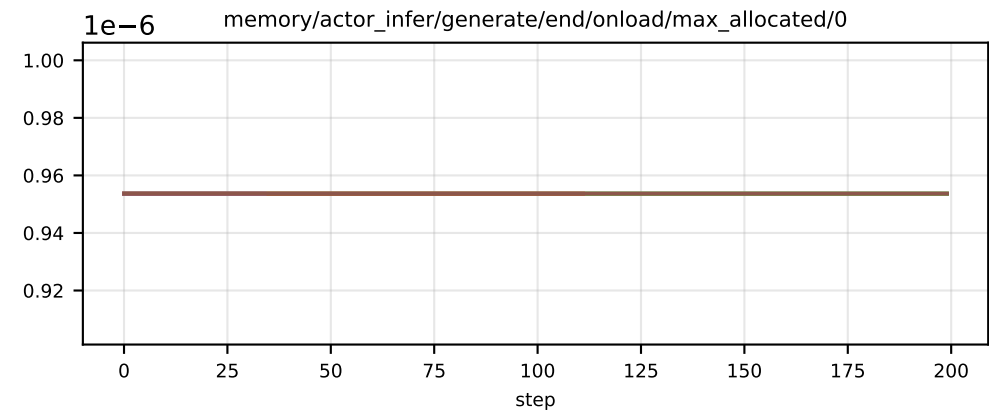
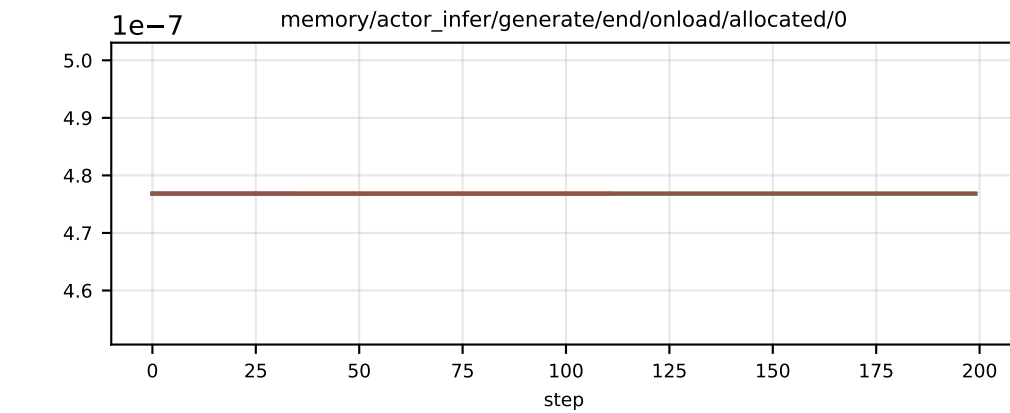
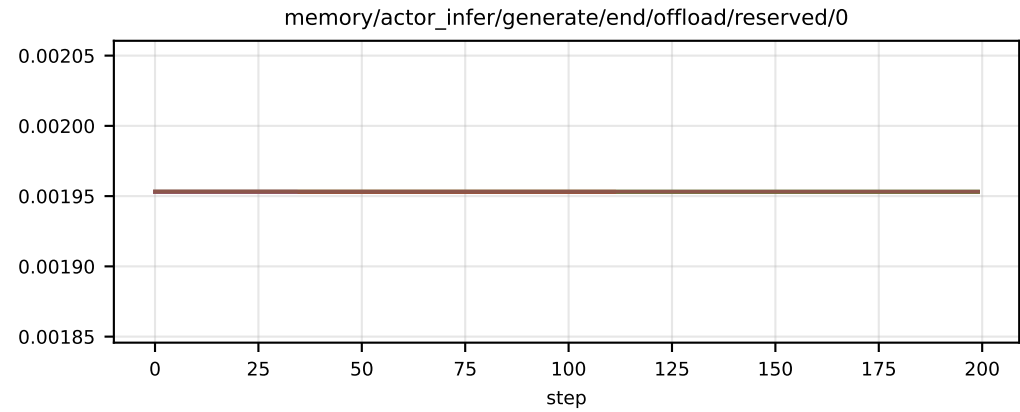
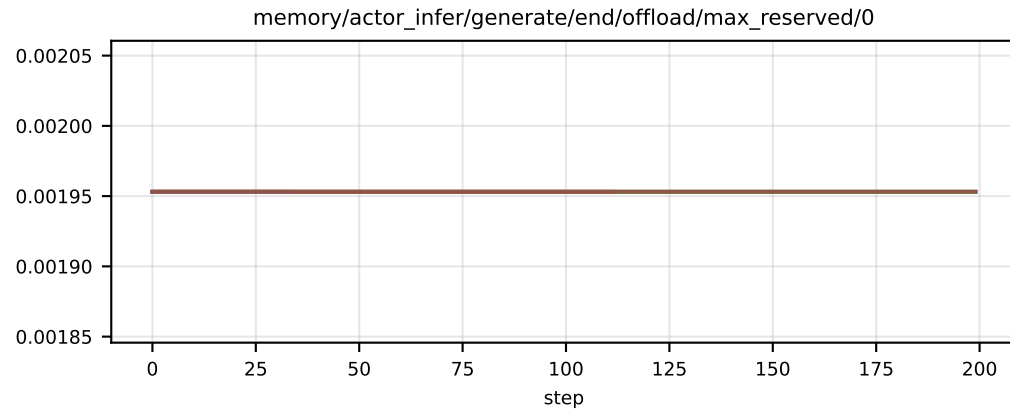
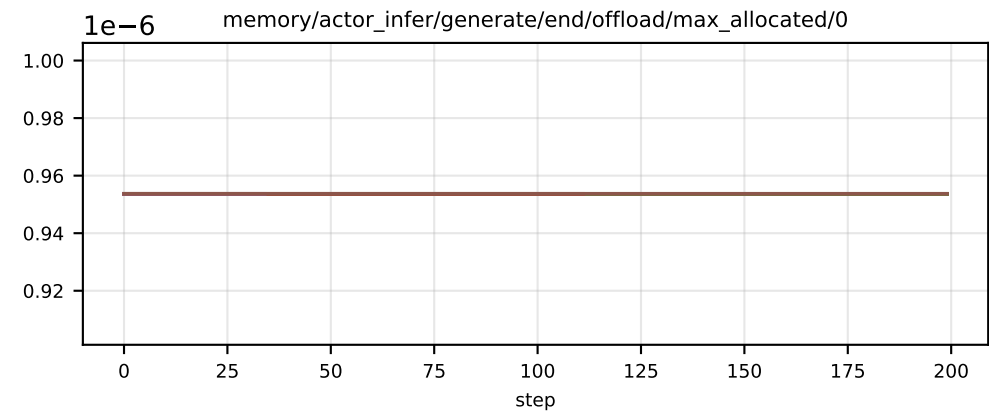
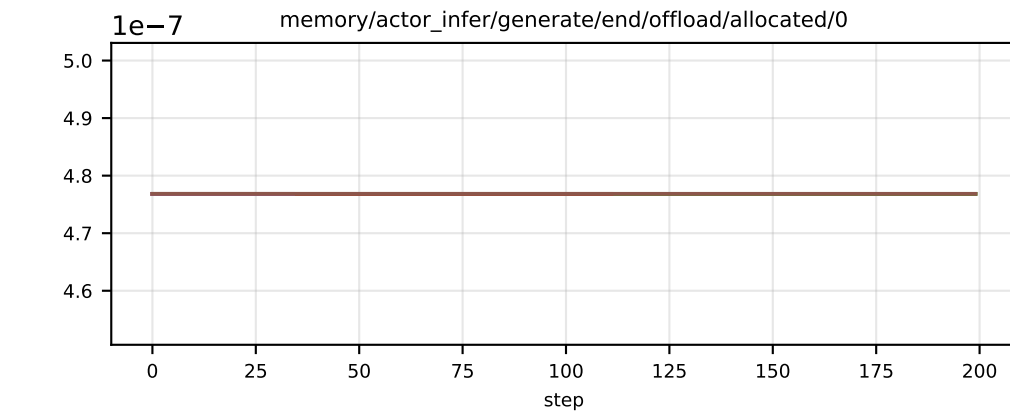
env/TicTacToe/response_length_player_1



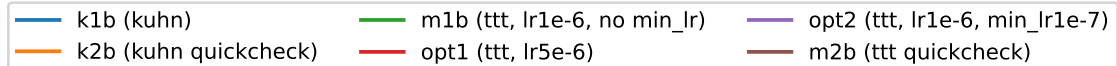
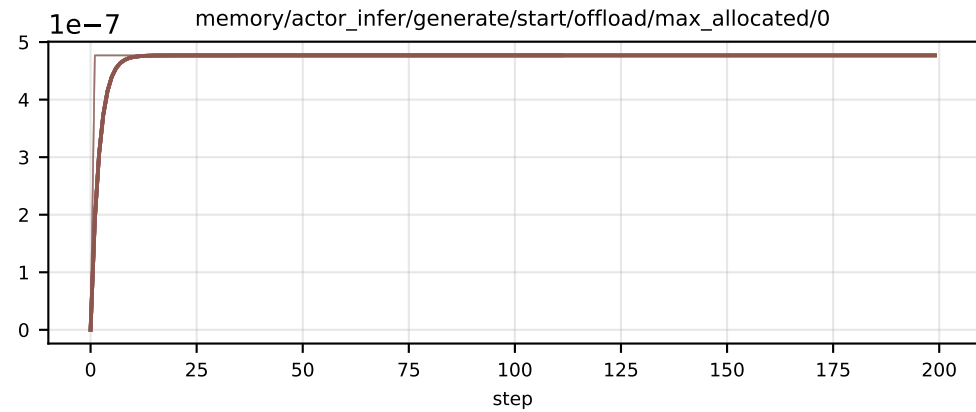
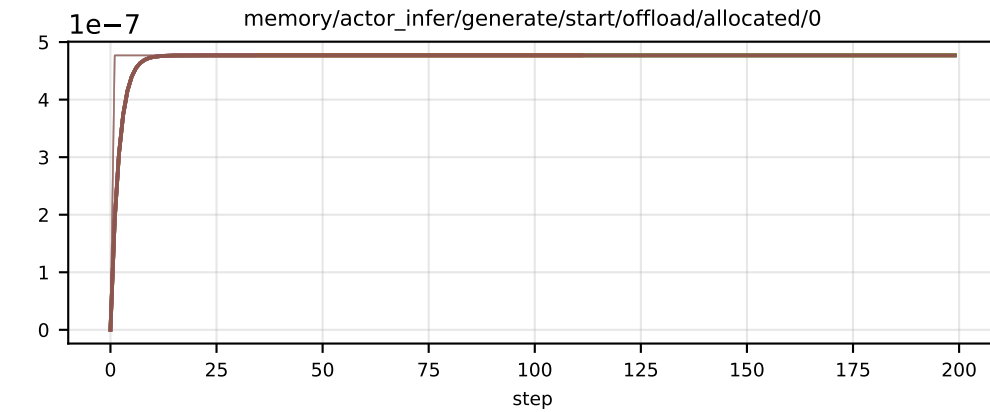
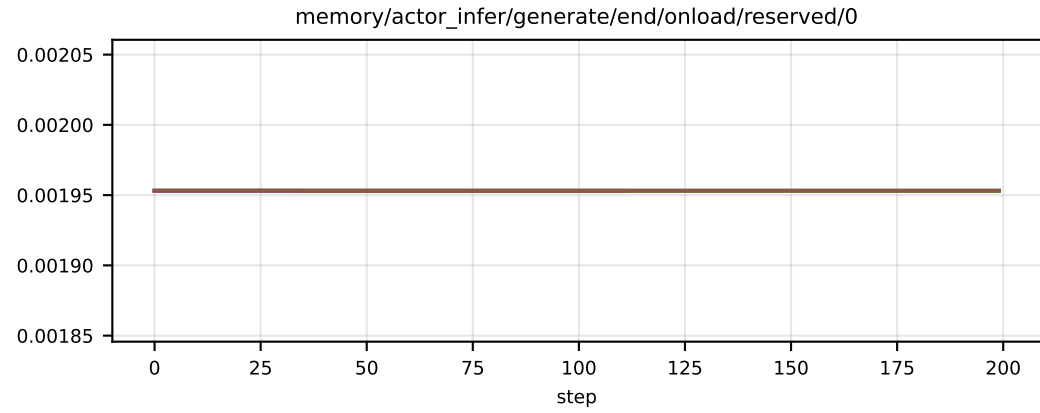
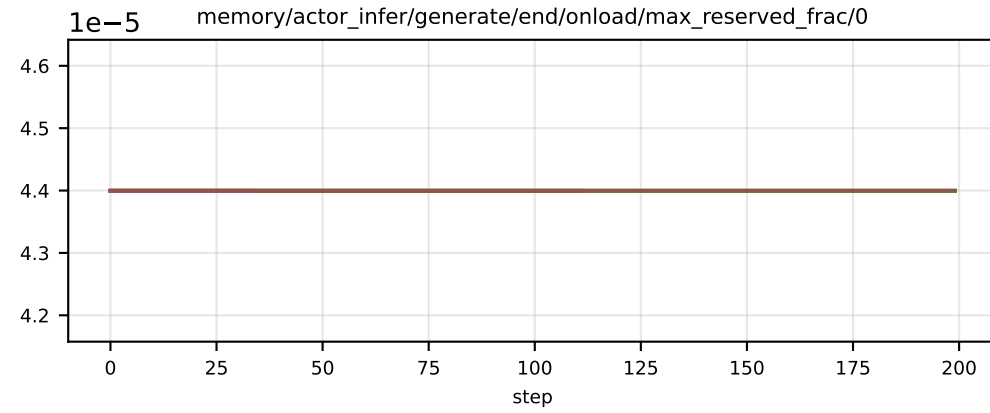
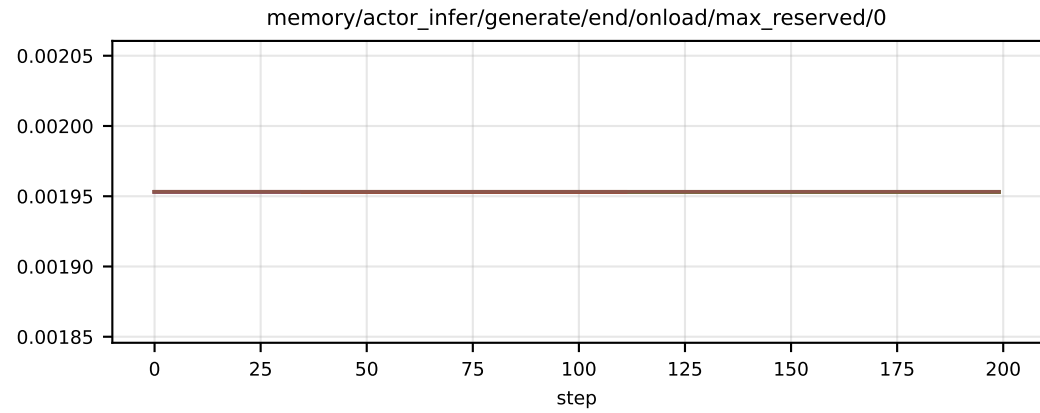
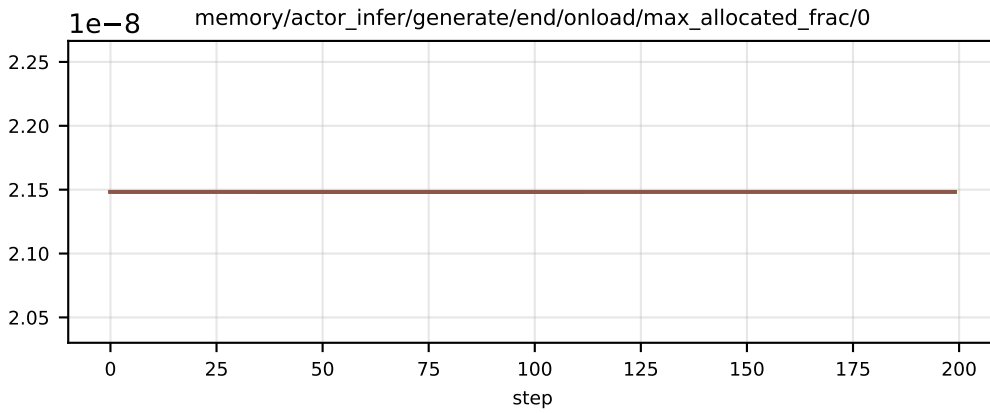
— m1b (ttt, lr1e-6, no min_lr)
 — opt2 (ttt, lr1e-6, min_lr1e-7)
 — m2b (ttt quickcheck)

— opt1 (ttt, lr5e-6)

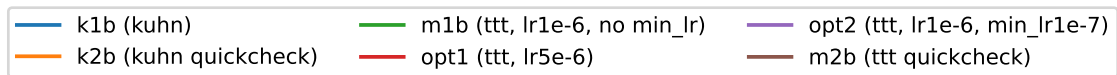
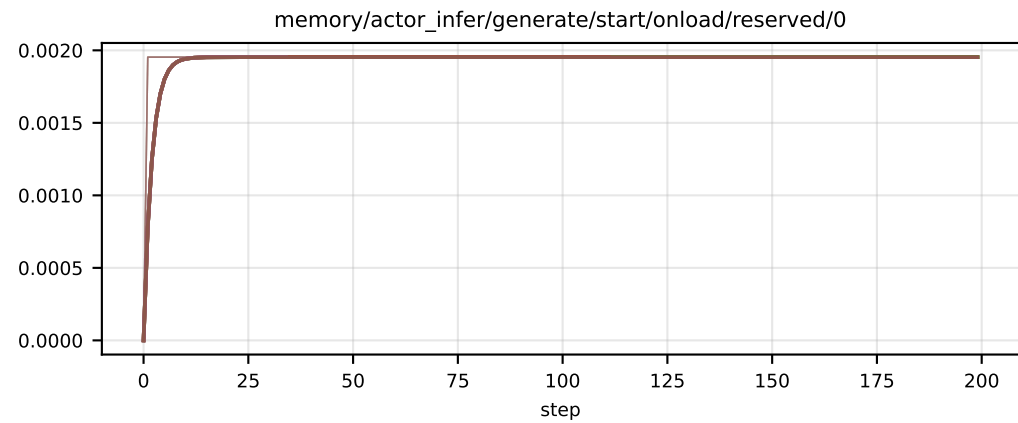
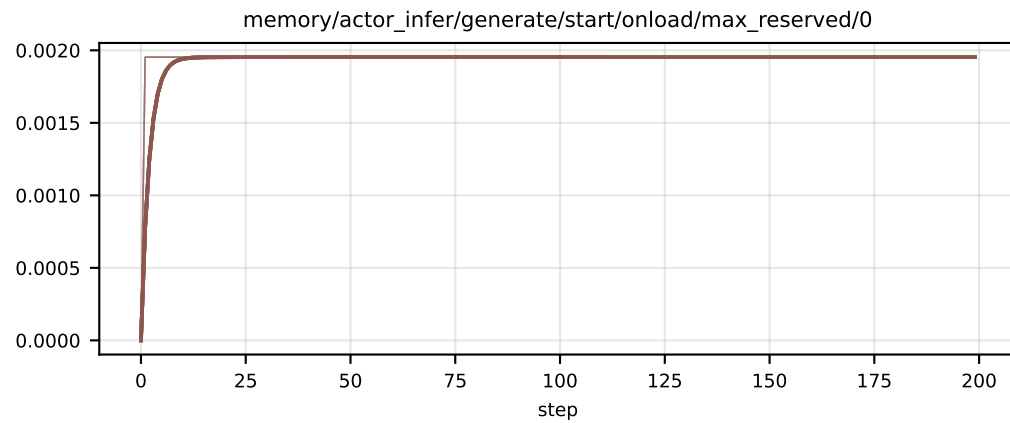
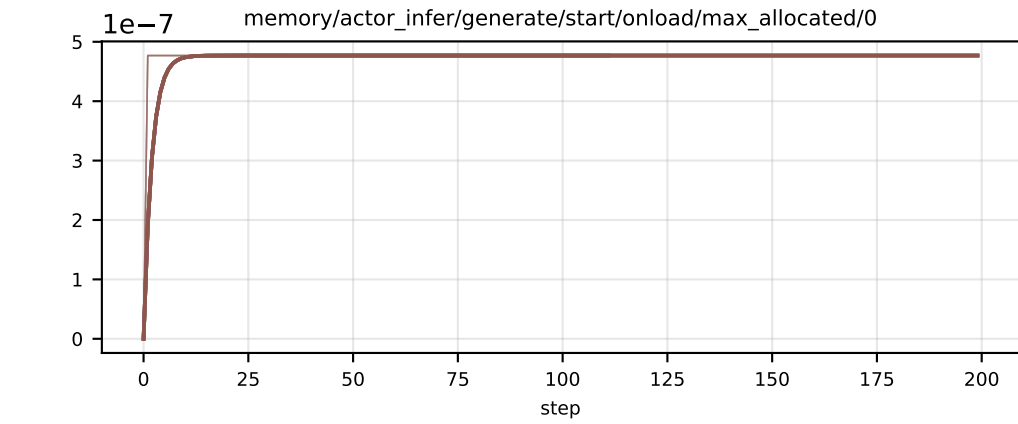
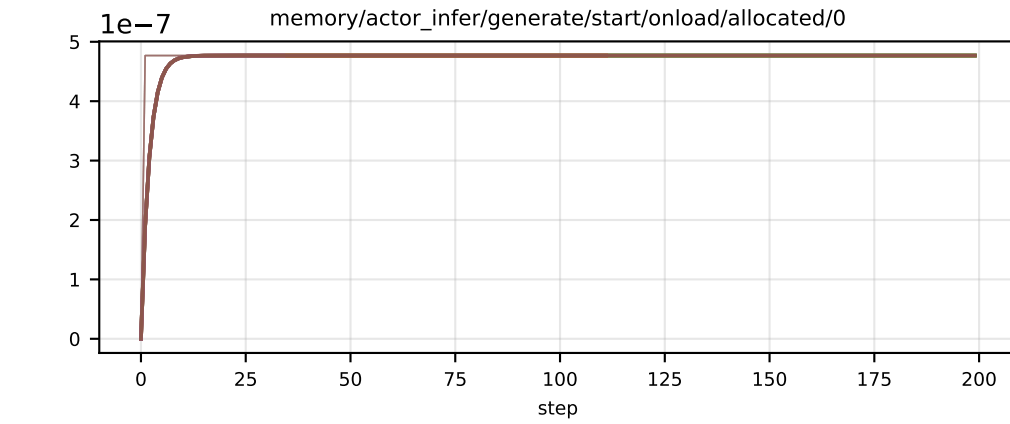
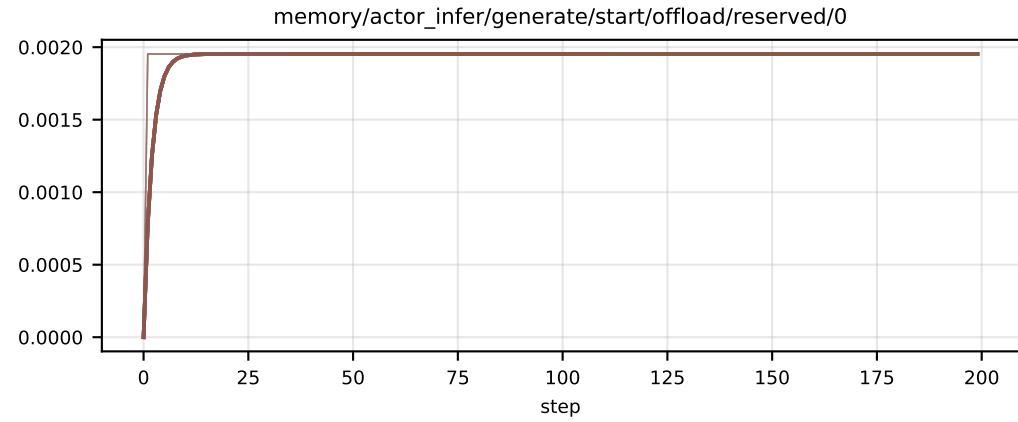
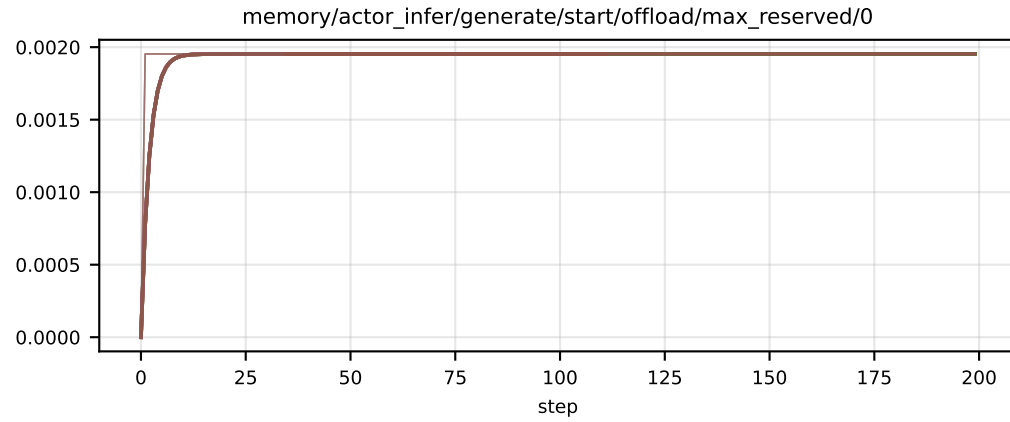
[Appendix] memory/* (1/19)



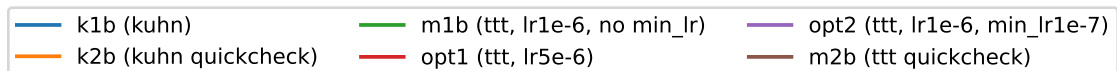
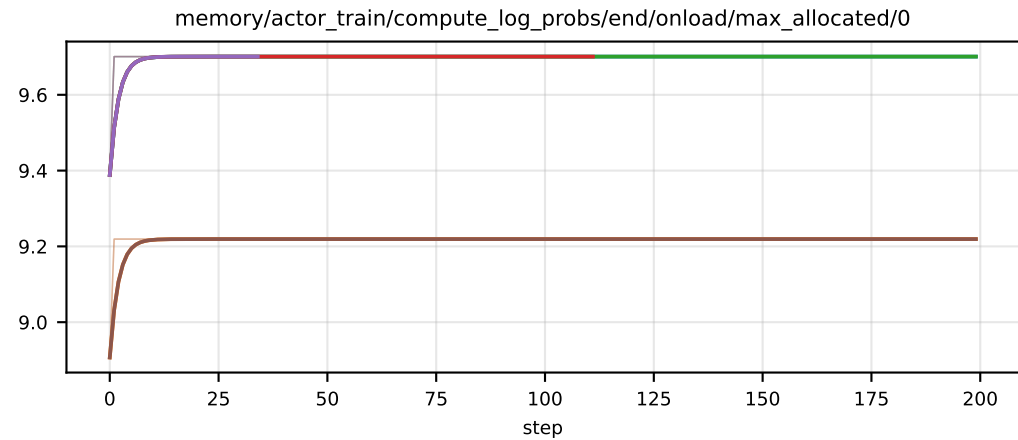
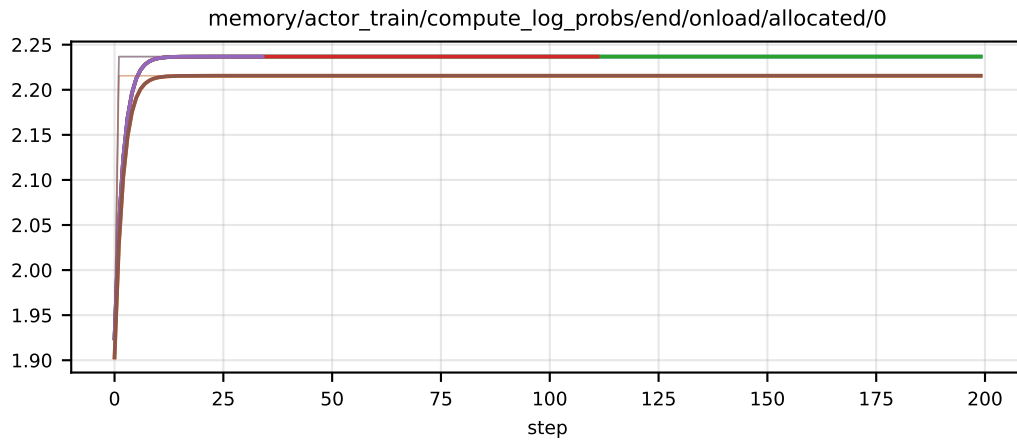
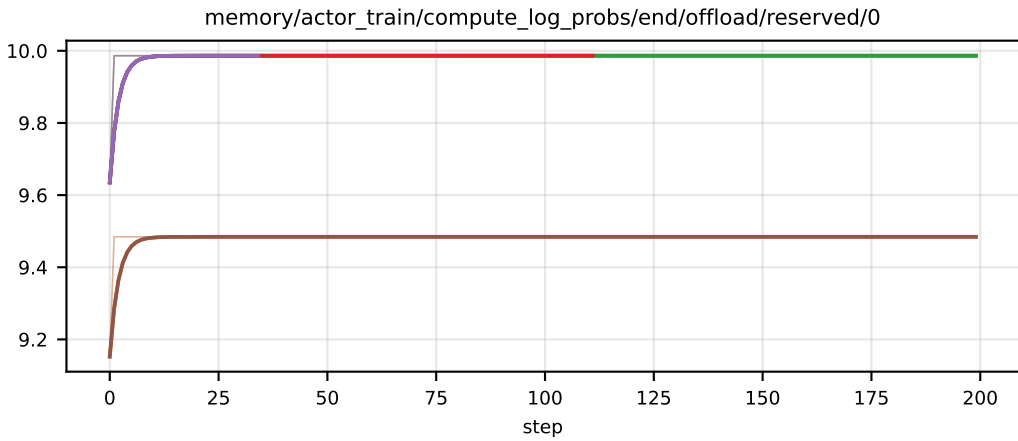
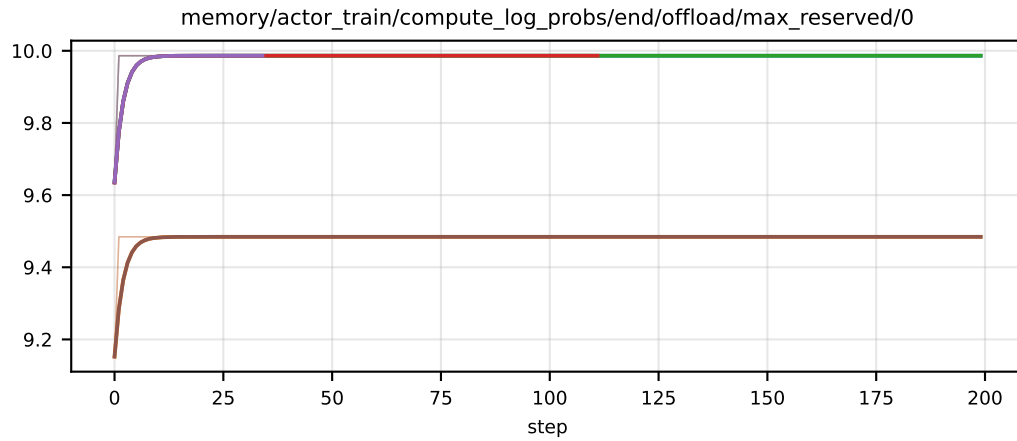
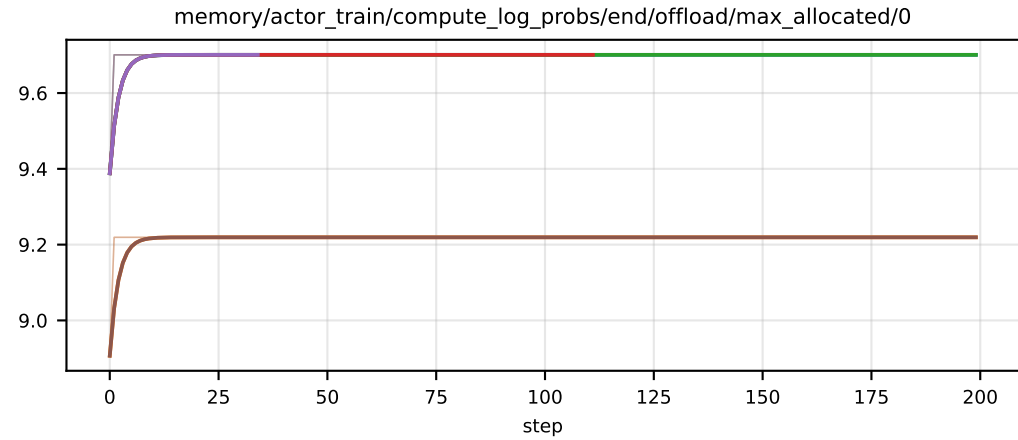
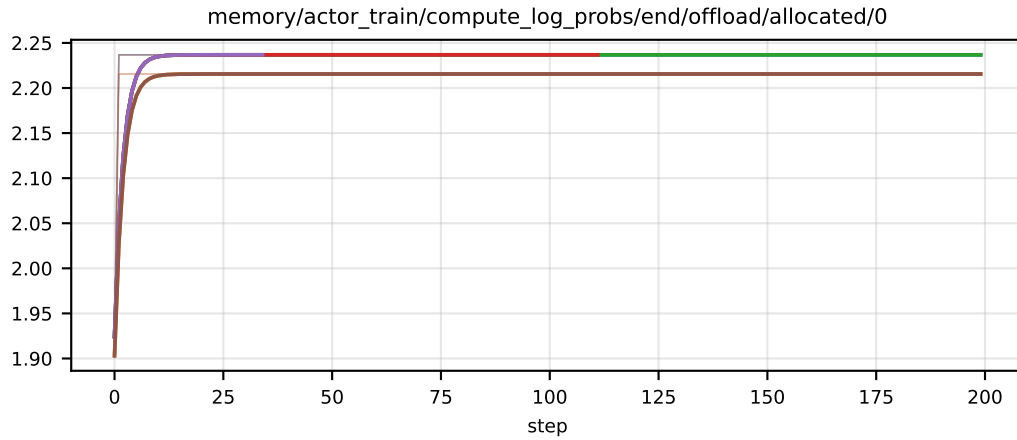
[Appendix] memory/* (2/19)



[Appendix] memory/* (3/19)

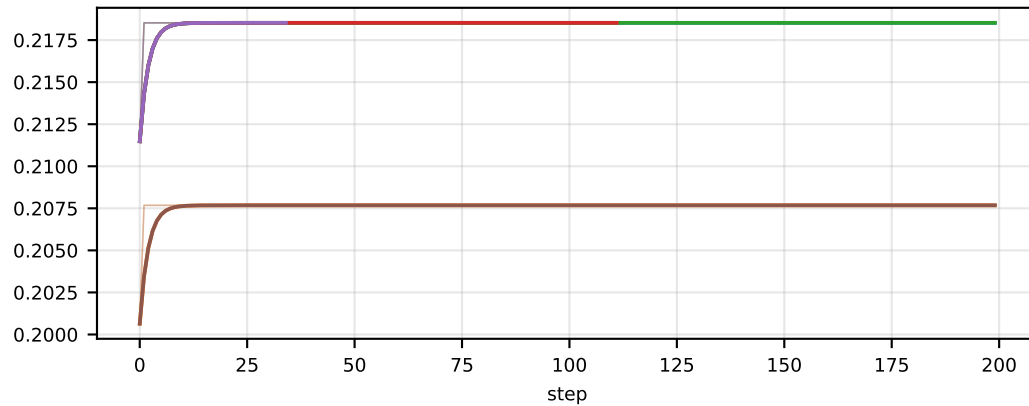


[Appendix] memory/* (4/19)

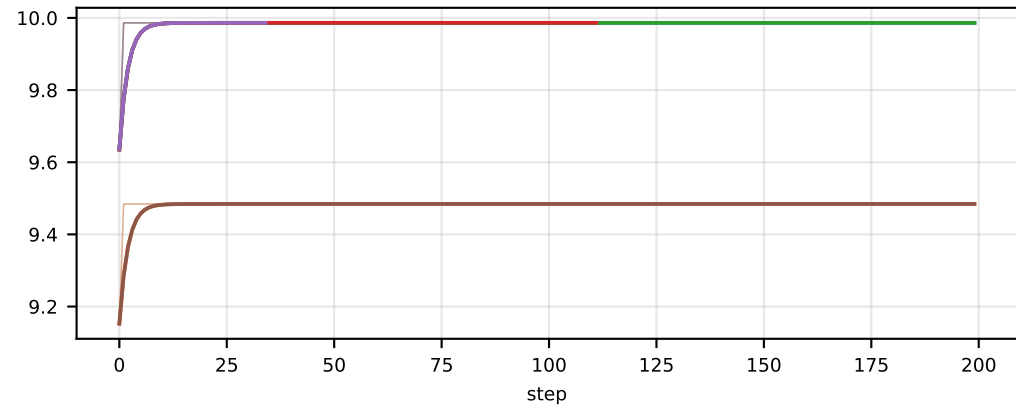


[Appendix] memory/* (5/19)

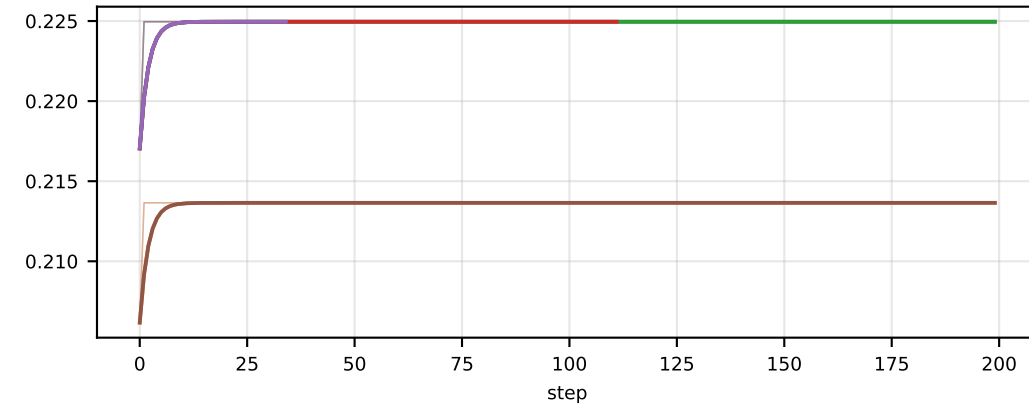
memory/actor_train/compute_log_probs/end/onload/max_allocated_frac/0



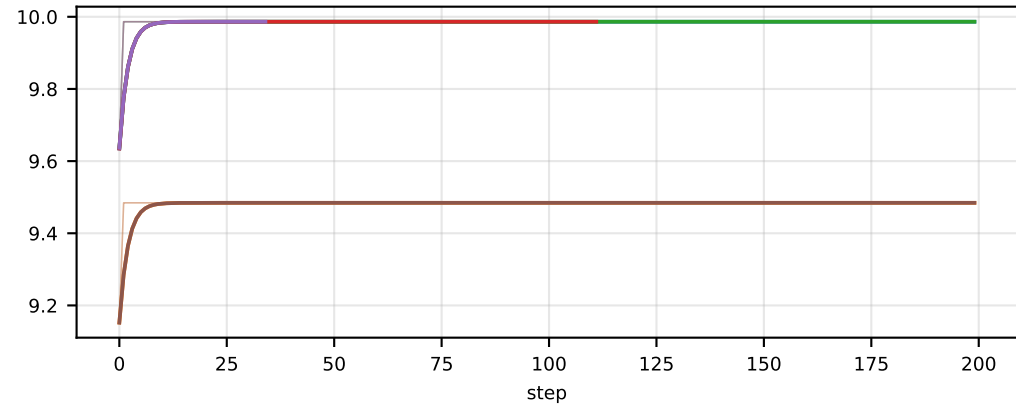
memory/actor_train/compute_log_probs/end/onload/max_reserved/0



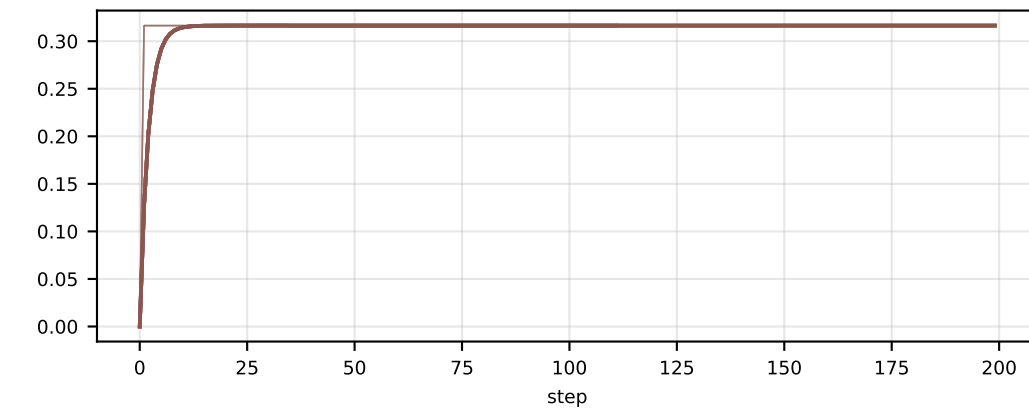
memory/actor_train/compute_log_probs/end/onload/max_reserved_frac/0



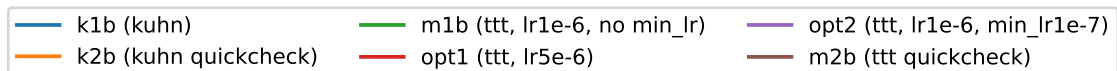
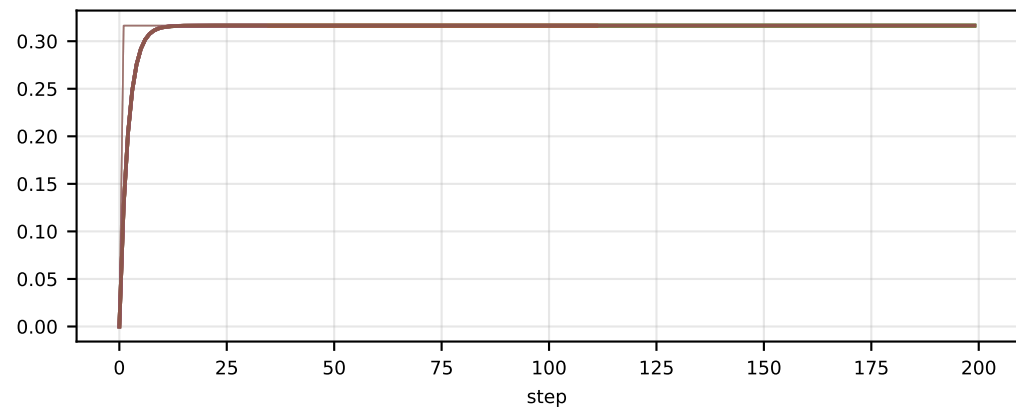
memory/actor_train/compute_log_probs/end/onload/reserved/0



memory/actor_train/compute_log_probs/start/offload/allocated/0

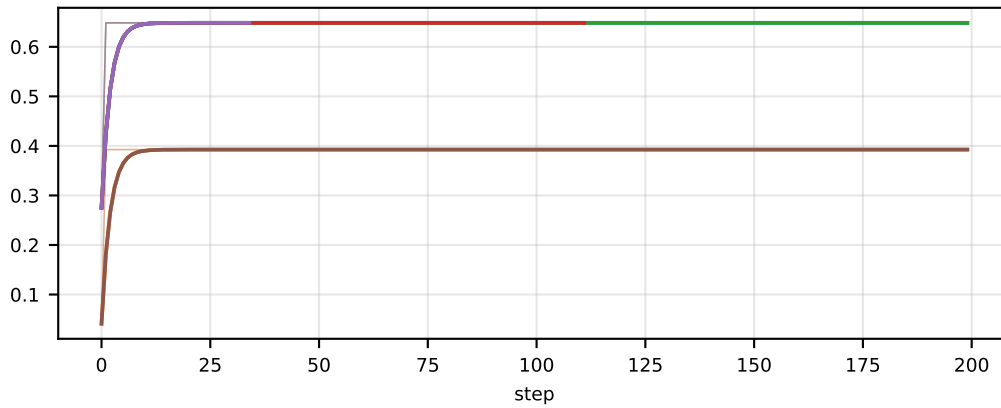


memory/actor_train/compute_log_probs/start/offload/max_allocated/0

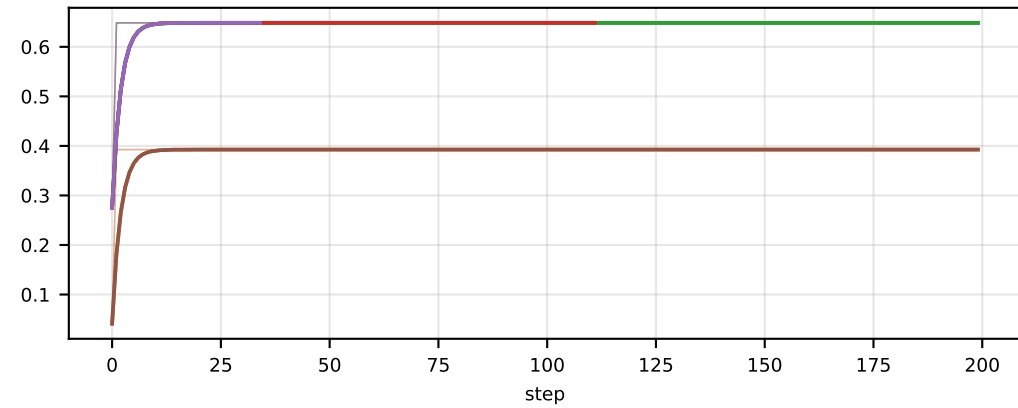


[Appendix] memory/* (6/19)

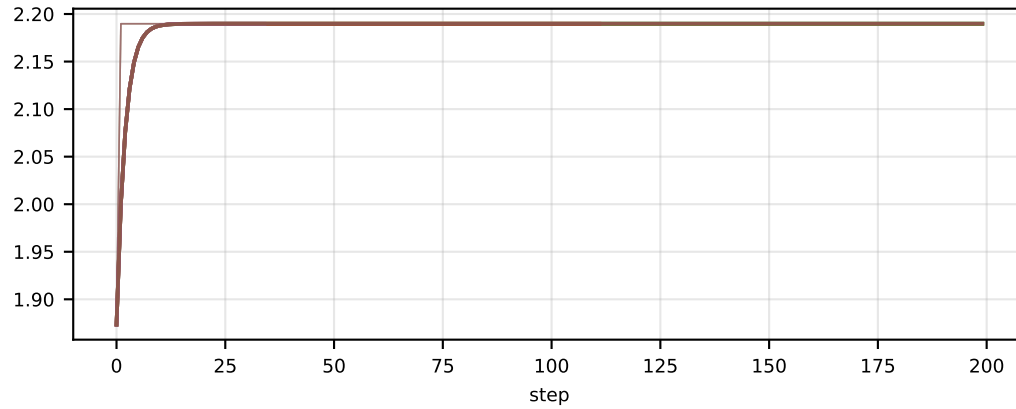
memory/actor_train/compute_log_probs/start/offload/max_reserved/0



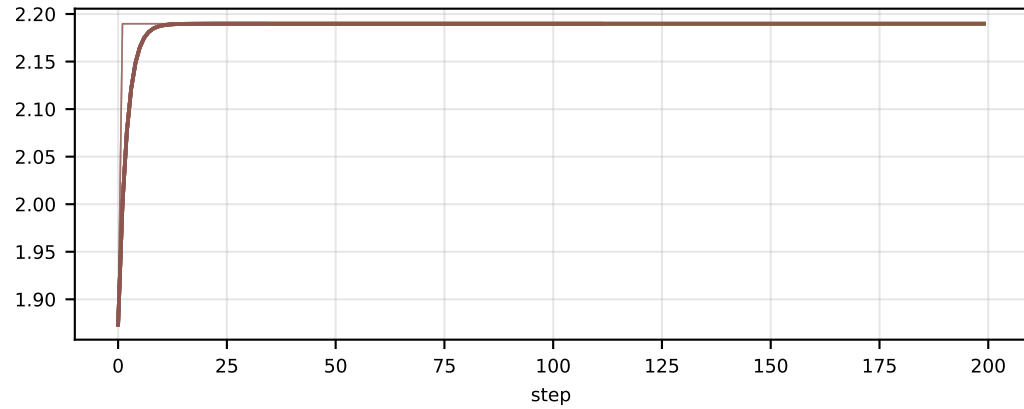
memory/actor_train/compute_log_probs/start/offload/reserved/0



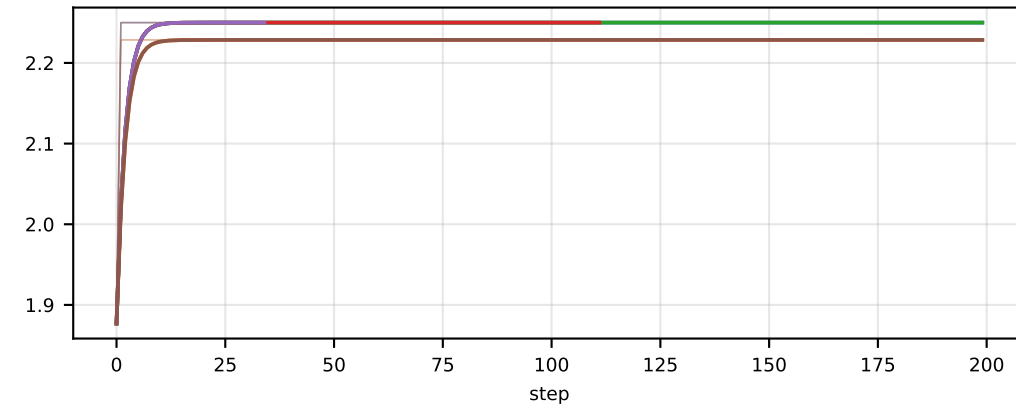
memory/actor_train/compute_log_probs/start/onload/allocated/0



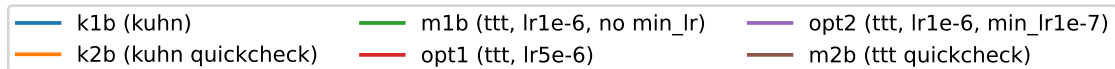
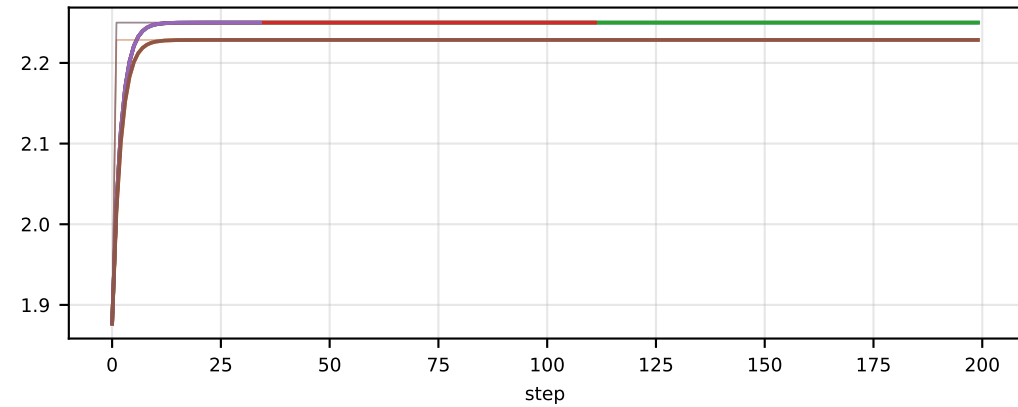
memory/actor_train/compute_log_probs/start/onload/max_allocated/0



memory/actor_train/compute_log_probs/start/onload/max_reserved/0

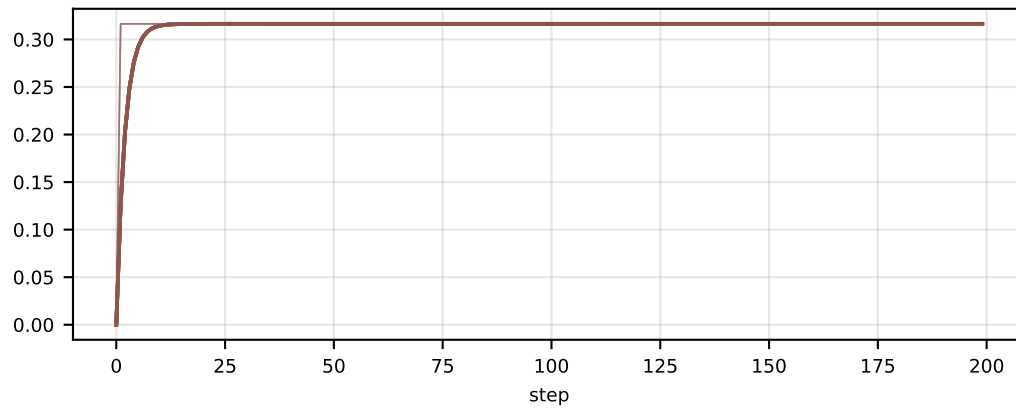


memory/actor_train/compute_log_probs/start/onload/reserved/0

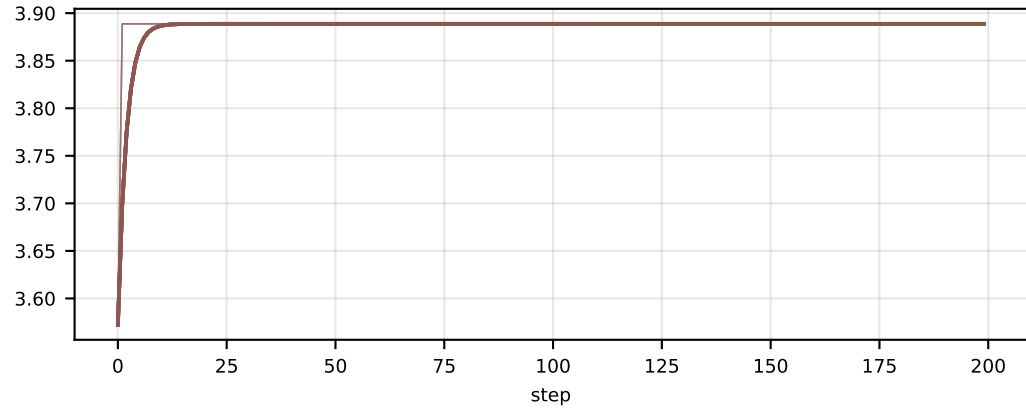


[Appendix] memory/* (7/19)

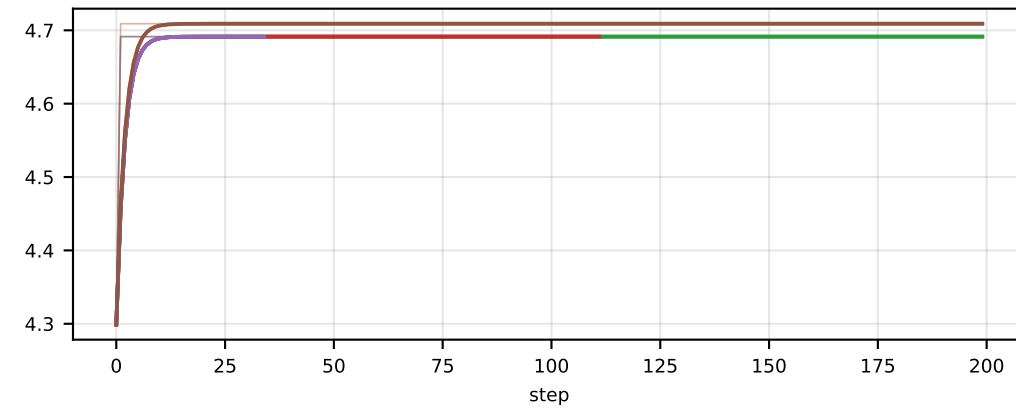
memory/actor_train/model_update/end/offload/allocated/0



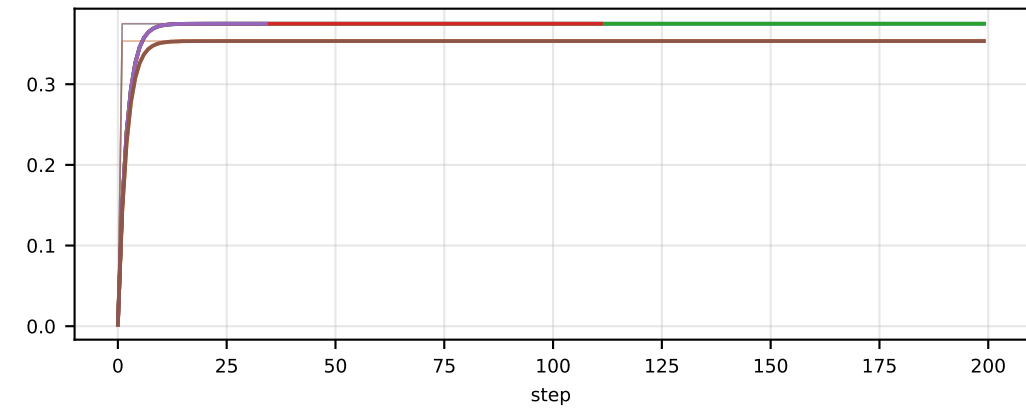
memory/actor_train/model_update/end/offload/max_allocated/0



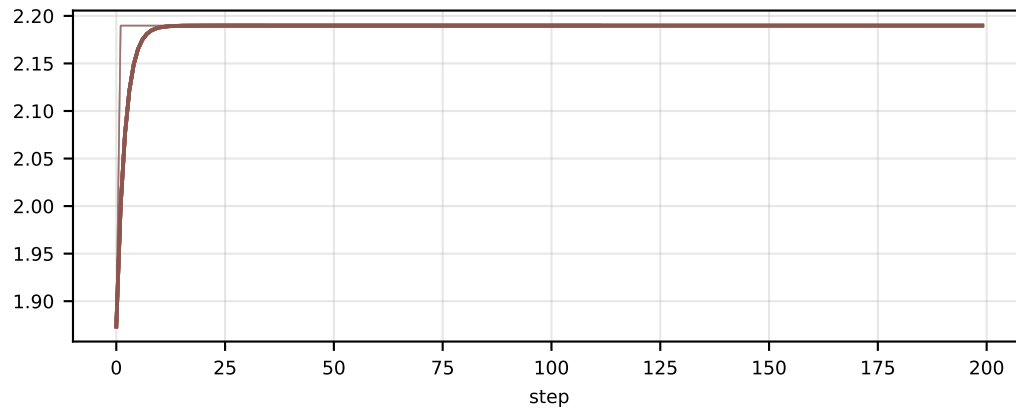
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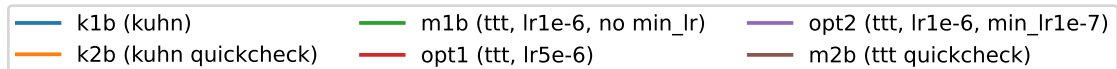
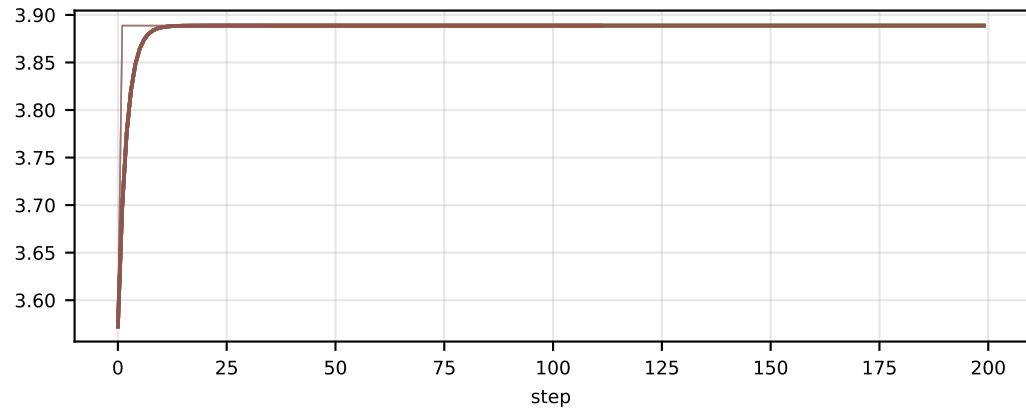
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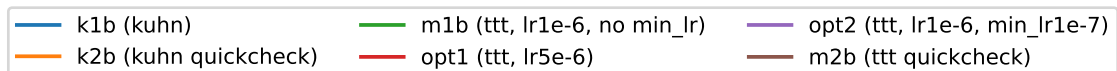
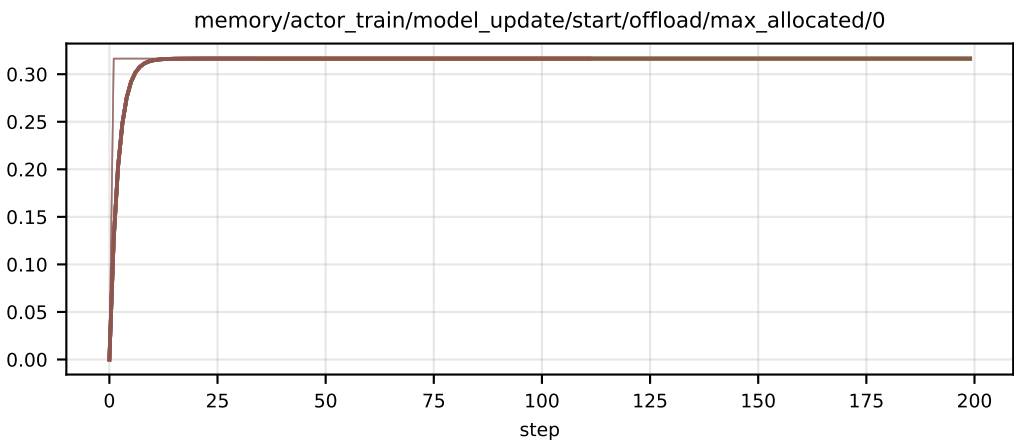
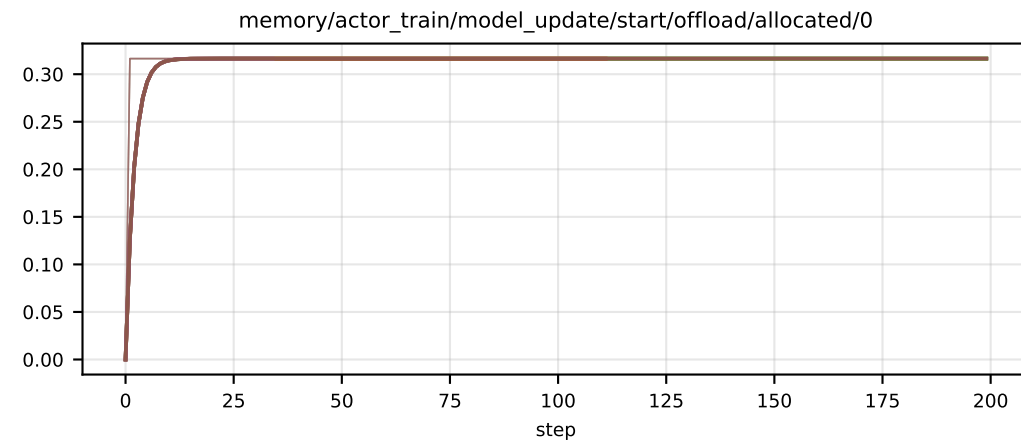
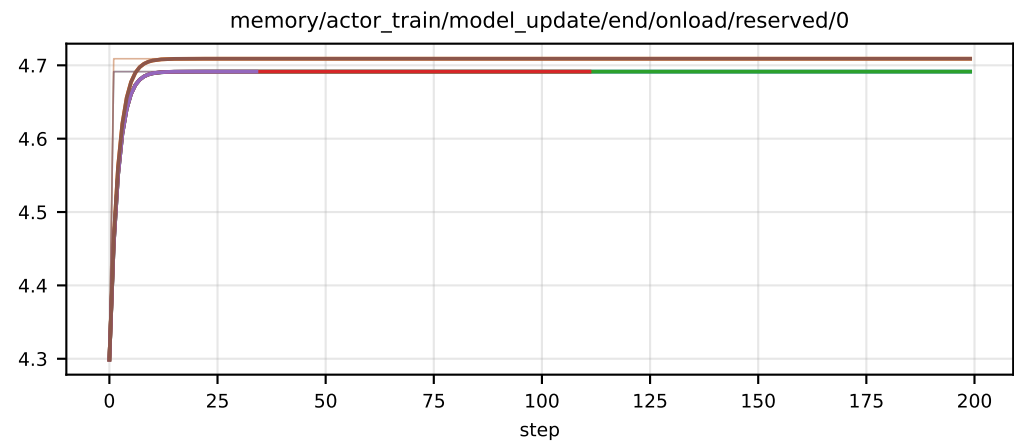
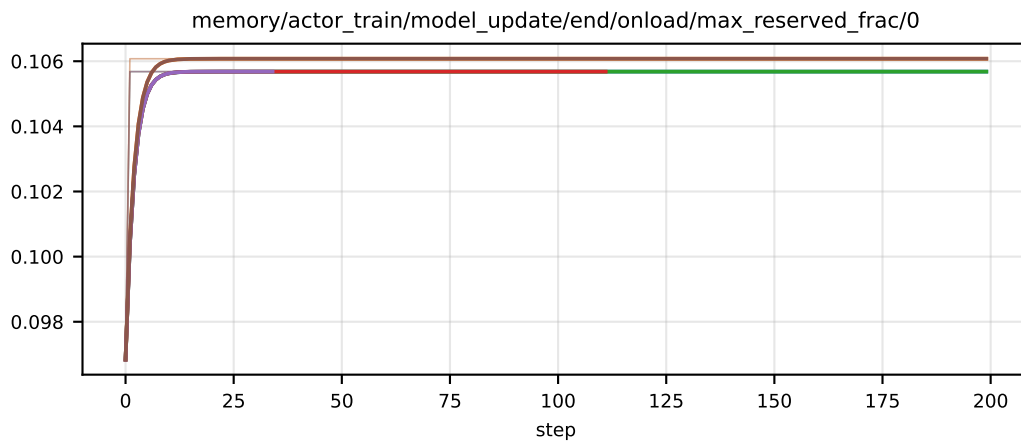
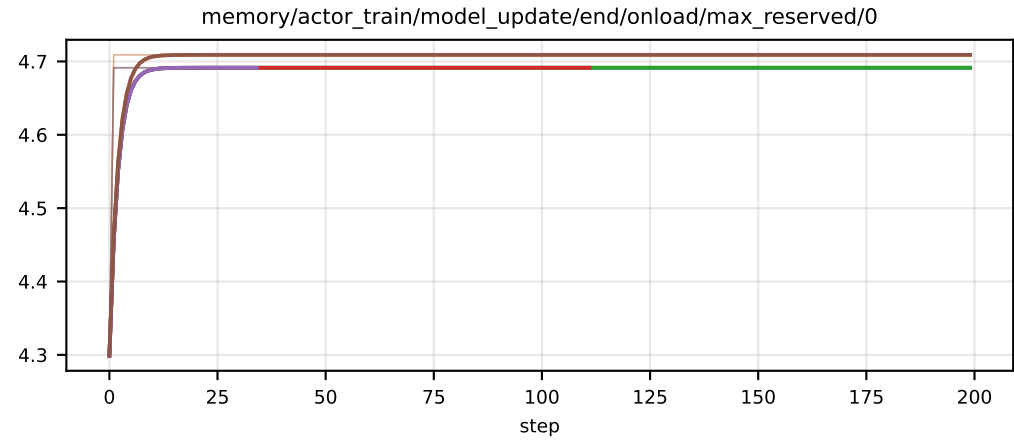
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memory/actor_train/model_update/end/onload/max_allocated/0

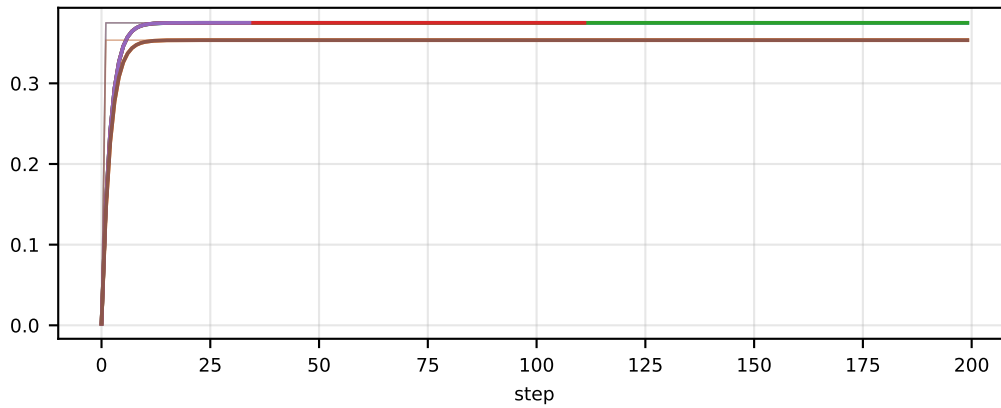


[Appendix] memory/* (8/19)

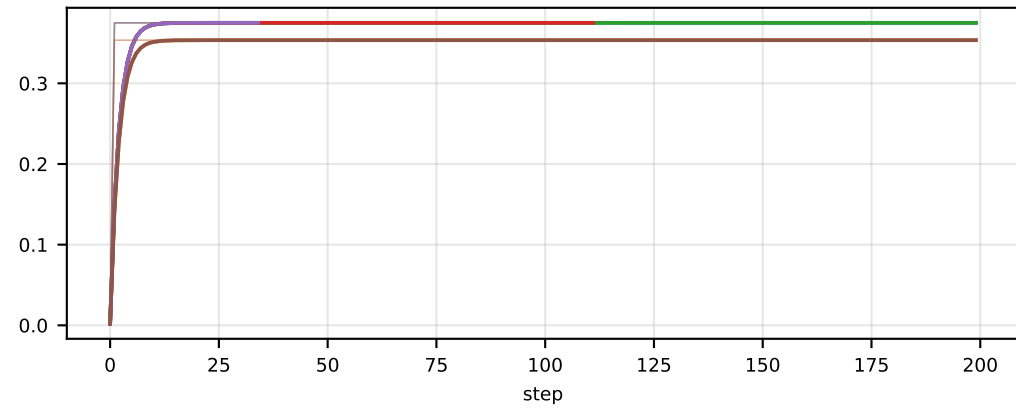


[Appendix] memory/* (9/19)

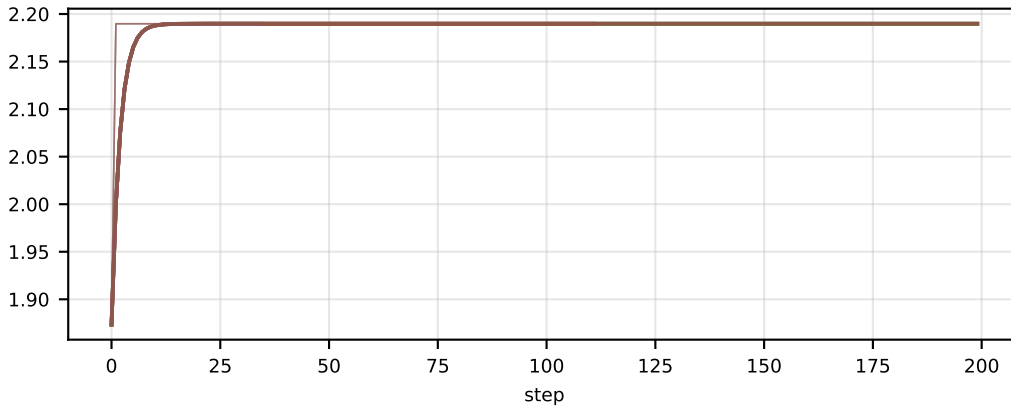
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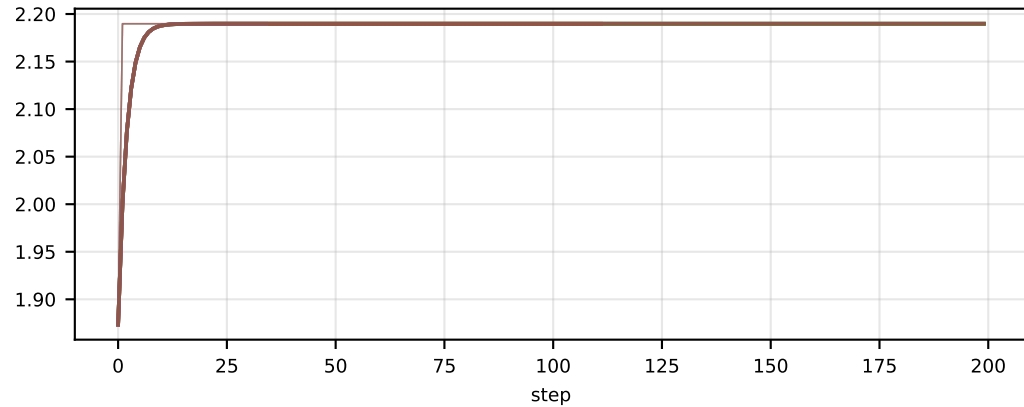
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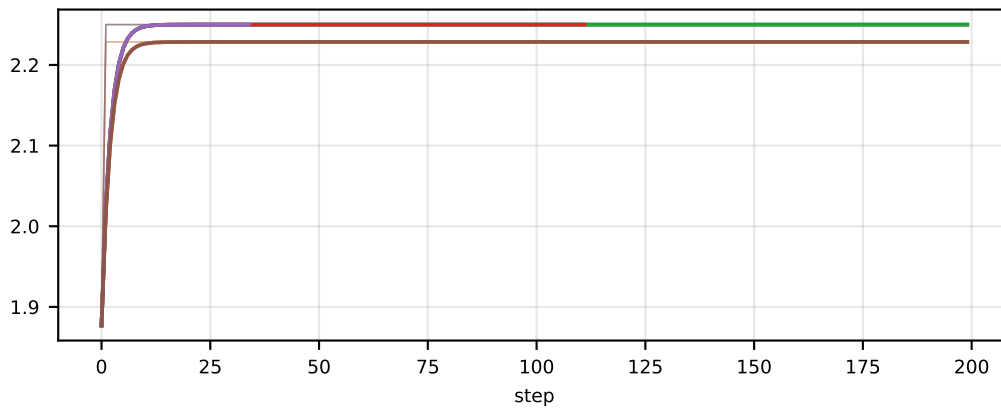
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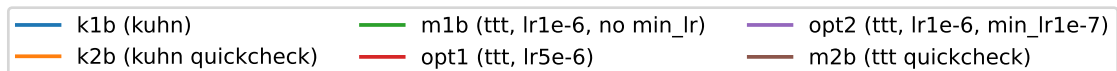
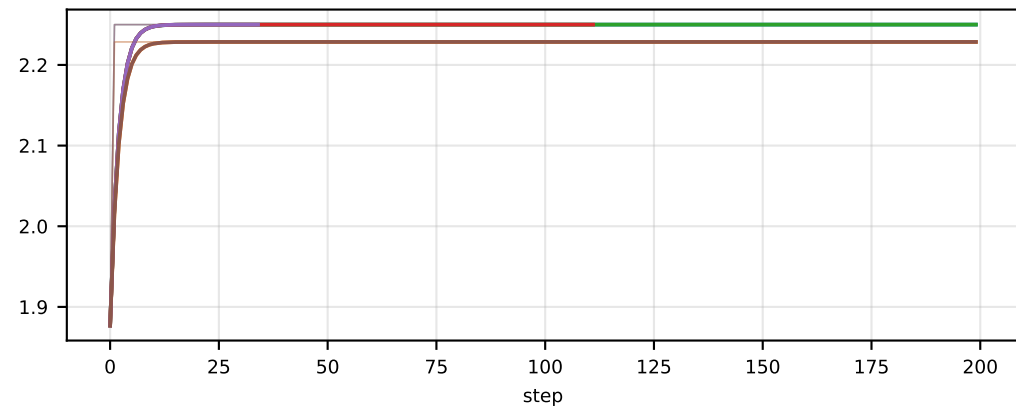
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memory/actor_train/model_update/start/onload/max_reserved/0

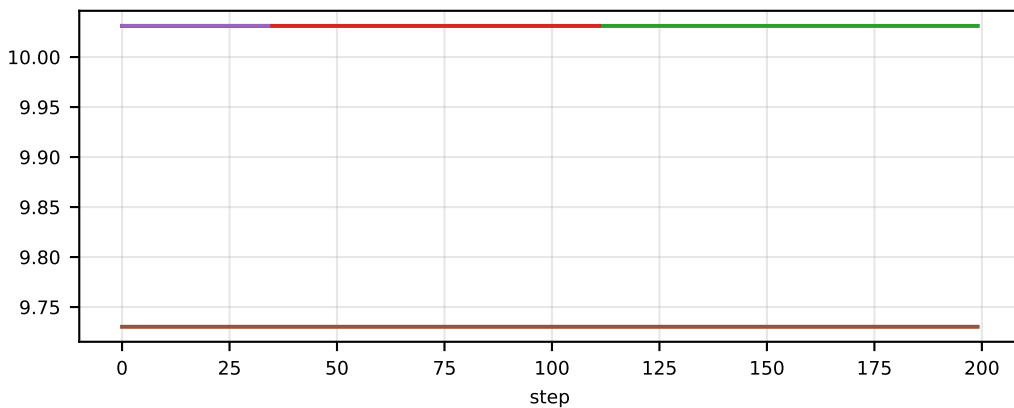


memory/actor_train/model_update/start/onload/reserved/0

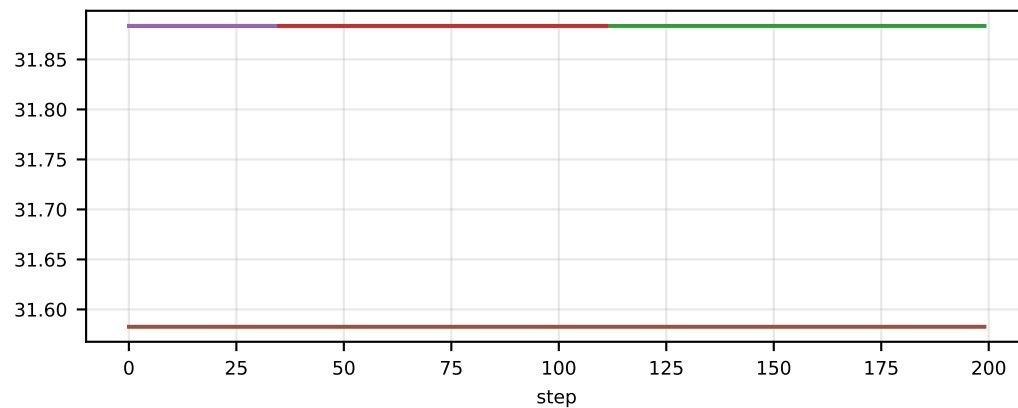


[Appendix] memory/* (10/19)

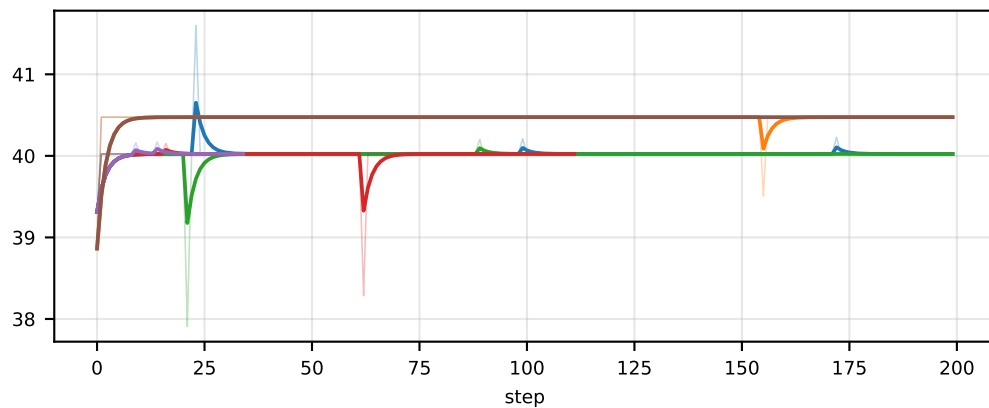
memory/actor_train/train_step/end/offload/allocated/0



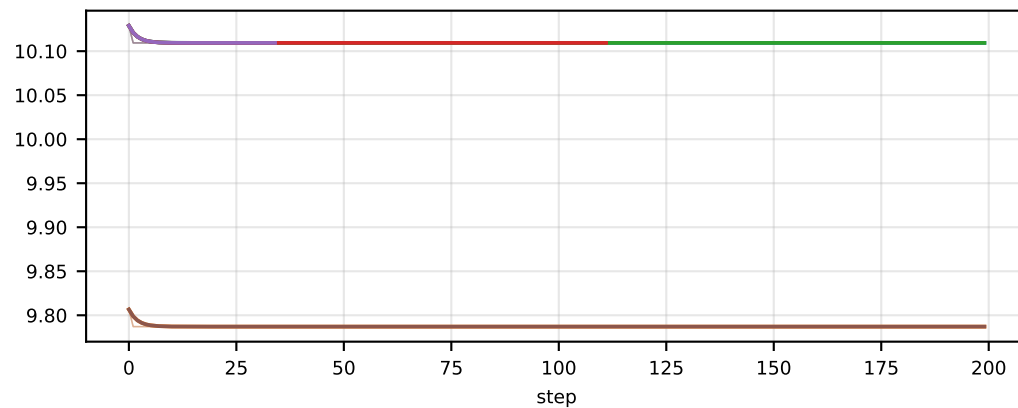
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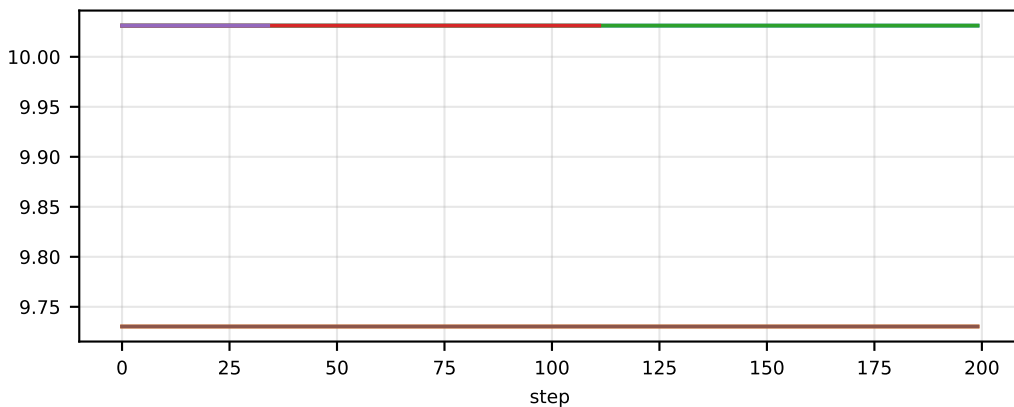
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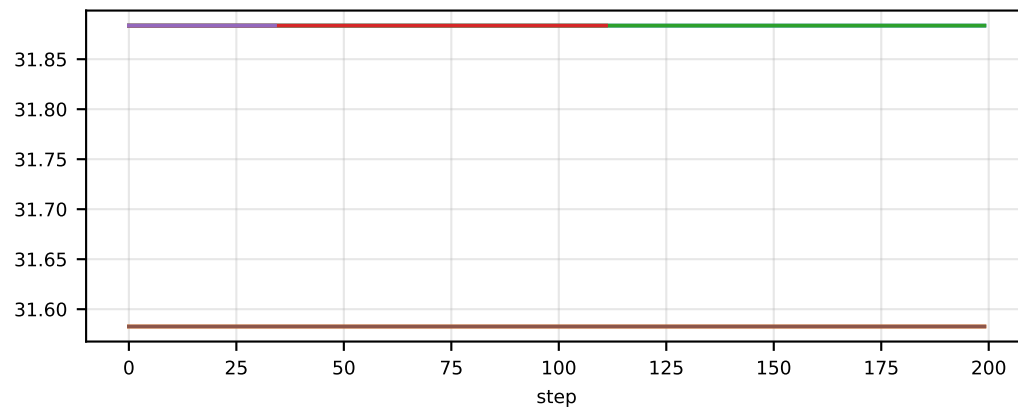
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memory/actor_train/train_step/end/onload/allocated/0



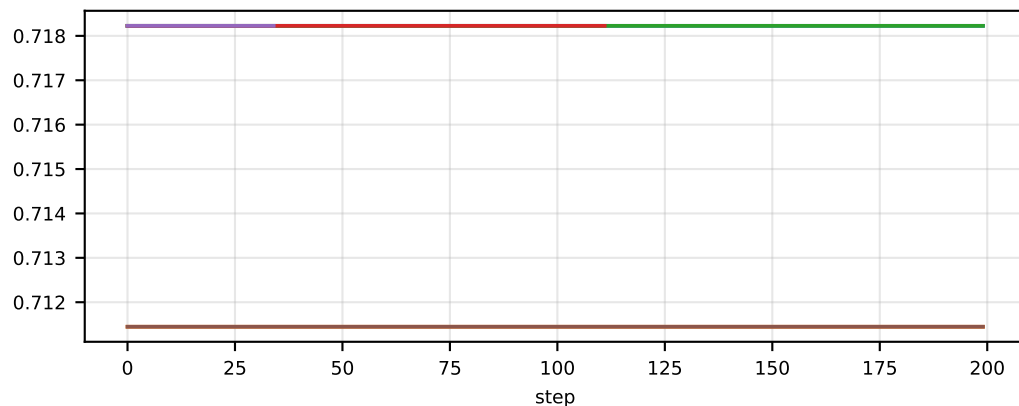
memory/actor_train/train_step/end/onload/max_allocated/0



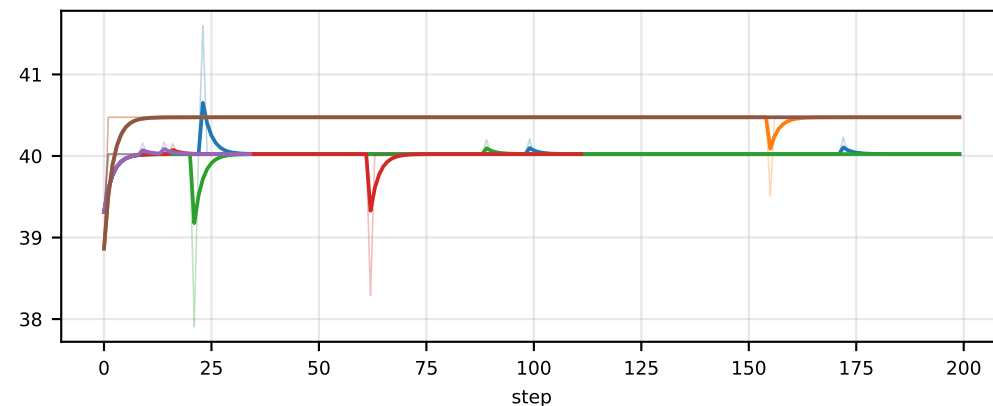
— k1b (kuhn) — m1b (ttr, lr1e-6, no min_lr) — opt2 (ttr, lr1e-6, min_lr1e-7)
— k2b (kuhn quickcheck) — opt1 (ttr, lr5e-6) — m2b (ttr quickcheck)

[Appendix] memory/* (11/19)

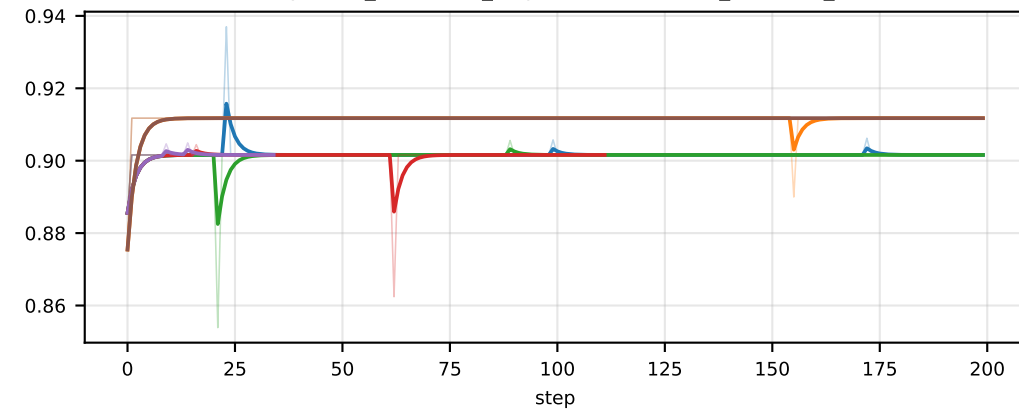
memory/actor_train/train_step/end/onload/max_allocated_frac/0



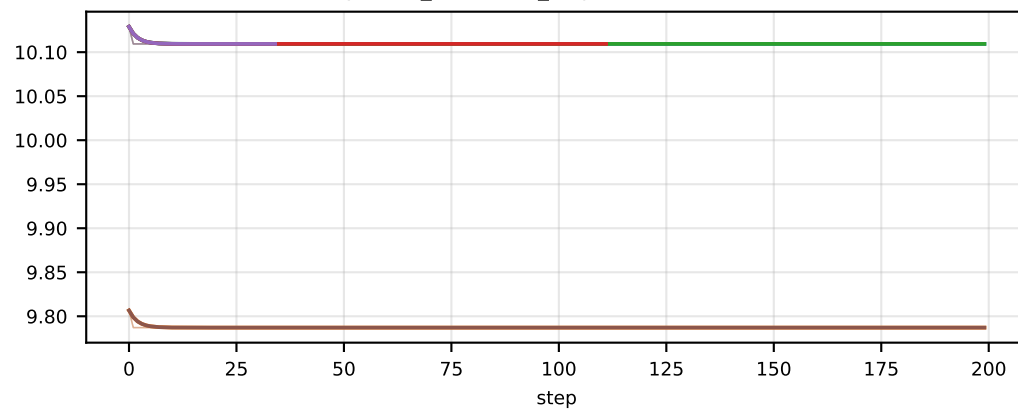
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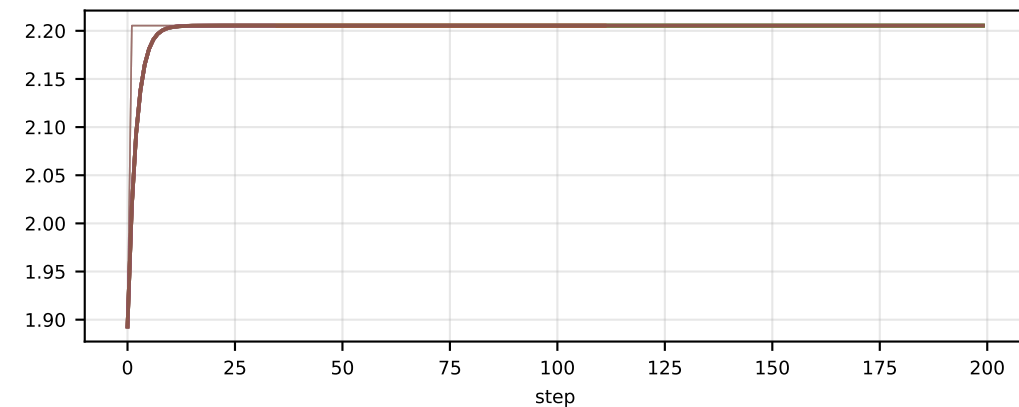
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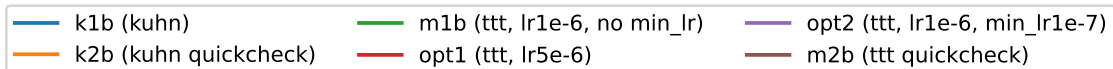
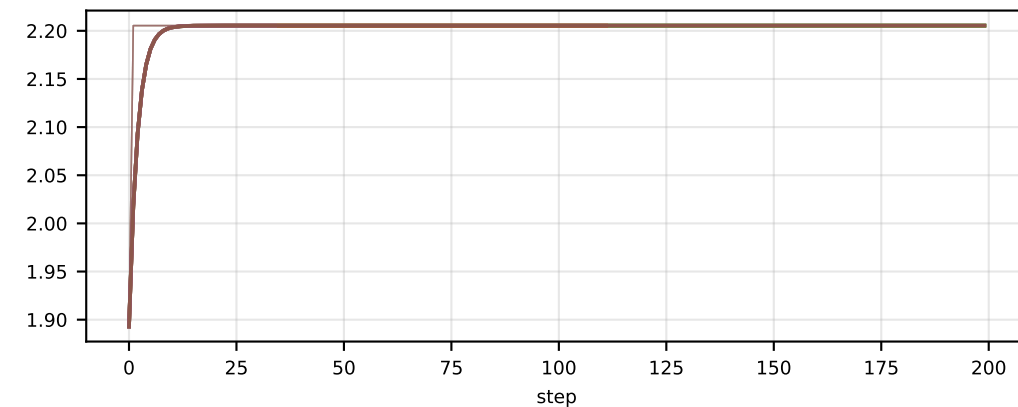
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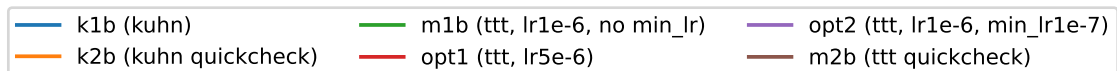
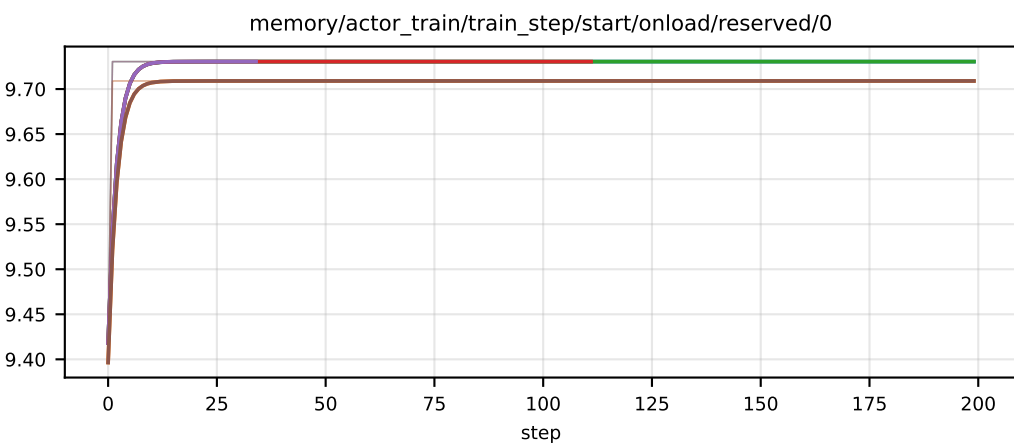
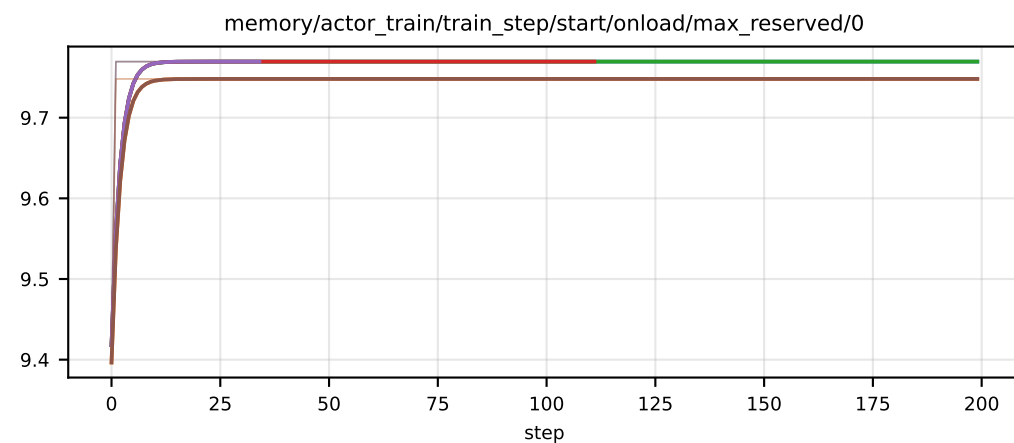
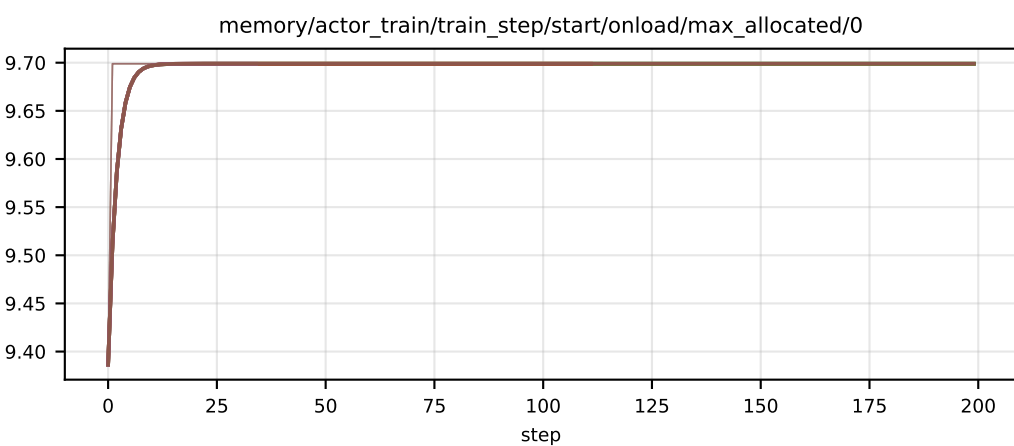
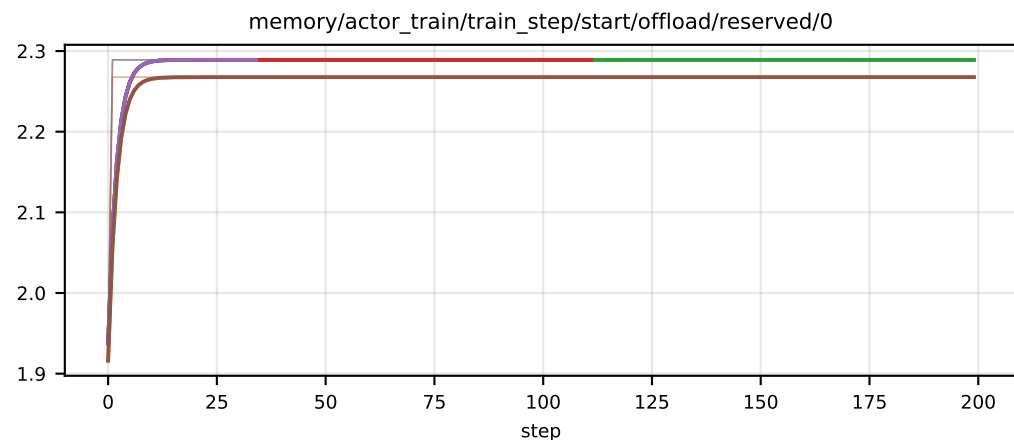
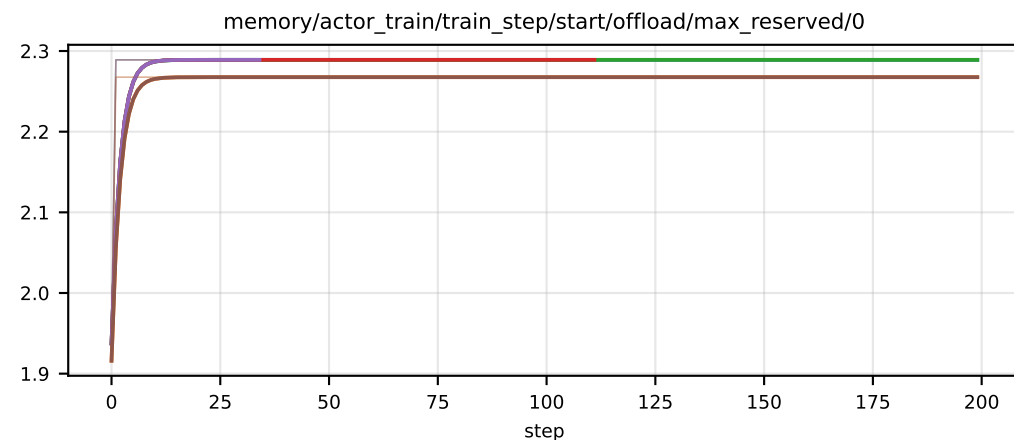
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memory/actor_train/train_step/start/offload/max_allocated/0

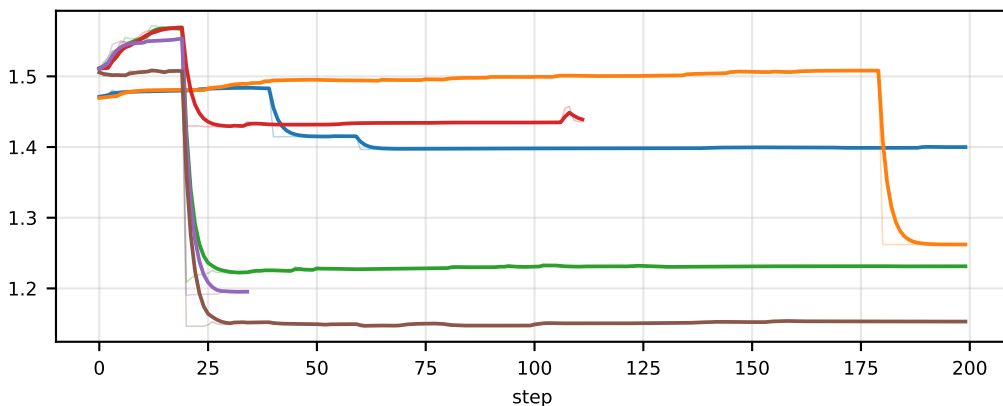


[Appendix] memory/* (12/19)

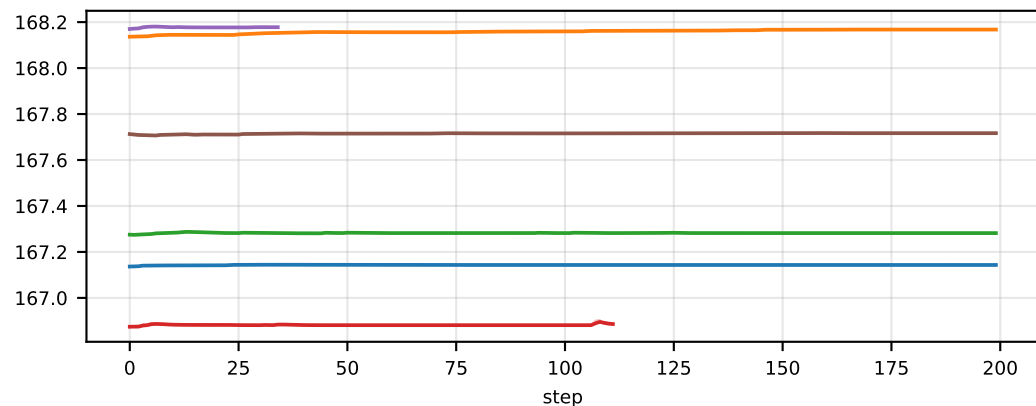


[Appendix] memory/* (13/19)

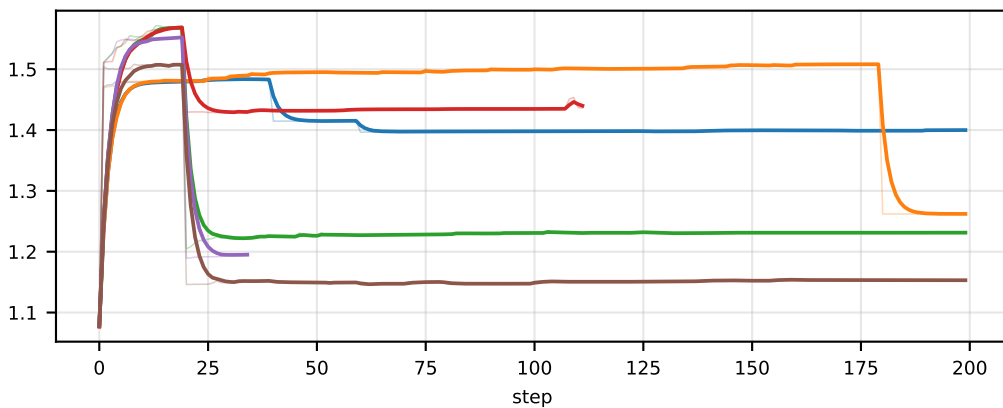
memory/cpu/actor_infer/generate/end/rss



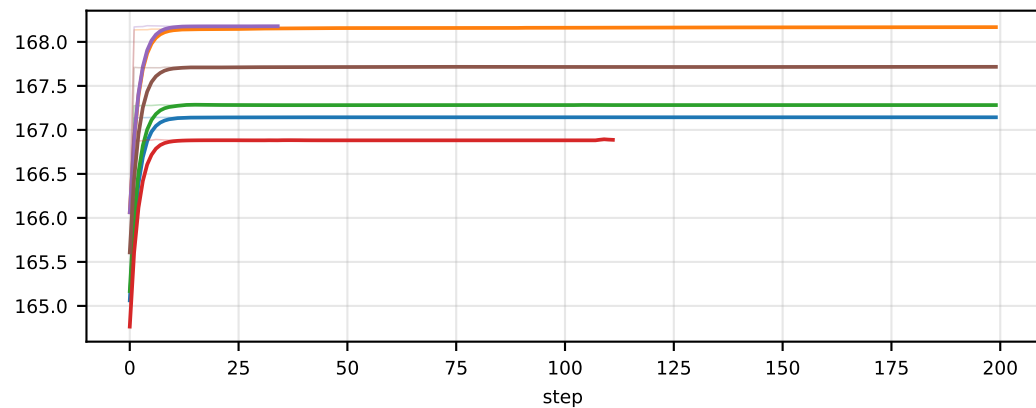
memory/cpu/actor_infer/generate/end/vms



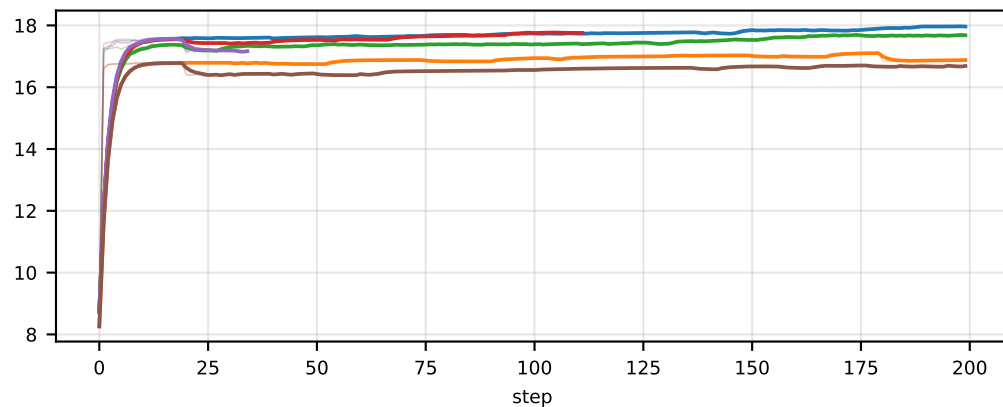
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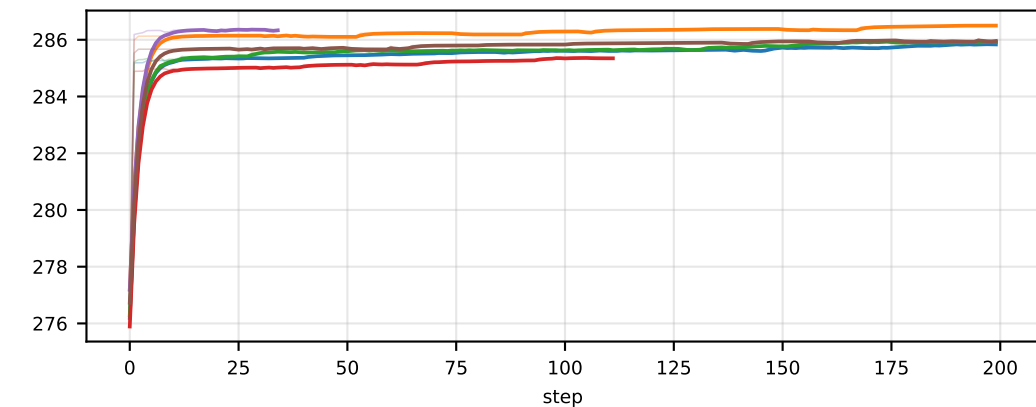
memory/cpu/actor_infer/generate/start/vms



memory/cpu/actor_train/compute_log_probs/end/rss



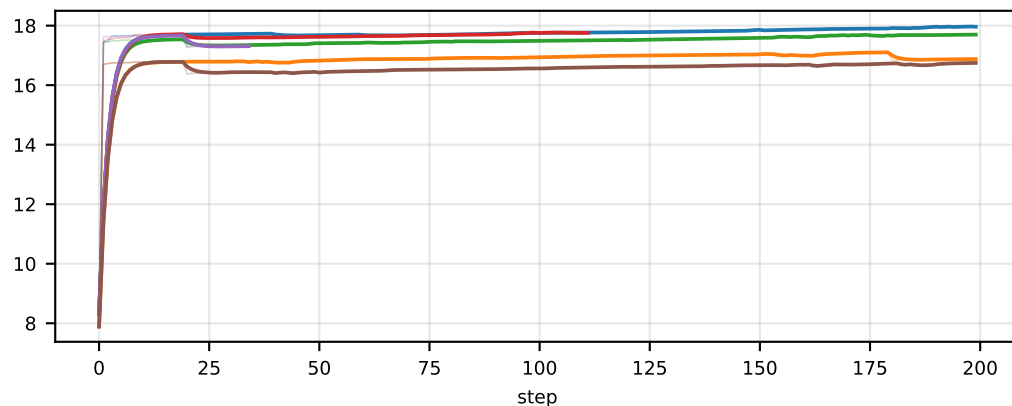
memory/cpu/actor_train/compute_log_probs/end/vms



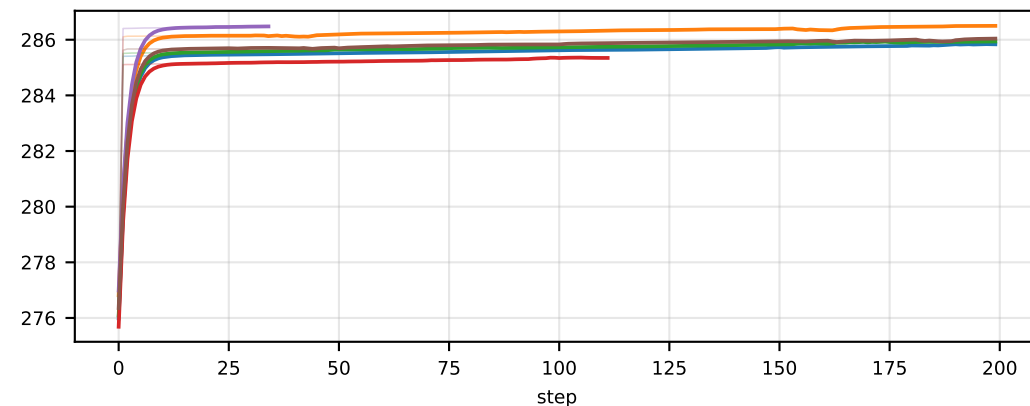
— k1b (kuhn) — m1b (ttt, lr1e-6, no min_lr) — opt2 (ttt, lr1e-6, min_lr1e-7)
— k2b (kuhn quickcheck) — opt1 (ttt, lr5e-6) — m2b (ttt quickcheck)

[Appendix] memory/* (14/19)

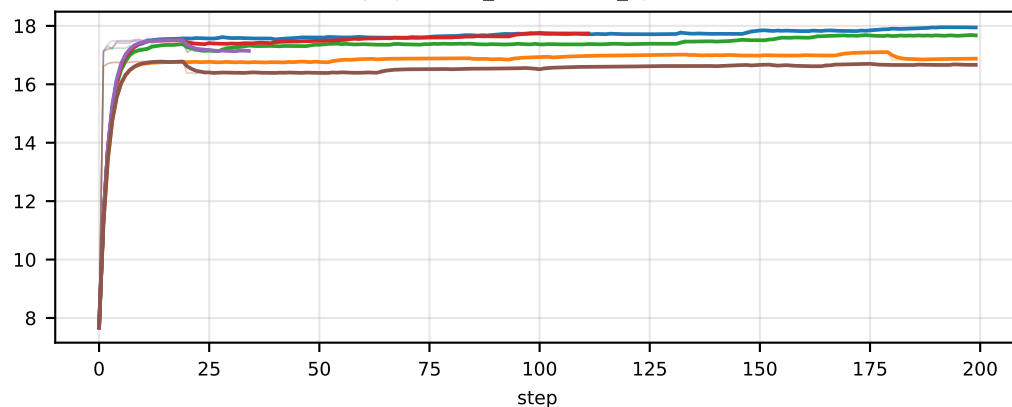
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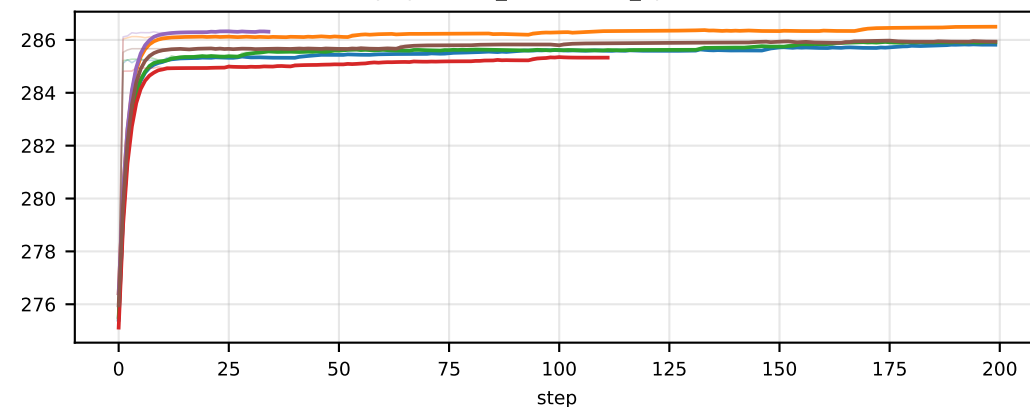
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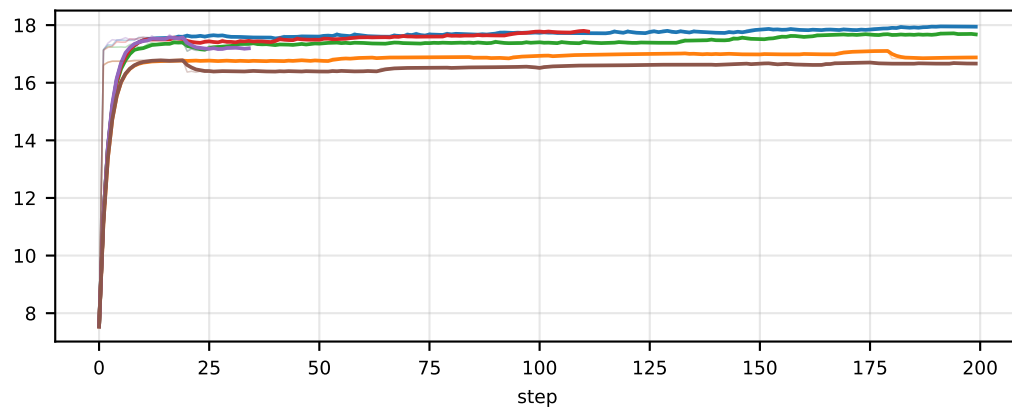
memory/cpu/actor_train/model_update/end/rss



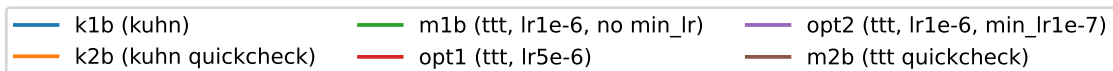
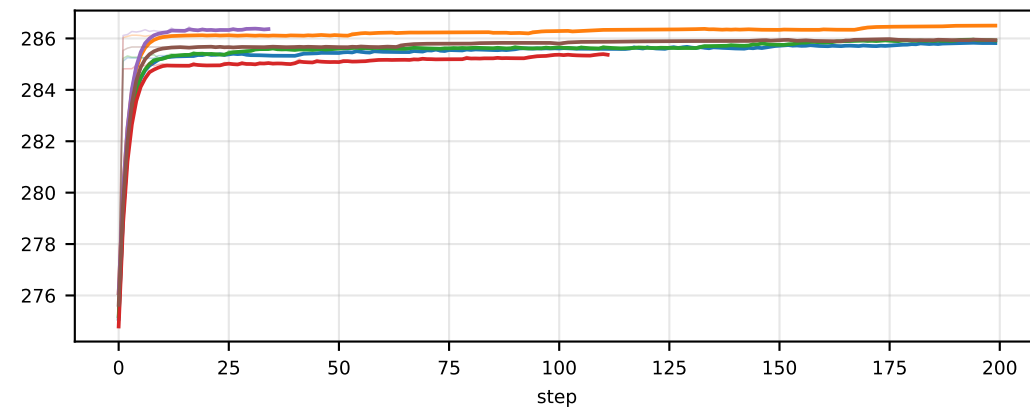
memory/cpu/actor_train/model_update/end/vms



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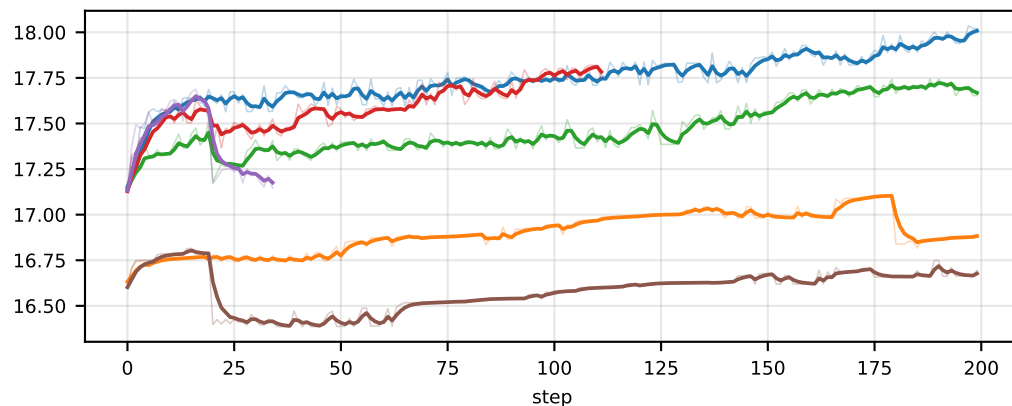


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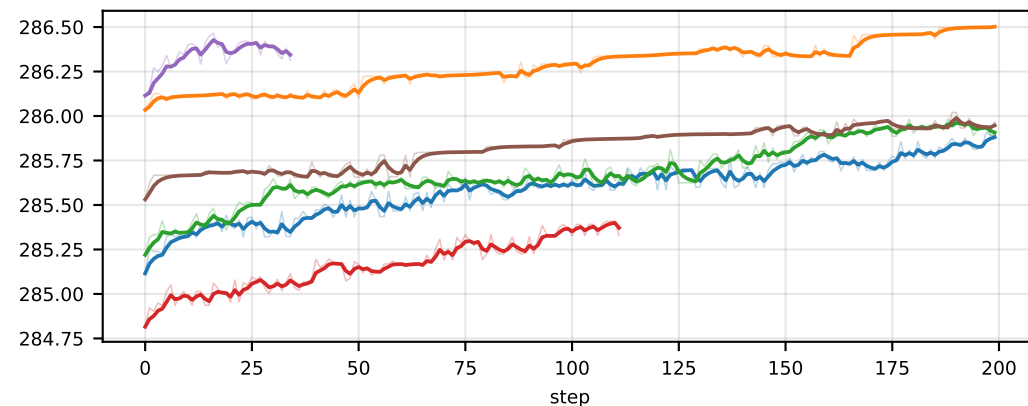


[Appendix] memory/* (15/19)

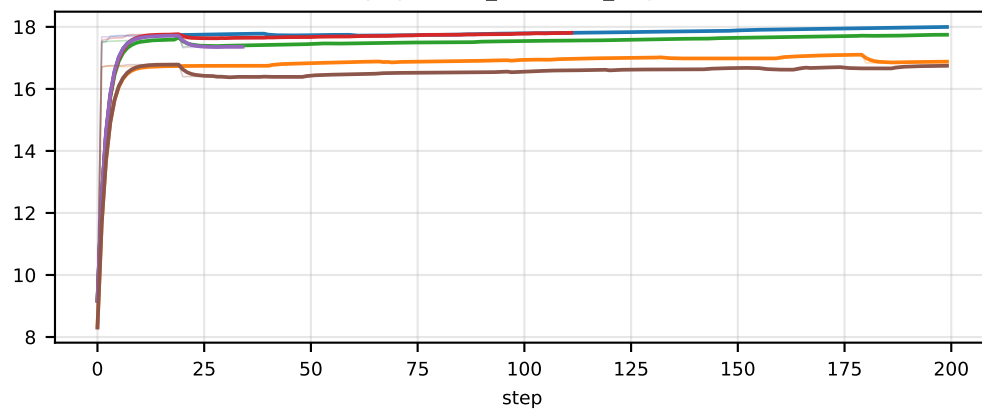
memory/cpu/actor_train/train_step/end/rss



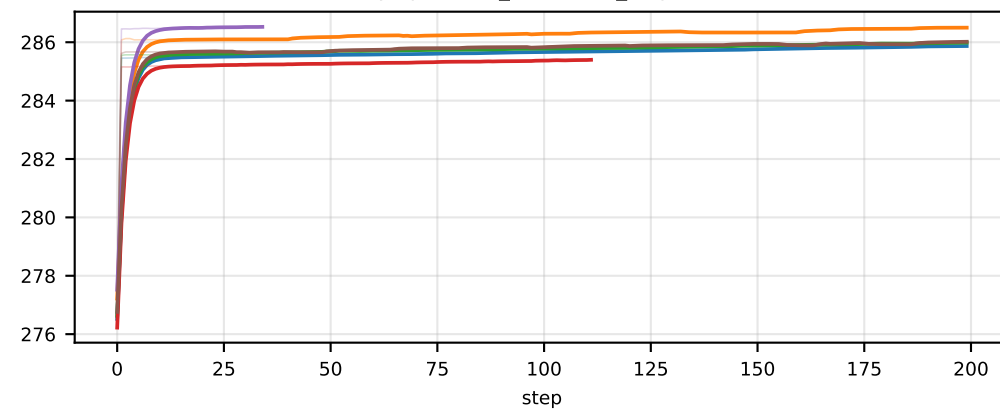
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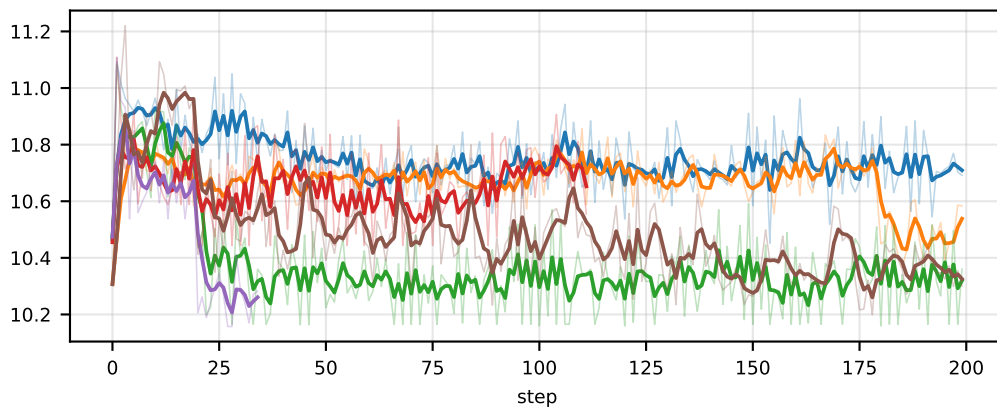
memory/cpu/actor_train/train_step/start/rss



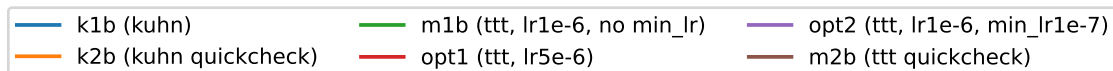
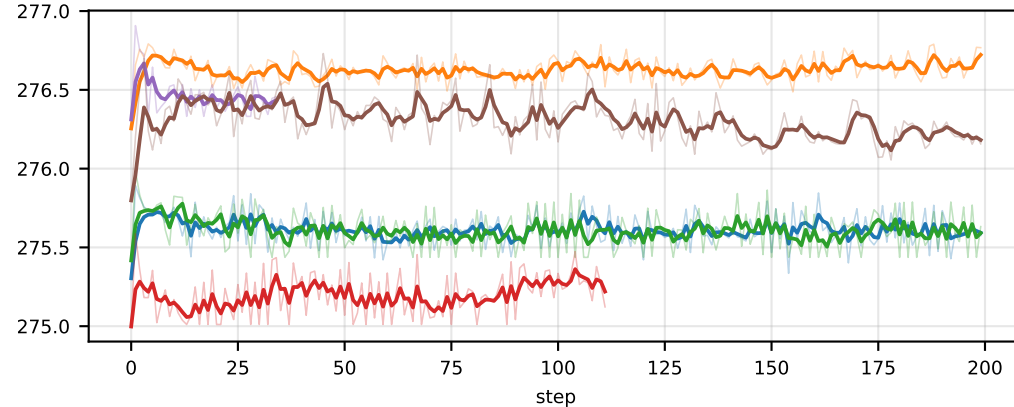
memory/cpu/actor_train/train_step/start/vms



memory/cpu/reference/compute_log_probs/end/rss

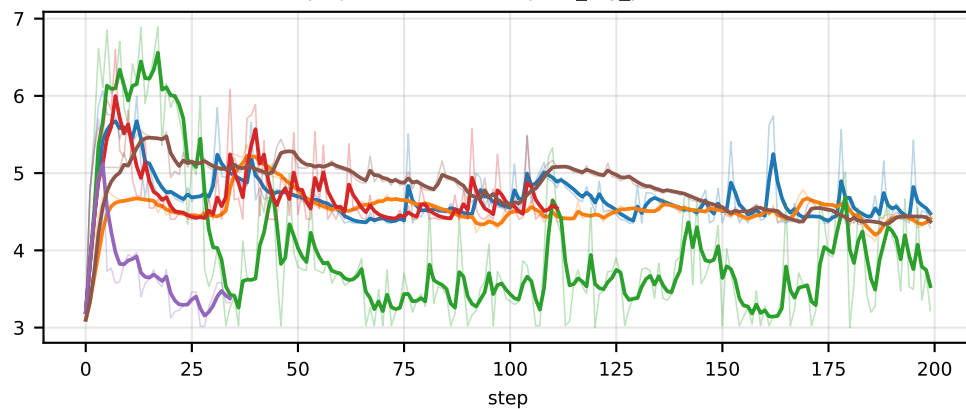


memory/cpu/reference/compute_log_probs/end/vms

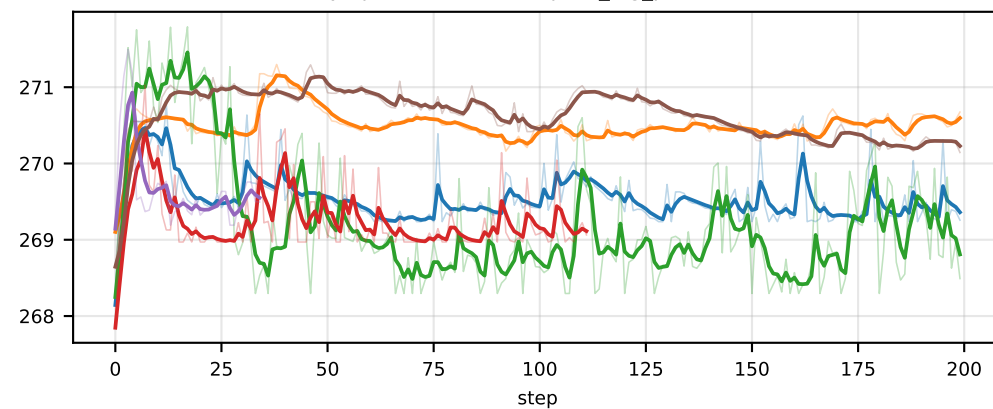


[Appendix] memory/* (16/19)

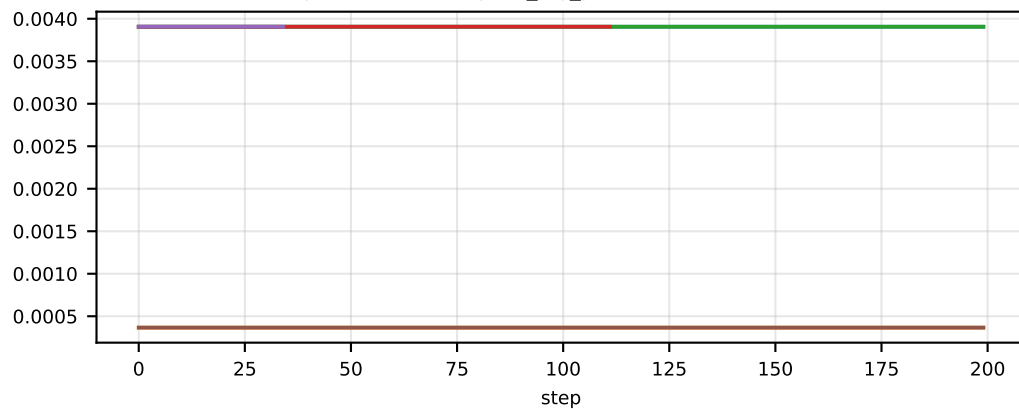
memory/cpu/reference/compute_log_probs/start/rss



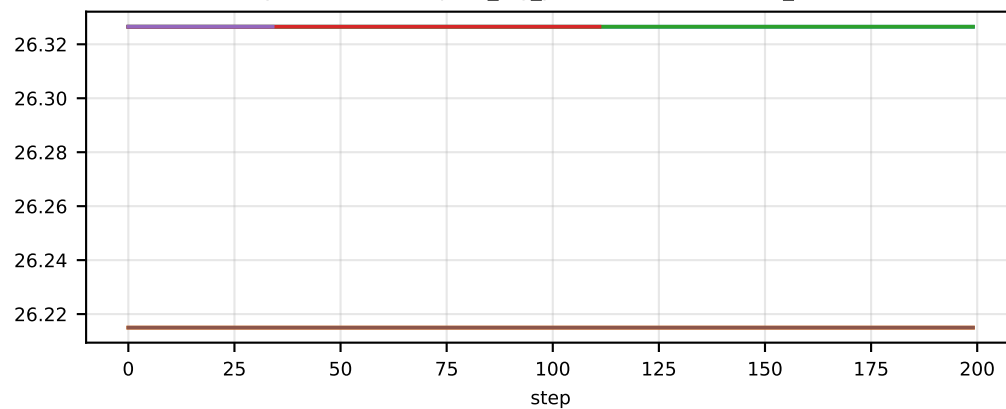
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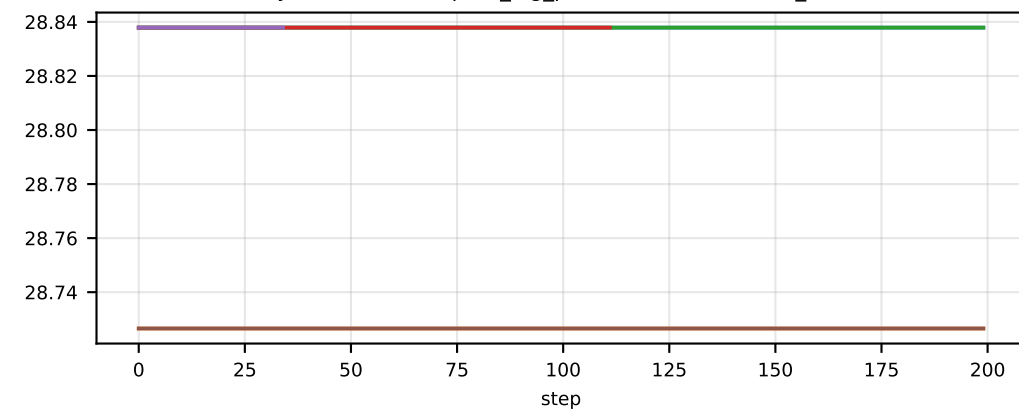
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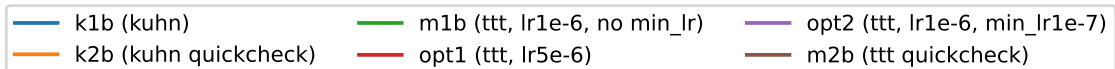
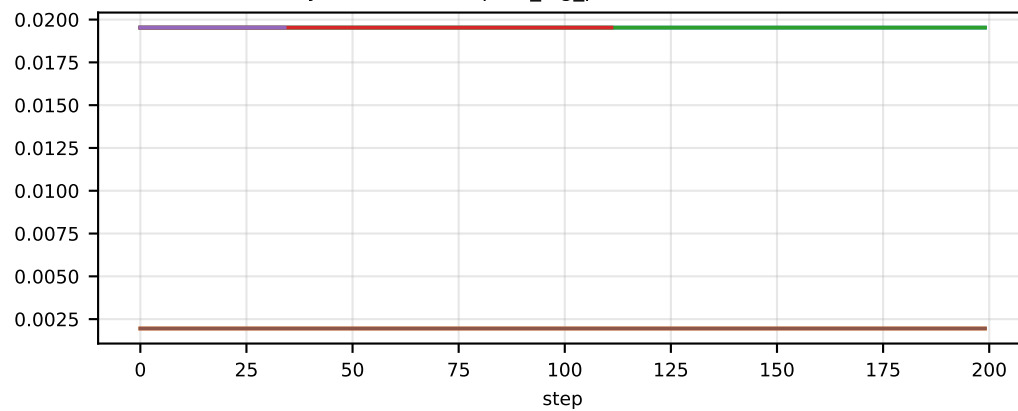
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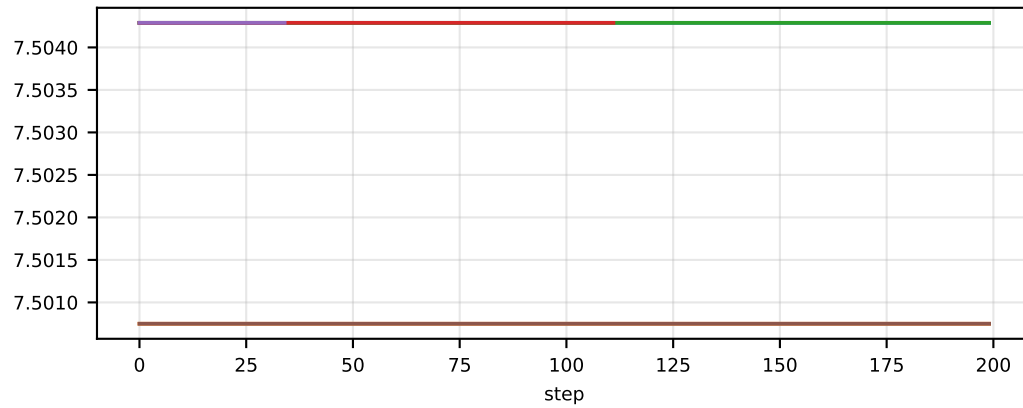


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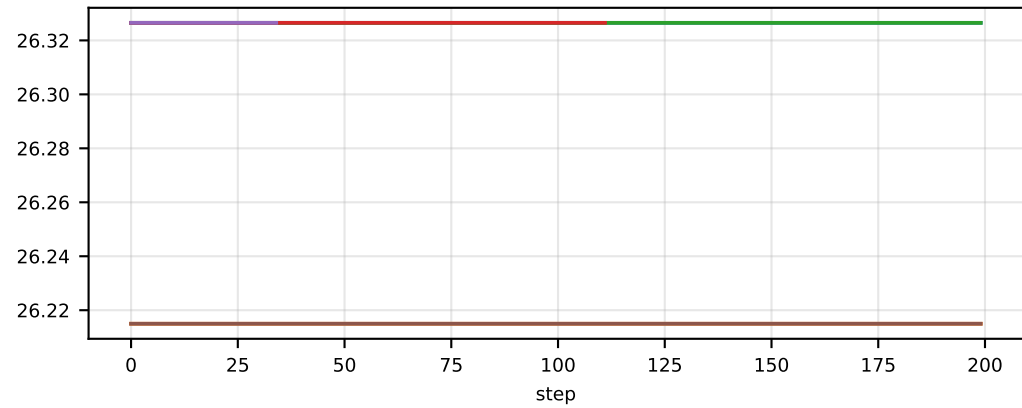


[Appendix] memory/* (17/19)

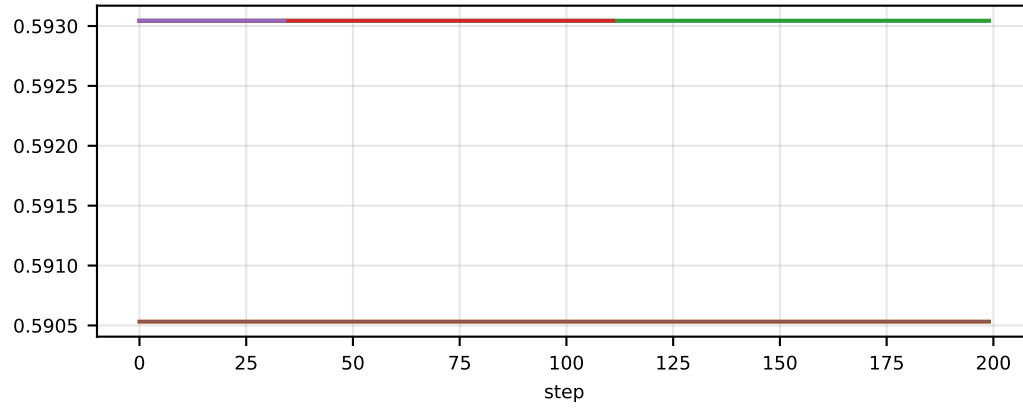
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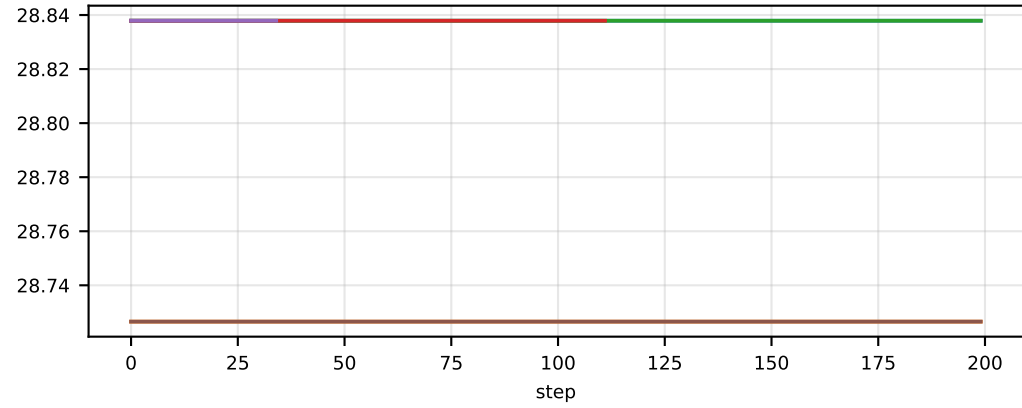
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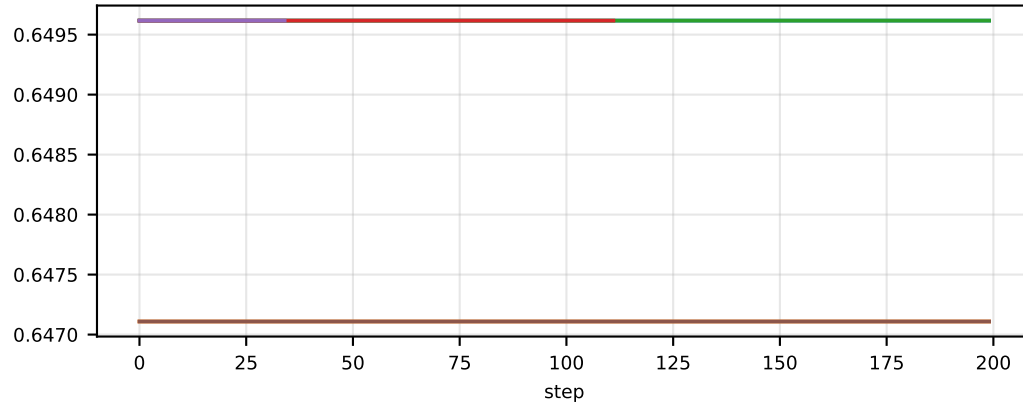
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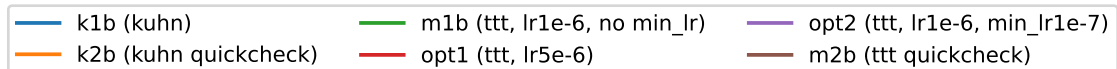
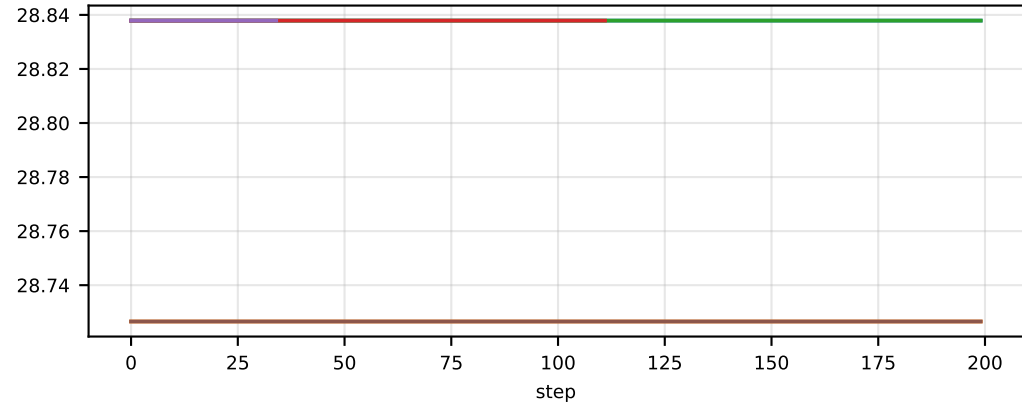
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memory/reference/compute_log_probs/end/onload/max_reserved_frac/0

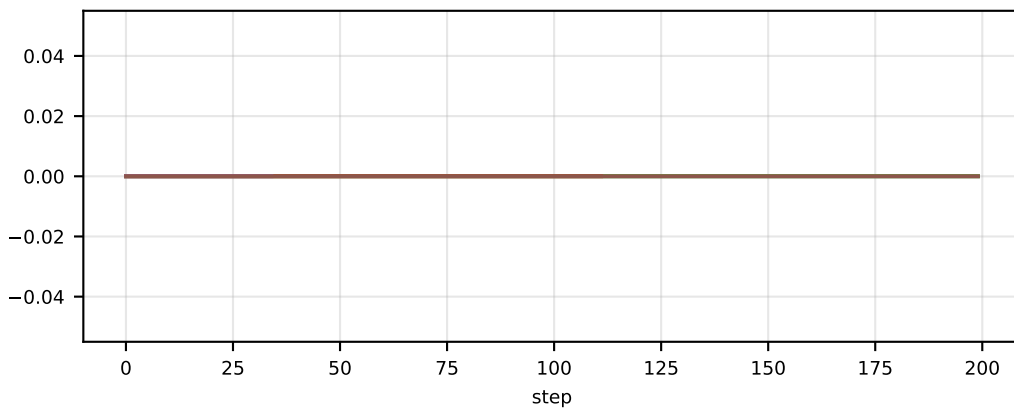


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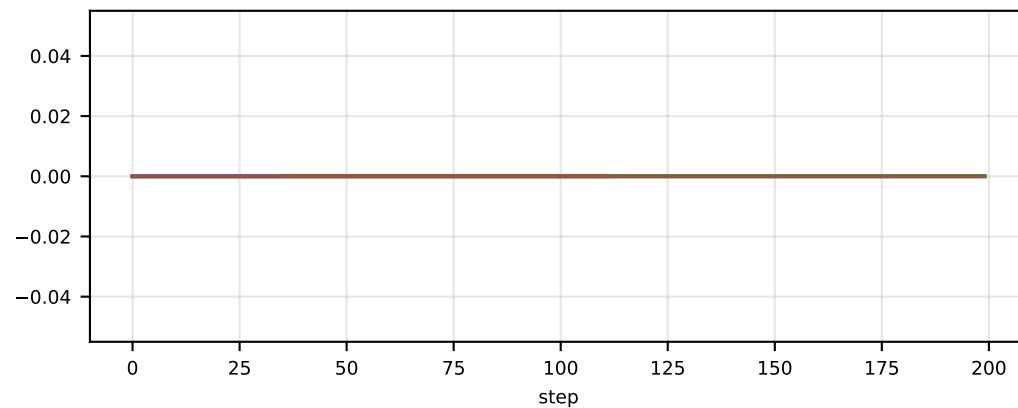


[Appendix] memory/* (18/19)

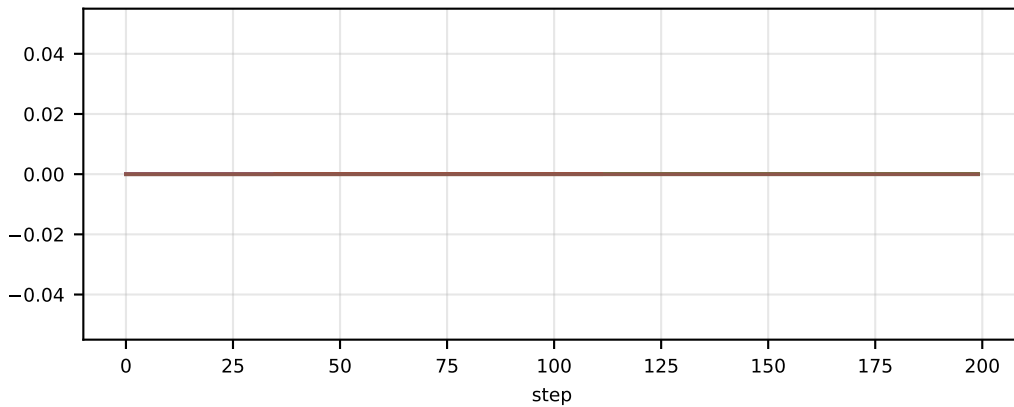
memory/reference/compute_log_probs/start/offload/allocated/0



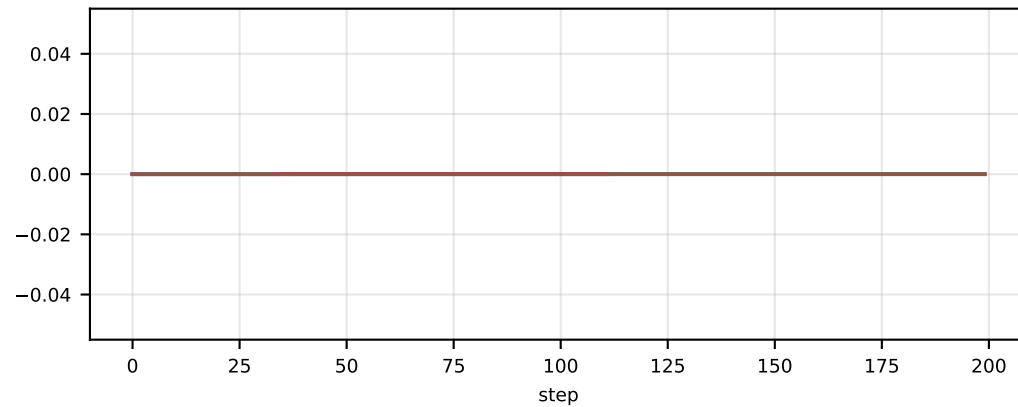
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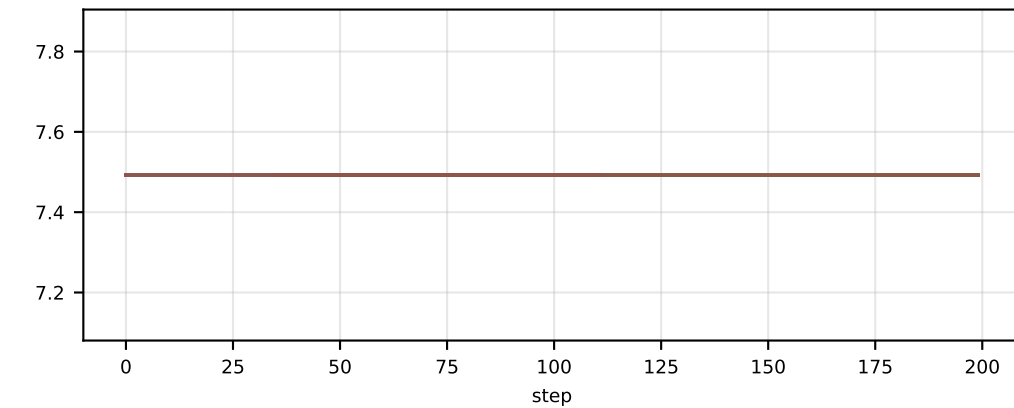
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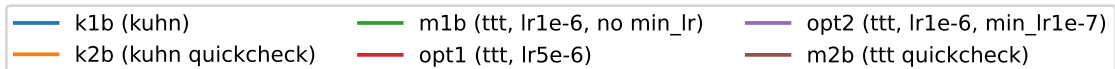
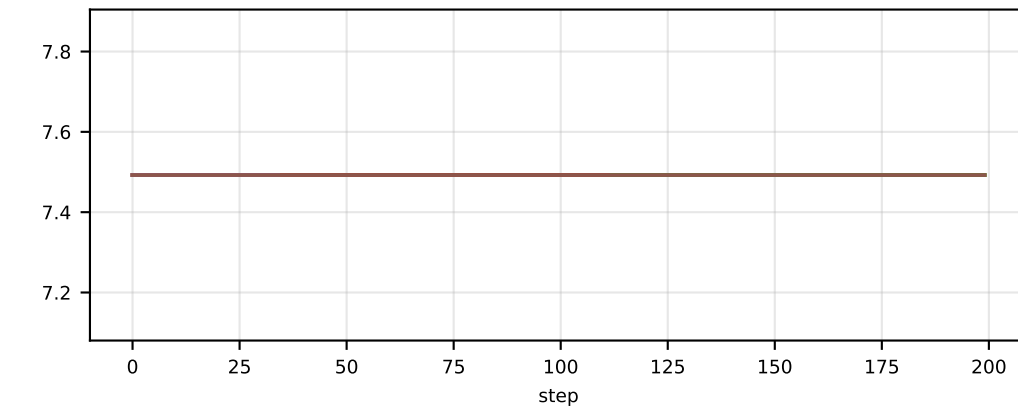
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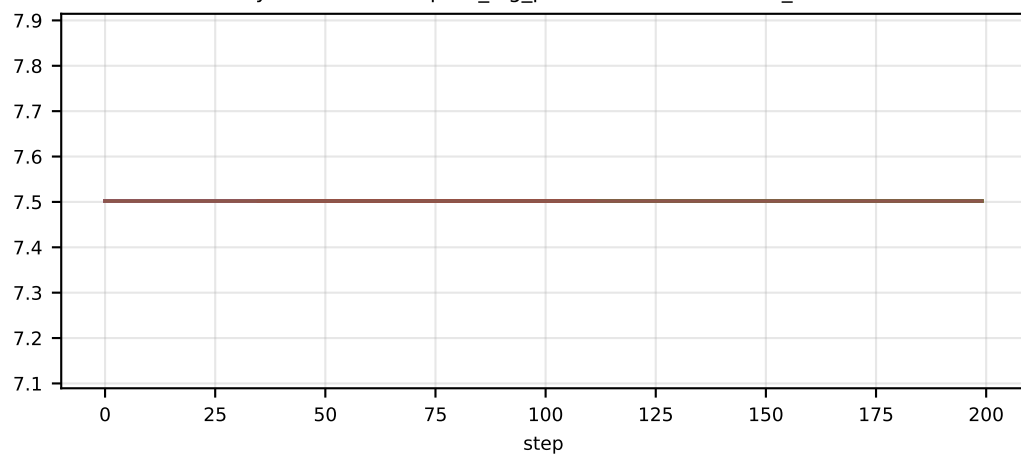


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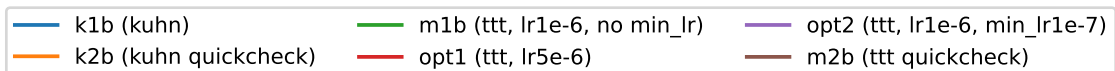
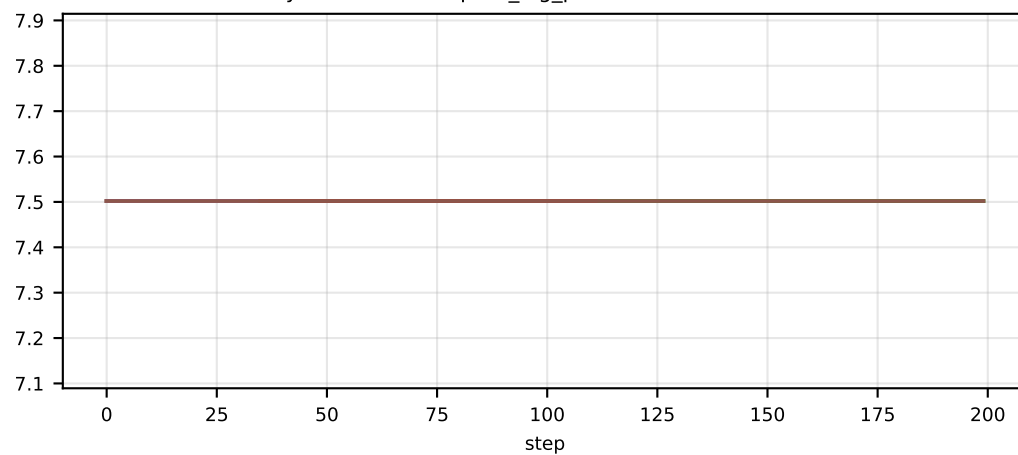


[Appendix] memory/* (19/19)

memory/reference/compute_log_probs/start/onload/max_reserved/0

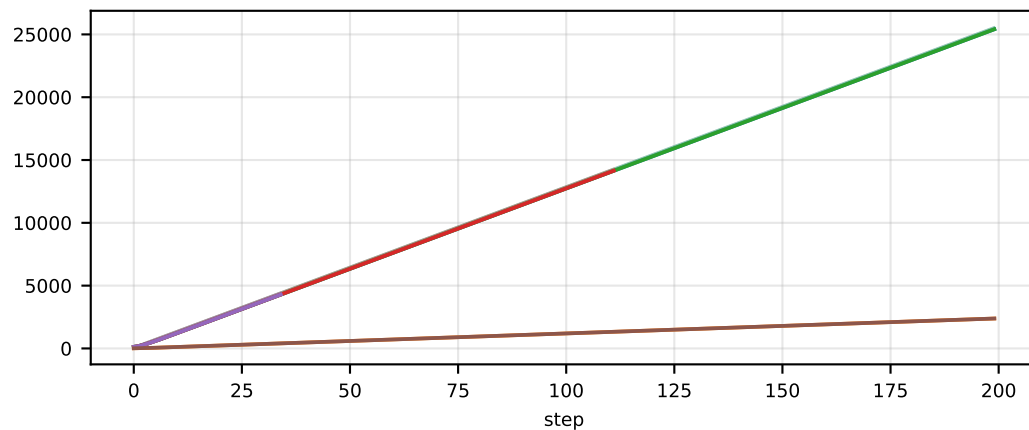


memory/reference/compute_log_probs/start/onload/reserved/0

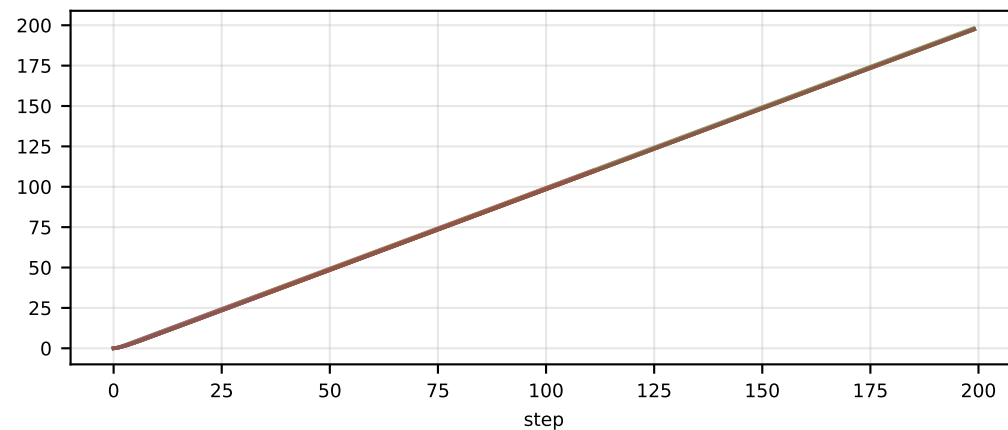


[Appendix] system/* (1/1)

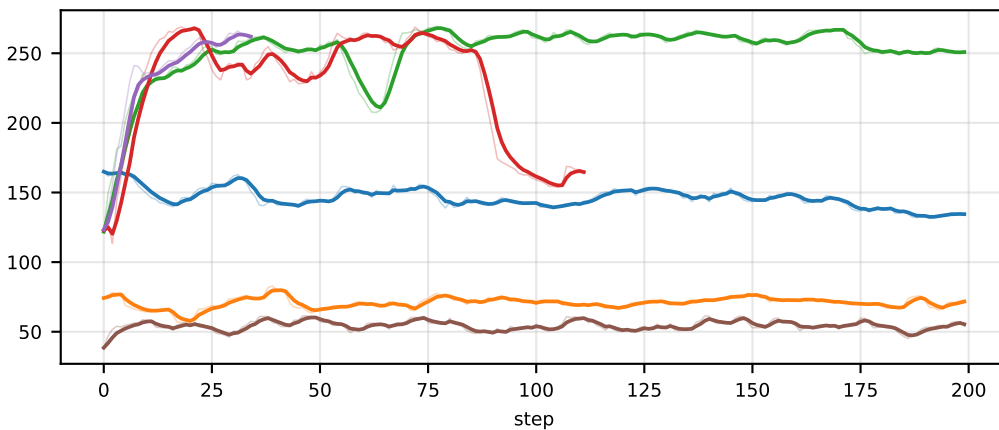
system/samples



system/step



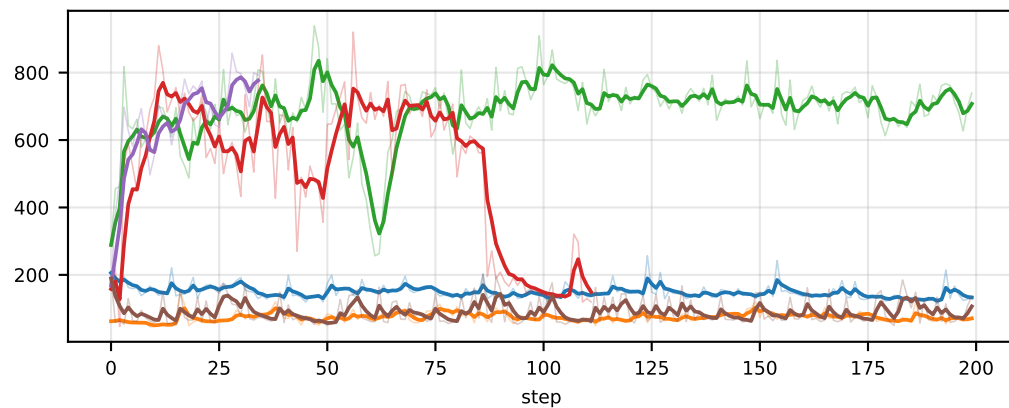
system/tps



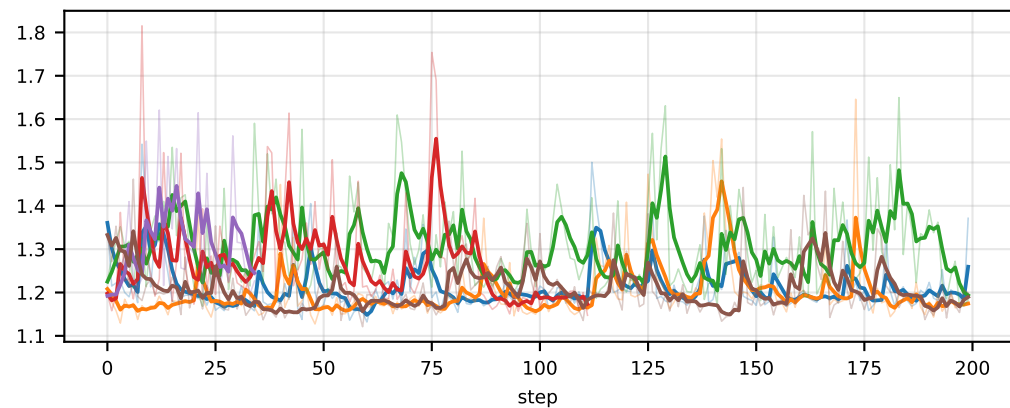
k1b (kuhn) m1b (ttt, lr1e-6, no min_lr) opt2 (ttt, lr1e-6, min_lr1e-7)
k2b (kuhn quickcheck) opt1 (ttt, lr5e-6) m2b (ttt quickcheck)

[Appendix] time/* (1/5)

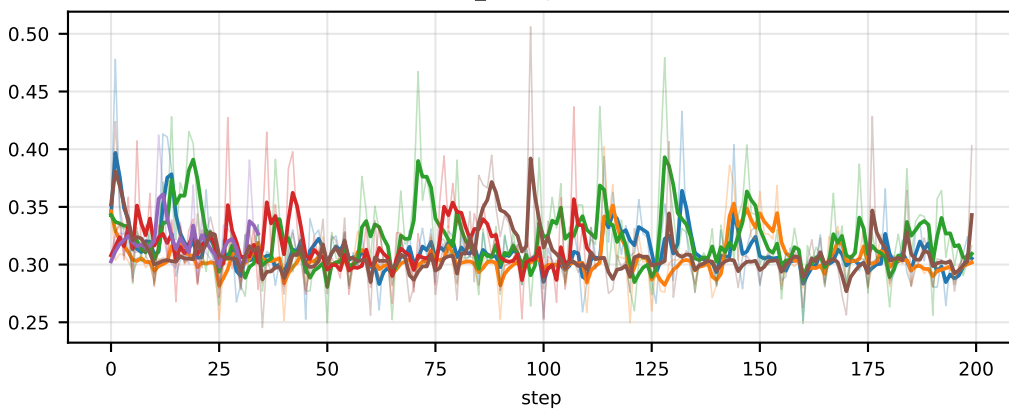
time/actor_infer/generate/execute



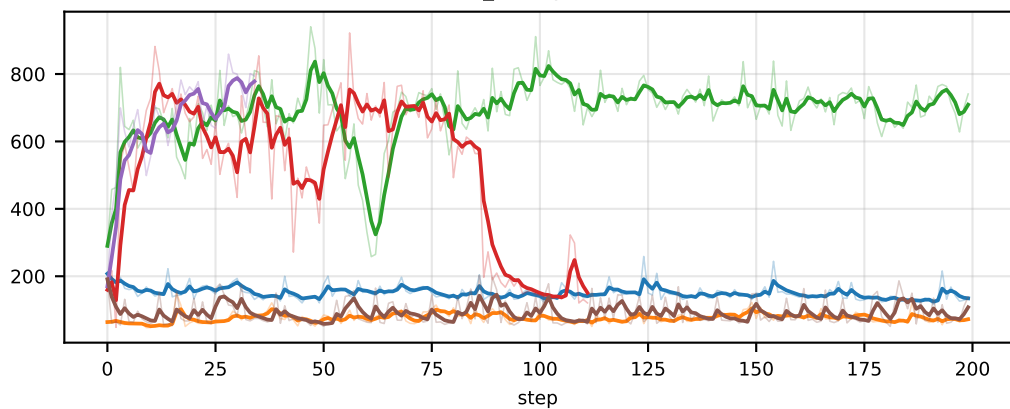
time/actor_infer/generate/offload



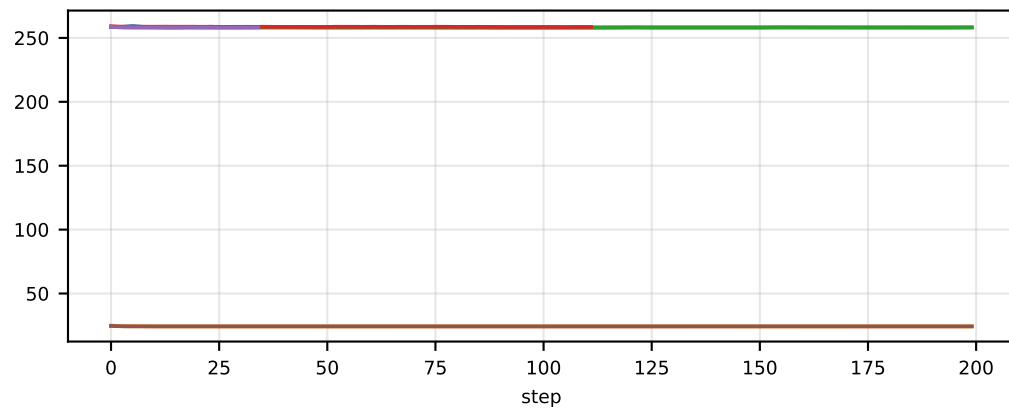
time/actor_infer/generate/onload



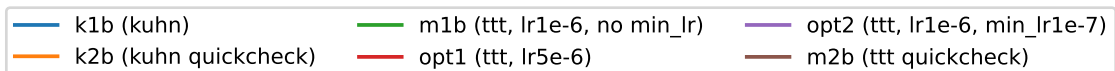
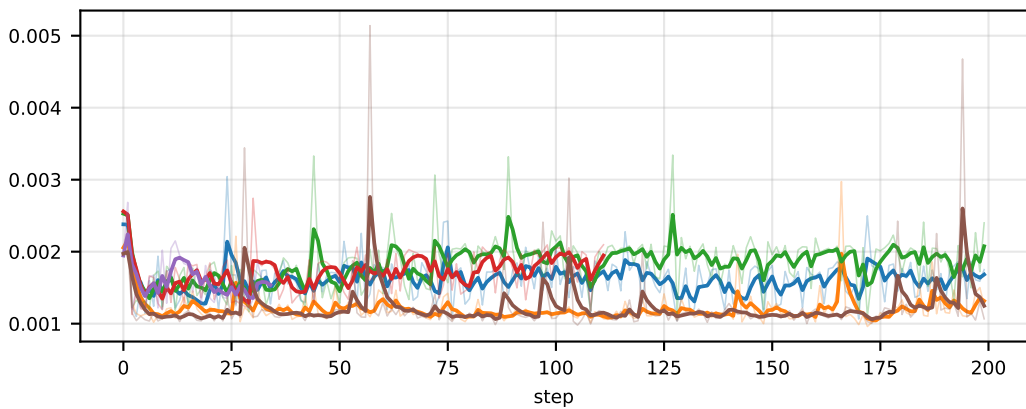
time/actor_infer/generate/total



time/actor_train/compute_log_probs/execute

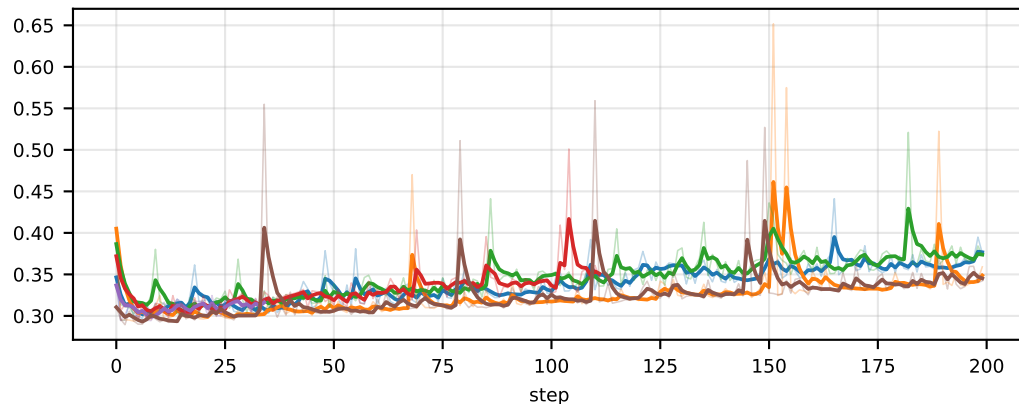


time/actor_train/compute_log_probs/offload

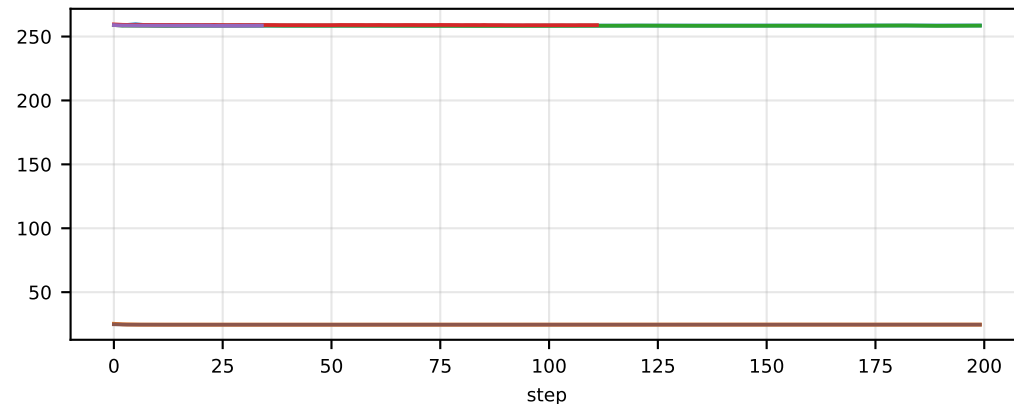


[Appendix] time/* (2/5)

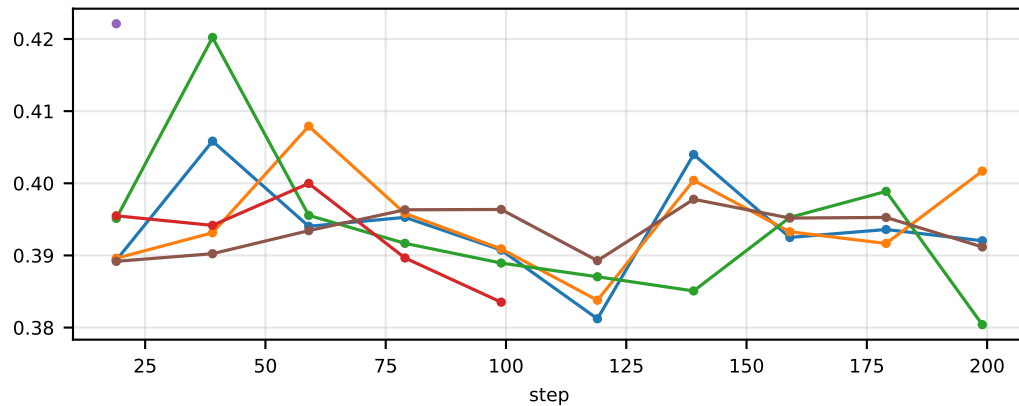
time/actor_train/compute_log_probs/onload



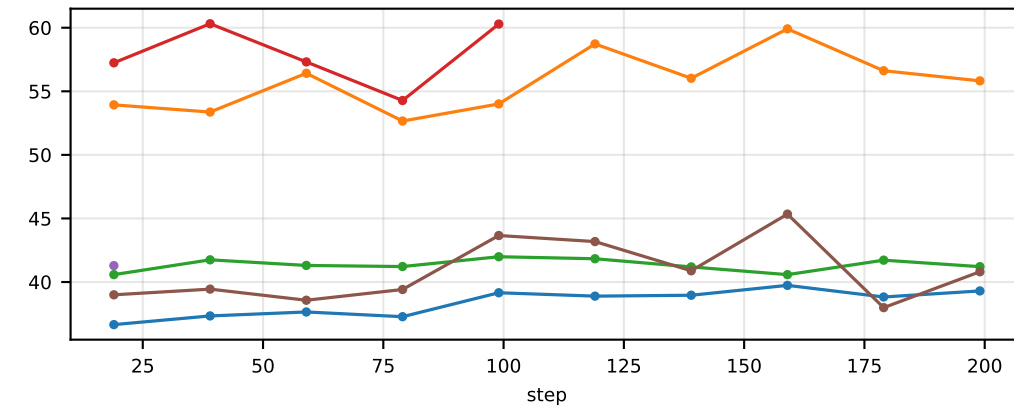
time/actor_train/compute_log_probs/total



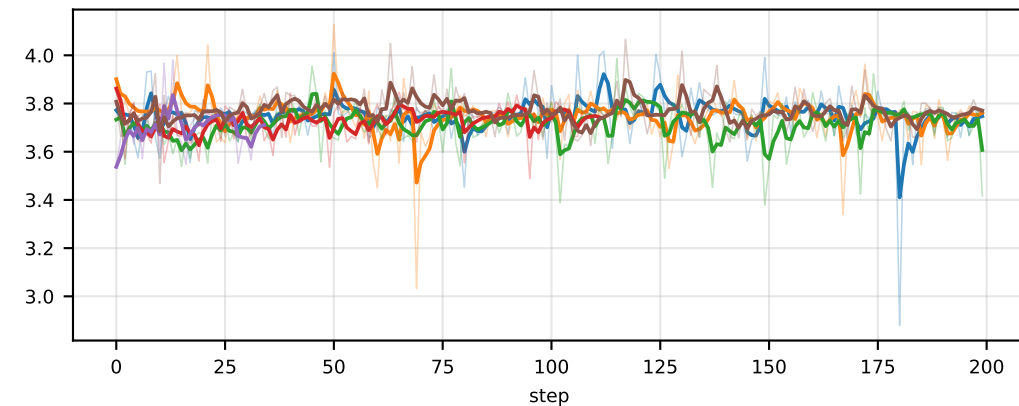
time/actor_train/do_checkpoint/load



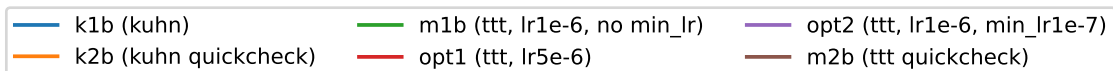
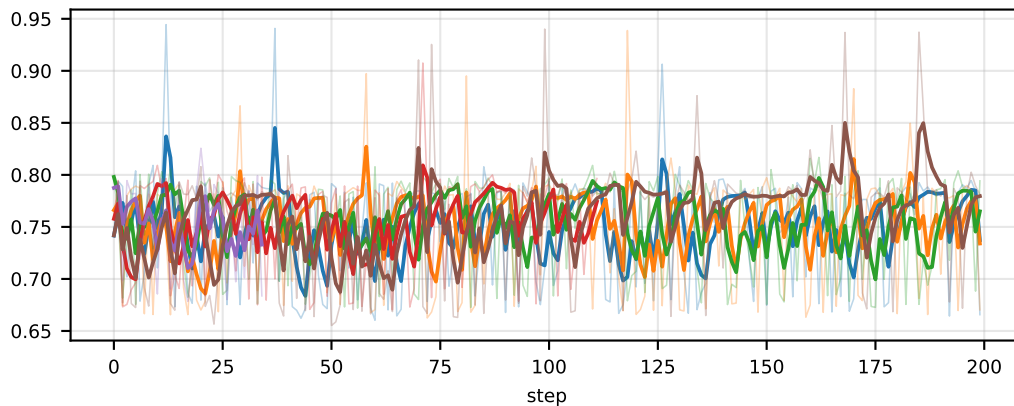
time/actor_train/do_checkpoint/total



time/actor_train/model_update/all_gather

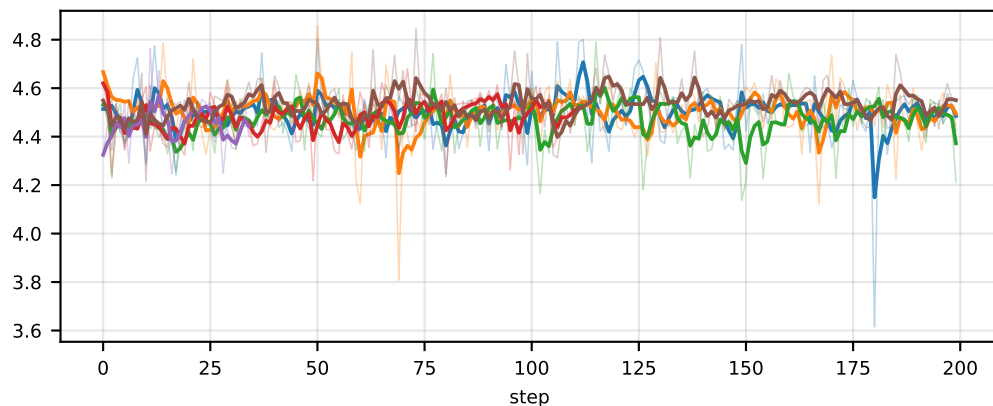


time/actor_train/model_update/broadcast

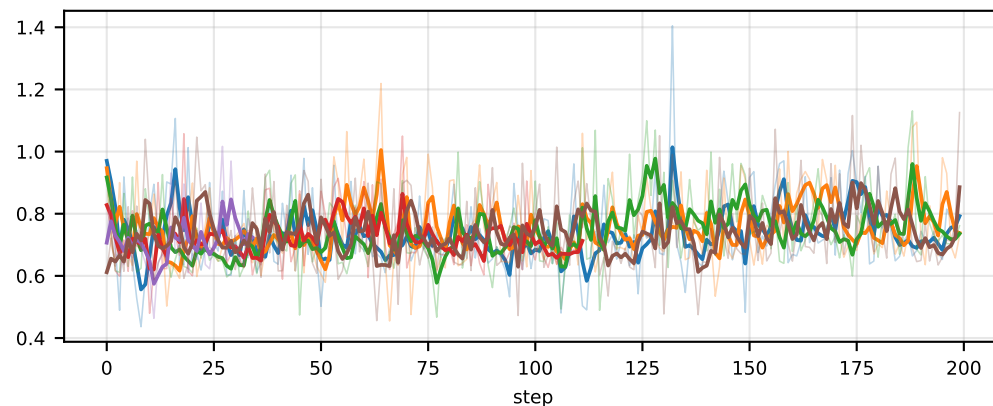


[Appendix] time/* (3/5)

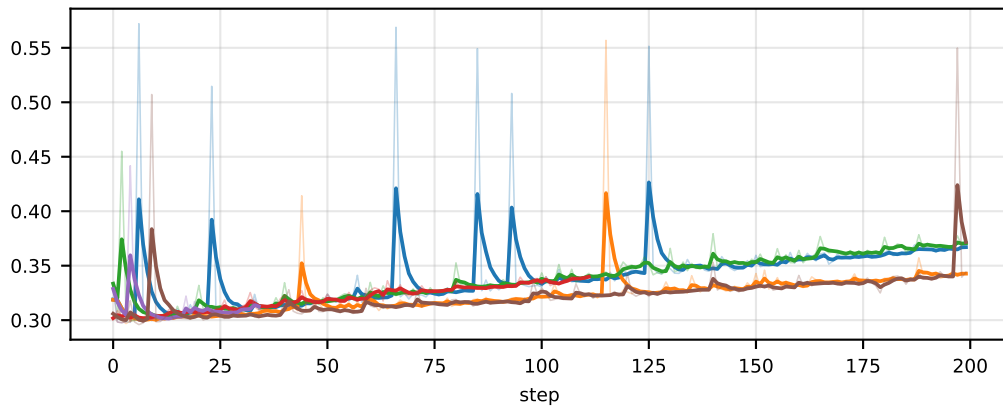
time/actor_train/model_update/execute



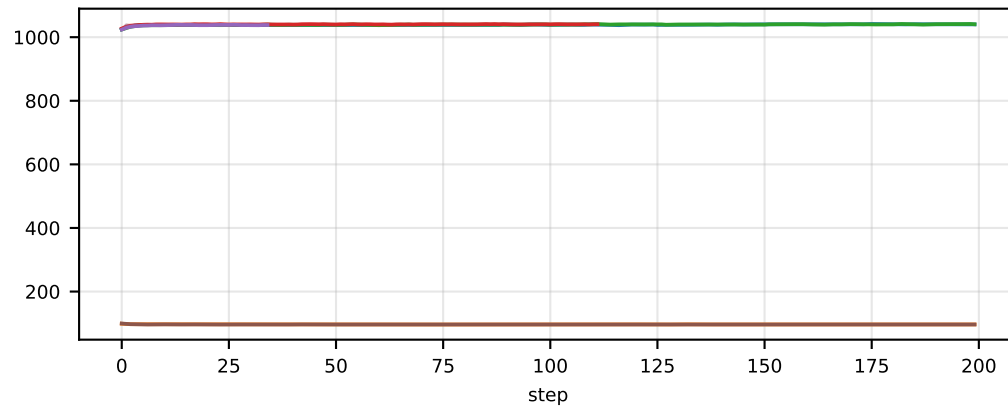
time/actor_train/model_update/offload



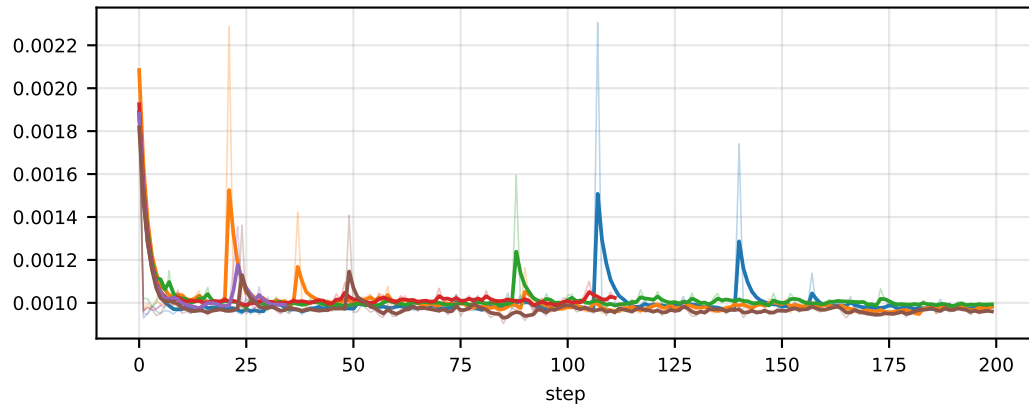
time/actor_train/model_update/onload



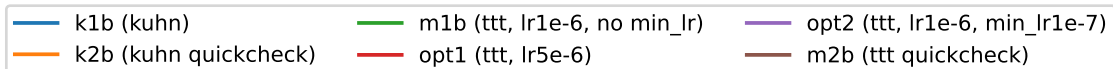
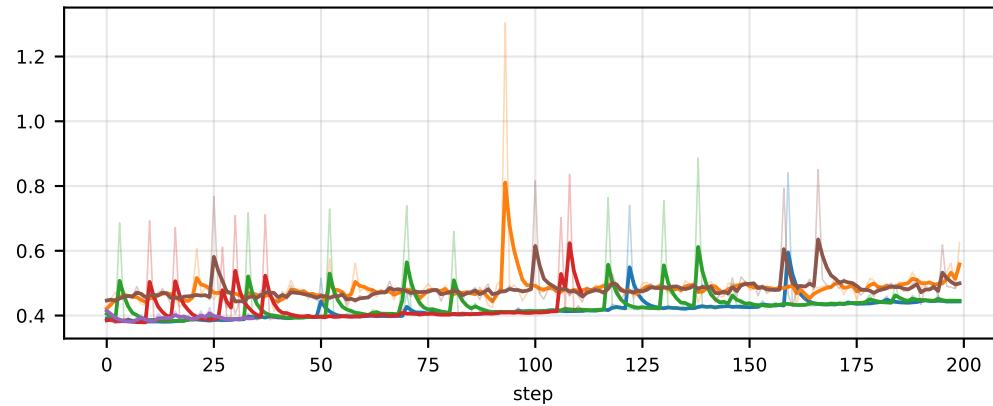
time/actor_train/train_step/execute



time/actor_train/train_step/offload

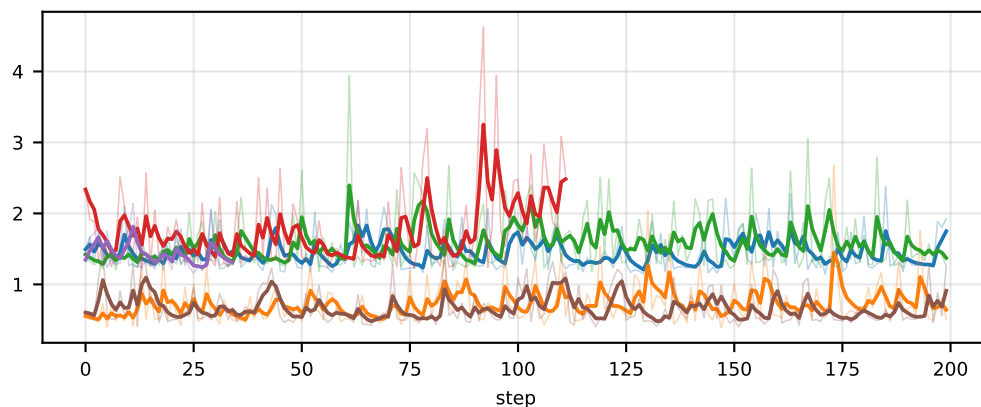


time/actor_train/train_step/onload

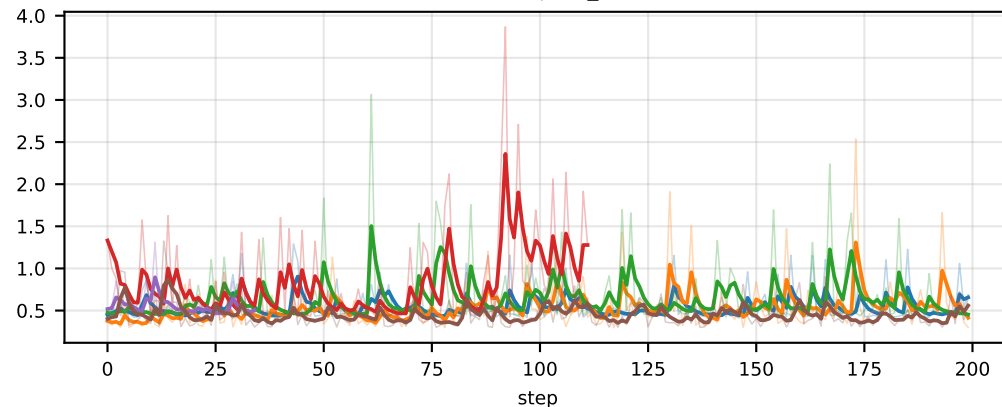


[Appendix] time/* (4/5)

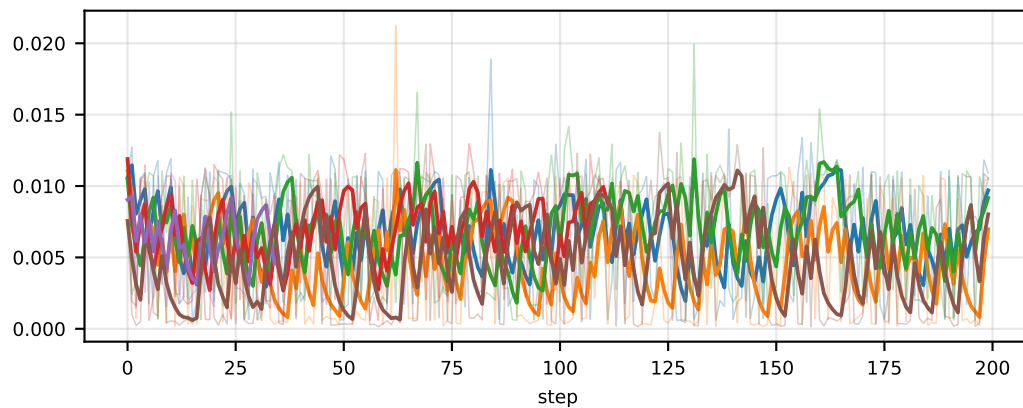
time/adv



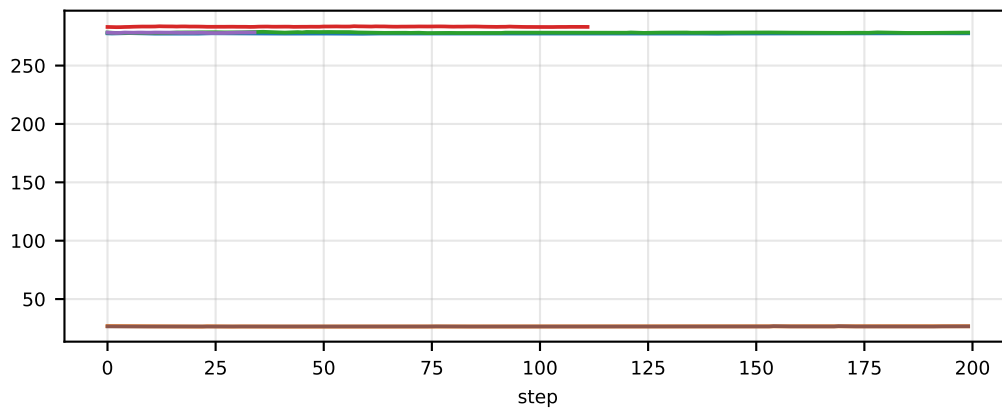
time/compute_adv



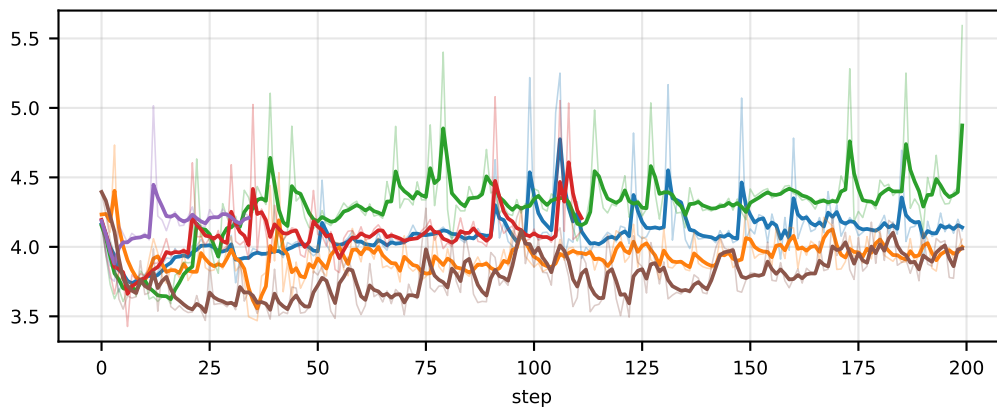
time/get_rewards



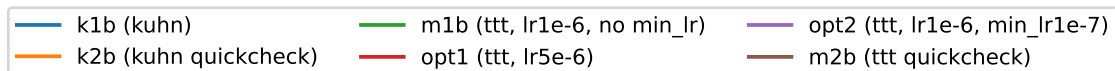
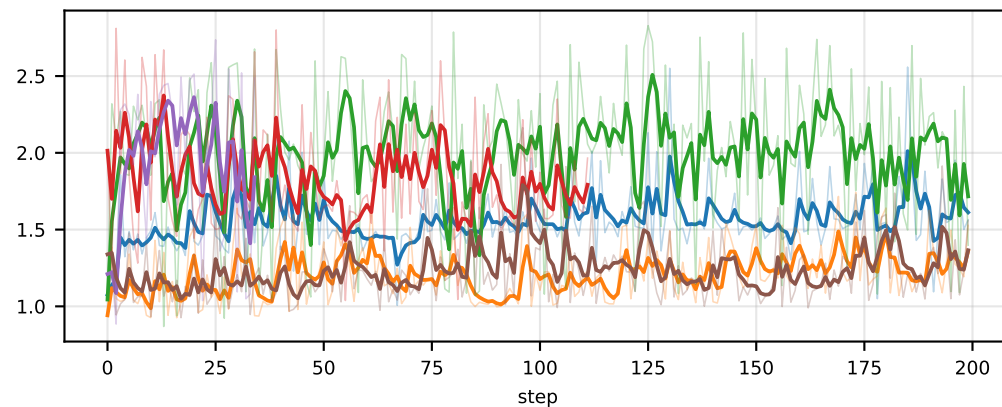
time/reference/compute_log_probs/execute



time/reference/compute_log_probs/offload

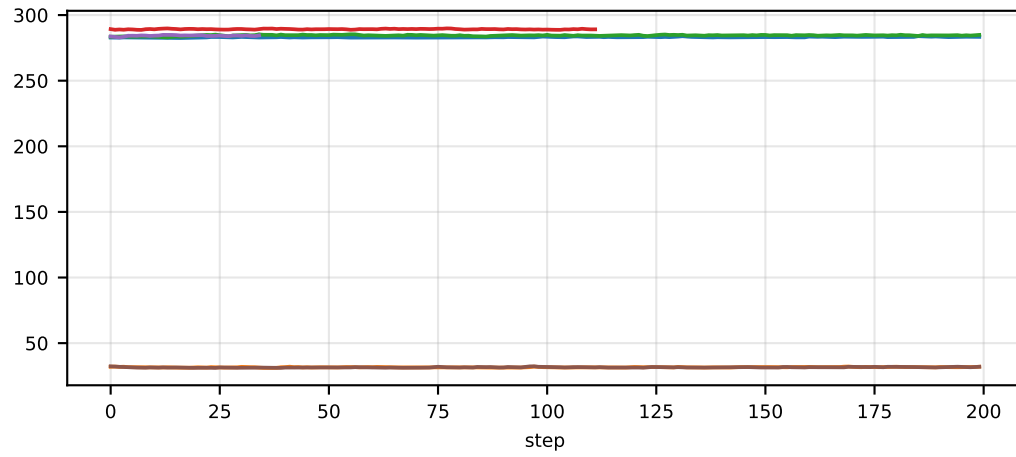


time/reference/compute_log_probs/onload

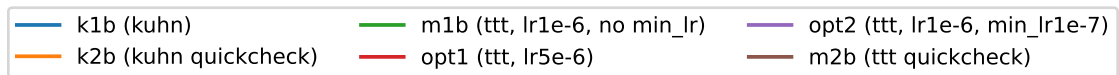
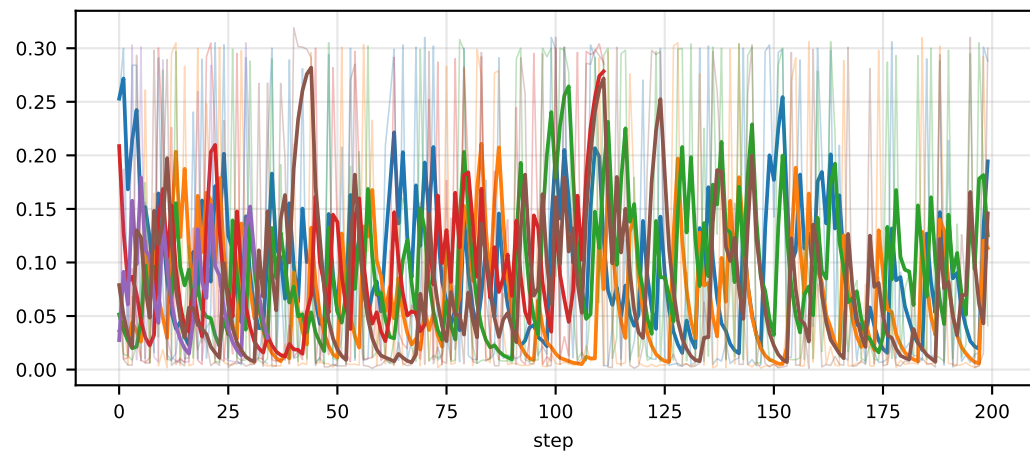


[Appendix] time/* (5/5)

time/reference/compute_log_probs/total

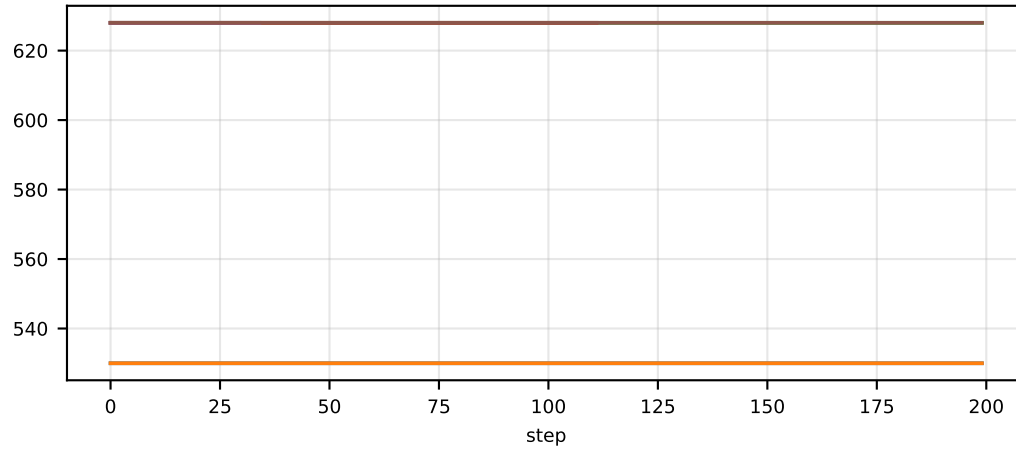


time/reward_postprocess

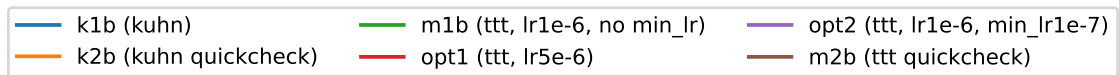
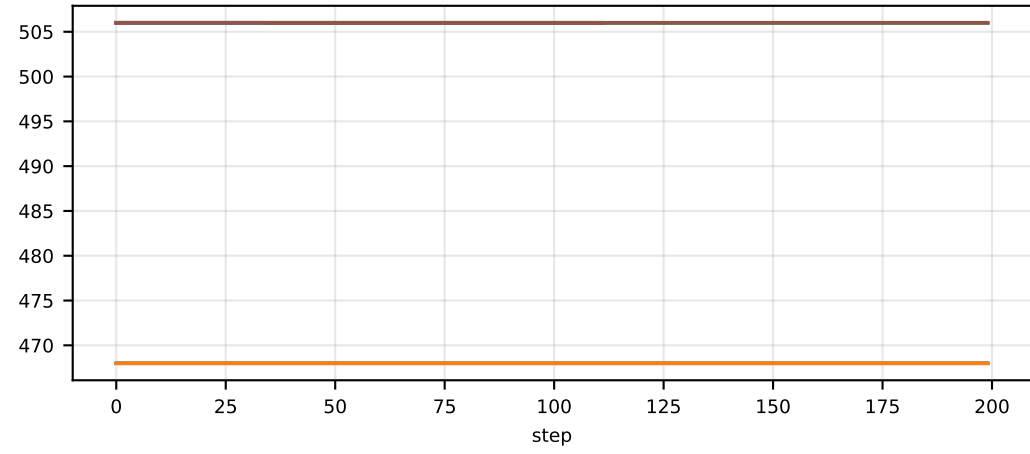


[Appendix] tokens/* (1/1)

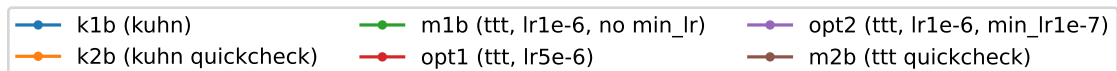
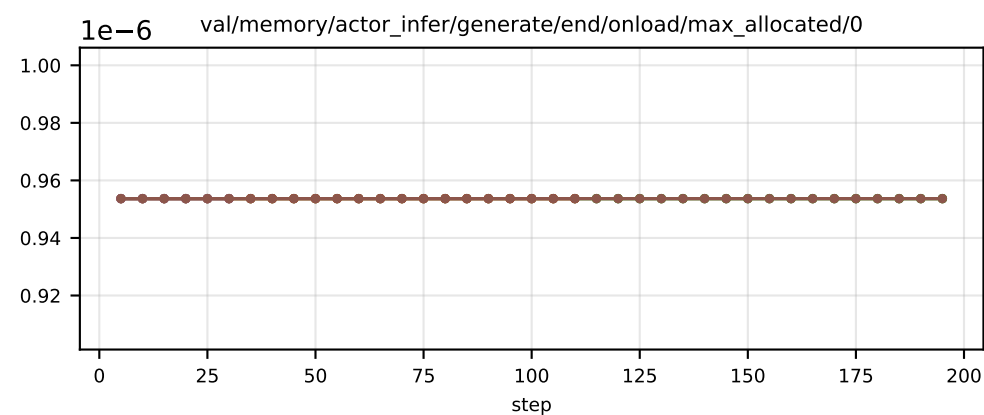
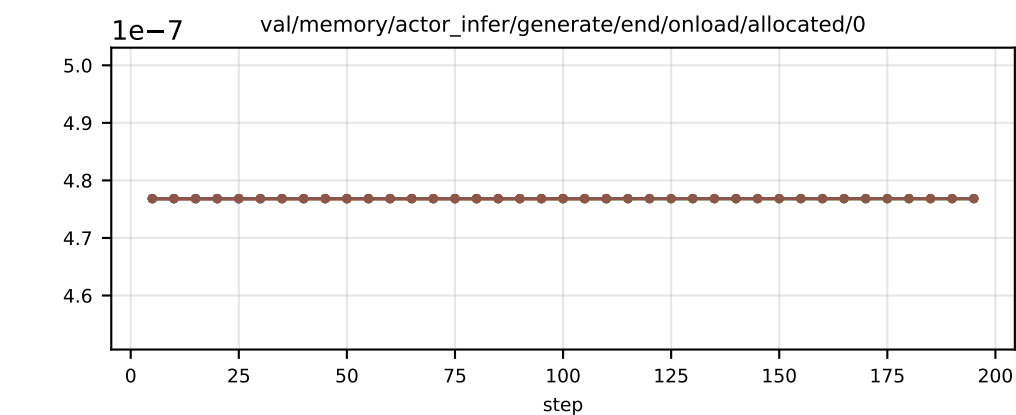
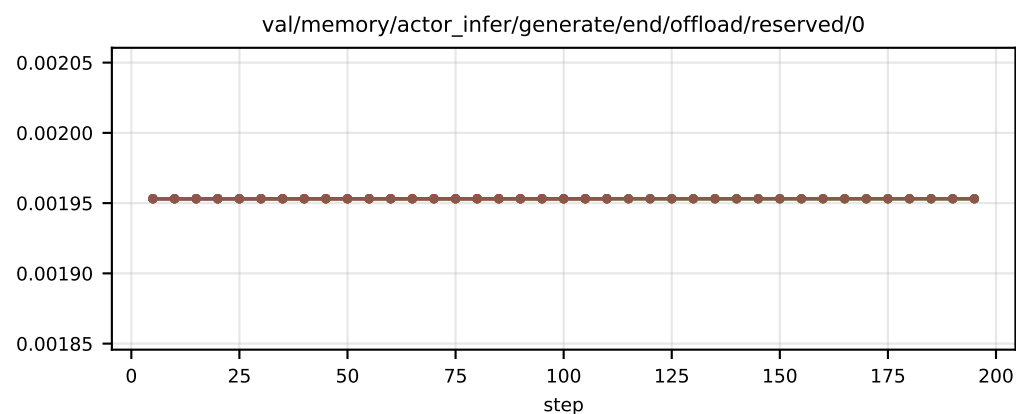
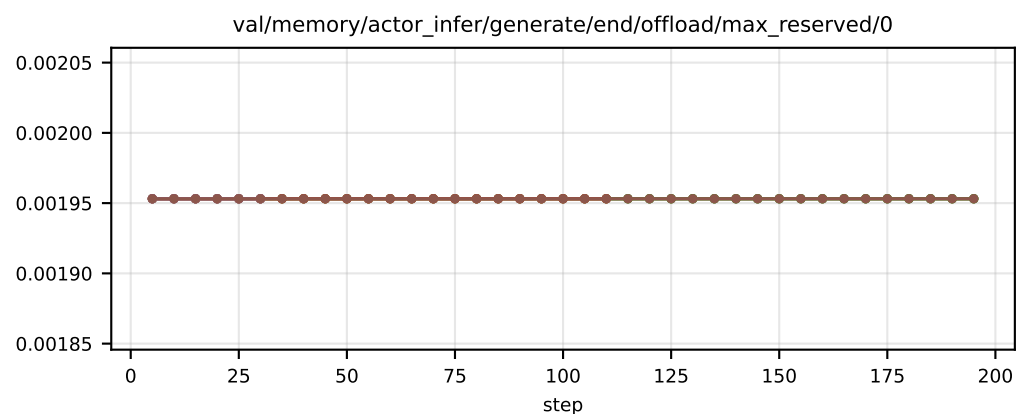
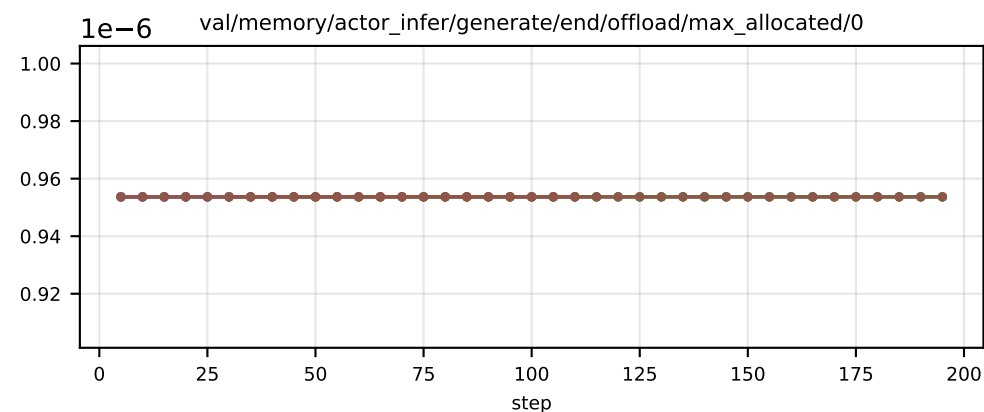
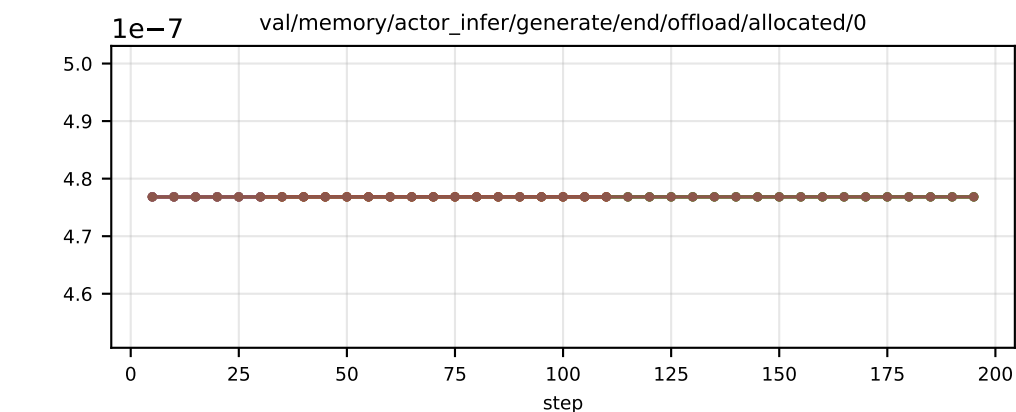
tokens/prompt_length/max



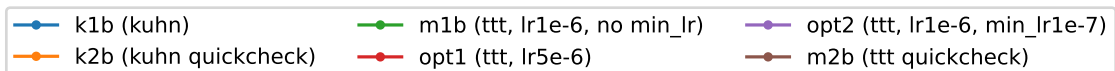
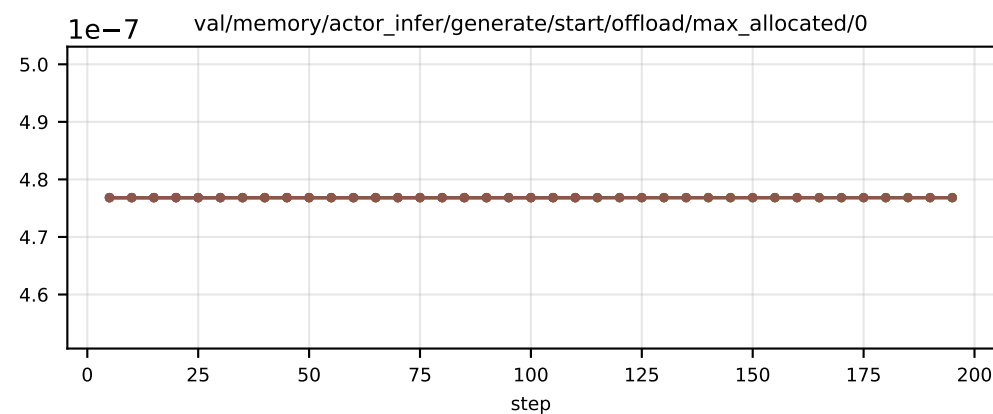
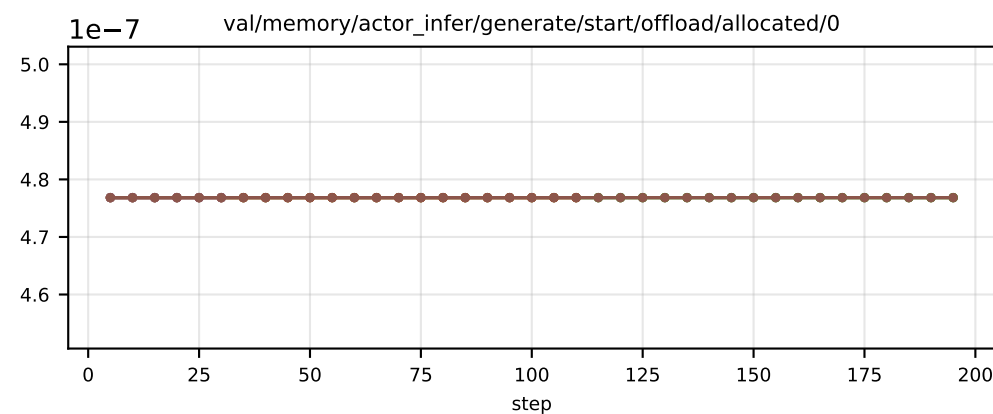
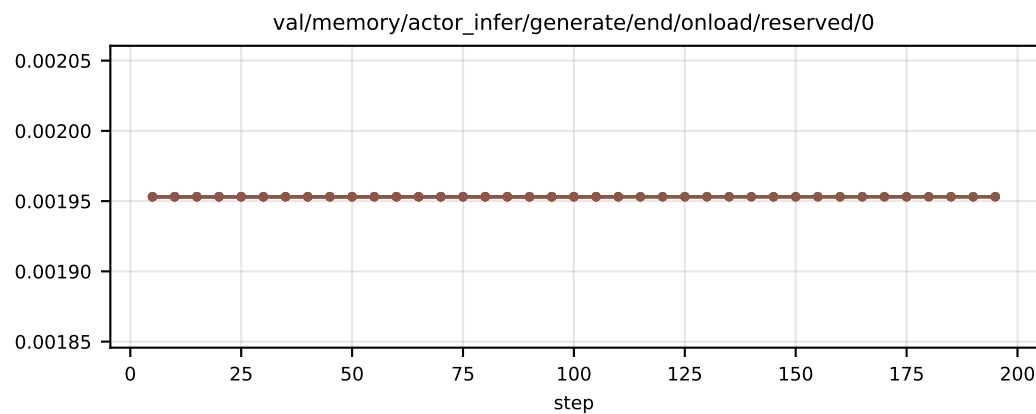
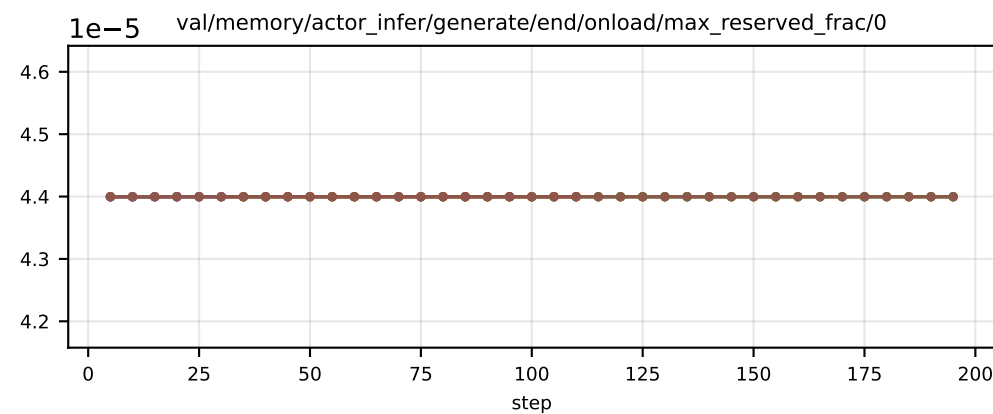
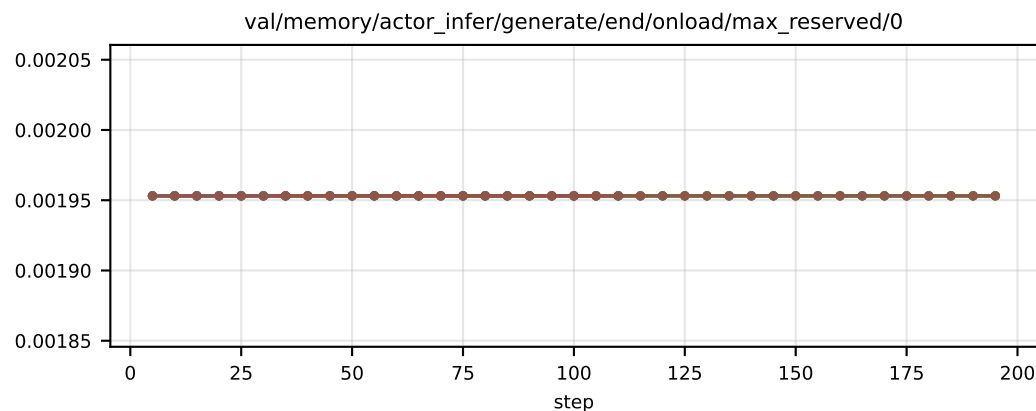
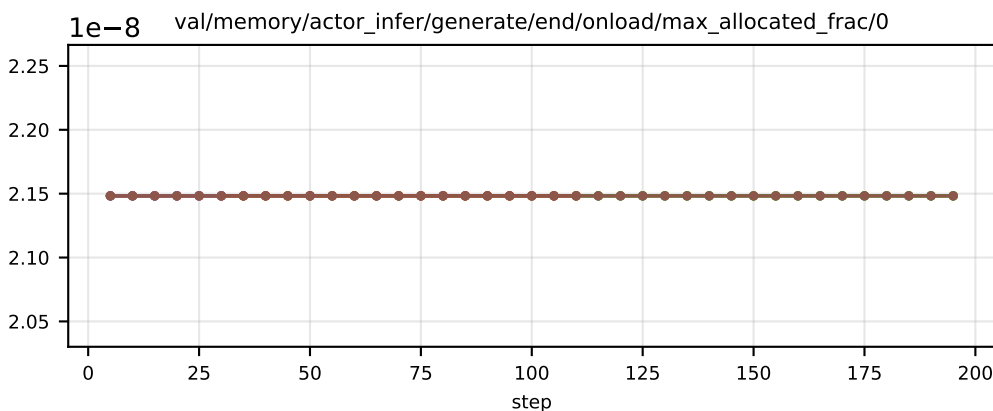
tokens/prompt_length/min



[Appendix] val/* (1/5)

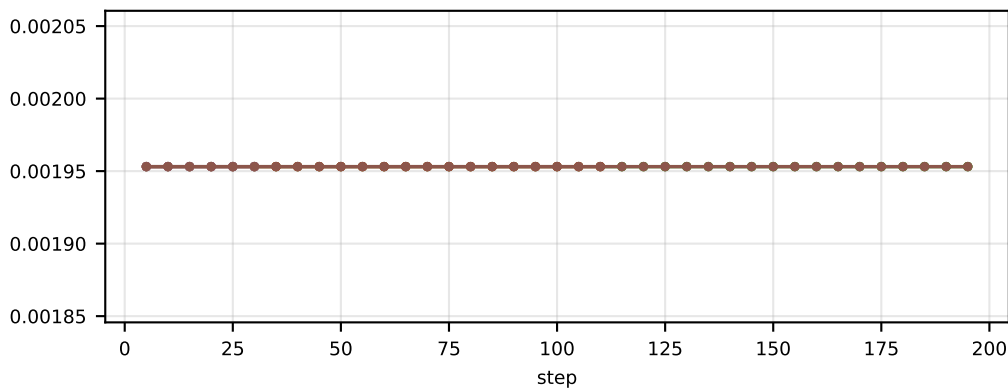


[Appendix] val/* (2/5)

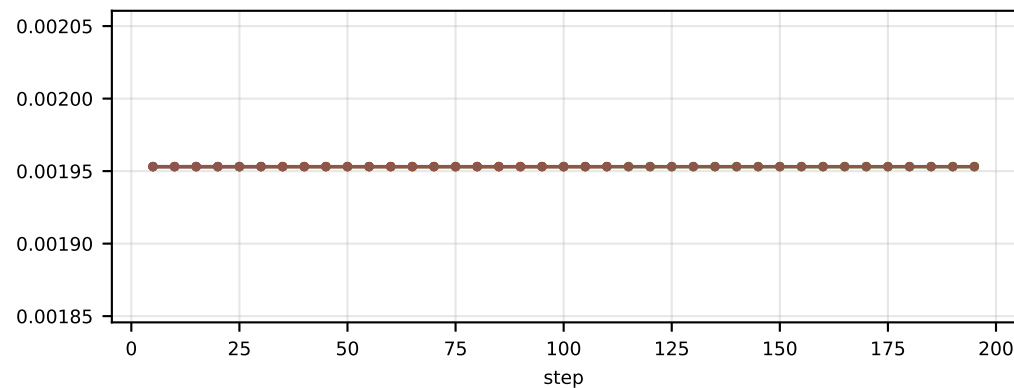


[Appendix] val/* (3/5)

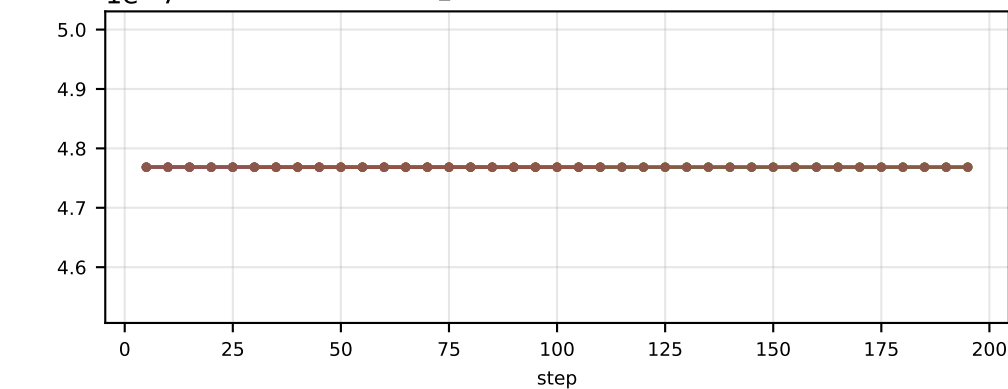
val/memory/actor_infer/generate/start/offload/max_reserved/0



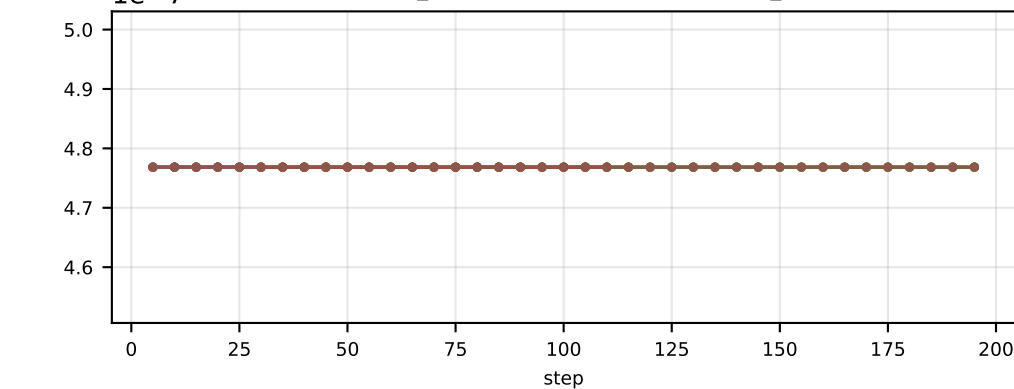
val/memory/actor_infer/generate/start/offload/reserved/0



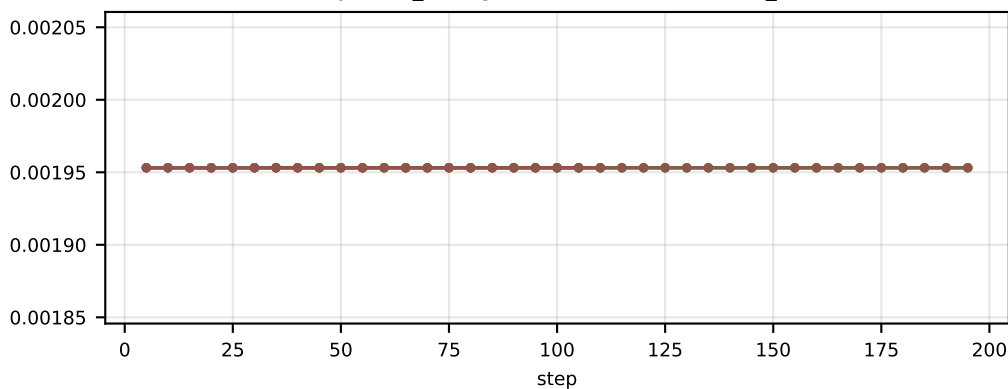
val/memory/actor_infer/generate/start/onload/allocated/0



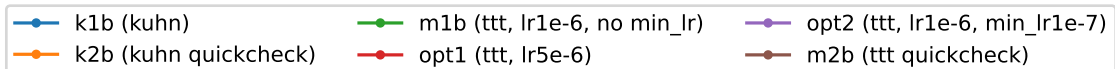
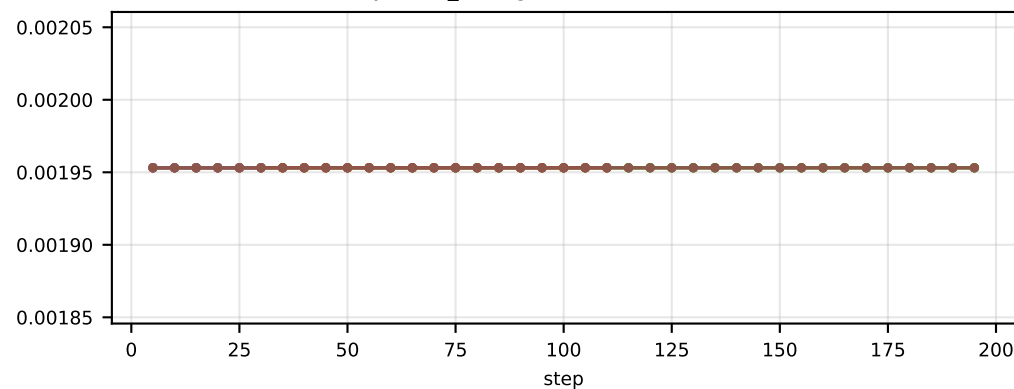
val/memory/actor_infer/generate/start/onload/max_allocated/0



val/memory/actor_infer/generate/start/onload/max_reserved/0

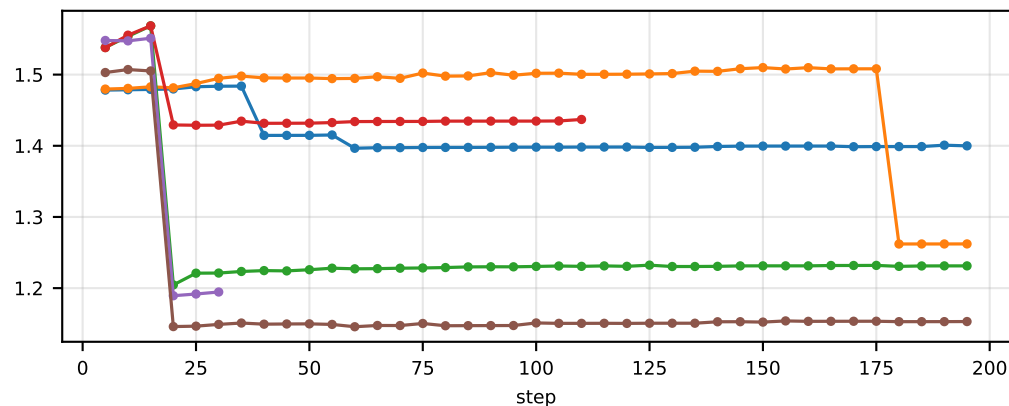


val/memory/actor_infer/generate/start/onload/reserved/0

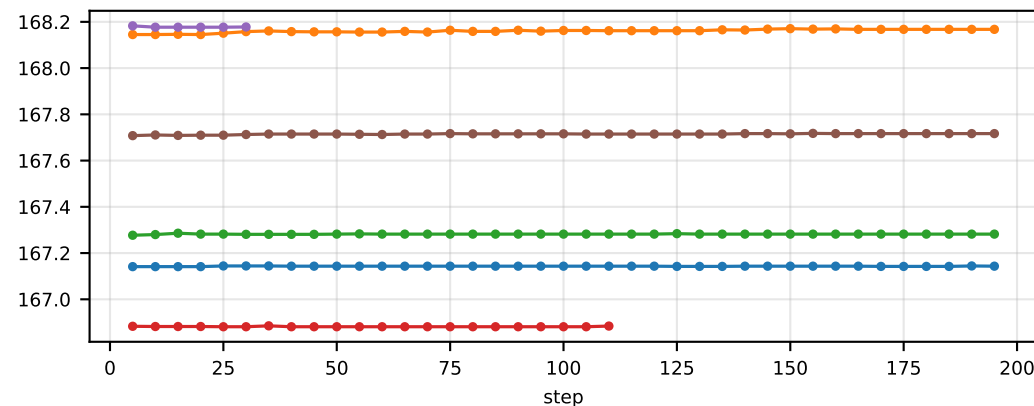


[Appendix] val/* (4/5)

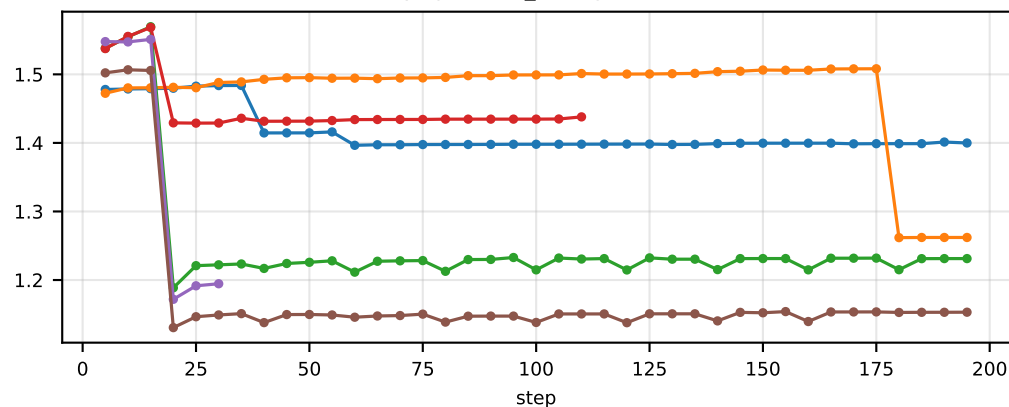
val/memory/cpu/actor_infer/generate/end/rss



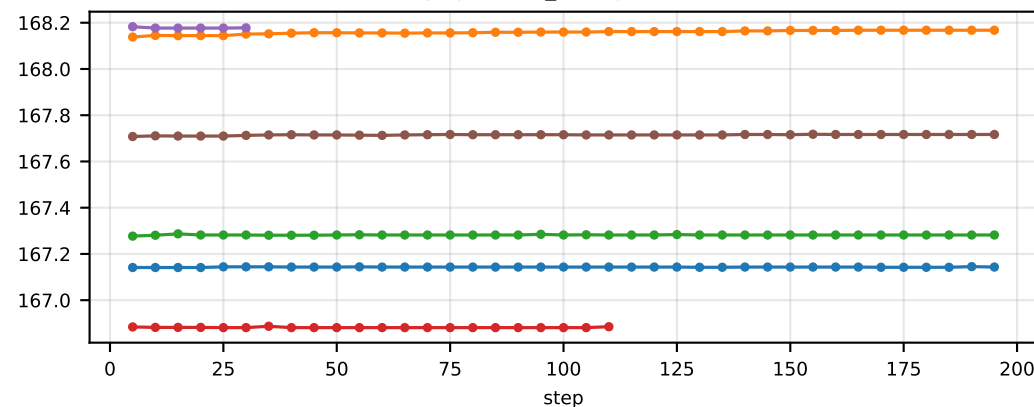
val/memory/cpu/actor_infer/generate/end/vms



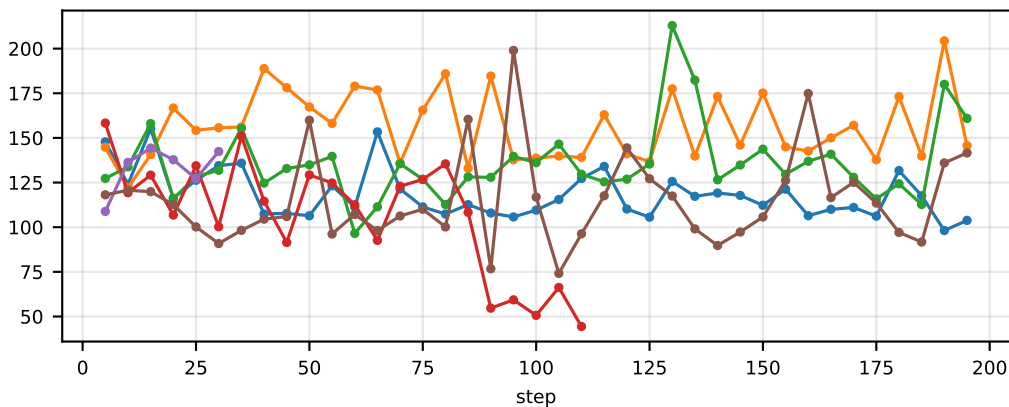
val/memory/cpu/actor_infer/generate/start/rss



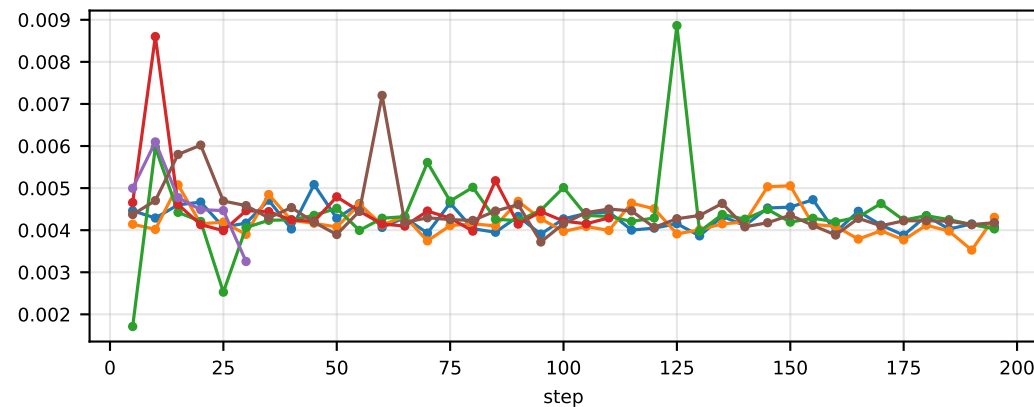
val/memory/cpu/actor_infer/generate/start/vms



val/time/actor_infer/generate/execute

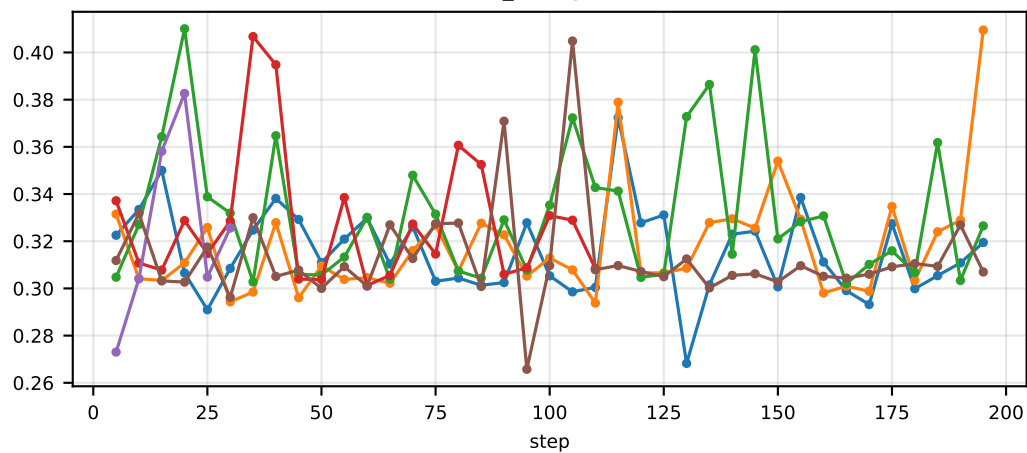


val/time/actor_infer/generate/offload



k1b (kuhn) m1b (ttt, lr1e-6, no min_lr) opt2 (ttt, lr1e-6, min_lr1e-7)
 k2b (kuhn quickcheck) opt1 (ttt, lr5e-6) m2b (ttt quickcheck)

val/time/actor_infer/generate/onload



val/time/actor_infer/generate/total

